Lie Groups and Their Representation Theory

Two parts of fourteen lectures

Part I: Global structure and compact representations

Part II: Noncompact representations, geometry and physics

Draft course notes
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Studying this course

This course has two parts of fourteen lectures. Each lecture is intended for about ninety minutes of study. The exercise sheet is separate. Allow another two to four hours for its core problems.

The prerequisites are *Linear Algebra for Physics*, *Finite Groups and Their Representations*, *Lie Algebras and Their Representations*, and basic category language. We use their algebraic results throughout. Ordinary multivariable calculus and elementary ordinary differential equations are also assumed.

Part I studies the global structure of Lie groups and the representations of compact groups. Part II studies selected noncompact groups and the geometry behind their representations. It includes the Heisenberg, Lorentz and Poincaré groups. The final lecture prepares for a later course on quantum groups.

Reading a lecture

Start with the objectives and outcomes. Work through the displayed calculations before reading the next paragraph. Then answer the self-checks without looking at their answers.

Attempt the core exercises before opening the worked solutions. Compare the reasoning as well as the final result. Some sheets include an optional extension. It is marked explicitly and is not needed for the next lecture.

The notes introduce the geometry and analysis needed at each stage. They do not replace courses on differential geometry, topology or functional analysis. Long existence theorems are stated with their hypotheses and a source. The main calculations are carried out here.

The course plan uses four labels. [P] means that the result is proved in the lecture. [S] means that the argument is sketched and the omitted step is identified. [C] means that a theorem is cited. [I] means that a result comes from a prerequisite course. A cited theorem is an input to the argument that follows it.

Conventions

Lie groups are finite-dimensional real Lie groups. Their manifolds are Hausdorff and second countable. Compactness and connectedness are stated when they are needed. Representations are continuous and act on complex vector spaces. A Hilbert-space unitary representation is strongly continuous. Inner products are antilinear in the first slot.

A real Lie algebra and its complexification are distinct objects. In particular, $\mathfrak{k} = \mathrm{Lie}(K)$ is real, while highest-weight theory applies to $\mathfrak{k}_{\mathbb{C}}$. For a torus with Lie algebra $\mathfrak{t}$, our period lattice is

$$\Lambda = \ker(\exp\colon \mathfrak{t} \to T).$$

We write a character as $\chi(\exp H) = e^{i\ell(H)}$. Thus its period condition is $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

The module V_m of $\mathrm{SU}(2)$ has dimension $m+1$ and spin $m/2$. The operator of a group representation is $\Pi(g)$, and its differential is $d\Pi(X)$. When a basis is chosen, D denotes an operator matrix. Every change of convention is explained where it is used.

Books used in the notes

The reading lists give a route into the books. The following abbreviations are used in some lectures.

H Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, Springer, 2015. This is the main route for matrix groups, compact groups and integration of representations.

BD Theodor Bröcker and Tammo tom Dieck, *Representations of Compact Lie Groups*, Springer, 1985. This supplies the general compact-group theory and reconstruction.

S Mark R. Sepanski, *Compact Lie Groups*, Springer, 2007. This is a second route through compact structure and Peter–Weyl theory.

Knapp Beyond Anthony W. Knapp, *Lie Groups Beyond an Introduction*, second edition. This supplies general Lie-group structure.

Knapp Representation Theory Anthony W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. This supplies noncompact examples, smooth vectors and induced representations.

Kirillov Jr. Alexander Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Cambridge University Press, 2008.

Kirillov Elements A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.

Kirillov Orbit Method A. A. Kirillov, *Lectures on the Orbit Method*, American Mathematical Society, 2004.

Warner Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Springer edition, 1983.

Further primary sources are named in the relevant lectures. Knapp's two books serve different purposes; their titles are distinguished throughout.

Part I

Global structure and compact representations

Lecture 1

Lie groups and their tangent algebras

Objectives. *A Lie group has both a group structure and a smooth geometry. We describe that geometry near the identity and recover its Lie algebra. The group* $\mathrm{SU}(2)$ *will give an example in which every step can be calculated.*

By the end of this lecture you will be able to:

- Identify the real dimension of a matrix Lie group.
- Find tangent spaces by differentiating a defining condition and exhibiting curves.
- Recover a matrix Lie bracket from left translations.
- Distinguish a real Lie algebra from its complexification.

The Lie algebras in the earlier course were complex. The tangent algebra of a real Lie group is a real Lie algebra. We will always state the field of scalars.

1.1 Smooth coordinates

A sphere cannot be described by one set of Euclidean coordinates without omitting a point or introducing a singularity. It can be described by several overlapping coordinate systems. A smooth manifold formalises this local description.

A topology specifies which subsets are open. For subsets of Euclidean space we use the usual relative topology. A *neighbourhood* of a point is a set containing an open set around that point. A topological space is *Hausdorff* if distinct points have disjoint open neighbourhoods. It is *second countable* if its topology has a countable *basis*, namely a countable collection of open sets such that every open set is a union of some of them.

A map between topological spaces is *continuous* if the inverse image of every open set is open. A *homeomorphism* is a continuous bijection whose inverse is continuous.

Definition 1.1 (Smooth manifold)**.** A smooth real manifold of dimension d is a topological space M covered by coordinate charts

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^d.$$

Each chart has open domain $U \subseteq M$ and is a homeomorphism onto an open set V. Where the domains of two charts φ and ψ overlap, the coordinate change $\psi \circ \varphi^{-1}$ is infinitely differentiable. We require M to be Hausdorff and second countable.

The Hausdorff and second-countability conditions exclude topological pathologies. All the matrix groups in this lecture satisfy them as subspaces of a Euclidean space. A smooth map between manifolds is a continuous map whose coordinate expressions are infinitely differentiable on their open domains.

Example 1.2 (A coordinate on the circle)**.** The unit circle is

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

On the open right semicircle, use y as the coordinate. Its inverse chart is

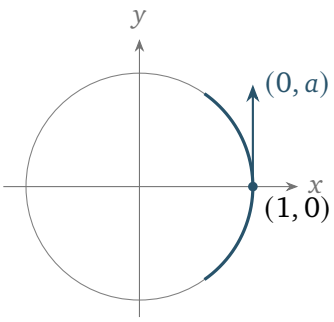
$$y \longmapsto \left(\sqrt{1 - y^2}, y\right), \qquad -1 < y < 1.$$

The other three semicircles have similar charts. The coordinate changes are smooth on their overlaps. Thus the circle is a one-dimensional real manifold, although it lies in a two-dimensional ambient space.

The coordinate y can also describe a velocity. For a real number a, the curve on the circle

$$\gamma_a(t) = \left(\sqrt{1 - a^2t^2}, at\right)$$

is defined near zero. It starts at $(1, 0)$ and has ambient derivative $(0, a)$. The derivative of its y-coordinate is a. Thus a one-dimensional tangent space can appear as a line inside a larger ambient space.



The coloured arc lies in the domain of one chart, in which y is the local coordinate. The tangent arrow is the velocity at $(1, 0)$ for positive a. Its drawn length is schematic.

An *embedded submanifold* has the relative topology and is locally a coordinate subspace in ambient smooth coordinates. We import its standard properties. Its inclusion is smooth with injective differential. Products of embedded submanifolds are embedded in the corresponding product. Thus smooth ambient maps restrict smoothly to these submanifolds. A smooth ambient map whose image lies in an embedded submanifold is smooth as a map into that submanifold. The differential of the inclusion identifies its tangent vectors with ambient velocities.

We also use one result from multivariable calculus. Let $U \subseteq \mathbb{R}^N$ be open, and suppose a smooth map $F\colon U \to \mathbb{R}^r$ has surjective differential at every point of $F^{-1}(c)$. The *regular level-set theorem* says that $F^{-1}(c)$ is an embedded submanifold of U of dimension $N - r$, with the relative topology. The tangent space of $F^{-1}(c)$ at a point p is $\ker dF_p$. The regular level-set theorem and the embedded-submanifold properties are cited from Warner, Chapter 1, sections “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem” and “Implicit Function Theorems” (Theorem 1.38).

1.2 Smooth groups

Product charts $\varphi \times \psi$ on $U \times V$ give $M \times N$ a smooth manifold structure of dimension $\dim M + \dim N$. A finite product of Hausdorff, second-countable spaces has both properties.

Definition 1.3 (Lie group)**.** A real Lie group is a group G that is also a smooth real manifold, with smooth multiplication and inversion:

$$G \times G \longrightarrow G, \quad (g,h) \longmapsto gh, \qquad G \longrightarrow G, \quad g \longmapsto g^{-1}.$$

A Lie group homomorphism is a smooth group homomorphism.

The additive group $\mathbb{R}^d$ is a Lie group. Any finite group with the discrete topology is also a Lie group, since each one-point subset is the domain of a chart modelled on $\mathbb{R}^0$. The tangent space of a finite group therefore has dimension zero. Hence a Lie algebra cannot distinguish finite groups.

Example 1.4 (General linear groups)**.** The group of invertible real matrices is

$$\mathrm{GL}_n(\mathbb{R}) = \{g \in M_{n\times n}(\mathbb{R}) : \det g \neq 0\}.$$

The determinant is continuous, so $\mathrm{GL}_n(\mathbb{R})$ is open in $\mathbb{R}^{n^2}$. Hence its dimension is n^2. The entries of a matrix product are polynomials in the entries of the factors, so matrix multiplication is smooth. Inversion is smooth because

$$g^{-1} = \frac{\operatorname{adj}(g)}{\det g}.$$

The same argument applies to $\mathrm{GL}_n(\mathbb{C})$. The real and imaginary parts of the entries give its coordinates, so its dimension as a real manifold is $2n^2$.

A *complex Lie group* has complex charts and holomorphic group operations. The group $\mathrm{GL}_n(\mathbb{C})$ has this extra structure. Its complex dimension is n^2. A group of complex matrices need not be a complex Lie group. Its defining conditions may involve complex conjugation.

For example, the unitary and special unitary groups are

$$\mathrm{U}(n) = \{U \in M_{n\times n}(\mathbb{C}) : U^*U = I\}, \qquad \mathrm{SU}(n) = \{U \in \mathrm{U}(n) : \det U = 1\}.$$

Here U^* denotes conjugate transpose. Their real Lie-group structures are a standard classical-group result, cited from Bröcker–tom Dieck, Chapter I, Section 1. We verify the manifold structure of $\mathrm{SU}(2)$ directly; Exercise Sheet 1 treats $\mathrm{U}(n)$.

A manifold is *connected* if it cannot be written as the union of two nonempty disjoint open subsets. For example, the orthogonal group is

$$\mathrm{O}(n) = \{g \in M_{n\times n}(\mathbb{R}) : g^{\mathsf{T}}g = I\}, \qquad n \geq 1.$$

Its determinant takes values in $\{1,-1\}$. Both values occur. The identity has determinant one, and $\operatorname{diag}(-1,1,\ldots,1)$ has determinant minus one. The inverse images of these two values are disjoint nonempty open subsets that cover the group. Thus $\mathrm{O}(n)$ is disconnected. Its subgroup with determinant one is $\mathrm{SO}(n)$.

1.3 Tangent vectors at the identity

Definition 1.5 (Tangent vector). Let $\varphi : U \to V$ be a chart of M with $p \in U$. A *smooth curve through* p is a smooth map γ from an open interval containing 0 to M with $\gamma(0) = p$. Two such curves γ and δ are *equivalent* when $(\varphi \circ \gamma)'(0) = (\varphi \circ \delta)'(0)$. A *tangent vector* at p is an equivalence class $[\gamma]$. The set of tangent vectors at p is denoted T_pM. The map $[\gamma] \mapsto (\varphi \circ \gamma)'(0)$ identifies T_pM with $\mathbb{R}^d$. Vector addition and real scalar multiplication on T_pM are transferred through this identification.

The class $[\gamma]$ is the *velocity* of γ, also written $\gamma'(0)$. The chain rule makes the relation between curves independent of the chart. The derivative of a coordinate change is linear, so the vector-space structure on T_pM is also independent of the chart.

A smooth map $f : M \to N$ sends tangent vectors to tangent vectors. If v is represented by γ, the image of v is the velocity of $f \circ \gamma$. This defines the *differential*

$$df_p : T_pM \longrightarrow T_{f(p)}N, \qquad df_p\big(\gamma'(0)\big) = (f \circ \gamma)'(0).$$

In charts this is the usual derivative, so it is real linear. The chain rule gives $d(f \circ h)_p = df_{h(p)} \circ dh_p$.

The kernel clause of the regular level-set theorem now follows from the chain rule and rank–nullity. Put $P = F^{-1}(c)$ and let $i : P \to U$ be the inclusion. For $p \in P$, since $F \circ i$ is constant, $dF_p \circ di_p = 0$. Thus $di_p(T_pP) \subseteq \ker dF_p$. The inclusion is an embedding, so di_p is injective. Both spaces have dimension $N - r$, because dF_p is surjective. Hence they are equal.

For an embedded manifold in a matrix space, we compute velocities by differentiating the matrix entries of curves. The identity element of a group is denoted e, or I for a matrix group. For a Lie group G, the vector space of interest is

$$\mathfrak{g} = T_eG.$$

For $\mathrm{GL}_n(\mathbb{R})$, every matrix X is a tangent vector at I. Indeed, the curve $\gamma(t) = I + tX$ is invertible for sufficiently small t, and $\gamma'(0) = X$.

Example 1.6 (Tangent vectors to the orthogonal group). Let $g(t)$ be a smooth curve in $\mathrm{O}(n)$ with $g(0) = I$. Put $X = g'(0)$. Differentiate the defining identity at zero by the product rule:

$$g(t)^{\mathsf{T}} g(t) = I \quad \Longrightarrow \quad X^{\mathsf{T}} + X = 0.$$

Thus every tangent vector at I is skew-symmetric. Differentiating a condition gives a necessary condition. We must also show that every such matrix occurs.

For a skew-symmetric matrix X, consider the Cayley curve

$$g_X(t) = \left(I + \frac{tX}{2}\right)\left(I - \frac{tX}{2}\right)^{-1}.$$

It is defined near zero. Set $A = I + tX/2$ and $B = I - tX/2$. Then $A^{\mathsf{T}} = B$, $B^{\mathsf{T}} = A$, and $AB = BA$. Therefore

$$g_X(t)^{\mathsf{T}} g_X(t) = A^{-1}BAB^{-1} = I.$$

The determinant is continuous, takes values in $\{1, -1\}$, and equals one at zero. It remains one near zero. Hence this curve lies in $\mathrm{SO}(n)$. The product rule gives

$$g_X'(0) = \frac{X}{2} + \frac{X}{2} = X.$$

Consequently both orthogonal groups have the same tangent space at I:

$$T_I \,\mathrm{SO}(n) = T_I \,\mathrm{O}(n) = \{X : X^{\mathsf{T}} = -X\}.$$

The example used that $\mathrm{O}(n)$ is a smooth manifold. To justify this assertion, apply the regular level-set theorem to $F(g) = g^{\mathsf{T}} g$, with target the real symmetric matrices. At an orthogonal matrix g,

$$dF_g(H) = H^{\mathsf{T}} g + g^{\mathsf{T}} H.$$

For any symmetric matrix S, choose $H = gS/2$. Since S is symmetric and g is orthogonal, $dF_g(H) = S$. Hence the surjectivity hypothesis holds. Subtracting the dimension of the space of real symmetric matrices from the ambient dimension gives

$$\dim_{\mathbb{R}} \mathrm{O}(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

The subgroup $\mathrm{SO}(n)$ is open in $\mathrm{O}(n)$, so it has the same local dimension.

Multiplication and inversion by transposition are restrictions of smooth matrix maps. By the embedded-submanifold properties, both orthogonal groups are Lie groups. Equal tangent spaces do not imply equal groups.

1.4 Translations and the bracket

Left multiplication by $g \in G$ is the smooth map

$$L_g : G \longrightarrow G, \qquad L_g(h) = gh.$$

Since L_g has inverse $L_{g^{-1}}$, the differential of L_g at e identifies tangent spaces:

$$(dL_g)_e : T_eG \longrightarrow T_gG.$$

For a matrix group this differential sends X to gX. For example, in $\mathrm{SO}(2)$ take the rotation and the tangent matrix

$$g = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Left translation gives the tangent vector at g:

$$gX = \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}.$$

This is the derivative of the rotation matrix with angle $\theta + t$ at $t = 0$.

A *vector field* assigns a tangent vector to each point. It is smooth if its components in each coordinate chart are smooth functions. The field determined by $X \in T_eG$ is

$$X^L(g) = (dL_g)_e X.$$

A vector field V on G is *left-invariant* if

$$(dL_h)_g\big(V(g)\big) = V(hg) \qquad \text{for all } g, h \in G.$$

The field X^L is left-invariant. Indeed, $L_h \circ L_g = L_{hg}$, so the chain rule gives $(dL_h)_g\big((dL_g)_e X\big) = (dL_{hg})_e X$.

Smooth vector fields act on smooth functions by directional differentiation. For a smooth vector field V and a smooth function f,

$$(Vf)(p) = df_p(V(p)).$$

Their commutator as differentiation operators is

$$[V, W]f = V(Wf) - W(Vf).$$

Theorem 1.7 (The tangent Lie algebra; cited)**.** *The fields X^L obtained by left translation are smooth. The commutator of two smooth vector fields is again a vector field. The commutator of left-invariant fields is left-invariant. The value of this commutator at the identity defines a real Lie bracket on T_eG. For smooth vector fields V and W, in coordinates,*

$$[V, W](p) = DW_p(V(p)) - DV_p(W(p)).$$

For an embedded submanifold, this formula may be evaluated using smooth ambient extensions of tangent fields. The result is tangent to the submanifold.

The construction of the bracket from left-invariant fields is cited from Bröcker–tom Dieck, Chapter I, Section 2. The coordinate formula and the statement for embedded submanifolds are cited from Warner, Chapter 1, sections “Vector Fields” and “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem”. The Jacobi identity is the commutator identity for the corresponding differentiation operators.

For an embedded matrix group, use the ambient extensions $X^L(g) = gX$ and $Y^L(g) = gY$. Since $Y^L(h) = hY$ is linear in h, its derivative in direction H is HY. Thus ambient calculus gives

$$D(Y^L)_g(gX) = gXY,$$
$$D(X^L)_g(gY) = gYX.$$

By the coordinate formula in the cited theorem, the vector-field bracket is the first of these derivatives minus the second. At the identity, this gives

$$[X, Y] = XY - YX.$$

This is the bracket studied in the earlier course. The cited submanifold compatibility shows that this ambient calculation is the intrinsic bracket on G. Hence $XY - YX$ belongs to T_IG. For skew-symmetric matrices, closure can also be seen directly:

$$(XY - YX)^{\mathsf{T}} = YX - XY = -(XY - YX).$$

The real tangent Lie algebra of $\mathrm{SO}(n)$ is denoted $\mathfrak{so}(n)$. For $\mathrm{GL}_n(\mathbb{C})$, viewed as a real group, the tangent algebra is all complex matrices with real scalars and the same commutator.

1.5 The group SU(2) and its real algebra

The two rows of a matrix in SU(2) are orthonormal. Write the first row as (α, β). The unit vectors orthogonal to this row are the multiples of $(-\overline{\beta}, \overline{\alpha})$ by complex numbers of modulus one. The determinant then equals that multiple, so determinant one forces it to be one. Hence every element of SU(2) has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Conversely, a matrix of the displayed form satisfies

$$\begin{pmatrix} \overline{\alpha} & -\beta \\ \overline{\beta} & \alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & 0 \\ 0 & |\alpha|^2 + |\beta|^2 \end{pmatrix}.$$

The matrix U has determinant $|\alpha|^2 + |\beta|^2$. Thus this formula identifies SU(2) with the unit sphere S^3 in $\mathbb{C}^2 = \mathbb{R}^4$. The regular level-set theorem gives real dimension three. The displayed map from (α, β) to U is a real-linear injection from $\mathbb{R}^4$ into $M_{2\times 2}(\mathbb{C})$. It is a linear coordinate isomorphism onto its image, so it carries the embedded sphere to an embedded submanifold of the full matrix space. Multiplication and inversion are smooth restrictions of the matrix operations.

Proposition 1.8 (The tangent algebra of SU(2)). *The tangent Lie algebra of* SU(2) *is the real space*

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} : x, y, z \in \mathbb{R} \right\},$$

with the matrix commutator.

Proof. First we prove the necessary conditions. Let $U(t)$ be a smooth curve in SU(2) with $U(0) = I$, and put $X = U'(0)$. Differentiating $U(t)^* U(t) = I$ at zero gives $X^* = -X$.

The determinant condition requires a separate derivative. Write $X = (x_{jk})$. To first order in t,

$$U(t) = \begin{pmatrix} 1 + tx_{11} + o(t) & tx_{12} + o(t) \\ tx_{21} + o(t) & 1 + tx_{22} + o(t) \end{pmatrix}.$$

Its determinant has expansion

$$\det U(t) = 1 + t(x_{11} + x_{22}) + o(t).$$

Here $o(t)$ denotes a quantity whose ratio to t tends to zero. The products of two off-diagonal entries have order t^2, so they do not contribute to the derivative. Differentiating $\det U(t) = 1$ now gives $\operatorname{tr} X = 0$.

A skew-Hermitian matrix has purely imaginary diagonal entries. Its entry below the diagonal is minus the conjugate of the entry above it. Since the trace is zero, the diagonal entries are ix and $-ix$ for some $x \in \mathbb{R}$. Writing the entry above the diagonal as $y + iz$ gives the displayed form.

Next we prove the converse. Every matrix of the displayed form is a real linear combination of

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Let $X = xA + yB + zC$ and $r^2 = x^2 + y^2 + z^2$. Direct multiplication gives

$$X^2 = -r^2 I, \qquad \det(I + tX) = 1 + t^2 r^2.$$

These identities provide a curve with the required velocity:

$$U_X(t) = \frac{I + tX}{\sqrt{1 + t^2 r^2}}.$$

The scalar factor appears squared in a 2×2 determinant. Hence the determinant of $U_X(t)$ is one. Since $X^* = -X$ and $X^2 = -r^2 I$,

$$U_X(t)^* U_X(t) = \frac{(I - tX)(I + tX)}{1 + t^2 r^2} = I.$$

Also $U_X(0) = I$ and $U_X'(0) = X$. Hence every matrix of the displayed form is tangent to $\mathrm{SU}(2)$ at I.

Finally, $\mathrm{SU}(2)$ is an embedded matrix group. By the calculation with ambient extensions in the previous section, its bracket is the matrix commutator. ∎

The multiplication of the basis matrices starts with $AB = C$ and $BA = -C$. The next products confirm a cyclic pattern:

$$BC = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = A, \qquad CB = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -A.$$

Similarly, $CA = B$ and $AC = -B$. The bracket table is

$$[A, B] = 2C, \qquad [B, C] = 2A, \qquad [C, A] = 2B.$$

The scalars in this basis expansion are real. Multiplying A by i gives a Hermitian matrix, which is not in $\mathfrak{su}(2)$. The fact that its elements have complex entries does not make $\mathfrak{su}(2)$ a complex vector space.

We now relate this real algebra to the earlier complex algebra $\mathfrak{sl}_2(\mathbb{C})$.

Definition 1.9 (Complexification). The complexification of a real Lie algebra $\mathfrak{k}$ is the complex vector space

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C},$$

with the complex-bilinear extension of its bracket. Thus $[X \otimes a, Y \otimes b] = [X, Y] \otimes ab$.

Let $\iota \colon \mathfrak{su}(2)_{\mathbb{C}} \to \mathfrak{sl}_2(\mathbb{C})$ be the complex-linear map given on pure tensors by $\iota(X \otimes a) = aX$. It extends the inclusion of $\mathfrak{su}(2)$ into complex matrices. Its values are traceless because the elements of $\mathfrak{su}(2)$ are traceless. The images of A, B, C span all traceless complex matrices, since

$$h = -iA, \qquad e = \frac{B - iC}{2}, \qquad f = -\frac{B + iC}{2}$$

give the three matrices

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

They are the usual basis matrices of $\mathfrak{sl}_2(\mathbb{C})$. These three matrices are complex linearly independent. The map ι preserves brackets by complex bilinearity. Its domain and image both have complex dimension three, so ι is a complex Lie algebra isomorphism:

$$\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C}).$$

The real algebra has dimension three over $\mathbb{R}$. Its complexification has dimension three over $\mathbb{C}$. The real algebra underlying $\mathfrak{sl}_2(\mathbb{C})$ has dimension six over $\mathbb{R}$; it is a different object.

Self-checks

1. Why is $\mathrm{GL}_n(\mathbb{C})$ open in $\mathbb{C}^{n^2}$ but $\mathrm{U}(n)$ is not?
2. Does differentiating $g(t)^{\mathsf{T}} g(t) = I$ prove that every skew-symmetric matrix is tangent to $\mathrm{SO}(n)$?
3. Is $\mathrm{diag}(1, -1)$ in $\mathfrak{su}(2)$? Is it in the complex span of $\mathfrak{su}(2)$?

Answers. (1) The determinant is continuous, so the set where it is nonzero is open in $\mathbb{C}^{n^2}$. For real ε with $0 < |\varepsilon| < 1$, a unitary matrix U satisfies

$$((1+\varepsilon)U)^*((1+\varepsilon)U) = (1+\varepsilon)^2 I \neq I.$$

These non-unitary matrices approach U, so no ambient ball about U lies in $\mathrm{U}(n)$. (2) No. Differentiation proves necessity. The Cayley curve proves sufficiency. (3) No. The matrix $\mathrm{diag}(1, -1)$ is traceless but Hermitian, so it is not skew-Hermitian. It does lie in the complex span, since it equals $-iA$.

Summary and reading

A Lie group has smooth multiplication and inversion. Its tangent space at the identity is a real Lie algebra. Left translation determines the bracket. For embedded matrix groups, the bracket is the matrix commutator. The algebra records local information. Distinct groups can have the same tangent algebra.

Read Hall, second edition, Sections 1.1–1.5 and 3.3–3.6, and Bröcker–tom Dieck, Chapter I, Sections 1–2. The regular level-set theorem and the embedded-submanifold properties are the geometric imports, cited from Warner, Chapter 1. Lecture 2 develops the exponential map and connects these tangent vectors to one-parameter subgroups.

Lecture 2

Exponential maps and local group structure

Objectives. *A tangent vector at the identity generates a one-parameter subgroup. The exponential map collects these subgroups into one map from the Lie algebra to the group. We ask how much of the multiplication is determined by the bracket. Near the identity, the bracket determines the whole multiplication. Far from the identity, it does not.*

By the end of this lecture you will be able to:

- Compute exponentials of rotation and boost generators and of elements of $\mathfrak{sl}_2(\mathbb{R})$.
- Identify continuous one-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$ with exponentials.
- Differentiate a homomorphism and use $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$.
- Compute the second-order BCH term and recognise elements that are not exponentials.

2.1 The matrix exponential

For a complex $n \times n$ matrix X, let $\|X\|$ be the operator norm for the standard Hermitian norm on $\mathbb{C}^n$. The operator norm satisfies $\|XY\| \le \|X\|\,\|Y\|$.

Definition 2.1 (Matrix exponential)**.** For $X \in M_{n\times n}(\mathbb{C})$, the exponential of X is

$$\exp X = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

Since $\|X^k\| \le \|X\|^k$, the series is dominated by the scalar series of $e^{\|X\|}$. Hence the series converges absolutely, and uniformly on bounded sets. Each entry of $\exp X$ is an everywhere convergent power series in the entries of X. So exp is smooth, and its series may be differentiated term by term.

Proposition 2.2 (Properties of the matrix exponential)**.** *Let $X, Y \in M_{n\times n}(\mathbb{C})$ and $g \in \mathrm{GL}_n(\mathbb{C})$.*

(i) *If $XY = YX$, then $\exp(X+Y) = \exp X \exp Y$.*

(ii) *The matrix $\exp X$ is invertible, with inverse $\exp(-X)$.*

(iii) *The curve $t \mapsto \exp(tX)$ has derivative $X\exp(tX) = \exp(tX)X$.*

(iv) $g\exp(X)g^{-1} = \exp(gXg^{-1})$.

(v) $\det\exp X = e^{\operatorname{tr} X}$.

Proof. (i) Absolute convergence lets us multiply the series and collect terms of total degree m. The binomial theorem holds because $XY = YX$:

$$\exp X \exp Y = \sum_{m=0}^{\infty} \sum_{k+l=m} \frac{X^k Y^l}{k!\,l!} = \sum_{m=0}^{\infty} \frac{(X+Y)^m}{m!}.$$

(ii) Apply the product rule (i) to the commuting pair X and $-X$. (iii) Differentiate $\sum_k t^k X^k/k!$ term by term and factor out X on either side. (iv) We have $(gXg^{-1})^k = gX^k g^{-1}$. Conjugation by g is linear and continuous, so it commutes with the sum of the series. (v) Schur triangularisation gives $X = PTP^{-1}$ with T upper triangular. Let t_{jj} be the diagonal entries of T. Then $\exp T$ is upper triangular with diagonal entries $e^{t_{jj}}$. By the conjugation formula (iv),

$$\det \exp X = \det \exp T = \prod_{j} e^{t_{jj}} = e^{\operatorname{tr} T} = e^{\operatorname{tr} X}. \qquad \blacksquare$$

The hypothesis $XY = YX$ in the product rule, Proposition 2.2(i), cannot be dropped. Exercise 2.3 gives nilpotent matrices E and F with $\exp E \exp F \neq \exp(E+F)$. The Baker–Campbell–Hausdorff formula in § 2.5 shows that the bracket governs the correction.

2.2 One-parameter subgroups

Definition 2.3 (One-parameter subgroup)**.** A one-parameter subgroup of a Lie group G is a continuous homomorphism $\varphi\colon \mathbb{R} \to G$ from the additive group of real numbers.

For every matrix X, the curve $t \mapsto \exp(tX)$ is a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$, by the product rule, Proposition 2.2(i). Conversely, continuity alone forces this form.

Theorem 2.4 (One-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$)**.** *Let $\varphi\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{C})$ be a one-parameter subgroup. Then φ is differentiable, and $\varphi(t) = \exp(tX)$ for the unique matrix $X = \varphi'(0)$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then X is a real matrix.*

Proof. Step 1: differentiability. By continuity of φ, choose $\varepsilon > 0$ with $\|\varphi(s) - I\| \le \tfrac{1}{2}$ for $0 \le s \le \varepsilon$. Integrate entrywise and set

$$M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(s)\, ds.$$

Then $\|M - I\| \le \tfrac{1}{2}$, so the Neumann series $\sum_{k\ge 0} (I-M)^k$ inverts M. The homomorphism property gives

$$\varphi(t) M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(t+s)\, ds = \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \varphi(u)\, du.$$

The right side is continuously differentiable in t, by the fundamental theorem of calculus. Multiplication by the constant matrix M^{-1} preserves this property, so φ is continuously differentiable too.

Step 2: the equation. Put $X = \varphi'(0)$. Differentiate $\varphi(t+s) = \varphi(t)\varphi(s)$ in s at $s = 0$. This gives $\varphi'(t) = \varphi(t) X$.

Step 3: uniqueness. Let $\psi(t) = \varphi(t)\exp(-tX)$. By the Leibniz rule and the derivative formula, Proposition 2.2(iii),

$$\psi'(t) = \varphi(t)X\exp(-tX) - \varphi(t)X\exp(-tX) = 0.$$

Hence $\psi(t) = \psi(0) = I$, and $\varphi(t) = \exp(tX)$. Any Y with $\varphi(t) = \exp(tY)$ satisfies $Y = \varphi'(0) = X$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then its derivative $X = \varphi'(0)$ is a real matrix, as a limit of real difference quotients. ∎

Example 2.5 (Rotations)**.** The algebra $\mathfrak{so}(2)$ is spanned by

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Separating even and odd powers in the series gives

$$\exp(\theta J) = \cos\theta\, I + \sin\theta\, J = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The first column of an element of $\mathrm{SO}(2)$ is a unit vector, so it is $(\cos\theta, \sin\theta)$ for some θ. Orthogonality and determinant one then fix the second column. So $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ is onto. It is not injective, since $\exp(2\pi J) = I$.

Example 2.6 (Boosts)**.** In special relativity on a line, with the speed of light set to 1, an event has a time coordinate x_0 and a position x_1. Observers in uniform relative motion use coordinates related by linear maps preserving $x_0^2 - x_1^2$. With $\eta = \mathrm{diag}(1, -1)$, these maps form the group

$$\mathrm{O}(1,1) = \{g \in M_{2\times 2}(\mathbb{R}) : g^{\mathsf{T}}\eta g = \eta\}.$$

Let $g(t)$ be a smooth curve in $\mathrm{O}(1,1)$ with $g(0) = I$ and $g'(0) = X$. Differentiating $g(t)^{\mathsf{T}}\eta g(t) = \eta$ at $t = 0$ gives the necessary condition $X^{\mathsf{T}}\eta + \eta X = 0$. In terms of the entries of X,

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad X^{\mathsf{T}}\eta + \eta X = \begin{pmatrix} 2a & b - c \\ b - c & -2d \end{pmatrix}.$$

The condition therefore forces $a = d = 0$ and $b = c$. So X is a multiple of

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

The even and odd parts of the series now give hyperbolic functions:

$$B(u) = \exp(uK) = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix}.$$

Since $\cosh^2 u - \sinh^2 u = 1$, we get $B(u)^{\mathsf{T}}\eta B(u) = \eta$. So K is the velocity of a curve in the group, and the tangent algebra is exactly $\mathbb{R}K$.

The parameter u is the rapidity. The boost $B(u)$ sends the events $(\tau, 0)$ of an observer at rest to $(\tau\cosh u, \tau\sinh u)$, a motion with velocity $\tanh u$. Rapidities add, since $B(u)B(u') = B(u+u')$. Velocities do not add. Unlike the parametrisation of rotations, the map $u \mapsto B(u)$ is injective. Every boost has trace $2\cosh u \geq 2$.

2.3 The exponential map of a Lie group

On a general Lie group, the exponential power series is not defined. Instead we use the left-invariant vector fields $X^L(g) = (dL_g)_e X$ of Lecture 1, with $X \in \mathfrak{g} = T_eG$.

Definition 2.7 (Integral curve). An integral curve of a vector field V on a manifold M is a smooth curve $\gamma : I \to M$, on an open interval I, with $\gamma'(t) = V(\gamma(t))$ for all $t \in I$.

In a chart, this condition is a first-order system of ODEs with smooth coefficients. The Picard–Lindelöf theorem of the ODE background gives local existence and uniqueness. Two integral curves with the same initial point agree on their common interval. The agreement set is open by local uniqueness, closed because M is Hausdorff, and nonempty. An interval is connected, so it is the whole interval.

Left translation maps integral curves of X^L to integral curves of X^L. For such a curve γ and $g \in G$, the chain rule and $L_g \circ L_h = L_{gh}$ give

$$(L_g \circ \gamma)'(t) = (dL_g)_{\gamma(t)}(dL_{\gamma(t)})_e X = (dL_{g\gamma(t)})_e X = X^L\big(g\gamma(t)\big).$$

Theorem 2.8 (Exponential map of a Lie group; cited). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $X \in \mathfrak{g}$. The integral curve γ_X of X^L with $\gamma_X(0) = e$ is defined on all of $\mathbb{R}$. The map*

$$\exp : \mathfrak{g} \longrightarrow G, \qquad \exp X = \gamma_X(1),$$

is smooth, and $\gamma_X(t) = \exp(tX)$ for all real t.

This is proved in Kirillov Jr., Chapter 3, in the section on the exponential map, and in Warner, Chapter 3. We sketch the algebraic part. The omitted, cited step is that γ_X exists for all t and that $\gamma_X(1)$ depends smoothly on X.

For fixed s, the curves $t \mapsto \gamma_X(s+t)$ and $t \mapsto \gamma_X(s)\gamma_X(t)$ are integral curves of X^L starting at $\gamma_X(s)$. By uniqueness,

$$\gamma_X(s+t) = \gamma_X(s)\gamma_X(t).$$

For a real number c, the curve $t \mapsto \gamma_X(ct)$ is the integral curve of $(cX)^L$ through e. So $\gamma_{cX}(t) = \gamma_X(ct)$. At $t = 1$ this gives $\exp(cX) = \gamma_X(c)$.

Let $G \subseteq \mathrm{GL}_n(\mathbb{C})$ be an embedded matrix Lie group, as in Lecture 1. Left translation is matrix multiplication, so γ_X solves $\gamma' = \gamma X$ with $\gamma(0) = I$ in the matrix space. Step 3 of the proof of the one-parameter subgroup theorem, Theorem 2.4, shows that $\exp(tX)$ is the only solution. Hence the two exponentials agree, and $\exp(tX) \in G$ for every $X \in \mathfrak{g}$.

Lemma 2.9 (Smooth one-parameter subgroups). *Let $\beta : \mathbb{R} \to G$ be a smooth homomorphism with $\beta'(0) = X$. Then $\beta(t) = \exp(tX)$ for all t.*

Proof. Differentiate $\beta(t+s) = L_{\beta(t)}\beta(s)$ in s at $s = 0$. This gives $\beta'(t) = X^L(\beta(t))$. A homomorphism sends the identity to the identity, so $\beta(0) = e$. Hence β and γ_X are both integral curves of X^L through e. By uniqueness of integral curves, they agree. ∎

Proposition 2.10 (Local invertibility). *The differential of* exp *at* 0 *is the identity of* $\mathfrak{g}$. *Consequently there are open sets $U \ni 0$ in $\mathfrak{g}$ and $V \ni e$ in G such that* exp *restricts to a diffeomorphism $U \to V$. We may take $U = -U$.*

Proof. Identify $T_0\mathfrak{g}$ with $\mathfrak{g}$. The line $t \mapsto tX$ has velocity X. Its image under exp is γ_X, whose velocity at 0 is $X^L(e) = X$. Hence $d\exp_0(X) = X$.

Read exp in a chart around e. The inverse function theorem of multivariable calculus, which we cite, says the following. A smooth map between open subsets of $\mathbb{R}^d$ with invertible derivative at a point restricts to a diffeomorphism between open neighbourhoods of that point and its image. This gives U and V. Replace U by $U \cap (-U)$, and replace V by the image of $U \cap (-U)$ under exp. Both sets remain open. ∎

Write $\log\colon V \to U$ for the inverse. Local invertibility, Proposition 2.10, is a local statement. Indeed, the rotation example, Example 2.5, shows that exp need not be injective on all of $\mathfrak{g}$.

Proposition 2.11 (Generation by exponentials)**.** *If G is connected, every element of G is a finite product $\exp X_1 \exp X_2 \cdots \exp X_k$.*

Proof. Let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. Since $\exp(-X) = (\exp X)^{-1}$ and $U = -U$, we have $V^{-1} = V$. Let H be the set of finite products of elements of V. It is a subgroup. It is open, since $H = \bigcup_{h \in H} hV$. Its complement is a union of cosets gH, so it is open too. As $e \in H$ and G is connected, $H = G$. ∎

The proof uses only that V is open, contains e and satisfies $V^{-1} = V$.

2.4 Conjugation and homomorphisms

Conjugation $C_g(h) = ghg^{-1}$ is a smooth automorphism of G fixing e. The adjoint action of g is the linear map $\mathrm{Ad}_g = (dC_g)_e\colon \mathfrak{g} \to \mathfrak{g}$. For a matrix group, differentiating $g\exp(tX)g^{-1}$ at $t = 0$ gives $\mathrm{Ad}_g X = gXg^{-1}$. Hence $gXg^{-1} \in \mathfrak{g}$.

Theorem 2.12 (Differentiating homomorphisms)**.** *Let $\Phi\colon G \to H$ be a Lie group homomorphism and $\phi = d\Phi_e$.*

(i) $\Phi(\exp X) = \exp(\phi(X))$ *for all* $X \in \mathfrak{g}$.

(ii) $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ *for all* $g \in G$ *and* $X \in \mathfrak{g}$.

(iii) If G is connected, then Φ is determined by ϕ.

(iv) If G and H are matrix Lie groups, then $\phi([X,Y]) = [\phi(X), \phi(Y)]$.

Proof. (i) The curve $\beta(t) = \Phi(\exp tX)$ is a smooth homomorphism $\mathbb{R} \to H$ with velocity $\phi(X)$ at 0. Apply the lemma on smooth one-parameter subgroups, Lemma 2.9, in H and set $t = 1$.

(ii) Apply the intertwining formula (i) to $\Phi = C_g$, whose differential is Ad_g.

(iii) By generation by exponentials, Proposition 2.11, each element of G is a product $\exp X_1 \cdots \exp X_k$. By the intertwining formula (i), the image of this product under Φ is $\exp\phi(X_1) \cdots \exp\phi(X_k)$, which depends only on ϕ.

(iv) For $g \in G$ and $Y \in \mathfrak{g}$, the matrix $gYg^{-1} \in \mathfrak{g}$ is the velocity of $g\exp(tY)g^{-1}$ at $t = 0$. Differentiate $\Phi(g\exp(tY)g^{-1}) = \Phi(g)\Phi(\exp tY)\Phi(g)^{-1}$ at $t = 0$:

$$\phi(gYg^{-1}) = \Phi(g)\,\phi(Y)\,\Phi(g)^{-1}.$$

Put $g = \exp(sX)$ and use the intertwining formula (i) on the right:

$$\phi\big(e^{sX} Y e^{-sX}\big) = e^{s\phi(X)} \phi(Y) e^{-s\phi(X)}.$$

Differentiate at $s = 0$, using the derivative formula, Proposition 2.2(iii). Since ϕ is linear, it commutes with differentiation. The result is $\phi(XY - YX) = \phi(X)\phi(Y) - \phi(Y)\phi(X)$. This is the bracket identity $\phi([X,Y]) = [\phi(X), \phi(Y)]$. ∎

The bracket identity, Theorem 2.12(iv), holds for every Lie group homomorphism. The general proof uses the vector-field bracket of Lecture 1. We cite Kirillov Jr., Chapter 3.

2.5 The Baker–Campbell–Hausdorff formula

By local invertibility, Proposition 2.10, a product of exponentials of small elements has a logarithm in $\mathfrak{g}$. We compute its leading terms in $\mathrm{GL}_n(\mathbb{C})$. Write $O(t^k)$ for a quantity bounded in norm by a constant times $|t|^k$ for small $|t|$.

Proposition 2.13 (Second-order term)**.** *Let $X, Y \in M_{n\times n}(\mathbb{C})$. For small $|t|$, put $Z(t) = \log\big(\exp(tX)\exp(tY)\big)$. Then*

$$Z(t) = t(X+Y) + \frac{t^2}{2}[X,Y] + O(t^3).$$

Proof. The map Z is smooth with $Z(0) = 0$. Taylor's theorem gives matrices A, B with $Z(t) = tA + t^2 B + O(t^3)$. Hence $\|Z(t)\| = O(t)$, so the tail $\sum_{k\ge 3} Z^k/k!$ has norm at most $\|Z\|^3 e^{\|Z\|} = O(t^3)$. Therefore

$$\exp Z(t) = I + Z + \tfrac{1}{2}Z^2 + O(t^3) = I + tA + t^2\big(B + \tfrac{1}{2}A^2\big) + O(t^3).$$

Multiplying the two exponential series to second order gives

$$\exp(tX)\exp(tY) = I + t(X+Y) + t^2\big(\tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2\big) + O(t^3).$$

The two left sides are equal. If $tP + t^2 Q = O(t^3)$, dividing by t and letting $t \to 0$ gives $P = 0$, and then $Q = 0$. Comparing coefficients,

$$A = X + Y, \qquad B = \tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2 - \tfrac{1}{2}(X+Y)^2 = \tfrac{1}{2}(XY - YX).$$ ∎

Exercise 2.7 computes the third-order term. The general statement is the following.

Theorem 2.14 (Convergent BCH formula; cited)**.** *Let G be a Lie group with finite-dimensional real Lie algebra $\mathfrak{g}$. There is a neighbourhood W of 0 in $\mathfrak{g}$ with the following property. For $X, Y \in W$, the series grouped by total degree,*

$$H(X,Y) = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) + \cdots,$$

converges in $\mathfrak{g}$, depends continuously on (X,Y), and satisfies $\exp X \exp Y = \exp H(X,Y)$. The degree-$m$ term is a combination of iterated brackets of m entries from X and Y, with universal rational coefficients independent of G.

Hall, Sections 5.3–5.5, proves the integral form of the formula for matrix Lie groups and X, Y near 0. Hall, Section 5.6, computes the series only through degree three.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves the convergent series for connected groups. This suffices, because the identity component of G is open, has Lie algebra $\mathfrak{g}$ and contains each curve $t \mapsto \exp(tX)$. Knapp's proof reduces to matrices with $\|X\| + \|Y\| < \log 2$. There the degree-n term of his logarithm series (B.24) has norm at most $c_n(\|X\| + \|Y\|)^n$, where $-\log(2 - e^s) = \sum_n c_n s^n$ and $c_n \geq 0$. Hence the series converges uniformly for $\|X\| + \|Y\| \leq r < \log 2$, and H is continuous. Knapp's Proposition B.43 converts the rational coefficients of (B.24) into rational bracket coefficients. Kirillov Jr., Section 3.7, Theorem 3.37, states the same series.

The theorem answers the main question near the identity. Let G and G' be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{g}'$. Let $\psi\colon \mathfrak{g} \to \mathfrak{g}'$ be an isomorphism of real Lie algebras. Since it is linear and preserves brackets, ψ maps each term of $H(X, Y)$ to the corresponding term of $H(\psi X, \psi Y)$. Since it is continuous, it maps the convergent sum to the sum.

Write $\exp'$ for the exponential of G' and W' for its BCH neighbourhood. Let $\log\colon V \to U$ be the local inverse of exp on G. Define $f = \exp' \circ \psi \circ \log$ on V.

Choose $a, b \in V$ so close to e that $X = \log a$ and $Y = \log b$ satisfy three conditions:

(a) $X, Y \in W$;
(b) $\psi X, \psi Y \in W'$;
(c) $H(X, Y) \in U$, the set on which exp is inverted.

Continuity of log, ψ and H, with $H(0, 0) = 0$, makes this possible. Since $X, Y \in W$ by condition (a), $ab = \exp H(X, Y)$. Since $H(X, Y) \in U$ by condition (c), $ab \in V$ and $\log(ab) = H(X, Y)$. Therefore

$$f(ab) = \exp'\big(\psi H(X, Y)\big) = \exp'\big(H(\psi X, \psi Y)\big) = \exp'(\psi X)\exp'(\psi Y) = f(a)f(b).$$

The third equality uses condition (b), $\psi X, \psi Y \in W'$.

The map f is also a local diffeomorphism. Write e' for the identity of G'. Then $f(e) = \exp'(\psi(0)) = \exp'(0) = e'$. By local invertibility, Proposition 2.10, $d\exp_0$ and $d\exp'_0$ are identity maps, so $d\log_e$ is the identity too. The chain rule gives

$$df_e = d\exp'_0 \circ \psi \circ d\log_e = \psi.$$

Since ψ is invertible, the inverse function theorem gives an open neighbourhood $V_0 \subseteq V$ of e that f maps diffeomorphically onto an open neighbourhood of e'. For a and b chosen as above, f carries the product ab to $f(a)f(b)$. So the bracket determines multiplication near the identity.

The bracket does not determine the global group. For example, $\mathrm{O}(n)$ and $\mathrm{SO}(n)$ have the same Lie algebra. The next section shows a subtler effect.

2.6 Exponentials in $\mathrm{SL}(2,\mathbb{R})$

For $g \in \mathrm{SL}(2,\mathbb{R})$, expanding a 2×2 determinant gives

$$\det(g+tM) = 1 + t\,\mathrm{tr}(g^{-1}M) + t^2 \det(g^{-1}M).$$

Hence $d\det_g(M) = \mathrm{tr}(g^{-1}M)$. This differential is onto $\mathbb{R}$, since $d\det_g(g) = \mathrm{tr}\, I = 2$. By the regular level-set theorem of Lecture 1, $\mathrm{SL}(2,\mathbb{R})$ is an embedded matrix Lie group. Its algebra $\mathfrak{sl}_2(\mathbb{R})$ is the kernel of $d\det_I$, so it consists of the real traceless 2×2 matrices.

Let $X \in \mathfrak{sl}_2(\mathbb{R})$. Cayley–Hamilton gives $X^2 = \delta I$ with $\delta = -\det X$. Separating even and odd powers, as for rotations and boosts,

$$\exp X = c(\delta) I + s(\delta) X, \qquad c(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k)!}, \quad s(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k+1)!}.$$

Comparison with the hyperbolic and circular series gives

$$c(\delta) = \begin{cases} \cosh\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \cos\sqrt{-\delta}, & \delta < 0, \end{cases} \qquad s(\delta) = \begin{cases} \sinh\sqrt{\delta}/\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \sin\sqrt{-\delta}/\sqrt{-\delta}, & \delta < 0. \end{cases}$$

Proposition 2.15 (A trace bound). *For every $X \in \mathfrak{sl}_2(\mathbb{R})$, $\mathrm{tr}\exp X = 2c(\delta) \ge -2$. Equality holds only when $\exp X = -I$. Consequently, an element of $\mathrm{SL}(2,\mathbb{R})$ is not an exponential if its trace is less than -2, or if its trace is -2 and it differs from $-I$.*

Proof. Since $\mathrm{tr}\, X = 0$, we have $\mathrm{tr}\exp X = 2c(\delta)$. For $\delta \ge 0$, $c(\delta) \ge 1$. For $\delta < 0$, $c(\delta)$ is a cosine, so it is at least -1.

Suppose $c(\delta) = -1$. Then $\delta < 0$, and $\sqrt{-\delta}$ is an odd multiple of π. Hence $\sin\sqrt{-\delta} = 0$ and $s(\delta) = 0$. Therefore $\exp X = c(\delta)I + s(\delta)X = -I$. ∎

Example 2.16 (A non-exponential). Let $a > 0$ with $a \ne 1$, and set

$$g_a = \begin{pmatrix} -a & 0 \\ 0 & -1/a \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}).$$

Its eigenvalues are negative and unequal, and

$$\mathrm{tr}\, g_a + 2 = -\Big(a + \frac{1}{a} - 2\Big) = -\frac{(a-1)^2}{a} < 0.$$

By the trace bound, Proposition 2.15, the element g_a is not $\exp X$ for any $X \in \mathfrak{sl}_2(\mathbb{R})$. For $a = 2$, the trace is $-5/2$.

The element g_a is a product of two exponentials. With $h = \mathrm{diag}(1,-1)$,

$$\exp(\pi J) = -I, \qquad \exp\big((\log a)h\big) = \begin{pmatrix} a & 0 \\ 0 & 1/a \end{pmatrix},$$

so $g_a = \exp(\pi J)\exp\big((\log a)h\big)$.

Exercise 2.4 proves that $\mathrm{SL}(2,\mathbb{R})$ is connected. So generation by exponentials, Proposition 2.11, applies, but one exponential does not suffice. Local invertibility, Proposition 2.10, is not contradicted. For a near 1, the element g_a is near $-I$, far from I. Lecture 6 shows that in a compact connected group every element is a single exponential.

Self-checks

1. Is $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ injective? Is $-I \in \mathrm{O}(1,1)$ of the form $\exp(uK)$?
2. Can a real matrix with negative determinant be $\exp X$ for a real matrix X? Does this contradict generation by exponentials, Proposition 2.11?
3. If $X, Y \in W$ and $[X,Y] = 0$, what is $H(X,Y)$? Is $\log(\exp X \exp Y) = X + Y$ for all commuting X, Y?

Answers. (1) No, since $\exp(2\pi J) = I = \exp 0$. Also no: $\operatorname{tr}(-I) = -2$, while $\operatorname{tr}\exp(uK) = 2\cosh u \geq 2$. (2) No: for real X, $\det\exp X = e^{\operatorname{tr} X} > 0$. There is no contradiction, because $\mathrm{GL}_n(\mathbb{R})$ is disconnected by the determinant sign. (3) Every later term is an iterated bracket, so $H(X,Y) = X + Y$. Still, take $G = \mathrm{SO}(2)$ and $X = Y = \pi J$. The product $\exp X \exp Y$ is I. For the local inverse $\log\colon V \to U$ of exp on $\mathrm{SO}(2)$, $\log I = 0 \neq 2\pi J$. Here $2\pi J \notin U$, since $\exp(2\pi J) = \exp 0$, $0 \in U$ and exp is injective on U.

Summary and reading

Every continuous one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$ is $t \mapsto \exp(tX)$. For a general Lie group, flows of left-invariant fields define exp, a local diffeomorphism at 0. Homomorphisms intertwine exponentials, and connected groups are generated by exponentials. Near the identity, the BCH series computes multiplication from the bracket. Globally, exp need not be injective or surjective, and the Lie algebra $\mathfrak{g}$ does not determine the components of G.

Read Hall, second edition, Chapter 2 on the matrix exponential, Chapter 3 on matrix Lie algebras and homomorphisms, and Sections 5.3–5.7 on BCH and group versus algebra homomorphisms. Kirillov Jr., Chapter 3, treats the general exponential map and the Campbell–Hausdorff formula. Lecture 3 uses the exponential to study subgroups and quotients.

Lecture 3

Subgroups, quotients and homogeneous spaces

Objectives. *A subgroup of a Lie group need not be a submanifold, and a coset space need not be a manifold. Closedness of the subgroup makes the coset space a manifold. Normality is a separate condition, needed for a group law. We test both conditions on an irrational line in a torus and on two descriptions of the sphere S^2.*

By the end of this lecture you will be able to:

- State the closed-subgroup theorem and the subgroup–subalgebra correspondence with their hypotheses.
- Prove that an irrational line in the two-torus is dense and not embedded.
- Identify a homogeneous space with a coset space by computing a stabiliser.
- Compute tangent spaces and isotropy actions for $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$.

3.1 Closed and immersed subgroups

A subset S of a manifold M carries the *subspace topology*: its open sets are the sets $U \cap S$ with U open in M. The subset S is *closed* if $M \setminus S$ is open. In a manifold, S is closed exactly when S contains the limit of each convergent sequence of its points.

The *closure* $\overline{S}$ consists of the points whose every open neighbourhood meets S. It is the smallest closed set containing S. The set S is *dense* if $\overline{S} = M$. By the definition of closure, S is dense exactly when S meets every nonempty open subset of M.

In a manifold, every point of $\overline{S}$ is a limit of a sequence in S. To obtain the sequence, choose points of S in chart balls of radii tending to zero.

Definition 3.1 (Immersions and embeddings)**.** A smooth map $f\colon N \to M$ is an *immersion* if df_p is injective for every $p \in N$. It is an *embedding* if it is an injective immersion and a homeomorphism from N onto $f(N)$ with the subspace topology.

The regular level sets of Lecture 1 are images of embeddings. An injective immersion can fail to be an embedding. The reason is that the topology carried over from N to $f(N)$ can be finer than the subspace topology.

Definition 3.2 (Immersed and embedded subgroups)**.** Let G be a Lie group. An *immersed subgroup* of G is a subgroup H with a Lie group structure for which the inclusion $\iota\colon H \to G$ is an immersion. It is an *embedded subgroup* if ι is an embedding.

The differential $d\iota_e$ identifies $\mathrm{Lie}(H)$ with a subspace $\mathfrak{h} \subseteq \mathfrak{g}$. By Lecture 2, the differential of a Lie group homomorphism preserves brackets. Hence $\mathfrak{h}$ is a real Lie subalgebra of $\mathfrak{g}$.

Theorem 3.3 (Closed-subgroup theorem; cited)**.** *Let G be a Lie group and let $H \subseteq G$ be a subgroup that is a closed subset. Then H is an embedded submanifold of G, and with this structure H is a Lie group. Its Lie algebra is*

$$\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}.$$

The hypothesis concerns only the subset H. Smoothness of H is a conclusion. For example, $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are closed in $\mathrm{GL}_n(\mathbb{C})$, because they are defined by continuous equations.

A related regularity theorem removes a hypothesis from homomorphisms.

Theorem 3.4 (Automatic smoothness; cited)**.** *Every continuous homomorphism between finite-dimensional real Lie groups is smooth.*

See Bröcker–tom Dieck, Chapter I, Section 3, Proposition (3.12). In particular, finite-dimensional continuous representations have differentials. We use this fact from Lecture 4 onwards.

Theorem 3.5 (Subgroups and subalgebras; cited)**.** *Let G be a Lie group and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a real Lie subalgebra. There is exactly one connected immersed subgroup $H \subseteq G$ with $\mathrm{Lie}(H) = \mathfrak{h}$. As a set, H is the subgroup generated by $\exp(\mathfrak{h})$. Every connected immersed subgroup arises in this way from its own Lie algebra.*

Warner, Chapter 3, proves Theorems 3.3 and 3.5 for general Lie groups. Hall, Section 5.9, proves Theorem 3.5 for matrix Lie groups. Kirillov Jr., Chapters 2 and 3, states these two subgroup theorems.

The two theorems run in opposite directions. The first starts with a subset and concludes that it is a manifold. The second starts with a subalgebra and constructs a group. That group need not be closed or embedded, so the word “immersed” in Theorem 3.5 cannot be replaced by “embedded”. The next section gives an example of a connected immersed subgroup that is neither closed nor embedded.

3.2 An irrational line in the two-torus

The two-torus $T^2 = \mathrm{U}(1) \times \mathrm{U}(1)$ is a compact abelian Lie group. Identify its Lie algebra $i\mathbb{R} \oplus i\mathbb{R}$ with $\mathbb{R}^2$ by $(2\pi i a, 2\pi i b) \mapsto (a, b)$. The exponential map then becomes the surjective homomorphism

$$E\colon \mathbb{R}^2 \longrightarrow T^2, \qquad E(a, b) = (e^{2\pi i a}, e^{2\pi i b}),$$

with kernel $\mathbb{Z}^2$. For a real number α, the subalgebra $\mathfrak{h}_\alpha = \mathbb{R}(1, \alpha)$ gives the homomorphism and image

$$\varphi_\alpha(t) = E(t, \alpha t) = (e^{2\pi i t}, e^{2\pi i \alpha t}), \qquad L_\alpha = \varphi_\alpha(\mathbb{R}).$$

The velocity of φ_α at t is the left translate of $(2\pi i, 2\pi i\alpha) \neq 0$, so φ_α is an immersion. For rational α the image is a closed circle subgroup (Exercise Sheet 3). Irrational α behaves differently.

Lemma 3.6 (Small multiples). *Let α be irrational and $n \geq 1$ an integer. There are integers q and m with $1 \leq q \leq n$ and*

$$0 < |q\alpha - m| < \frac{1}{n}.$$

Proof. For $j = 0, 1, \ldots, n$, put $f_j = j\alpha - \lfloor j\alpha \rfloor \in [0, 1)$. These $n + 1$ numbers lie in the n intervals $[s/n, (s+1)/n)$. By the pigeonhole principle, two of them, f_{j_1} and f_{j_2} with $j_1 < j_2$, lie in one interval. Set $q = j_2 - j_1$ and $m = \lfloor j_2\alpha \rfloor - \lfloor j_1\alpha \rfloor$. Then $q\alpha - m = f_{j_2} - f_{j_1}$, which has absolute value less than $1/n$. It is nonzero, since otherwise $\alpha = m/q$ would be rational. ∎

Proposition 3.7 (The irrational line). *Let α be irrational. Then:*

(a) φ_α is injective;
(b) L_α is dense in T^2;
(c) $L_\alpha \neq T^2$, so L_α is not closed;
(d) φ_α is not a homeomorphism onto L_α with the subspace topology.

Proof. (a) If $\varphi_\alpha(t) = \varphi_\alpha(s)$, then $t - s$ and $\alpha(t - s)$ are integers. If $t \neq s$, then α is the ratio of these two integers, so it is rational. Hence $t = s$.

(b) Fix a point $(e^{2\pi ix}, e^{2\pi iy})$ and an integer $n \geq 1$. First, take q and m from the small-multiples lemma, Lemma 3.6, and put $\delta = q\alpha - m \neq 0$. Let k be an integer nearest to $(y - \alpha x)/\delta$. Multiplying by $|\delta|$,

$$|k\delta - (y - \alpha x)| \leq \tfrac{1}{2}|\delta|.$$

Next, put $t = x + kq$. Here kq is an integer, and $\alpha kq = k\delta + km$ with km an integer. Therefore

$$\varphi_\alpha(t) = \left(e^{2\pi ix},\ e^{2\pi i(\alpha x + k\delta)}\right).$$

Finally, compare coordinates. The first coordinates of $\varphi_\alpha(t)$ and the chosen point agree. A chord of the unit circle is no longer than its arc, so $|e^{2\pi iu} - e^{2\pi iv}| \leq 2\pi|u - v|$ for all $u \in \mathbb{R}$ and $v \in \mathbb{R}$. The second coordinates therefore differ by at most $\pi|\delta| < \pi/n$. As n is arbitrary, L_α meets every neighbourhood of the chosen point. Since the chosen point is arbitrary, L_α is dense.

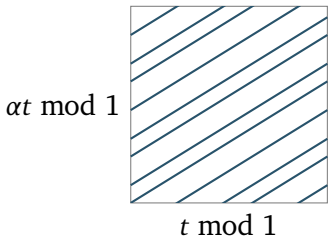
(c) Suppose $\varphi_\alpha(t) = (1, -1)$. Then t is an integer r, and $r\alpha = m + \frac{1}{2}$ for an integer m. Here $r \neq 0$, so $\alpha = (2m + 1)/(2r)$ would be rational. Thus $(1, -1) \notin L_\alpha$. A dense closed subset is the whole space, so L_α is not closed.

(d) For each n, take q_n and m_n from the small-multiples lemma, Lemma 3.6. Then

$$\varphi_\alpha(q_n) = \left(1, e^{2\pi i(q_n\alpha - m_n)}\right) \longrightarrow (1, 1) = \varphi_\alpha(0),$$

but $q_n \geq 1$ for every n. If the inverse map $L_\alpha \to \mathbb{R}$ were continuous at $(1, 1)$, then q_n would converge to 0. This is impossible. Hence the inverse map is not continuous at $(1, 1)$, and φ_α is not a homeomorphism onto L_α. ∎

The following figure illustrates the density of L_α.



The figure shows L_α for $0 \le t \le 8$ and $\alpha = (\sqrt{5}-1)/2$, drawn in the unit square with opposite sides identified. Each new unit-time piece is shifted upwards by α modulo one.

Give L_α the Lie group structure transported from $\mathbb{R}$ by the bijection φ_α. It is then the connected immersed subgroup with Lie algebra $\mathfrak{h}_\alpha$ given by the subgroup–subalgebra correspondence, Theorem 3.5. By the irrational-line proposition, Proposition 3.7, the map φ_α is not a homeomorphism onto L_α with the subspace topology, so this subgroup is not embedded. The same proposition shows that L_α is not closed, so the closed-subgroup theorem, Theorem 3.3, does not apply to it. Since L_α is dense by the same proposition, its closure is the closed subgroup T^2. The Lie algebra $\mathbb{R}^2$ of T^2 is strictly larger than $\mathfrak{h}_\alpha$.

3.3 Coset spaces, actions and quotient groups

Let H be a subgroup of a Lie group G. The *coset space* $G/H = \{gH : g \in G\}$ has the quotient map π given by $\pi(g) = gH$. Its *quotient topology* declares $V \subseteq G/H$ open when $\pi^{-1}(V)$ is open in G. The group G acts on G/H by left translation, $\ell_g(xH) = gxH$.

Closedness is necessary for G/H to be Hausdorff. In a Hausdorff space each point is closed, because every other point has an open neighbourhood avoiding it. So if G/H is Hausdorff, then $H = \pi^{-1}(eH)$ is closed. For irrational α, the space T^2/L_α is therefore not a manifold.

Theorem 3.8 (Quotients by closed subgroups; cited). *Let G be a Lie group and H a closed subgroup with Lie algebra $\mathfrak{h}$. Then:*

1. *G/H is Hausdorff and second countable. It has a unique smooth manifold structure for which π is a* submersion*, that is, smooth with surjective differential at every point;*
2. *$\dim G/H = \dim G - \dim H$, and $d\pi_e$ induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_{eH}(G/H)$;*
3. *the action $G \times G/H \to G/H$ is smooth;*
4. *a map f from G/H to a manifold is smooth exactly when $f \circ \pi$ is smooth.*

This theorem is in Bröcker–tom Dieck, Chapter I, Section 4, and in Warner, Chapter 3.

Definition 3.9 (Smooth actions). A *smooth action* of G on a manifold M is a smooth map $G \times M \to M$, $(g, p) \mapsto g \cdot p$, with $e \cdot p = p$ and $g \cdot (g' \cdot p) = (gg') \cdot p$. The *stabiliser* of p is $G_p = \{g \in G : g \cdot p = p\}$, and the *orbit* of p is $G \cdot p$. The action is *transitive* if it has only one orbit. A manifold with a transitive smooth action is a *homogeneous space*.

Proposition 3.10 (Stabilisers and orbit maps)**.** *Let G act smoothly on M, fix $p \in M$, and put $H = G_p$.*

(a) H is a closed subgroup, and $G_{g\cdot p} = gHg^{-1}$.

(b) The orbit map $\beta(gH) = g \cdot p$ is a well-defined, injective, smooth map $G/H \to M$ with image $G \cdot p$. It is equivariant: $\beta(\ell_{g'}(gH)) = g' \cdot \beta(gH)$.

Proof. (a) Since $e \cdot p = p$, we have $e \in H$. If $g, g' \in H$, then $(gg') \cdot p = g \cdot (g' \cdot p) = p$ and $g^{-1} \cdot p = g^{-1} \cdot (g \cdot p) = p$. The point p is closed, because M is Hausdorff. So H is closed, as the preimage of p under the continuous map $g \mapsto g \cdot p$. Finally, $x \cdot (g \cdot p) = g \cdot p$ exactly when $(g^{-1}xg) \cdot p = p$, that is, when $x \in gHg^{-1}$.

(b) We have $g \cdot p = g' \cdot p$ exactly when $g^{-1}g' \in H$, that is, when $gH = g'H$. One direction gives well-definedness, the other injectivity. Since H is closed by part (a), the quotient theorem, Theorem 3.8, applies to G/H. The composite $\beta \circ \pi(g) = g \cdot p$ is smooth, so β is smooth by the smoothness criterion of Theorem 3.8(4). Equivariance is the identity $(g'g) \cdot p = g' \cdot (g \cdot p)$. ∎

Theorem 3.11 (Homogeneous spaces; cited)**.** *Let a Lie group G act smoothly and transitively on a manifold M, and let $p \in M$. Then the orbit map $\beta : G/G_p \to M$ is a G-equivariant diffeomorphism.*

The proof uses the second countability of G, which is part of the definition of a manifold in Lecture 1; see Warner, Chapter 3, or Kirillov Jr., Chapter 2.

Two derivatives of the orbit map are used below. First, for $X \in \mathfrak{g}$, differentiate $\beta(\pi(\exp tX)) = \exp(tX) \cdot p$ at $t = 0$. This shows that $d\beta_{eH}(d\pi_e X)$ is the velocity of $t \mapsto \exp(tX) \cdot p$. This holds for every smooth action.

Second, assume now that the action is transitive. By the homogeneous-space theorem, Theorem 3.11, $d\beta_{eH}$ is an isomorphism $T_{eH}(G/H) \to T_pM$. Combining this with the tangent-space isomorphism of Theorem 3.8(2), we see that the composite $d\beta_{eH} \circ d\pi_e$ has kernel $\mathfrak{h}$ and induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_pM$.

As before, $H = G_p$. Let $k \in H$. The translation ℓ_k fixes eH. Its differential $d(\ell_k)_{eH}$ is called the *isotropy action* of k. The map $a_k : M \to M$, given by $a_k(p') = k \cdot p'$ for $p' \in M$, fixes p. Equivariance of β says $\beta \circ \ell_k = a_k \circ \beta$. The chain rule at eH therefore gives

$$d\beta_{eH} \circ d(\ell_k)_{eH} = d(a_k)_p \circ d\beta_{eH}.$$

So, transported by the isomorphism $d\beta_{eH}$, the isotropy action of k is the differential of a_k at p.

A coset space is not automatically a group.

Proposition 3.12 (Normality)**.** *The rule $(xH)(yH) = xyH$ is well defined on G/H exactly when H is normal. It then makes G/H a group.*

Proof. Suppose the rule is well defined. Let $k \in H$ and $y \in G$. The representatives e and k of the coset H give the same product with yH, so $kyH = yH$. Hence $y^{-1}ky \in H$, and H is normal. Conversely, if H is normal, then for $k, k' \in H$,

$$xk\,yk' = xy\,(y^{-1}ky)\,k' \in xyH.$$

The group axioms are inherited from G. ∎

Theorem 3.13 (Quotient groups; cited)**.** *Let H be a closed normal subgroup of a Lie group G. With the structure of Theorem 3.8, G/H is a Lie group and π is a homomorphism. The subalgebra $\mathfrak{h}$ is an ideal, and $\operatorname{Lie}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ as Lie algebras.*

This theorem is in Warner, Chapter 3. Closedness of H makes G/H a manifold, and normality of H makes G/H a group. Since T^2 is abelian, L_α is normal, so T^2/L_α is a group. For irrational α it is not a manifold. The two models of the sphere below are manifolds without a coset group law.

3.4 The sphere as SO(3)/SO(2)

The group $\mathrm{SO}(3)$ acts smoothly on $S^2 = \{v \in \mathbb{R}^3 : |v| = 1\}$ by matrix multiplication, with base point e_3. Let $J_1, J_2, J_3 \in \mathfrak{so}(3)$ be the matrices of Exercise Sheet 1, so that $J_k v = e_k \times v$. The rotations about the third axis are

$$R_\theta = \exp(\theta J_3) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Proposition 3.14 (Stabiliser of the north pole)**.** *The action of $\mathrm{SO}(3)$ on S^2 is transitive. The stabiliser of e_3 is $H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2)$.*

Proof. Let $v \in S^2$. Extend v to an orthonormal basis (u, w, v) of $\mathbb{R}^3$. The matrix g with columns u, w, v is orthogonal and $ge_3 = v$. Replacing u by $-u$ if necessary, we get $\det g = 1$. Thus the action is transitive.

Now let $ge_3 = e_3$. The third column of g is e_3. The rows of g are also orthonormal, because $gg^{\mathsf{T}} = I$. The third row is a unit vector with third entry one, so its other entries vanish. Hence

$$g = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^{\mathsf{T}} A = I, \qquad \det A = \det g = 1.$$

The first column of A is a unit vector, so it equals $(\cos\theta, \sin\theta)$ for some θ. The second column is a unit vector orthogonal to the first, and $\det A = 1$ selects $(-\sin\theta, \cos\theta)$. So $g = R_\theta$. Conversely, $R_\theta e_3 = e_3$. ∎

By the homogeneous-space theorem, Theorem 3.11, the map $gH \mapsto ge_3$ is an equivariant diffeomorphism $\mathrm{SO}(3)/\mathrm{SO}(2) \to S^2$. The dimensions agree: $3 - 1 = 2$.

The sphere is the level set $|v|^2 = 1$, so $T_{e_3} S^2 = e_3^\perp$ by Lecture 1. By the first derivative computation of the previous section, $d\beta_{eH} \circ d\pi_e$ sends $X \in \mathfrak{so}(3)$ to the velocity of $\exp(tX)e_3$, which is Xe_3. On the basis,

$$J_1 e_3 = e_1 \times e_3 = -e_2, \qquad J_2 e_3 = e_2 \times e_3 = e_1, \qquad J_3 e_3 = 0.$$

So $X \mapsto Xe_3$ has kernel $\mathbb{R}J_3 = \operatorname{Lie}(H)$ and image $e_3^\perp$. It induces the isomorphism $\mathfrak{so}(3)/\mathbb{R}J_3 \cong T_{e_3} S^2$, which is $d\beta_{eH}$ composed with the tangent-space isomorphism of Theorem 3.8(2).

For the isotropy action, the map $a_{R_\theta}(v) = R_\theta v$ is the restriction to S^2 of a linear map of $\mathbb{R}^3$. Its differential at e_3 is therefore R_θ restricted to $e_3^\perp$. In the coordinates (x, y) for $xe_1 + ye_2$, this is the rotation of the tangent plane by θ. Only $R_\theta = I$ acts trivially.

The stabiliser H is not normal. Take the rotation by $\pi/2$ about e_1,

$$g_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad g_0 e_3 = -e_2.$$

By the conjugation formula of Proposition 3.10(a), $g_0 H g_0^{-1}$ is the stabiliser of $-e_2$. But $R_{\pi/2}(-e_2) = e_1$. Hence $R_{\pi/2} \in H$ and $R_{\pi/2} \notin g_0 H g_0^{-1}$. By the normality proposition, Proposition 3.12, there is no coset multiplication on $\mathrm{SO}(3)/\mathrm{SO}(2)$.

3.5 The sphere as SU(2)/U(1)

Let V be the real vector space of traceless Hermitian 2×2 matrices. It has the coordinates

$$\xi(x, y, z) = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}, \qquad -\det \xi(x, y, z) = x^2 + y^2 + z^2.$$

The group $\mathrm{SU}(2)$ acts on V by $U \cdot \xi = U \xi U^*$. The matrix $U\xi U^*$ is Hermitian. Since $U^* = U^{-1}$, it has the same trace and determinant as ξ. Thus $\mathrm{SU}(2)$ acts smoothly on $S = \{\xi \in V : \det \xi = -1\}$, which the coordinates identify with S^2. The base point is $N = \xi(0, 0, 1) = \mathrm{diag}(1, -1)$. With $A = \mathrm{diag}(i, -i)$ from Lecture 1, put $k_\theta = \exp(\theta A) = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ and $T = \{k_\theta : \theta \in \mathbb{R}\} \cong \mathrm{U}(1)$.

Proposition 3.15 (Stabiliser in SU(2))**.** *The action of* $\mathrm{SU}(2)$ *on* S *is transitive, and the stabiliser of* N *is* T*.*

Proof. Let $\xi \in S$. Its characteristic polynomial is $s^2 - (\operatorname{tr} \xi)s + \det \xi = s^2 - 1$, so its eigenvalues are ± 1. By the spectral theorem, $\xi = WNW^*$ for a unitary W whose columns are unit eigenvectors of ξ for the eigenvalues 1 and -1. Since W is unitary, its determinant has modulus one. Choose $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $\lambda^2 = \overline{\det W}$. Then $U = \lambda W$ has determinant one, and $UNU^* = |\lambda|^2 WNW^* = \xi$.

For the stabiliser, $UNU^* = N$ is equivalent to $UN = NU$. Write U with entries a, b, c, d. Then

$$UN = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \qquad NU = \begin{pmatrix} a & b \\ -c & -d \end{pmatrix}.$$

Equality forces $b = c = 0$. A diagonal unitary matrix has $|a| = 1$, and $ad = 1$ gives $d = \overline{a}$. Thus $U = k_\theta$ for some θ. Conversely, k_θ commutes with N. ∎

By the homogeneous-space theorem, Theorem 3.11, the map $UT \mapsto UNU^*$ is an equivariant diffeomorphism $\mathrm{SU}(2)/T \to S \cong S^2$. Again the dimensions are $3 - 1 = 2$.

The map $d\beta_{eT} \circ d\pi_e$ sends $X \in \mathfrak{su}(2)$ to the velocity of $\exp(tX)N\exp(tX)^*$ at $t = 0$. By the product rule and since $X^* = -X$, this velocity is $XN + NX^* = [X, N]$. Use the basis A, B, C of $\mathfrak{su}(2)$ from Lecture 1, where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Direct multiplication gives $[A, N] = 0$ and

$$[B, N] = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = \xi(-2, 0, 0), \qquad [C, N] = \begin{pmatrix} 0 & -2i \\ 2i & 0 \end{pmatrix} = \xi(0, 2, 0).$$

The kernel of $X \mapsto [X, N]$ is $\mathbb{R}A = \mathrm{Lie}(T)$. The image of this map is $\{\xi(x, y, 0)\}$. In the coordinates, S is the unit sphere and N is its north pole, so this image is $T_N S$. Thus $X \mapsto [X, N]$ induces the isomorphism $\mathfrak{su}(2)/\mathbb{R}A \cong T_N S$, which is $d\beta_{eT}$ composed with the tangent-space isomorphism of Theorem 3.8(2).

For the isotropy action, the map $a_{k_\theta}(\xi) = k_\theta \xi k_\theta^*$ is the restriction to S of a linear map of V. Its differential at N is the same conjugation, acting on $T_N S$:

$$k_\theta\, \xi(x, y, 0)\, k_\theta^* = \begin{pmatrix} 0 & e^{2i\theta}(x - iy) \\ e^{-2i\theta}(x + iy) & 0 \end{pmatrix}.$$

Thus $x + iy \mapsto e^{-2i\theta}(x + iy)$, a rotation by -2θ. It is trivial exactly when $\theta \in \pi\mathbb{Z}$, that is, when $k_\theta = \pm I$.

The two models have the same space, circle stabilisers and tangent dimension. They differ in the isotropy action. With the parametrisations R_θ and k_θ and the coordinates (x, y) used above, let θ run over $[0, 2\pi]$. The SO(3) isotropy rotation goes once round, while the SU(2) isotropy rotation goes twice round in the opposite sense.

The sense of rotation depends on the choice of parametrisation, because replacing k_θ by $k_{-\theta}$ reverses it. The factor two does not depend on this choice. The SU(2) isotropy action has kernel $\{\pm I\}$. Moreover, $-I$ acts trivially on all of V, since $(-I)\xi(-I)^* = \xi$ for every $\xi \in V$. Lecture 4 uses the element $-I$ to relate the two groups.

The subgroup T is not normal in SU(2) (Exercise 3.6(ii)), so SU(2)/U(1) has no coset group law.

Self-checks

1. For which real α is L_α closed in T^2?
2. Is $(gH)(g'H) = gg'H$ a well-defined product on SO(3)/SO(2)?
3. Why is the stabiliser in SU(2) of every point of S a compact subgroup conjugate to T?

Answers. (1) Exactly for rational α. Let $\alpha = p/q$ with integers p and $q \geq 1$. Since q and $\alpha q = p$ are integers, $\varphi_\alpha(q) = (1, 1)$. So φ_α has period q, and $L_\alpha = \varphi_\alpha([0, q])$. This is a continuous image of a compact interval, hence compact in $\mathbb{C}^2$. It is therefore closed in $\mathbb{C}^2$, and so closed in T^2. For irrational α, the subgroup L_α is not closed by the irrational-line proposition, Proposition 3.7(c). (2) No. The stabiliser SO(2) is not normal, as g_0 shows. By the normality proposition, Proposition 3.12, the product is therefore not well defined.

(3) By transitivity each point is $U \cdot N$. Its stabiliser is UTU^{-1} by the conjugation formula of Proposition 3.10(a). This is the image of the compact circle T under the continuous map $k \mapsto UkU^{-1}$, so it is compact.

Summary and reading

A closed subgroup of a Lie group is an embedded subgroup. A real Lie subalgebra determines a connected immersed subgroup, which need not be closed or embedded. For example, the irrational line is dense and not embedded. For closed H, the coset space G/H is a manifold whose tangent space at eH is isomorphic to $\mathfrak{g}/\mathfrak{h}$. If a Lie group G acts smoothly and transitively on a manifold M, then M is equivariantly diffeomorphic to G/G_p for each $p \in M$. Normality is a further condition, needed for G/H to be a group. The sphere is diffeomorphic to both $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$. The two models have different isotropy actions.

Read Bröcker–tom Dieck, Chapter I, Section 4; Hall, Section 5.9; and Kirillov Jr., Chapter 2. Warner, Chapter 3, is the reference for the cited closed-subgroup, correspondence, quotient and homogeneous-space theorems. Lecture 4 studies the element $-I$ of $\mathrm{SU}(2)$ through covering groups.

Lecture 4

Covering groups, integration and descent

Objectives. *We ask when a finite-dimensional Lie algebra representation is the differential of a representation of a given connected Lie group. On a simply connected group it always is. On other connected groups we integrate on the universal covering group and then test a central kernel.*

By the end of this lecture you will be able to:

- Describe the universal covering group by path classes and use the lifting theorem.
- Prove that a discrete normal subgroup of a connected group is central.
- State Lie's integration theorem and Lie III with their hypotheses, and explain path transport.
- Prove the descent criterion and apply it to $\mathbb{R} \to \mathrm{U}(1)$ and $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Lie groups are finite-dimensional, real, Hausdorff and second countable. From Lecture 1–Lecture 3 we use three facts. The map exp is a diffeomorphism near 0. A connected Lie group is generated by any neighbourhood of e. A continuous homomorphism Φ of Lie groups is smooth and satisfies $\Phi(\exp X) = \exp d\Phi(X)$. Hence two continuous homomorphisms on a connected Lie group with equal differentials are equal.

4.1 Loops, homotopy and covering groups

Definition 4.1. A *path* in a space X is a continuous map $\gamma\colon [0,1] \to X$. It is a *loop at* x if $\gamma(0) = \gamma(1) = x$. Paths γ_0, γ_1 from x to y are *homotopic* if there is a continuous map $F\colon [0,1]^2 \to X$ such that

$$F(0,t) = \gamma_0(t), \qquad F(1,t) = \gamma_1(t), \qquad F(s,0) = x, \qquad F(s,1) = y.$$

The space X is *simply connected* if it is path connected and every loop is homotopic to a constant loop.

If σ starts where γ ends, the path $\gamma * \sigma$ traverses γ on $[0, \tfrac{1}{2}]$ and then σ on $[\tfrac{1}{2}, 1]$, each at double speed. Homotopy classes of loops at x form the *fundamental group* $\pi_1(X, x)$ under this product.

Definition 4.2. Let X and Y be path connected. A continuous surjection $p\colon Y \to X$ is a *covering map* if each point of X has an open neighbourhood U such that $p^{-1}(U)$ is a disjoint union of open sets, each mapped homeomorphically onto U by p.

Theorem 4.3 (Lifting theorem)**.** *Let $p\colon Y \to X$ be a covering map.*

(a) Let Z be path connected, locally path connected and simply connected. Let $f\colon Z \to X$ be continuous, and let $p(y_0) = f(z_0)$. Then there is a unique continuous $\tilde{f}\colon Z \to Y$ with $p \circ \tilde{f} = f$ and $\tilde{f}(z_0) = y_0$.

(b) Let γ_0, γ_1 be homotopic paths from x to x', and let $p(y_0) = x$. Then the lifts of γ_0 and γ_1 that start at y_0 have the same endpoint.

We cite this theorem from H §13.1 and S §1.2. The endpoint statement (b) follows from the lifting statement (a), applied to a homotopy F with $Z = [0,1]^2$ and $\tilde{F}(0,0) = y_0$. Each fibre of p is discrete. So if $\tilde{F}$ maps a connected set into one fibre, then $\tilde{F}$ maps that set to a single point.

The lift $\tilde{F}$ maps the edge $t = 0$ into the fibre over x, so $\tilde{F}$ maps this edge to y_0. By uniqueness of lifts in (a) with $Z = [0,1]$, the restrictions of $\tilde{F}$ to the edges $s = 0$ and $s = 1$ are the lifts of γ_0 and γ_1 from y_0.

The lift $\tilde{F}$ maps the edge $t = 1$ into the fibre $p^{-1}(x')$, so $\tilde{F}$ also maps this edge to a single point. Hence the two lifts have the same endpoint.

By the endpoint statement (b), a loop whose lift has distinct endpoints is not homotopic to a constant loop.

Definition 4.4. A *universal covering group* of a connected Lie group G is a connected, simply connected Lie group $\tilde{G}$ with a smooth homomorphism $p\colon \tilde{G} \to G$ that is a covering map.

Path classes. Let G be connected. Let $\tilde{G}$ be the set of homotopy classes $[\gamma]$ of paths in G starting at e. Define

$$p[\gamma] = \gamma(1), \qquad [\gamma][\delta] = [\, t \mapsto \gamma(t)\delta(t)\,], \qquad [\gamma]^{-1} = [\, t \mapsto \gamma(t)^{-1}\,].$$

If F is a homotopy from γ to γ' and E one from δ to δ', then $(s,t) \mapsto F(s,t)E(s,t)$ is a homotopy from $\gamma\delta$ to $\gamma'\delta'$. So the product is well defined. The group axioms hold pointwise in t. The kernel of p consists of the classes of loops at e.

It remains to give $\tilde{G}$ a topology. We sketch the construction. The details are cited after the theorem on smooth consequences, Theorem 4.5. Take a simply connected open neighbourhood U of $\gamma(1)$, for example a translate of a small exponential ball.

The basic open set $\mathcal{U}([\gamma], U)$ consists of the classes $[\gamma * \sigma]$, where σ is a path in U starting at $\gamma(1)$. The map p sends this set bijectively onto U, because paths in U with the same endpoints are homotopic in U.

The cited details show that p is a covering map, that the group operations are continuous and that $\tilde{G}$ is simply connected. Charts on G pulled back through p give the smooth structure on $\tilde{G}$.

An *isomorphism over G* from a universal covering group $p\colon \tilde{G} \to G$ to another one $p'\colon \tilde{G}' \to G$ is an isomorphism of Lie groups $\alpha\colon \tilde{G} \to \tilde{G}'$ with $p' \circ \alpha = p$.

Theorem 4.5 (Smooth consequences)**.** *Let G be a connected Lie group.*

(a) The path-class construction gives a universal covering group $p\colon \tilde{G} \to G$, unique up to isomorphism over G. Its kernel is discrete and isomorphic to $\pi_1(G, e)$, and dp is an isomorphism.

(b) Let G' be a connected Lie group and $q\colon G' \to G$ a surjective smooth homomorphism with dq bijective. Then q is a covering map and $\ker q$ is discrete.

We cite this from H §5.8 and from Knapp Beyond, Section I.11, *Covering groups*. Hall assumes that G is a matrix group, but $\tilde{G}$ need not be one.

Proposition 4.6. *Let G be a connected topological group. Let N be a normal subgroup of G that is discrete in the subspace topology. Then N lies in the centre of G.*

Proof. Fix $n \in N$ and put $c(g) = gng^{-1}$. The map $c\colon G \to G$ is continuous, because multiplication and inversion are continuous. Normality gives $c(G) \subseteq N$. Since G is connected, $c(G)$ is connected. It contains $c(e) = n$.

Since N is discrete, every point of N is open in N. In particular, $\{n\}$ is open in N. Its complement $N \setminus \{n\}$ is a union of open singletons, so $\{n\}$ is also closed in N. Hence $\{n\}$ is open and closed in the subset $c(G)$ of N. Since $c(G)$ is connected and contains n, $c(G) = \{n\}$. In other words, $gng^{-1} = n$ for every $g \in G$. ∎

By the theorem on smooth consequences, Theorem 4.5(a), $\ker p$ is a discrete normal subgroup of the connected group $\tilde{G}$. By the proposition on discrete normal subgroups, Proposition 4.6, $\ker p$ is central. Hence $\pi_1(G, e)$, which is isomorphic to $\ker p$, is abelian. Exercise 4.1 shows that connectedness cannot be dropped.

4.2 Lie's integration theorem and Lie III

Theorem 4.7 (Lie's integration theorem)**.** *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Assume that G is connected and simply connected. Let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of real Lie algebras. Then there is a unique smooth homomorphism $\Phi\colon G \to H$ with $d\Phi = \varphi$. It satisfies $\Phi(\exp X) = \exp \varphi(X)$.*

We cite this from H §5.7 for matrix groups. For general Lie groups we cite Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*. For representations take $H = \mathrm{GL}(V)$ with $\dim_{\mathbb{C}} V < \infty$.

Path transport. The existence proof has three steps and a conclusion. Only the conclusion uses simple connectivity.

Step 1: a local homomorphism. Choose a neighbourhood B of 0 on which exp is a diffeomorphism. Define $f(\exp X) = \exp \varphi(X)$ for $X \in B$. For small X, Y we have $\exp X \exp Y =$

$\exp Z(X,Y)$. The Baker–Campbell–Hausdorff series is the series of iterated brackets

$$Z(X,Y) = X + Y + \frac{1}{2}[X,Y] + \cdots .$$

Since φ preserves brackets, it sends $Z(X,Y)$ to $Z(\varphi X, \varphi Y)$. Hence, for some neighbourhood W of e,

$$f(xy) = f(x)f(y) \qquad (x, y \in W). \tag{4.1}$$

Hall proves the series for matrix groups in H §§5.3–5.6. In general we cite Knapp Beyond, Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*.

Step 2: transport along a path. Let γ start at e. Choose $0 = t_0 < \cdots < t_n = 1$ with $\gamma(s)^{-1}\gamma(t) \in W$ whenever s, t lie in one interval $[t_{j-1}, t_j]$. Define

$$\Phi_\gamma = f\big(\gamma(t_0)^{-1}\gamma(t_1)\big) \cdots f\big(\gamma(t_{n-1})^{-1}\gamma(t_n)\big).$$

Inserting a point s into $[t_{j-1}, t_j]$ replaces one factor $f(ab)$ by $f(a)f(b)$, where $a = \gamma(t_{j-1})^{-1}\gamma(s)$ and $b = \gamma(s)^{-1}\gamma(t_j)$. By the local homomorphism property (4.1), applied to a and b, the product is unchanged. Any two such partitions have a common refinement. Hence Φ_γ depends only on γ, not on the partition.

Step 3: homotopy invariance. We sketch the topological step. Let F be a homotopy from γ to γ', both from e to g. Cut $[0,1]^2$ into squares so small that $F(q)^{-1}F(q') \in W$ for any two points q, q' of one square. Give an edge from q to q' the factor $f(F(q)^{-1}F(q'))$. The transport of an edge path is the product of its factors in order. Let one square have corners a, b, c, d in cyclic order, with images x_a, x_b, x_c, x_d. By the local homomorphism property (4.1),

$$f(x_a^{-1}x_b)\, f(x_b^{-1}x_c) = f(x_a^{-1}x_c) = f(x_a^{-1}x_d)\, f(x_d^{-1}x_c).$$

So the two routes $a \to b \to c$ and $a \to d \to c$ around one square have equal transport.

Each edge path we use runs in the grid from $(0,0)$ to $(1,1)$. Start with the column $s = 0$ followed by the edge $t = 1$. End with the edge $t = 0$ followed by the column $s = 1$. Pass from the first to the last one square at a time. Each move swaps the routes $a \to b \to c$ and $a \to d \to c$ around one square, so it keeps the transport. The edges at $t = 0$ and $t = 1$ contribute $f(e) = e$, because F is constant there. By Step 2 the two columns give Φ_γ and $\Phi_{\gamma'}$. Hence $\Phi_\gamma = \Phi_{\gamma'}$. This step holds in every Lie group.

Conclusion. As G is simply connected, any two paths from e to g are homotopic. So $\Phi(g) = \Phi_\gamma$ does not depend on the path γ. The *increments* of a path δ are the elements $\delta(s)^{-1}\delta(t)$ that enter the factors of Step 2. They do not change when δ is replaced by its left translate $g\delta$, given by $t \mapsto g\delta(t)$. So if δ runs from e to h, then

$$\Phi(gh) = \Phi_{\gamma * g\delta} = \Phi_\gamma\, \Phi_\delta = \Phi(g)\, \Phi(h).$$

Extending a path from e to g by a single short step gives $\Phi(g \exp X) = \Phi(g) \exp \varphi(X)$ for small X. Hence Φ is smooth and $d\Phi = \varphi$.

Without simple connectivity, Steps 1–3 still hold, but transport may depend on the homotopy class of the path. Self-check 2 gives an example.

> **Theorem 4.8** (Lie III). *Every finite-dimensional real Lie algebra is the Lie algebra of a connected, simply connected Lie group. This group is unique up to isomorphism.*

We cite existence from H §5.10. There Ado's theorem and H §5.9 give a connected Lie subgroup of a general linear group with the given Lie algebra. This subgroup need not be closed. The universal covering group of this subgroup is the required group. Uniqueness follows by integrating an isomorphism and its inverse with Lie's integration theorem, Theorem 4.7. The universal covering group of $\mathrm{SL}(2,\mathbb{R})$ is not isomorphic to any matrix Lie group (H §4.8). Hence Lie III needs abstract Lie groups.

4.3 Descent through a covering kernel

Theorem 4.9 (Descent). *Let $p\colon \tilde G \to G$ be a surjective smooth homomorphism of Lie groups with dp bijective, and let $\Gamma = \ker p$. Let H be a topological group and $\tilde\Phi\colon \tilde G \to H$ a continuous homomorphism.*

(a) A homomorphism $\Phi\colon G \to H$ with $\Phi \circ p = \tilde\Phi$ exists if and only if $\Gamma \subseteq \ker\tilde\Phi$.

(b) Such Φ is unique and continuous. If H is a Lie group and $\tilde\Phi$ is smooth, then Φ is smooth and $d\Phi = d\tilde\Phi \circ (dp)^{-1}$.

Proof. Local sections. Since p is a homomorphism, left translations satisfy $p \circ L_x = L_{p(x)} \circ p$. Differentiating at e gives

$$dp_x \circ (dL_x)_e = (dL_{p(x)})_e \circ dp.$$

Both translations have invertible derivatives, and dp is bijective. Hence dp_x is bijective at every $x \in \tilde G$. By the inverse function theorem, each x has an open neighbourhood V_x that p maps diffeomorphically onto an open set. Let $s_x\colon p(V_x) \to V_x$ be the inverse of this restriction of p.

Necessity. If $\Phi \circ p = \tilde\Phi$ and $\gamma \in \Gamma$, then $\tilde\Phi(\gamma) = \Phi(e) = e$.

Sufficiency. Assume $\Gamma \subseteq \ker\tilde\Phi$. Let $g \in G$. Since p is surjective, we can write $g = p(x)$. Set $\Phi(g) = \tilde\Phi(x)$. If also $g = p(y)$, then $x^{-1}y \in \Gamma$ and

$$\tilde\Phi(y) = \tilde\Phi(x)\,\tilde\Phi(x^{-1}y) = \tilde\Phi(x).$$

Hence Φ is well defined and $\Phi \circ p = \tilde\Phi$. Moreover, Φ is a homomorphism, because

$$\Phi\big(p(x)p(y)\big) = \Phi\big(p(xy)\big) = \tilde\Phi(xy) = \tilde\Phi(x)\tilde\Phi(y).$$

Regularity. On $p(V_x)$ we have $\Phi = \Phi \circ p \circ s_x = \tilde\Phi \circ s_x$. This is continuous, and smooth when $\tilde\Phi$ is smooth. The open sets $p(V_x)$ cover G. Differentiating $\Phi \circ p = \tilde\Phi$ at e gives $d\Phi \circ dp = d\tilde\Phi$. Uniqueness holds because p is surjective. ∎

The proof uses only that H is a topological group. This matters in Lecture 15, where H is a unitary group of a Hilbert space.

Corollary 4.10. *Let G be a connected Lie group with universal covering group $p\colon \tilde G \to G$, and let $\Gamma = \ker p$. Let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $\dim_{\mathbb{C}} V < \infty$. Let $\tilde\Pi$ be the representation of $\tilde G$ with $d\tilde\Pi = \varphi \circ dp$. Then $\varphi = d\Pi$ for a representation Π of G if and only if $\tilde\Pi(\Gamma) = \{I\}$. In that case Π is unique and $\Pi \circ p = \tilde\Pi$.*

Proof. Since $\tilde{G}$ is simply connected, Lie's integration theorem, Theorem 4.7, gives $\tilde{\Pi}$, and $\tilde{\Pi}$ is unique. If Π exists, then $\Pi \circ p$ has differential $\varphi \circ dp$. Hence, by the uniqueness of $\tilde{\Pi}$, $\Pi \circ p = \tilde{\Pi}$, and $\tilde{\Pi}(\gamma) = \Pi(e) = I$ for $\gamma \in \Gamma$. Conversely, if $\tilde{\Pi}(\Gamma) = \{I\}$, then the Descent Theorem, Theorem 4.9, gives Π with $d\Pi = \varphi \circ dp \circ (dp)^{-1} = \varphi$. Uniqueness holds because G is connected. ∎

For disconnected groups the differential need not determine the homomorphism. For example, Exercise 4.6 gives two characters of $\mathrm{O}(2)$ with the same differential.

4.4 The line over the circle

Let the surjection $p\colon \mathbb{R} \to \mathrm{U}(1)$ be given by $p(t) = e^{it}$. Here $\operatorname{Lie}\mathrm{U}(1) = i\mathbb{R}$ and $\exp(i\theta) = e^{i\theta}$. Then

$$p(s+t) = p(s)\,p(t), \qquad dp(s) = \frac{d}{du}\Big|_{u=0} e^{ius} = is, \qquad \Gamma = \ker p = 2\pi\mathbb{Z}.$$

The line is simply connected, because $F(s,t) = (1-s)\gamma(t) + s\gamma(0)$ contracts any loop γ. By the theorem on smooth consequences, Theorem 4.5(b), p is a covering map, so it is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{U}(1), 1) \cong 2\pi\mathbb{Z} \cong \mathbb{Z}$.

Concretely, the loop $\omega(t) = e^{2\pi i t}$ lifts from 0 to $t \mapsto 2\pi t$, which ends at $2\pi \neq 0$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), ω is not homotopic to a constant loop.

A *scalar infinitesimal weight* on $\mathbb{R}$ is a number $\lambda \in \mathbb{C}$, viewed as the representation $\varphi_\lambda(s) = \lambda s$ on $\mathbb{C}$. It integrates to $\tilde{\chi}_\lambda(t) = e^{\lambda t}$.

The *period* of λ is $e^{2\pi\lambda} = \tilde{\chi}_\lambda(2\pi)$, the value at the generator 2π of Γ.

Say that λ *descends* if $\varphi_\lambda \circ (dp)^{-1}$ is the differential of a character of $\mathrm{U}(1)$. The descent criterion, Corollary 4.10, gives

$$\lambda \text{ descends} \iff e^{2\pi\lambda} = 1 \iff \lambda \in i\mathbb{Z}.$$

For $\lambda = in$ the descended character is $\chi(z) = z^n$, because $\chi(e^{it}) = e^{int}$. Its differential is $d\chi(i\theta) = in\theta$. In the course convention $d\chi = i\ell$ with $\ell(i\theta) = n\theta$. On $\Lambda = \ker\exp = 2\pi i\mathbb{Z}$ we get $\ell(2\pi i k) = 2\pi n k$, so $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

Success. For $\lambda = 3i$ the period is $e^{6\pi i} = 1$. The character $\chi(z) = z^3$ satisfies $\chi(p(t)) = e^{3it} = \tilde{\chi}_{3i}(t)$.

Failure. For $\lambda = i/2$ the period is $e^{i\pi} = -1$. Suppose a character χ had $d\chi(i\theta) = i\theta/2$. The corollary gives $\chi \circ p = \tilde{\chi}_{i/2}$, so

$$1 = \chi(1) = \chi\big(p(2\pi)\big) = \tilde{\chi}_{i/2}(2\pi) = -1.$$

This contradiction shows that $\lambda = i/2$ does not descend.

4.5 SU(2) over SO(3)

Let $\sigma_1, \sigma_2, \sigma_3$ be the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Put $X_j = -\frac{i}{2}\sigma_j$ for $j = 1, 2, 3$. Then $[X_1, X_2] = X_3$, with cyclic permutations.

Each element of SU(2) has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So SU(2) is homeomorphic to the unit sphere in $\mathbb{C}^2$, which is connected and simply connected (H §13.2).

The covering. The basis (X_1, X_2, X_3) is orthonormal for the form $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Each $\mathrm{Ad}(g)$, given by $\mathrm{Ad}(g)X = gXg^{-1}$, is a linear map of $\mathfrak{su}(2)$ to itself. It preserves this form, because the trace is invariant under conjugation. Hence the matrix of $\mathrm{Ad}(g)$ in the basis (X_1, X_2, X_3) lies in O(3). We identify $\mathrm{Ad}(g)$ with this matrix, so Ad maps SU(2) into O(3). Its image lies in SO(3), because $\det \circ \mathrm{Ad}$ is continuous on the connected group SU(2), takes values ± 1, and equals 1 at I.

The operator $\mathrm{ad}\, X_3$ sends $X_1 \mapsto X_2$, $X_2 \mapsto -X_1$ and $X_3 \mapsto 0$. Hence

$$\mathrm{Ad}(\exp \theta X_3) = e^{\theta\, \mathrm{ad}\, X_3} = R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The differential $d\,\mathrm{Ad} = \mathrm{ad}$ is injective, since $[X_1, X] = [X_2, X] = 0$ forces $X = 0$. By dimension it is bijective onto $\mathfrak{so}(3)$.

If $\mathrm{Ad}(g) = I$, then g commutes with σ_3 and σ_1. The first condition makes g diagonal. The second makes it scalar. So $g = zI$ with $z^2 = \det g = 1$, and $\ker \mathrm{Ad} = \{\pm I\}$.

The image of Ad contains exp of a neighbourhood of 0 in $\mathfrak{so}(3)$, because $d\,\mathrm{Ad}$ is bijective. Hence the image of Ad is an open subgroup of SO(3). It is also closed, because its complement is a union of open cosets.

The elementary axis-angle form from *Linear Algebra for Physics* says that every rotation has the form $kR_3(\theta)k^{-1}$ with $k \in \mathrm{SO}(3)$. The path $t \mapsto kR_3(t\theta)k^{-1}$ joins this rotation to I. Hence SO(3) is connected. The image of Ad is open, closed and nonempty, so Ad is onto.

Let $\Gamma = \ker \mathrm{Ad}$. By the theorem on smooth consequences, Theorem 4.5(b), Ad is a covering map. Since SU(2) is connected and simply connected, Ad is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{SO}(3), I) \cong \Gamma = \{\pm I\}$.

The loop $t \mapsto R_3(2\pi t)$ lifts to $t \mapsto \exp(2\pi t X_3) = \mathrm{diag}(e^{-i\pi t}, e^{i\pi t})$, which ends at $-I$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), this loop is not homotopic to a constant loop.

Representations. Let V_m be the irreducible $\mathfrak{sl}(2,\mathbb{C})$-module of dimension $m+1$ and spin $m/2$. It has a basis $v_0,\dots,v_m$ with $\sigma_3 v_k = (m-2k)v_k$. Restriction to $\mathfrak{su}(2)$ gives φ_m, which integrates to Π_m on SU(2). Since $X_3 = -\frac{i}{2}\sigma_3$,

$$\Pi_m(\exp tX_3)v_k = e^{\lambda_k t}v_k, \qquad \lambda_k = -\tfrac{i}{2}(m-2k), \qquad \Pi_m(-I)v_k = e^{2\pi\lambda_k}v_k = (-1)^m v_k.$$

By the descent criterion, Corollary 4.10, $\varphi_m \circ (d\,\mathrm{Ad})^{-1}$ is the differential of a representation of SO(3) exactly when m is even. The homomorphism $t \mapsto R_3(t) = \mathrm{Ad}(\exp tX_3)$ from $\mathbb{R}$ has kernel $2\pi\mathbb{Z}$, and $\exp(2\pi X_3) = -I$. So the test is the scalar test of § 4.4 for each weight λ_k: its period $e^{2\pi\lambda_k}$ must equal 1.

Success: $m = 2$. Here $\Pi_2(-I) = I$. Hence there is a representation Π of SO(3) with $\Pi_2 = \Pi\circ\mathrm{Ad}$. This Π is equivalent to the defining representation. To see this, note that the matrices $d\,\mathrm{Ad}(X)$ also describe $\mathrm{ad}\,X$ on $\mathfrak{sl}(2,\mathbb{C})$ in the complex basis (X_1, X_2, X_3). This adjoint module has σ_3-weights $2, 0, -2$. Hence there is an invertible $A\colon \mathbb{C}^3 \to V_2$ with

$$A\,d\,\mathrm{Ad}(X)A^{-1} = \varphi_2(X).$$

The representation $g \mapsto A\,\mathrm{Ad}(g)A^{-1}$ of SU(2) has differential φ_2. By uniqueness on the connected group SU(2), it equals Π_2. Since Ad is onto, $\Pi(R) = ARA^{-1}$.

Failure: $m = 1$. Here $V_1 = \mathbb{C}^2$ and $\Pi_1(g) = g$. Suppose a representation Π of SO(3) had $d\Pi = \varphi_1 \circ (d\,\mathrm{Ad})^{-1}$. The corollary gives $\Pi \circ \mathrm{Ad} = \Pi_1$, so

$$I = \Pi\big(R_3(2\pi)\big) = \Pi\big(\mathrm{Ad}(-I)\big) = \Pi_1(-I) = -I.$$

Hence no representation of SO(3) has differential $\varphi_1 \circ (d\,\mathrm{Ad})^{-1}$.

For a spin-$\frac{1}{2}$ particle, $\mathbb{C}^2$ describes the internal spin degree of freedom. Rotations act on this space only up to sign. Lecture 13 treats such projective representations.

Self-checks

Check 1. Does the character $t \mapsto e^{i\sqrt{2}t}$ of $\mathbb{R}$ descend through $p(t) = e^{it}$ to U(1)?

Answer. No. Its period is $e^{2\pi i\sqrt{2}} \neq 1$, because $\sqrt{2} \notin \mathbb{Z}$.

Check 2. Take $G = \mathrm{U}(1)$ and $\varphi(i\theta) = i\theta/2$, so $f(e^{i\theta}) = e^{i\theta/2}$ for $|\theta| < \pi$. Use $W = \{e^{i\theta} : |\theta| < \pi/2\}$. Compute the transport around $\omega(t) = e^{2\pi i t}$. Which part of the path-transport argument fails?

Answer. The partition $t_j = j/5$ is admissible, since $2\pi/5 < \pi/2$. So $\Phi_\omega = f(e^{2\pi i/5})^5 = (e^{\pi i/5})^5 = -1$. The constant loop gives 1. Steps 1–3 remain valid. The Conclusion fails, because ω is not homotopic to the constant loop, by the lift computed in § 4.4. So $\Phi(1)$ is not well defined.

Check 3. Is φ_3 the differential of a representation of SU(2)? Is $\varphi_3 \circ (d\,\mathrm{Ad})^{-1}$ the differential of a representation of SO(3)?

Answer. Yes for SU(2), by Lie's integration theorem, Theorem 4.7. No for SO(3), since $\Pi_3(-I) = -I$. Equivalently, the weight $\lambda_0 = -\frac{3i}{2}$ has period $e^{-3\pi i} = -1$.

Summary

A connected Lie group G has a universal covering group $p\colon \tilde{G} \to G$. Its kernel $\Gamma \cong \pi_1(G, e)$ is discrete and central. Let φ be a representation of the Lie algebra of G on a finite-dimensional complex vector space. Then $\varphi \circ dp$ integrates to a representation $\tilde{\Pi}$ of $\tilde{G}$. The representation φ is the differential of a representation of G exactly when Γ acts trivially through $\tilde{\Pi}$. For $\mathbb{R} \to \mathrm{U}(1)$ this is the period condition $\lambda \in i\mathbb{Z}$. For $\mathrm{SU}(2) \to \mathrm{SO}(3)$ the condition holds exactly for even m, that is, for integer spin.

Reading

H is Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, 2015. S is Sepanski, *Compact Lie Groups*, 2007. Knapp Beyond is Knapp, *Lie Groups Beyond an Introduction*, digital second edition, 2023.

- H §§4.7–4.8 and 5.7–5.10: group versus Lie algebra homomorphisms, universal covers, subgroups and subalgebras, Lie's third theorem, and a nonmatrix Lie group.
- H §§13.1–13.2: the fundamental group and the fundamental groups of the compact classical groups.
- S §§1.2, 4.2: topological background and the related material for compact groups.
- Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*; Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*: general versions of the cited results.

Lecture 5

Haar Measure, Averaging and Orthogonality

Objectives. *For a finite group G, averaging with $|G|^{-1}\sum_{g\in G}$ proves unitarisability, complete reducibility and orthogonality of matrix coefficients. In this lecture we build an integral on a compact Lie group that replaces this average. We then record which property of the integral each argument uses.*

By the end of this lecture you will be able to:

- Integrate continuous functions on a compact manifold against a density.
- Construct the normalised Haar integral of a compact Lie group, and prove that it is right invariant and strictly positive.
- Average inner products and operators to prove unitarisability, complete reducibility and Schur orthogonality for finite-dimensional representations of compact Lie groups.
- Compute Haar integrals, an invariant-vector projector and a coefficient norm on $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Lie groups are real, finite-dimensional, Hausdorff and second countable. Representations are complex, finite-dimensional and continuous. Inner products are antilinear in the first slot.

5.1 Densities and integration on manifolds

Let M be a smooth n-manifold with $n \geq 1$. Charts and tangent spaces are as in Lecture 1. In a chart, the Euclidean integral $\int f\,du$ changes by a Jacobian factor when the chart changes. A density supplies a factor that cancels this change. We cite the routine analytic parts of this construction.

Definition 5.1. Let V be a real vector space of dimension n. A *density* on V is a function $\delta\colon V^n \to \mathbb{R}$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \quad \text{for all linear } T\colon V\to V \text{ and all } v_j \in V.$$

It is *positive* if it is positive on every basis.

Fix a basis $b = (b_j)$ and write $v_j = Tb_j$. Then $\delta(v_1, \dots, v_n) = |\det T|\, \delta(b_1, \dots, b_n)$, so δ is determined by one value. Conversely, let $[v]_b$ be the matrix whose jth column holds the coordinates of v_j. Then

$$\delta_b(v_1, \dots, v_n) = |\det [v]_b|$$

is a density, since $[Tv]_b = [T]_b [v]_b$. Hence the densities on V form the real line $\mathbb{R}\,\delta_b$, and $c\,\delta_b$ is positive exactly when $c > 0$.

Definition 5.2. A *continuous density* δ on M is a density δ_p on each T_pM such that, for every chart $x : U \to \Omega \subset \mathbb{R}^n$, the function ρ_x given by

$$\rho_x(x(p)) = \delta_p\big(\partial_{x_1}|_p, \dots, \partial_{x_n}|_p\big), \qquad p \in U,$$

is continuous on Ω.

Let $y : U' \to \Omega'$ be another chart, and put $\phi = y \circ x^{-1}$ on $x(U \cap U')$. The chain rule gives $\partial_{x_j} = \sum_i (\partial y_i / \partial x_j)\, \partial_{y_i}$. The defining property of a density, applied to this change of basis, gives

$$\rho_x(u) = |\det D\phi(u)|\, \rho_y(\phi(u)), \qquad u \in x(U \cap U'). \tag{5.1}$$

Let $f : M \to \mathbb{C}$ be continuous and vanish outside a compact subset of U. Define

$$\int_M f\,\delta = \int_\Omega f(x^{-1}(u))\, \rho_x(u)\, du. \tag{5.2}$$

If f vanishes outside a compact subset of $U \cap U'$, then the change-of-variables theorem and the transformation rule (5.1) show that the chart y gives the same value.

Now let M be compact. A finite *partition of unity* is a finite family of continuous functions $\chi_k \geq 0$ with $\sum_k \chi_k = 1$, each vanishing outside a compact subset of a chart domain U_k. We cite the existence of a finite partition of unity on M. For continuous f put

$$\int_M f\,\delta = \sum_k \int_M \chi_k f\,\delta.$$

Let (ψ_l) be a second partition of unity, with chart domains U'_l. Each $\chi_k \psi_l f$ vanishes outside a compact subset of $U_k \cap U'_l$, so its integral is the same in the charts on U_k and on U'_l. Since each partition sums to 1, linearity of the chart integral (5.2) shows that both sums equal $\sum_{k,l} \int_M \chi_k \psi_l f\,\delta$. Hence the integral is well defined. Moreover, it is linear, and it is nonnegative when $f \geq 0$ and δ is positive.

For a diffeomorphism $F : M \to M$, the *pullback* density is

$$(F^*\delta)_p(v_1, \dots, v_n) = \delta_{F(p)}(dF_p v_1, \dots, dF_p v_n).$$

The chain rule gives $(F_1 \circ F_2)^* = F_2^* F_1^*$.

Lemma 5.3. *$F^*\delta$ is a continuous density, positive if δ is. For $f \in C(M)$,*

$$\int_M (f \circ F)\, F^*\delta = \int_M f\,\delta.$$

Proof. Let x be a chart on U. Then $x \circ F$ is a chart on $F^{-1}(U)$, and dF sends its coordinate vectors to those of x. So the coefficient of $F^*\delta$ in the chart $x \circ F$ is the continuous function

ρ_x. The transformation rule (5.1) then gives continuity in every chart. If f vanishes outside a compact subset of U, both integrals are the same Euclidean integral (5.2). For general f, apply this to each $\chi_k f$ and use the partition of unity $(\chi_k \circ F)$. ∎

In examples one chart often covers M apart from a set contained in finitely many submanifolds of lower dimension. A set $Z \subset \mathbb{R}^n$ is *Lebesgue-null* if, for every $\varepsilon > 0$, countably many boxes of total volume less than ε cover Z. Changing a function on a Lebesgue-null set does not change its Lebesgue integral. We cite the following consequence of Lebesgue theory from Knapp, Section VIII.1. The integrability statement in the lemma, and the term *measurable* below, belong to that cited theory.

Lemma 5.4. *Let M be compact with a positive continuous density δ. Let $x : U \to \Omega$ be a chart such that $M \setminus U$ lies in a finite union of smooth submanifolds of dimension less than n. Then for $f \in C(M)$ the function $(f \circ x^{-1})\rho_x$ is Lebesgue integrable on Ω, and*

$$\int_M f\,\delta = \int_\Omega f(x^{-1}(u))\,\rho_x(u)\,du.$$

The lemma rests on two cited facts. A submanifold of dimension less than n has Lebesgue-null image in every chart. The change-of-variables formula holds for every nonnegative measurable function. For $f \geq 0$ one applies both facts to each $\chi_k f$, using the transformation rule (5.1), and sums over k. A general $f \in C(M)$ is a linear combination of four nonnegative continuous functions, namely the positive and negative parts of the real and imaginary parts of f.

5.2 The Haar integral

Let G be a compact Lie group with Lie algebra $\mathfrak{g} = T_eG$. If $\dim G = 0$, then G is discrete and compact, hence finite. Until the definition of the Haar integral, Definition 5.6, we assume $n = \dim G \geq 1$. Write $L_h(g) = hg$ and $R_h(g) = gh$. Fix a positive density δ_e on $\mathfrak{g}$ and define

$$\delta_g(v_1, \ldots, v_n) = \delta_e(dL_{g^{-1}}v_1, \ldots, dL_{g^{-1}}v_n), \qquad v_j \in T_gG. \tag{5.3}$$

Lemma 5.5. *δ is a positive continuous density, and $L_h^*\delta = \delta$ for all $h \in G$. Every continuous density δ' with $L_h^*\delta' = \delta'$ for all h is a constant multiple of δ.*

Proof. Each $dL_{g^{-1}} : T_gG \to \mathfrak{g}$ is a linear isomorphism, so δ_g is positive. Fix a basis (X_j) of $\mathfrak{g}$. The fields $X_j^L(g) = dL_g X_j$ are smooth and form a basis of each tangent space (Lecture 1). In a chart write $\partial_{x_i} = \sum_j a_{ji} X_j^L$. The matrix (a_{ji}) is the inverse of the coefficient matrix of the fields X_j^L in this chart. That coefficient matrix is continuous, hence so is (a_{ji}). Since $dL_{g^{-1}}X_j^L(g) = X_j$, we get $\rho_x = |\det(a_{ji})|\,\delta_e(X_1, \ldots, X_n)$, which is continuous. With $v = (v_1, \ldots, v_n)$,

$$(L_h^*\delta)_g(v) = \delta_{hg}(dL_h v) = \delta_e(dL_{(hg)^{-1}}dL_h v) = \delta_e(dL_{g^{-1}}v) = \delta_g(v).$$

Now let δ' be left invariant. Taking $h = g^{-1}$ gives $\delta'_g(v) = \delta'_e(dL_{g^{-1}}v)$. The densities on $\mathfrak{g}$ form a real line, so $\delta'_e = c\,\delta_e$ for a real constant c. Hence $\delta' = c\,\delta$. ∎

Definition 5.6. The *normalised Haar integral* of a compact Lie group G is the functional on $C(G)$ given by

$$\int_G f(g)\,dg = \begin{cases} \dfrac{\int_G f\,\delta}{\int_G 1\,\delta}, & \dim G \geq 1, \\ \dfrac{1}{|G|}\sum_{g\in G} f(g), & G \text{ finite.} \end{cases}$$

The denominator is positive because every ρ_x is positive. Rescaling δ_e does not change the quotient.

Theorem 5.7. *Let G be a compact Lie group, $f \in C(G)$ and $h \in G$.*

(a) The integral is linear, $\int_G f(g)\,dg \geq 0$ if $f \geq 0$, and $\int_G 1\,dg = 1$.
(b) Left invariance: $\int_G f(hg)\,dg = \int_G f(g)\,dg$.
(c) Right invariance: $\int_G f(gh)\,dg = \int_G f(g)\,dg$.
(d) Strict positivity: if $f \geq 0$ and $f(g_0) > 0$ for some g_0, then $\int_G f(g)\,dg > 0$.

Proof. For finite G all parts are clear, since translations permute G. Let $n \geq 1$. Linearity, positivity and normalisation are immediate. Left invariance follows from the pullback lemma, Lemma 5.3, with $F = L_h$, because $L_h^*\delta = \delta$ by the left-invariant density lemma, Lemma 5.5.

(c) By the pullback lemma, Lemma 5.3, $R_h^*\delta$ is a continuous density. Left and right translations commute, so

$$L_k^* R_h^* \delta = (R_h \circ L_k)^*\delta = (L_k \circ R_h)^*\delta = R_h^* L_k^* \delta = R_h^*\delta.$$

By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $R_h^*\delta = c(h)\,\delta$ for a real constant $c(h)$. We compute it at e. The map $L_{h^{-1}} \circ R_h$ is $g \mapsto h^{-1}gh$, whose differential at e is $\mathrm{Ad}(h^{-1})$ (Lecture 2). So for $X = (X_1, \dots, X_n)$ in $\mathfrak{g}$,

$$(R_h^*\delta)_e(X) = \delta_h(dR_h X) = \delta_e(dL_{h^{-1}} dR_h X) = \delta_e(\mathrm{Ad}(h^{-1})X) = j(h)^{-1}\delta_e(X),$$

where $j(h) = |\det \mathrm{Ad}(h)|$. Thus $R_h^*\delta = j(h)^{-1}\delta$. Since Ad is a continuous homomorphism, so is $j\colon G \to (0,\infty)$. Its image is a compact subgroup. If it contained some $t \neq 1$, the powers t^m, $m \in \mathbb{Z}$, would be unbounded. Hence $j = 1$ and $R_h^*\delta = \delta$. Now apply the pullback lemma, Lemma 5.3, with $F = R_h$.

(d) By continuity of f, there exist $c > 0$ and an open set $U \ni g_0$ with $f > c$ on U. The translates aU with $a \in G$ cover G, so compactness gives $G = a_1U \cup \cdots \cup a_NU$. Every $g \in G$ has some $a_i^{-1}g \in U$. Since $f > c$ on U and $f \geq 0$ everywhere, we get

$$\Phi(g) = \sum_{i=1}^{N} f(a_i^{-1}g) \geq c \qquad \text{for all } g \in G.$$

By linearity and left invariance, $N\int_G f(g)\,dg = \int_G \Phi(g)\,dg$. By positivity and normalisation, $\int_G \Phi(g)\,dg \geq c > 0$. Hence $\int_G f(g)\,dg > 0$. ∎

We cite uniqueness of the normalised Haar integral from BD I.5. Let $J\colon C(G) \to \mathbb{C}$ be linear with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all $f \in C(G)$ and $h \in G$. Then J is the normalised Haar integral. We also cite the Riesz representation theorem for positive linear functionals on $C(X)$, with X compact Hausdorff. It shows that the Haar integral is

integration against a unique regular Borel probability measure. This is the *normalised Haar measure*. We integrate only continuous functions here.

5.3 Averaging inner products and operators

Let W be a finite-dimensional complex vector space, and let $F\colon G \to W$ be continuous. We integrate F coordinatewise in a basis of W. The result does not depend on the basis. It satisfies $A\int_G F\,dg = \int_G AF\,dg$ for linear A. It also inherits left and right invariance from the Haar integral, Theorem 5.7.

Theorem 5.8 (Unitarisability). *Every representation (π, V) of a compact Lie group G admits an inner product for which all $\pi(g)$ are unitary.*

Proof. Choose any inner product $\langle\cdot,\cdot\rangle_0$ on V and set

$$\langle v, w\rangle = \int_G \langle \pi(g)v, \pi(g)w\rangle_0\, dg.$$

The integrand is continuous because π is continuous. By linearity of the integral, the form is antilinear in v and linear in w. It is Hermitian, since the integral commutes with complex conjugation. For $v \neq 0$, the function $g \mapsto \|\pi(g)v\|_0^2$ is nonnegative and positive at e. Strict positivity gives $\langle v, v\rangle > 0$. Right invariance, applied to $g \mapsto \langle \pi(g)v, \pi(g)w\rangle_0$, gives

$$\langle \pi(h)v, \pi(h)w\rangle = \int_G \langle \pi(gh)v, \pi(gh)w\rangle_0\, dg = \langle v, w\rangle.$$

∎

Corollary 5.9 (Complete reducibility). *Every representation of a compact Lie group is a direct sum of irreducible subrepresentations.*

Proof. Let (π, V) be a representation, and induct on $\dim V$. If V is irreducible, there is nothing to prove. Otherwise let U be a proper nonzero invariant subspace. Use the invariant inner product of unitarisability, Theorem 5.8. For $u \in U$ and $w \in U^{\perp}$,

$$\langle u, \pi(g)w\rangle = \langle \pi(g)^{*}u, w\rangle = \langle \pi(g^{-1})u, w\rangle = 0.$$

So $U^{\perp}$ is invariant and $V = U \oplus U^{\perp}$. Apply induction to both summands. ∎

The *invariant vectors* of a representation (π, V) form the subspace

$$V^G = \{v \in V : \pi(g)v = v \text{ for all } g \in G\}.$$

Proposition 5.10 (Invariant-vector projector). *Let (π, V) be a representation of a compact Lie group G and $P = \int_G \pi(g)\,dg$. Then $\pi(h)P = P = P\pi(h)$ for all h, and P is a projection with image V^G. If all $\pi(g)$ are unitary, P is the orthogonal projection onto V^G.*

Proof. Left invariance gives $\pi(h)P = \int_G \pi(hg)\,dg = P$. Right invariance gives $P\pi(h) = P$. The first identity shows $\operatorname{im} P \subset V^G$. For $v \in V^G$ we have $Pv = \int_G v\,dg = v$. Hence $P^2 = P$ and $\operatorname{im} P = V^G$. Now let π be unitary. As in the proof of complete reducibility, Corollary 5.9, the subspace $U = (V^G)^{\perp}$ is invariant. Since $\pi(g)U \subset U$ for all g, also $PU \subset U \cap V^G = 0$. So $U \subset \ker P$. By rank–nullity, $\dim\ker P = \dim V - \dim V^G = \dim U$. Hence $\ker P = (V^G)^{\perp}$. ∎

Let (σ, W) be a second representation. Then $\operatorname{Hom}(V,W)$ carries the representation in which g acts by $E \mapsto \sigma(g) E \pi(g)^{-1}$. The invariant vectors of this representation are the intertwiners. By the invariant-vector projector, Proposition 5.10, applied to this representation, for every linear $E\colon V \to W$,

$$E^{\natural} = \int_G \sigma(g) E \pi(g)^{-1}\, dg \in \operatorname{Hom}_G(V,W). \tag{5.4}$$

5.4 Orthogonality of matrix coefficients

Definition 5.11. Let π act by unitary operators on a finite-dimensional inner product space V. The *matrix coefficient* of $v, w \in V$ is the function $c^{\pi}_{v,w}(g) = \langle v, \pi(g) w \rangle$ on G.

It is antilinear in v and linear in w. Let $D(g)$ be the matrix of $\pi(g)$ in an orthonormal basis (e_i), so $\pi(g) e_j = \sum_i D_{ij}(g) e_i$. Then $D_{ij} = c^{\pi}_{e_i, e_j}$. On $C(G)$ we use

$$\langle f_1, f_2 \rangle_{L^2} = \int_G \overline{f_1(g)}\, f_2(g)\, dg.$$

This inner product is also antilinear in the first slot. It is positive definite by strict positivity.

We import Schur's lemma from finite group theory. Its proof is algebraic and holds for any group. For irreducible finite-dimensional complex representations, $\operatorname{Hom}_G(V,W) = 0$ if $V \not\cong W$, and $\operatorname{Hom}_G(V,V) = \mathbb{C}\,\mathrm{id}$.

Theorem 5.12 (Schur orthogonality). *Let (π, V) and (σ, W) be irreducible unitary representations of a compact Lie group G, with $d = \dim V$.*

(a) If $\pi \not\cong \sigma$, then $\langle c^{\pi}_{v,w}, c^{\sigma}_{v',w'} \rangle_{L^2} = 0$ for $v, w \in V$ and $v', w' \in W$.

(b) For $v, w, v', w' \in V$,

$$\int_G \overline{\langle v, \pi(g) w \rangle}\, \langle v', \pi(g) w' \rangle\, dg = \frac{1}{d}\, \langle v', v \rangle\, \langle w, w' \rangle.$$

Proof. The key step writes the integrand as a coefficient of an averaged operator. In case (b) put $\sigma = \pi$ and $W = V$. Define $E\colon V \to W$ by $Eu = \langle w, u \rangle w'$. Here $w \in V$ and $w' \in W$, so $E \in \operatorname{Hom}(V,W)$. Since $\pi(g)^{-1} = \pi(g)^{*}$,

$$\langle v', \sigma(g) E \pi(g)^{-1} v \rangle = \langle w, \pi(g)^{*} v \rangle \langle v', \sigma(g) w' \rangle = \overline{\langle v, \pi(g) w \rangle}\, \langle v', \sigma(g) w' \rangle.$$

The map $u \mapsto \langle v', u \rangle$ is linear, so it commutes with the integral. Integrating over G therefore gives

$$\int_G \langle v', \sigma(g) E \pi(g)^{-1} v \rangle\, dg = \langle v', E^{\natural} v \rangle,$$

with the averaged operator $E^{\natural}$ from (5.4). So the left sides in both parts equal $\langle v', E^{\natural} v \rangle$. In case (a), where $\pi \not\cong \sigma$, Schur's lemma gives $E^{\natural} = 0$. In case (b), where $\sigma = \pi$, Schur's lemma gives $E^{\natural} = \lambda\, \mathrm{id}$ for some $\lambda \in \mathbb{C}$. Trace is linear and conjugation invariant, so in an orthonormal basis (e_i)

$$d\lambda = \int_G \operatorname{tr}\!\big(\pi(g) E \pi(g)^{-1}\big)\, dg = \operatorname{tr} E = \sum_i \langle w, e_i \rangle \langle e_i, w' \rangle = \langle w, w' \rangle.$$

Hence $\langle v', E^{\natural} v \rangle = d^{-1} \langle w, w' \rangle \langle v', v \rangle$. ∎

Corollary 5.13. *Let (π, V) and (σ, W) be as in Schur orthogonality, Theorem 5.12, with $d = \dim V$. Let D^π and D^σ be their matrices in orthonormal bases of V and W. Then*

$$\int_G \overline{D^\pi_{ij}(g)}\, D^\pi_{kl}(g)\, dg = \frac{\delta_{ik}\delta_{jl}}{d}, \qquad \int_G \overline{D^\pi_{ij}(g)}\, D^\sigma_{kl}(g)\, dg = 0 \quad (\pi \not\cong \sigma).$$

In particular $\|c^\pi_{v,w}\|^2_{L^2} = \|v\|^2\|w\|^2/d$ for $v, w \in V$.

Proof. Take $v = e_i$, $w = e_j$, $v' = e_k$ and $w' = e_l$ in Schur orthogonality, Theorem 5.12. For the norm identity, take $v' = v$ and $w' = w$ in the relation for a single representation. ∎

Put $j = i$ and $l = k$, and sum over i and k. The character $\chi_\pi = \operatorname{tr}\pi$ is the sum of the diagonal entries D^π_{ii}. Hence the characters of inequivalent irreducible representations are orthonormal.

5.5 Examples: the groups U(1) and SU(2)

In each example the single-chart lemma, Lemma 5.4, lets us integrate over one chart.

Example 5.14 (The circle)**.** For $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ use the parametrisation $\theta \mapsto e^{i\theta}$, $0 < \theta < 2\pi$. Its inverse is a chart, whose domain misses only the point 1. At $z = e^{i\theta}$ we have $\partial_\theta = iz$ and $dL_{z^{-1}}(iz) = i$. Taking $\delta_e(i) = 1$ gives $\rho = 1$, so the integral of 1 against δ is 2π. Hence

$$\int_{\mathrm{U}(1)} f(z)\, dz = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\, d\theta.$$

The characters $\chi_n(z) = z^n$, $n \in \mathbb{Z}$, are irreducible unitary representations, and

$$\int_{\mathrm{U}(1)} \overline{z^m}\, z^n\, dz = \frac{1}{2\pi}\int_0^{2\pi} e^{i(n-m)\theta}\, d\theta = \delta_{mn}.$$

Example 5.15 (SU(2))**.** Every $g \in \mathrm{SU}(2)$ has the form

$$g = \begin{pmatrix} a & -\bar b \\ b & \bar a \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So $g \mapsto (a, b)$ identifies $\mathrm{SU}(2)$ with $S^3 \subset \mathbb{C}^2 = \mathbb{R}^4$. This map and its inverse are restrictions of real linear maps, so the identification is a diffeomorphism. Differentiating $|\omega|^2 = 1$ gives $T_\omega S^3 \subset \omega^\perp$, and both have dimension 3. So $T_\omega S^3 = \omega^\perp$.

The first column of hg is $h(a, b)^T$. Thus L_h acts on S^3 by the unitary matrix h, which is orthogonal on $\mathbb{R}^4$. For $\omega \in S^3$ and $v_j \in \omega^\perp$ set

$$\nu_\omega(v_1, v_2, v_3) = \big|\det[\,\omega\ v_1\ v_2\ v_3\,]\big|,$$

a real 4×4 determinant. Extending $T\colon \omega^\perp \to \omega^\perp$ by $T\omega = \omega$ shows that ν_ω is a positive density. In a chart with inverse $u \mapsto \omega(u)$, the coefficient of ν is $|\det[\,\omega\ \partial_1\omega\ \partial_2\omega\ \partial_3\omega\,]|$, which is continuous.

For orthogonal A we get $\nu_{A\omega}(Av) = |\det A|\,\nu_\omega(v) = \nu_\omega(v)$. Hence $L_h^*\nu = \nu$. By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $\nu = c\,\delta$ for the left-invariant density δ of (5.3), with $c > 0$.

Use Hopf coordinates $a = \cos\eta\, e^{i\xi_1}$ and $b = \sin\eta\, e^{i\xi_2}$, with $0 < \eta < \pi/2$ and $0 < \xi_1, \xi_2 < 2\pi$. This parametrisation is injective. The points it misses lie in the circles $\{a = 0\}$ and $\{b = 0\}$ and in the 2-spheres $\{\operatorname{Im} a = 0\}$ and $\{\operatorname{Im} b = 0\}$. In $\mathbb{R}^4$,

$$\begin{aligned}
\omega &= (\cos\eta\cos\xi_1,\ \cos\eta\sin\xi_1,\ \sin\eta\cos\xi_2,\ \sin\eta\sin\xi_2),\\
\partial_\eta\omega &= (-\sin\eta\cos\xi_1,\ -\sin\eta\sin\xi_1,\ \cos\eta\cos\xi_2,\ \cos\eta\sin\xi_2),\\
\partial_{\xi_1}\omega &= (-\cos\eta\sin\xi_1,\ \cos\eta\cos\xi_1,\ 0,\ 0),\\
\partial_{\xi_2}\omega &= (0,\ 0,\ -\sin\eta\sin\xi_2,\ \sin\eta\cos\xi_2).
\end{aligned}$$

These vectors are mutually orthogonal, with lengths $1, 1, \cos\eta, \sin\eta$. So the parametrisation is an injective immersion of an open box. By the inverse function theorem its inverse is a chart. The coefficient of ν in this chart is the product of these lengths, $\cos\eta\sin\eta$. So the integral of 1 against ν is $\frac{1}{2}(2\pi)^2 = 2\pi^2$. Therefore

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\int_0^{2\pi}\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2. \tag{5.5}$$

The monomial $a^p\bar a^q b^r \bar b^s$ carries the phase $e^{i(p-q)\xi_1}e^{i(r-s)\xi_2}$. So its integral vanishes unless $p = q$ and $r = s$. Also

$$\int_{\mathrm{SU}(2)} |a|^2\,dg = 2\int_0^{\pi/2}\cos^3\eta\sin\eta\,d\eta = 2\Big[-\frac{\cos^4\eta}{4}\Big]_0^{\pi/2} = \frac{1}{2}, \qquad \int_{\mathrm{SU}(2)} |b|^2\,dg = 1 - \frac{1}{2} = \frac{1}{2}.$$

A coefficient norm. The defining representation $V_1 = \mathbb{C}^2$ is irreducible of dimension 2. In the standard basis $D(g) = g$, so $D_{11} = a$ and $D_{12} = -\bar b$. The matrix-entry relations of Corollary 5.13 predict $\int |D_{11}|^2\,dg = \frac{1}{2}$ and $\int \overline{D_{11}}D_{12}\,dg = 0$. Both agree with the direct computation, since $\overline{D_{11}}D_{12} = -\bar a\bar b$ has a nonzero phase.

An invariant-vector projector. Let $\mathrm{SU}(2)$ act on $V_1 \otimes V_1$ by $g \otimes g$. Use the orthonormal basis $e_1\otimes e_1$, $e_1\otimes e_2$, $e_2\otimes e_1$, $e_2\otimes e_2$. The entry of $g\otimes g$ in row (i,k) and column (j,l) is $g_{ij}g_{kl}$. The entries are quadratic monomials in $a, \bar a, b, \bar b$. By the vanishing rule for monomials above, only the products $a\bar a$ and $b\bar b$ have nonzero integral. These products occur only in rows and columns $(1,2)$ and $(2,1)$:

$$g_{11}g_{22} = |a|^2, \qquad g_{12}g_{21} = -|b|^2, \qquad g_{21}g_{12} = -|b|^2, \qquad g_{22}g_{11} = |a|^2.$$

Let P be the averaging projector for this action. Writing D for matrices in this basis, we obtain

$$D(P) = \int_{\mathrm{SU}(2)} g\otimes g\,dg = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0\\ 0 & 1 & -1 & 0\\ 0 & -1 & 1 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}.$$

With $\varepsilon = e_1\otimes e_2 - e_2\otimes e_1$ we get $Px = \frac{1}{2}\langle\varepsilon, x\rangle\,\varepsilon$. This is the orthogonal projection onto $\mathbb{C}\varepsilon$. By the invariant-vector projector, Proposition 5.10, $(V_1\otimes V_1)^{\mathrm{SU}(2)} = \mathbb{C}\varepsilon$.

Self-checks

1. Which property of the Haar integral makes the form in the proof of unitarisability, Theorem 5.8, positive definite? Which makes it invariant?

   Answer. Strict positivity gives $\langle v, v\rangle > 0$ for $v \neq 0$. Right invariance gives invariance. Left invariance enters only through the proof of strict positivity.

2. Let π be a nontrivial irreducible representation of a compact Lie group. What is $\int_G \pi(g)\, dg$?

   Answer. It is 0. By the invariant-vector projector, Proposition 5.10, its image is the subrepresentation V^G. Irreducibility gives $V^G = 0$ or $V^G = V$. If $V^G = V$, then every $\pi(g)$ is the identity, so every line in V is invariant. Irreducibility then forces $\dim V = 1$, so π is the trivial representation, contrary to the hypothesis. Hence $V^G = 0$, and the integral is 0.

3. Explain $\int_{\mathrm{SU}(2)} a\, dg = 0$ in two ways.

   Answer. First, in Hopf coordinates the integrand carries the phase $e^{i\xi_1}$, so the integral over ξ_1 vanishes. Second, $a = D^{V_1}_{11}$, and the constant 1 is the matrix coefficient of the trivial representation V_0. Since $V_1 \not\cong V_0$, the relation for inequivalent representations in Corollary 5.13 gives $\int_G \overline{1}\, a\, dg = 0$.

Summary

On a compact Lie group G of positive dimension, left translation of one positive density on $\mathfrak{g}$ gives the Haar integral. Compactness forces $|\det \mathrm{Ad}| = 1$, which gives right invariance. A covering argument by translates gives strict positivity. With these properties, the finite-group proofs of unitarisability, complete reducibility and Schur orthogonality carry over with the sum replaced by the integral. By the matrix-entry relations, Corollary 5.13, the matrix entries of an irreducible unitary representation of dimension d in an orthonormal basis are pairwise L^2-orthogonal, and each has squared norm $1/d$. Completeness of matrix coefficients is the Peter–Weyl theorem of Lecture 10.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (BD), Sections I.5 and II.4.
- Sepanski, *Compact Lie Groups*, §1.4, §2.2 and §3.1.
- Knapp, *Lie Groups Beyond an Introduction*, §VIII.1, for integration and sets of measure zero.

Lecture 6

Compact Structure and Maximal Tori

Objectives. *This lecture explains why conjugacy in a compact connected group can be studied on a torus. First, an averaged inner product splits the Lie algebra. Next, maximal tori meet every conjugacy class. Finally, a finite Weyl group records which elements of the torus are conjugate.*

By the end of this lecture you will be able to:

- Average an inner product and prove $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple.
- Prove that maximal tori are conjugate, explain the strategy of the torus theorem, and deduce that exp is surjective for compact connected groups.
- Compute $N_K(T)/T$ for $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ and match it with the algebraic Weyl group.
- Locate where these statements fail for $\mathrm{SL}(2, \mathbb{R})$.

Throughout, K is a compact Lie group with Lie algebra $\mathfrak{k}$. Its Haar probability measure dg is left and right invariant (Lecture 5). Connectedness of K is assumed only where a statement says so.

6.1 An invariant inner product and the splitting

Lemma 6.1. *There is an inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$ such that for all $g \in K$ and $X, Y, Z \in \mathfrak{k}$*

$$\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle = \langle X, Y \rangle, \qquad \langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle. \tag{6.1}$$

Proof. Choose any inner product $\langle \cdot, \cdot \rangle_0$ on $\mathfrak{k}$ and set

$$\langle X, Y \rangle = \int_K \langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle_0 \, dg.$$

The integrand is continuous in g. So the form is well defined, symmetric and bilinear. For $X \neq 0$ the continuous function $g \mapsto \| \mathrm{Ad}(g)X \|_0^2$ is positive on the compact set K. It has a positive minimum m, so $\langle X, X \rangle \geq m > 0$. Right invariance of dg gives

$$\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle = \int_K \langle \mathrm{Ad}(gh)X, \mathrm{Ad}(gh)Y \rangle_0 \, dg = \langle X, Y \rangle.$$

For the second identity, recall $\mathrm{Ad}(\exp tZ) = e^{t \, \mathrm{ad} Z}$ from Lecture 2. By the first identity, the function $t \mapsto \langle e^{t \, \mathrm{ad} Z} X, e^{t \, \mathrm{ad} Z} Y \rangle$ is constant. By the product rule, its derivative at $t = 0$ is

$$0 = \langle [Z, X], Y \rangle + \langle X, [Z, Y] \rangle.$$

∎

In other words, every $\operatorname{ad} Z$ is skew-symmetric for $\langle \cdot, \cdot \rangle$. From now on we fix such an inner product.

Proposition 6.2. *Let $\mathfrak{z} = \mathfrak{z}(\mathfrak{k})$ be the centre and $\mathfrak{s} = [\mathfrak{k}, \mathfrak{k}]$.*

(a) $\mathfrak{s}^{\perp} = \mathfrak{z}$. Hence $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{s}$, an orthogonal sum of ideals.

(b) The Killing form B of $\mathfrak{s}$ is negative definite. Hence $\mathfrak{s}$ is semisimple.

Proof. (a) For $Z, X, Y \in \mathfrak{k}$, the skew-symmetry identity in (6.1) gives

$$\langle Z, [X, Y] \rangle = -\langle [X, Z], Y \rangle.$$

So $Z \perp \mathfrak{s}$ exactly when $[X, Z] \perp Y$ for all X, Y. This means $[X, Z] = 0$ for all X, that is, $Z \in \mathfrak{z}$. Both $\mathfrak{z}$ and $\mathfrak{s}$ are ideals. Since $\mathfrak{k}$ is finite-dimensional, $\mathfrak{k} = \mathfrak{s}^{\perp} \oplus \mathfrak{s} = \mathfrak{z} \oplus \mathfrak{s}$.

(b) First, $\mathfrak{z}(\mathfrak{s}) = 0$. An element of $\mathfrak{z}(\mathfrak{s})$ commutes with $\mathfrak{s}$ and with $\mathfrak{z}$, hence with $\mathfrak{k}$ by the splitting in (a). So it lies in $\mathfrak{z} \cap \mathfrak{s} = 0$.

Now let $X \in \mathfrak{s}$ and $A = \operatorname{ad} X|_{\mathfrak{s}}$. Since $\mathfrak{s}$ is an ideal, A maps $\mathfrak{s}$ to itself. The map A is skew-symmetric, since every $\operatorname{ad} X$ is. So in an orthonormal basis $e_1, \dots, e_m$ of $\mathfrak{s}$ the matrix of A satisfies $A^{\mathsf{T}} = -A$. Hence

$$B(X, X) = \operatorname{tr}(A^2) = -\operatorname{tr}(A^{\mathsf{T}} A) = -\sum_{i=1}^{m} \|Ae_i\|^2 \le 0.$$

Equality forces $A = 0$, so $X \in \mathfrak{z}(\mathfrak{s}) = 0$. Thus B is negative definite.

The complexification of B is the Killing form of $\mathfrak{s}_{\mathbb{C}}$, and it is nondegenerate. By the complex Cartan criterion from [I], $\mathfrak{s}_{\mathbb{C}}$ is semisimple. A nonzero solvable real ideal of $\mathfrak{s}$ would complexify to a nonzero solvable ideal of $\mathfrak{s}_{\mathbb{C}}$. Hence $\mathfrak{s}$ is semisimple. ∎

For example, let $\mathfrak{k} = \mathfrak{u}(n)$. The form $-\operatorname{tr}(XY)$ satisfies both identities in (6.1) without averaging (Exercise 6.1). The proof of the splitting proposition, Proposition 6.2, uses only these identities, so it applies to this form. Here $\mathfrak{z} = i\mathbb{R}I$ and $\mathfrak{s} = \mathfrak{su}(n)$. The splitting of $X \in \mathfrak{u}(n)$ is

$$X = \tfrac{1}{n}(\operatorname{tr} X)\, I + \Big(X - \tfrac{1}{n}(\operatorname{tr} X)\, I \Big) \in i\mathbb{R}I \oplus \mathfrak{su}(n).$$

6.2 Maximal tori and their conjugacy

Definition 6.3. A *torus* is a compact connected abelian Lie group. A *maximal torus* of K is a torus subgroup $T \subseteq K$ not properly contained in another torus subgroup of K.

Let T be a torus with Lie algebra $\mathfrak{t}$. For $t \in T$ and $Y \in \mathfrak{t}$, differentiating $t \exp(sY) t^{-1} = \exp(sY)$ at $s = 0$ gives $\operatorname{Ad}(t)Y = Y$. Putting $t = \exp(rZ)$ with $Z \in \mathfrak{t}$ and differentiating at $r = 0$ gives $[Z, Y] = 0$. So $\mathfrak{t}$ is abelian, and $\exp(X + Y) = \exp X \exp Y$ on $\mathfrak{t}$ (Lecture 2).

Thus $\exp(\mathfrak{t})$ is a subgroup containing a neighbourhood of e (Lecture 2). Such a subgroup is open. It is also closed, because its complement is a union of open cosets. As T is connected, $\exp\colon \mathfrak{t} \to T$ is surjective.

We cite two further facts (H §11.1; S Ch. 5):

$$T \cong \mathfrak{t}/\Lambda \text{ with } \Lambda = \ker \exp \text{ a lattice}; \qquad \overline{\exp(\mathbb{R}H)} = T \text{ for some } H \in \mathfrak{t}. \tag{6.2}$$

The second is Kronecker's theorem. We call such an H *generic*. Kronecker's theorem also gives $t_0 \in T$ whose powers are dense in T.

Lemma 6.4. *A torus $T \subseteq K$ is maximal if and only if $\mathfrak{t}$ is a maximal abelian subalgebra of $\mathfrak{k}$. Maximal tori exist.*

Proof. Let $\mathfrak{t}$ be maximal abelian, and let $T \subseteq T'$ with T' a torus. Then $\mathfrak{t} \subseteq \mathfrak{t}'$ with $\mathfrak{t}'$ abelian, so $\mathfrak{t} = \mathfrak{t}'$. Hence $T' = \exp(\mathfrak{t}') = \exp(\mathfrak{t}) = T$.

Conversely, let T be maximal, and let $\mathfrak{a}$ be an abelian subalgebra with $\mathfrak{t} \subseteq \mathfrak{a}$. Since $\mathfrak{a}$ is abelian, $\exp(\mathfrak{a})$ is a subgroup, as for $\mathfrak{t}$ above. This subgroup is connected and abelian. Its closure A is a Lie group by the closed subgroup theorem (Lecture 3). The group A is compact, connected and abelian, so it is a torus containing T. Since T is maximal, $A = T$. Hence $\mathfrak{a} \subseteq \mathrm{Lie}(A) = \mathfrak{t}$.

For existence, take a maximal abelian subalgebra $\mathfrak{a}$. As in the second paragraph, $A = \overline{\exp(\mathfrak{a})}$ is a torus. Its Lie algebra is abelian and contains $\mathfrak{a}$, so it equals $\mathfrak{a}$. By the first paragraph, A is a maximal torus. ∎

Theorem 6.5 (Torus theorem; BD §IV.1; S Ch. 5; H §§11.2–11.5)**.** *Let K be compact and connected, and let T be a maximal torus.*

(a) Every $g \in K$ lies in xTx^{-1} for some $x \in K$.

(b) Every maximal torus of K equals xTx^{-1} for some $x \in K$.

Hall (H) proves this for compact matrix Lie groups, and Bröcker–tom Dieck (BD) for general compact connected Lie groups. Part (b), the conjugacy of maximal tori, and a Lie algebra form of part (a) have a short complete proof.

Proposition 6.6. *Let T be a maximal torus of K. For every $X \in \mathfrak{k}$ there is $x \in K$ with $\mathrm{Ad}(x)X \in \mathfrak{t}$. Moreover, any two maximal tori of K are conjugate.*

Proof. Choose a generic $H \in \mathfrak{t}$. Define

$$f \colon K \to \mathbb{R}, \qquad f(x) = \langle \mathrm{Ad}(x)X, H \rangle.$$

It is continuous, so it attains its maximum on the compact group K at some x_0. Put $X_0 = \mathrm{Ad}(x_0)X$ and fix $Y \in \mathfrak{k}$. By Lecture 2,

$$f\big(\exp(sY)x_0\big) = \langle \mathrm{Ad}(\exp sY)X_0, H \rangle = \langle e^{s\,\mathrm{ad}\,Y}X_0, H \rangle.$$

Since f is maximal at x_0, this function of s is maximal at $s = 0$. Hence its derivative $\langle [Y, X_0], H \rangle$ at $s = 0$ vanishes. Using the second identity in (6.1) twice, together with antisymmetry of the bracket, we get

$$0 = \langle [Y, X_0], H \rangle = -\langle X_0, [Y, H] \rangle = \langle X_0, [H, Y] \rangle = -\langle [H, X_0], Y \rangle.$$

Since Y is arbitrary, $[H, X_0] = 0$.

Next we pass from the single element H to all of $\mathfrak{t}$. This step uses genericity of H. Consider

$$S = \{t \in T : \mathrm{Ad}(t)X_0 = X_0\}.$$

It is a closed subgroup of T, since $t \mapsto \mathrm{Ad}(t)X_0$ is continuous. From $[H, X_0] = 0$, every term after the first in the exponential series vanishes. So

$$\mathrm{Ad}(\exp sH)X_0 = e^{s\,\mathrm{ad}\,H}X_0 = X_0 \qquad (s \in \mathbb{R}).$$

Thus S contains $\exp(\mathbb{R}H)$, and hence its closure T. Now let $Z \in \mathfrak{t}$. Differentiating $\mathrm{Ad}(\exp sZ)X_0 = X_0$ at $s = 0$ gives $[Z, X_0] = 0$. So $\mathfrak{t} + \mathbb{R}X_0$ is an abelian subalgebra containing $\mathfrak{t}$. By the characterisation of maximal tori in Lemma 6.4, $\mathfrak{t}$ is maximal abelian, so $X_0 \in \mathfrak{t}$.

Now let T' be a maximal torus with generic element H'. Choose x with $\mathrm{Ad}(x)H' \in \mathfrak{t}$. Then

$$x \exp(sH')x^{-1} = \exp\left(s\,\mathrm{Ad}(x)H'\right) \in T \qquad (s \in \mathbb{R}).$$

Conjugation by x is a homeomorphism and T is closed. Taking closures, $xT'x^{-1} \subseteq T$. The left side is a maximal torus, so equality holds. ∎

Definition 6.7. Let $T \subseteq K$ be a maximal torus. Its *normaliser* is $N_K(T) = \{n \in K : nTn^{-1} = T\}$. The *Weyl group* is $W(K, T) = N_K(T)/T$.

The group $W(K, T)$ acts on T by $nT \cdot t = ntn^{-1}$ and on $\mathfrak{t}$ by $\mathrm{Ad}(n)$. For $n \in N_K(T)$ and $H \in \mathfrak{t}$, the curve $s \mapsto n \exp(sH)n^{-1}$ lies in T. Differentiating it at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. Since $\mathrm{Ad}(n)$ is invertible and $\mathfrak{t}$ is finite-dimensional, $\mathrm{Ad}(n)$ maps $\mathfrak{t}$ onto itself. These actions are well defined on cosets because T is abelian.

Lemma 6.8. *Let T be a maximal torus of K and $N = N_K(T)$. Then the identity component N_0 equals T, and $W(K, T)$ is finite.*

Proof. The group N is closed in K, so it is a compact Lie group (Lecture 3). Let $n \in N$ and $H \in \mathfrak{t}$. Then

$$n \exp(sH)n^{-1} = \exp\left(s\,\mathrm{Ad}(n)H\right) \in T \qquad (s \in \mathbb{R}).$$

Differentiating at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. If $\exp H = e$, the display gives $\exp(\mathrm{Ad}(n)H) = e$. So $A_n = \mathrm{Ad}(n)|_{\mathfrak{t}}$ maps the lattice Λ into itself.

First, N_0 centralises T. Choose a basis $\lambda_1, \dots, \lambda_r$ of $\mathfrak{t}$ inside Λ. Each map $n \mapsto A_n\lambda_i$ is continuous with values in the discrete set Λ. Since N_0 is connected, this map is constant on N_0, equal to its value λ_i at e. So $A_n = I$ for $n \in N_0$. Then $n \exp(H)n^{-1} = \exp H$ for all $H \in \mathfrak{t}$. Since $\exp(\mathfrak{t}) = T$, every $n \in N_0$ commutes with every element of T.

Second, $\mathrm{Lie}(N) = \mathfrak{t}$. Let $X \in \mathrm{Lie}(N)$. Then $\exp(sX) \in N_0$ for all s. By the first step, $\mathrm{Ad}(\exp sX)H = H$ for $H \in \mathfrak{t}$. Differentiating at $s = 0$ gives $[X, H] = 0$. So $\mathfrak{t} + \mathbb{R}X$ is abelian, and $X \in \mathfrak{t}$ because $\mathfrak{t}$ is maximal abelian (Lemma 6.4). The connected group N_0 is generated by $\exp(\mathrm{Lie}\,N)$ (Lecture 2). Hence N_0 is generated by $\exp(\mathfrak{t}) = T$, so $N_0 = T$.

Finally, N_0 is open in N (Lecture 3). The cosets of N_0 form a cover of the compact group N by disjoint open sets. Hence there are finitely many cosets. Since $N_0 = T$, the group $W(K, T) = N/N_0$ is finite. ∎

6.3 An outline of the torus theorem via mapping degree

Since $x\exp(X)x^{-1} = \exp(\mathrm{Ad}(x)X)$, the conjugacy statement of Proposition 6.6 shows that every element of $\exp(\mathfrak{k})$ lies in a conjugate of T. Conversely, every conjugate of an element of $T = \exp(\mathfrak{t})$ is an exponential. So part (a) of the torus theorem, Theorem 6.5(a), is equivalent to surjectivity of exp. Surjectivity of exp needs a global argument. We outline the proof in Hall (H §§11.3–11.5).

For connected K define

$$\Phi\colon K/T \times T \to K, \qquad \Phi(xT, t) = xtx^{-1}.$$

This is well defined because T is abelian. The source $K/T \times T$ and the target K are compact connected manifolds of dimension $\dim K$ (Lecture 3).

1. Let $t_0 \in T$ be an element whose powers are dense in T, as after (6.2). Then
$$\Phi^{-1}(t_0) = \{(nT,\ n^{-1}t_0 n) : nT \in W(K,T)\}.$$
Each listed point maps to t_0. Conversely, let $xtx^{-1} = t_0$. Then $x^{-1}t_0^m x = t^m \in T$ for all $m \in \mathbb{Z}$. Taking closures, $x^{-1}Tx \subseteq T$. Both are maximal tori, so equality holds and $x \in N_K(T)$. The set is finite by the finiteness of the Weyl group, Lemma 6.8, which was proved without the torus theorem.
2. The quotient K/T is orientable. Fix orientations of K/T, T and K, and give $K/T \times T$ the product orientation. At each point of $\Phi^{-1}(t_0)$ the differential of Φ is invertible. Its determinant, computed in oriented bases, has the same sign at all of these points.
3. A point $y \in K$ is a *regular value* of Φ if $d\Phi$ is invertible at every point of $\Phi^{-1}(y)$. By step 2, t_0 is a regular value. A smooth map between compact connected oriented manifolds of equal dimension has an integer degree. It is the number of preimages of any regular value, each counted with the sign of the determinant of $d\Phi$ there.
4. So $\deg\Phi = \pm|W(K,T)| \neq 0$. A point outside the image would be a regular value with no preimages, giving degree 0. Hence Φ is surjective.

The omitted step is the degree theory in steps 2–4. It consists of the definition of degree, its independence of the regular value, the orientation of K/T, and the sign comparison at all preimages.

Step 2 rests on a local computation, shown here at (eT, t). By Lecture 3, $Y \mapsto \exp(Y)T$ identifies $\mathfrak{t}^\perp$ with the tangent space of K/T at eT. We identify the tangent spaces of T and of K at t with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation by t. So $Z \in \mathfrak{t}$ corresponds to the curve $s \mapsto t\exp(sZ)$. For $Y \in \mathfrak{t}^\perp$ and $Z \in \mathfrak{t}$,

$$t^{-1}\Phi\big(\exp(sY)T,\ t\exp(sZ)\big) = \exp\big(s\,\mathrm{Ad}(t^{-1})Y\big)\exp(sZ)\exp(-sY).$$

By the product rule, the derivative at $s = 0$ is $\mathrm{Ad}(t^{-1})Y - Y + Z$. The orthogonal map $\mathrm{Ad}(t^{-1})$ fixes $\mathfrak{t}$ pointwise, since T is abelian. Hence it preserves $\mathfrak{t}^\perp$. In these identifications, the differential is block diagonal in $\mathfrak{t}^\perp \oplus \mathfrak{t}$, with blocks $\mathrm{Ad}(t^{-1})|_{\mathfrak{t}^\perp} - I$ and I. Hence

$$\det d\Phi_{(eT,t)} = \det\big(\mathrm{Ad}(t^{-1}) - I\big)\big|_{\mathfrak{t}^\perp}. \tag{6.3}$$

Hall (H §11.5) shows that this determinant is positive at $t = t_0$, and that the sign is the same at the other preimages of t_0. We do not reproduce that part.

6.4 Surjectivity of exp and the Weyl group

Corollary 6.9. *If K is compact and connected, then $\exp\colon \mathfrak{k} \to K$ is surjective.*

Proof. Let $g \in K$. By the torus theorem, Theorem 6.5(a), $g = xtx^{-1}$ with $x \in K$ and $t \in T$. As $\exp\colon \mathfrak{t} \to T$ is surjective, $t = \exp H$ with $H \in \mathfrak{t}$. Hence

$$g = x \exp(H) x^{-1} = \exp\big(\operatorname{Ad}(x)H\big). \qquad \blacksquare$$

Connectedness is needed. The path $s \mapsto \exp(sX)$ joins e to $\exp X$, so $\exp(\mathfrak{k})$ lies in the identity component. For example, no reflection in $\mathrm{O}(2)$ is an exponential.

The *centraliser* of T is $Z_K(T) = \{g \in K : gt = tg \text{ for all } t \in T\}$. It contains T, since T is abelian.

Theorem 6.10 (H §§11.2–11.5; BD §§IV.1–2)**.** *Let K be compact and connected with maximal torus T. Then $Z_K(T) = T$. Two elements of T are conjugate in K if and only if they lie in one $W(K,T)$-orbit.*

Together with the torus theorem, Theorem 6.5(a), we obtain a bijection

$$T/W(K,T) \longrightarrow \{\text{conjugacy classes of } K\}, \qquad W(K,T)\cdot t \longmapsto \text{class of } t.$$

Surjectivity is the torus theorem. Injectivity is the orbit statement of Theorem 6.10. By Lemma 6.8, the group $W(K,T)$ is finite. So conjugacy in K reduces to the action of a finite group on T.

The Weyl group also has an algebraic description. For $H \in \mathfrak{t}$ the operator $\operatorname{ad} H$ is skew-symmetric for our fixed $\operatorname{Ad}(K)$-invariant inner product. So its complex-linear extension to $\mathfrak{k}_{\mathbb{C}}$ is diagonalisable with imaginary eigenvalues.

A *root* is a nonzero $\alpha \in \mathfrak{t}_{\mathbb{C}}^{*}$ such that some nonzero $V \in \mathfrak{k}_{\mathbb{C}}$ satisfies

$$[H, V] = \alpha(H) V \qquad (H \in \mathfrak{t}_{\mathbb{C}}).$$

Let R be the set of roots.

For $H \in \mathfrak{t}$ the value $\alpha(H)$ is an eigenvalue of $\operatorname{ad} H$, so it is imaginary. Write $\alpha = ia$ with a real on $\mathfrak{t}$, and define $a^{\sharp} \in \mathfrak{t}$ by $\langle a^{\sharp}, H\rangle = a(H)$ for $H \in \mathfrak{t}$. The reflections and the *algebraic Weyl group* are

$$s_\alpha(H) = H - \frac{2a(H)}{\langle a^{\sharp}, a^{\sharp}\rangle}\, a^{\sharp}, \qquad W(R) = \langle s_\alpha : \alpha \in R\rangle \subseteq \mathrm{GL}(\mathfrak{t}).$$

Theorem 6.11 (H §11.7; BD Ch. V)**.** *Let K be compact and connected, with any $\operatorname{Ad}(K)$-invariant inner product on $\mathfrak{k}$. Then $nT \mapsto \operatorname{Ad}(n)|_{\mathfrak{t}}$ is an isomorphism from $W(K,T)$ onto $W(R)$.*

6.5 Unitary groups and a noncompact example

Proposition 6.12. *Let $T \subseteq \mathrm{U}(n)$ be the diagonal unitary matrices. For $\sigma \in S_n$ let $P_\sigma e_j = e_{\sigma(j)}$. Then T is a maximal torus,*

$$N_{\mathrm{U}(n)}(T) = \{DP_\sigma : D \in T,\ \sigma \in S_n\},$$

and $DP_\sigma T \mapsto \sigma$ is an isomorphism $W(\mathrm{U}(n), T) \to S_n$. The action on T is

$$P_\sigma \operatorname{diag}(z_1, \dots, z_n) P_\sigma^{-1} = \operatorname{diag}\left(z_{\sigma^{-1}(1)}, \dots, z_{\sigma^{-1}(n)}\right). \tag{6.4}$$

Proof. The group $T \cong \mathrm{U}(1)^n$ is a torus. Fix $t_* = \operatorname{diag}(c_1, \dots, c_n) \in T$ with distinct entries. A matrix commuting with t_* preserves each eigenline $\mathbb{C}e_j$, so it is diagonal. Hence $Z_{\mathrm{U}(n)}(T) = T$. A torus $T' \supseteq T$ is abelian, so $T' \subseteq Z_{\mathrm{U}(n)}(T) = T$. Thus T is maximal.

Formula (6.4) holds for every $t = \operatorname{diag}(z_1, \dots, z_n) \in T$, because

$$P_\sigma t P_\sigma^{-1} e_{\sigma(j)} = P_\sigma t e_j = z_j e_{\sigma(j)}.$$

So each DP_σ normalises T. Conversely, let $g \in N_{\mathrm{U}(n)}(T)$ and $t' = g t_* g^{-1} \in T$. Then

$$t'(g e_j) = g t_* e_j = c_j\, g e_j.$$

The diagonal matrix t' has the distinct eigenvalues $c_1, \dots, c_n$. So its c_j-eigenspace is a coordinate line $\mathbb{C}e_{\sigma(j)}$. Distinct eigenvalues have distinct eigenlines, so σ is a permutation. Thus $g e_j = d_{\sigma(j)} e_{\sigma(j)}$ with $|d_{\sigma(j)}| = 1$. That is, $g = DP_\sigma$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. Since $P_\sigma P_\tau = P_{\sigma\tau}$ and $P_\sigma D P_\sigma^{-1} \in T$, the map $DP_\sigma T \mapsto \sigma$ is a homomorphism. It is onto, since $P_\sigma \in \mathrm{U}(n)$. It is injective because $DP_\sigma \in T$ only when $\sigma = \mathrm{id}$. ∎

We now compare $W(\mathrm{U}(n), T)$ with the algebraic Weyl group, using the Ad-invariant form $-\operatorname{tr}(XY)$. On $\mathfrak{t} = \{i \operatorname{diag}(a_1, \dots, a_n) : a_j \in \mathbb{R}\}$ this form equals $\sum_j a_j b_j$. Here $\mathfrak{t}_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$. Let E_{jk} be the matrix unit with 1 in row j and column k. For $H = i \operatorname{diag}(a)$ and $j \neq k$,

$$[H, E_{jk}] = i(a_j - a_k) E_{jk}.$$

Let ε_j be the $\mathbb{C}$-linear form on $\mathfrak{t}_{\mathbb{C}}$ with $\varepsilon_j\big(i \operatorname{diag}(a_1, \dots, a_n)\big) = a_j$. The roots are $\alpha_{jk} = i(\varepsilon_j - \varepsilon_k)$ for $j \neq k$. So $a = \varepsilon_j - \varepsilon_k$, with $a^\sharp = i(E_{jj} - E_{kk})$ and $\langle a^\sharp, a^\sharp \rangle = 2$. Then

$$s_{\alpha_{jk}}\big(i \operatorname{diag}(a)\big) = i \operatorname{diag}(a) - (a_j - a_k)\, i(E_{jj} - E_{kk}).$$

This reflection swaps a_j and a_k. By the permutation formula (6.4), the reflection equals $\operatorname{Ad}(P_{(jk)})|_{\mathfrak{t}}$. Transpositions generate S_n. So $W(R)$ is the image of $W(\mathrm{U}(n), T)$, as the algebraic description of the Weyl group in Theorem 6.11 predicts. Hence conjugacy classes of $\mathrm{U}(n)$ correspond to T/S_n. In other words, a unitary matrix is determined up to conjugacy by its eigenvalues with multiplicity.

Corollary 6.13. *Let $n \geq 2$ and $T_1 = T \cap \mathrm{SU}(n)$. Then T_1 is a maximal torus of $\mathrm{SU}(n)$. The normaliser $N_{\mathrm{SU}(n)}(T_1)$ consists of the DP_σ with $D \in T$ and $\det(DP_\sigma) = 1$, and $DP_\sigma T_1 \mapsto \sigma$ is an isomorphism $W(\mathrm{SU}(n), T_1) \to S_n$.*

Proof. The map $(z_1, \dots, z_{n-1}) \mapsto \operatorname{diag}(z_1, \dots, z_{n-1}, (z_1 \cdots z_{n-1})^{-1})$ shows $T_1 \cong \mathrm{U}(1)^{n-1}$. Let $0 < \epsilon < \pi/(n-1)$. The entries $c_j = e^{i\epsilon(2j-n-1)}$ are distinct, and their product is 1 since $\sum_j (2j-n-1) = 0$. The proof of Proposition 6.12, the normaliser computation for $\mathrm{U}(n)$, applies with this t_*. It shows that T_1 is maximal. It also shows that the normaliser consists of the DP_σ with

$$\det(DP_\sigma) = \det D \cdot \operatorname{sgn}\sigma = 1.$$

Every σ occurs, with $D = \operatorname{diag}(\operatorname{sgn}\sigma, 1, \dots, 1)$. The injectivity argument is unchanged. ∎

The roots match again. The Lie algebra $\mathfrak{t}_1$ of T_1 consists of the $i\operatorname{diag}(a)$ with $\sum_j a_j = 0$. Since $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ is spanned by $\mathfrak{t}_{1,\mathbb{C}}$ and the E_{jk}, the roots are the restrictions of the α_{jk}. They are nonzero, because $a^\sharp = i(E_{jj} - E_{kk})$ lies in $\mathfrak{t}_1$ and $a(a^\sharp) = 2$. So $s_{\alpha_{jk}}$ still swaps a_j and a_k. Put $n_{(jk)} = \operatorname{diag}(-1, 1, \dots, 1)P_{(jk)}$, which lies in $N_{\mathrm{SU}(n)}(T_1)$. Diagonal matrices act trivially on $\mathfrak{t}$, so $\operatorname{Ad}(n_{(jk)})|_{\mathfrak{t}_1} = s_{\alpha_{jk}}$. Hence $W(R)$ is the image of $W(\mathrm{SU}(n), T_1) \cong S_n$.

For $\mathrm{SU}(2)$ the transposition is represented by

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \operatorname{diag}(-1, 1)P_{(12)}, \qquad w \operatorname{diag}(z, \bar{z}) w^{-1} = \operatorname{diag}(\bar{z}, z).$$

By the bijection between $T_1/W(\mathrm{SU}(2), T_1)$ and conjugacy classes, every conjugacy class of $\mathrm{SU}(2)$ contains exactly one matrix $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ with $\theta \in [0, \pi]$.

Now compare the connected noncompact group $\mathrm{SL}(2, \mathbb{R})$. Its connectedness was shown in Lecture 2 (Exercise 2.4). By the computation in Lecture 2, the element

$$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

is not in $\exp(\mathfrak{sl}(2, \mathbb{R}))$. So the surjectivity statement of Corollary 6.9 fails for this connected noncompact group. The lines $\mathbb{R}\operatorname{diag}(1, -1)$ and $\mathfrak{so}(2)$ are both maximal abelian. Their nonzero elements have real and imaginary eigenvalues respectively, so they are not Ad-conjugate.

The proof of the conjugacy statement in Proposition 6.6 breaks at the invariant inner product of Lemma 6.1. For $H = \operatorname{diag}(1, -1)$ the operator $\operatorname{ad} H$ has eigenvalues $2, 0, -2$, but a skew-symmetric operator has imaginary eigenvalues. Averaging is unavailable because $\mathrm{SL}(2, \mathbb{R})$ is not compact. Exercise 6.4 gives the details.

6.6 Finite central quotients

The next lecture uses the following global form of the splitting.

Theorem 6.14 (BD §V.8). *Let K be compact and connected. Let $T_0 = Z(K)_0$ be the identity component of the centre. It is a torus with Lie algebra $\mathfrak{z}(\mathfrak{k})$. Let K_{sc} be the simply connected Lie group with Lie algebra $[\mathfrak{k}, \mathfrak{k}]$ (Lecture 4). Then K_{sc} is compact, and the inclusion $[\mathfrak{k}, \mathfrak{k}] \to \mathfrak{k}$ integrates to a homomorphism $\varphi \colon K_{\mathrm{sc}} \to K$. The map*

$$\mu \colon T_0 \times K_{\mathrm{sc}} \to K, \qquad \mu(z, y) = z\,\varphi(y),$$

is a surjective homomorphism with finite central kernel F. Hence $K \cong (T_0 \times K_{\mathrm{sc}})/F$.

Compactness of K_{sc} is Weyl's theorem and is part of the citation.

For example, for $K = \mathrm{U}(n)$ we have $T_0 = \{zI : |z| = 1\}$ and $K_{\mathrm{sc}} = \mathrm{SU}(n)$, which is simply connected (H Ch. 13). Here $\mu(z, A) = zA$. If $zA = I$, then $A = z^{-1}I$ and $\det A = z^{-n} = 1$. Hence

$$F = \{(\zeta, \zeta^{-1}I) : \zeta^n = 1\} \cong \mathbb{Z}/n, \qquad \mathrm{U}(n) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(n)\big)/F.$$

Exercise 6.5 checks surjectivity for $n = 2$.

Self-checks

1. Split
$$X = \begin{pmatrix} 3i & 1 \\ -1 & i \end{pmatrix} \in \mathfrak{u}(2)$$
along $\mathfrak{z} \oplus \mathfrak{su}(2)$.
Answer. The centre component is $\frac{1}{2}(\operatorname{tr} X)I = 2iI$. The remainder is skew-Hermitian with trace zero:
$$X - 2iI = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}.$$
2. Let T be the diagonal torus of $\mathrm{U}(n)$. Without using the torus theorem, prove that every element of $\mathrm{U}(n)$ lies in a conjugate of T, as in Theorem 6.5(a).
Answer. A unitary U is normal. The spectral theorem for normal matrices gives a unitary V such that $V^{-1}UV$ is diagonal. The matrix $V^{-1}UV$ is also unitary, so it lies in T. Hence $U \in VTV^{-1}$.
3. Are $\operatorname{diag}(i, -i)$ and $\operatorname{diag}(-i, i)$ conjugate in $\mathrm{SU}(2)$? Are $\operatorname{diag}(i, -i)$ and I?
Answer. Yes. With w as above, $w \operatorname{diag}(i, -i)w^{-1} = \operatorname{diag}(-i, i)$. The second pair is not conjugate. Conjugate matrices have equal traces, but the traces of $\operatorname{diag}(i, -i)$ and I are 0 and 2.

Summary

An averaged inner product gives the central and semisimple splitting. Maximal tori are conjugate. For connected K, the cited global torus theorem makes exp surjective. Together with the cited orbit theorem for $W(K, T)$, it identifies the conjugacy classes of K with $T/W(K, T)$. The Weyl group is finite. For $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ it is isomorphic to S_n. Compactness is essential, as the connected group $\mathrm{SL}(2, \mathbb{R})$ shows.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, §§11.1–11.5 and 11.7.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1–3 and V.8.
- S: Sepanski, *Compact Lie Groups*, Chapter 5. [I]: prerequisite Lie Algebras.

Lecture 7

Lattices, centres and global forms

Objectives. *The groups* $\mathrm{SU}(2)$ *and* $\mathrm{SO}(3)$ *have isomorphic Lie algebras but are not isomorphic. This lecture asks which discrete data determine connected groups with the same Lie algebra. For compact groups, the answer is the kernel lattice* Λ *of a maximal torus, inside its Lie algebra* $\mathfrak{t}$*, read together with the root data.*

The lecture proves this for semisimple Lie algebras, using cited structure results. Different lattices can still give isomorphic groups, through automorphisms of the Lie algebra that preserve $\mathfrak{t}$*. When there is a central torus, the lecture shows how* Λ *determines a finite group* F*.*

By the end of this lecture you will be able to:

- Prove that a compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, with characters the ℓ satisfying $\ell(\Lambda) \subset 2\pi\mathbb{Z}$.
- Place Λ between the coroot and coweight lattices, and compute centres.
- Describe the connected compact groups (global forms) with a given semisimple Lie algebra $\mathfrak{k}$, and the groups $(T_0 \times K_{sc})/F$.
- Decide which representations descend to $\mathrm{SO}(3)$, $\mathrm{PSU}(3)$ and $\mathrm{U}(2)$.

The Lie groups denoted K or T in this lecture, with or without subscripts, are connected. The subgroups C, F and $Z(K)$ that appear later need not be connected. A Lie group and its identity component have the same Lie algebra, so we restrict to connected groups. Even connected groups with isomorphic Lie algebras need not be isomorphic, as $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ show.

7.1 Compact connected abelian groups

Definition 7.1. A *torus* is a Lie group isomorphic to $\mathrm{U}(1)^n$ for some $n \geq 0$. A *lattice* in a real vector space V is a subgroup $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_n$, where $v_1, \ldots, v_n$ is a basis of V.

Lemma 7.2 (Discrete subgroups)**.** *Every discrete subgroup of a finite-dimensional real vector space equals* $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_k$ *for some linearly independent* $v_1, \ldots, v_k$*.*

We cite this standard lemma from BD, Section I.3, Lemma (3.8).

Theorem 7.3. *Let* T *be a compact connected abelian Lie group with Lie algebra* $\mathfrak{t}$ *of dimension* n*. Then* $\exp\colon \mathfrak{t} \to T$ *is a surjective homomorphism, and* $\Lambda = \ker \exp$ *is a lattice*

in $\mathfrak{t}$. *For a basis* $v_1, \dots, v_n$ *of* Λ, *the map*

$$\Phi\colon \mathrm{U}(1)^n \to T, \qquad \Phi\big(e^{2\pi i x_1}, \dots, e^{2\pi i x_n}\big) = \exp(x_1 v_1 + \cdots + x_n v_n),$$

is an isomorphism of Lie groups.

Proof. Since T is abelian, so is $\mathfrak{t}$. Commuting elements satisfy $\exp(X+Y) = \exp X \exp Y$ (Lecture 2), so exp is a homomorphism. Choose an open set $U = -U$ containing 0 that exp maps diffeomorphically onto an open set (Lecture 2).

Surjectivity. The subgroup $\exp(\mathfrak{t})$ contains $g\exp(U)$ for each of its elements g, so it is open. Its complement is a union of open cosets, so it is also closed. Since T is connected, $\exp(\mathfrak{t}) = T$.

Discreteness. As exp is injective on U, we have $\Lambda \cap U = \{0\}$. If $H, H' \in \Lambda$ and $H' \in H + U$, then $H' - H \in \Lambda \cap U$, so $H' = H$. Hence Λ is discrete. The discrete-subgroup lemma, Lemma 7.2, gives independent generators $v_1, \dots, v_k$ of Λ.

Full rank. Suppose $k < n$. Choose a nonzero linear map $p\colon \mathfrak{t} \to \mathbb{R}$ with $p(v_1) = \cdots = p(v_k) = 0$, and define

$$f\colon T \to \mathbb{R}, \qquad f(\exp H) = p(H).$$

It is well defined because $p(\Lambda) = 0$. It is continuous: near $\exp H_1$ we have $f(\exp H_1 \exp Y) = p(H_1) + p(Y)$ for $Y \in U$, and $Y \mapsto \exp H_1 \exp Y$ is a homeomorphism of U onto a neighbourhood of $\exp H_1$. So $\mathbb{R} = p(\mathfrak{t}) = f(T)$ is compact, which is false. Hence $k = n$.

Isomorphism. Changing x_j by an integer changes $\sum_j x_j v_j$ by an element of Λ. So Φ is a well-defined homomorphism. It is continuous, because near any point it is the composite of a local inverse of $x \mapsto (e^{2\pi i x_j})_j$ with the continuous map $x \mapsto \exp(\sum_j x_j v_j)$. It is bijective because $\ker\exp = \bigoplus_j \mathbb{Z} v_j$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Continuous homomorphisms of Lie groups are smooth (Lecture 3), so Φ and Φ^{-1} are smooth. ∎

7.2 Characters and cocharacters

Definition 7.4. A *character* of a torus T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. A *cocharacter* is a continuous homomorphism $\lambda\colon \mathrm{U}(1) \to T$. Under pointwise multiplication, the characters form a group $X^*(T)$ and the cocharacters form a group $X_*(T)$.

These characters are one-dimensional. Trace characters return in Lecture 9.

Theorem 7.5 (Torus character criterion). *Let* T *be a torus with* $\Lambda = \ker\exp$, *and let* $\ell \in \mathfrak{t}^*$. *There is a character* χ_ℓ *with*

$$\chi_\ell(\exp H) = e^{i\ell(H)} \qquad (H \in \mathfrak{t})$$

if and only if $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. *Every character equals* χ_ℓ *for exactly one such* ℓ, *and* $d\chi_\ell = i\ell$. *Hence* $\ell \mapsto \chi_\ell$ *is an isomorphism from* $\{\ell \in \mathfrak{t}^* : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ *onto* $X^*(T)$.

Proof. Suppose $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. The homomorphism $H \mapsto e^{i\ell(H)}$ from $\mathfrak{t}$ to $\mathrm{U}(1)$ is constant on cosets of Λ. Since exp is onto with kernel Λ, it defines a homomorphism χ_ℓ on T. Continuity follows as for f above. Conversely, if χ_ℓ exists, then $1 = \chi_\ell(\exp H) = e^{i\ell(H)}$ for $H \in \Lambda$.

Now let χ be any character. It is smooth (Lecture 3). Its differential maps $\mathfrak{t}$ to $\operatorname{Lie}\mathrm{U}(1) = i\mathbb{R}$, so $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$. Naturality of exp (Lecture 2) gives

$$\chi(\exp H) = \exp\bigl(d\chi(H)\bigr) = e^{i\ell(H)}.$$

For $H \in \Lambda$ the left side is $\chi(1) = 1$, so $\ell(H) \in 2\pi\mathbb{Z}$. Thus $\chi = \chi_\ell$. The form ℓ is unique, because $i\ell(H)$ is the derivative of $s \mapsto \chi(\exp sH)$ at $s = 0$. Finally, $\chi_\ell\chi_{\ell'} = \chi_{\ell+\ell'}$ by the formula. ∎

For $H \in \Lambda$ define the cocharacter

$$\lambda_H(e^{i\theta}) = \exp\Bigl(\frac{\theta}{2\pi}H\Bigr).$$

Replacing θ by $\theta+2\pi$ multiplies the right side by $\exp H = 1$, so λ_H is well defined. Exercise 7.1 shows that $H \mapsto \lambda_H$ is an isomorphism $\Lambda \to X_*(T)$. Since $\chi_\ell(\lambda_H(e^{i\theta})) = e^{i\theta\ell(H)/2\pi}$,

$$\chi_\ell\bigl(\lambda_H(z)\bigr) = z^{\langle \ell, H\rangle}, \qquad \langle \ell, H\rangle = \frac{\ell(H)}{2\pi} \in \mathbb{Z}.$$

So $X^*(T)$ is the dual lattice of $X_*(T) \cong \Lambda$. From now on we identify $X^*(T)$ with $\{\ell : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ and $X_*(T)$ with Λ.

A finite-dimensional representation of T is unitary for an averaged inner product (Lecture 5). The operators of the representation commute and are unitary, so they diagonalise simultaneously. Hence the representation space is a direct sum of lines on which $\exp H$ acts by $e^{i\ell(H)}$. These $\ell \in X^*(T)$ are the *weights* of the representation.

Example 7.6. Let $T \subset \mathrm{U}(n)$ be the diagonal torus. Then $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, \dots, i\theta_n) : \theta \in \mathbb{R}^n\}$ and $\Lambda = \{2\pi i \operatorname{diag}(k_1, \dots, k_n) : k \in \mathbb{Z}^n\}$. Put $\ell_m(\operatorname{diag}(i\theta)) = \sum_j m_j\theta_j$. Then $\ell_m \in X^*(T)$ if and only if $m \in \mathbb{Z}^n$, and

$$\chi_{\ell_m}\bigl(\operatorname{diag}(z_1, \dots, z_n)\bigr) = z_1^{m_1} \cdots z_n^{m_n}.$$

For $H = 2\pi i \operatorname{diag}(k)$ we get $\lambda_H(z) = \operatorname{diag}(z^{k_1}, \dots, z^{k_n})$ and $\langle \ell_m, H\rangle = \sum_j m_j k_j$.

7.3 Roots, coroots and the centre

Let K be a compact connected Lie group with maximal torus T (Lecture 6). We import the root data. The real roots $\alpha \in \Delta \subset \mathfrak{t}^*$ are defined by

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha\in\Delta} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}(\exp H)X = e^{i\alpha(H)}X \quad (X \in \mathfrak{g}_\alpha,\ H \in \mathfrak{t}).$$

Each root α has a coroot H_α. It is the unique element

$$H_\alpha \in \mathfrak{t} \cap [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \qquad \text{with} \qquad \alpha(H_\alpha) = 2.$$

The condition $\alpha(H) = 2$ alone does not fix H once $\dim\mathfrak{t} \geq 2$, because then $\ker\alpha \neq 0$. There is a homomorphism $\varphi_\alpha\colon \mathrm{SU}(2) \to K$ with $d\varphi_\alpha(H_0) = H_\alpha$, where $H_0 = \operatorname{diag}(i, -i)$. Its

differential is the compact form of a root $\mathfrak{sl}_2$-triple (H, Chapter 11). It integrates because SU(2) is simply connected (Lecture 4).

Definition 7.7. Define subgroups Q and P of $\mathfrak{t}^*$, and $Q^\vee$ and $P^\vee$ of $\mathfrak{t}$, by

$$Q = \mathbb{Z}\Delta, \qquad P = \{\ell \in \mathfrak{t}^* : \ell(H_\alpha) \in \mathbb{Z} \text{ for all } \alpha\},$$
$$Q^\vee = \mathbb{Z}\{2\pi H_\alpha : \alpha \in \Delta\}, \quad P^\vee = \{H \in \mathfrak{t} : \alpha(H) \in 2\pi\mathbb{Z} \text{ for all } \alpha\}.$$

The background coroot h_α spans the line $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, and $\mathfrak{t} \cap \mathbb{C}h_\alpha = \mathbb{R}H_\alpha$. With $\mu = i\ell$ and $h_\alpha = -iH_\alpha$ we get $\mu(h_\alpha) = \ell(H_\alpha)$. When $\mathfrak{k}$ is semisimple, P is the background weight lattice, written in real form.

The factors 2π in $Q^\vee$ and $P^\vee$ fit the convention $\Lambda = \ker\exp$, in which characters satisfy $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. Some texts use $\{H : \exp(2\pi H) = 1\}$ in place of Λ, and write $\chi(\exp 2\pi H) = e^{2\pi i\ell(H)}$. This divides the lattices Λ, $Q^\vee$ and $P^\vee$ in $\mathfrak{t}$ by 2π. The weight lattices $X^*(T)$, Q and P are unchanged.

Proposition 7.8 (Lattice chain)**.**

$$Q^\vee \subset \Lambda \subset P^\vee, \qquad Q \subset X^*(T) \subset P.$$

Proof. Let $H \in \Lambda$ and $0 \neq X \in \mathfrak{g}_\alpha$. Then $X = \mathrm{Ad}(\exp H)X = e^{i\alpha(H)}X$, so $\alpha(H) \in 2\pi\mathbb{Z}$. Hence $\Lambda \subset P^\vee$. The same computation shows $\alpha \in X^*(T)$, so $Q \subset X^*(T)$. Next, naturality of exp gives

$$\exp(2\pi H_\alpha) = \varphi_\alpha\big(\exp(2\pi H_0)\big) = \varphi_\alpha\begin{pmatrix} e^{2\pi i} & 0 \\ 0 & e^{-2\pi i}\end{pmatrix} = \varphi_\alpha(I) = 1.$$

So $Q^\vee \subset \Lambda$. Finally, if $\ell \in X^*(T)$, then $\ell(2\pi H_\alpha) \in 2\pi\mathbb{Z}$, so $\ell(H_\alpha) \in \mathbb{Z}$ and $\ell \in P$. ∎

Theorem 7.9 (Centre)**.** *Let K be compact connected with maximal torus T. Then $Z(K) \subset T$, and* exp *restricts to a surjective homomorphism $P^\vee \to Z(K)$ with kernel Λ. Thus $Z(K) \cong P^\vee/\Lambda$.*

Proof. Let $z \in Z(K)$. By Lecture 6, $z = gtg^{-1}$ for some $g \in K$ and $t \in T$. Since z is central, $t = g^{-1}zg = z$, so $z \in T$.

Next we show that $g \in K$ is central if and only if $\mathrm{Ad}(g) = \mathrm{id}$. If g is central, differentiating $g\exp(sX)g^{-1} = \exp(sX)$ at $s = 0$ gives $\mathrm{Ad}(g) = \mathrm{id}$. If $\mathrm{Ad}(g) = \mathrm{id}$, then $g\exp(X)g^{-1} = \exp(\mathrm{Ad}(g)X) = \exp X$. The elements $\exp X$ generate the connected group K (Lecture 2), so g is central.

Now write $z = \exp H$ with $H \in \mathfrak{t}$, using the surjectivity of exp in the torus classification, Theorem 7.3. The operator $\mathrm{Ad}(\exp H) = e^{\mathrm{ad}\,H}$ is the identity on $\mathfrak{t}_\mathbb{C}$ and multiplication by $e^{i\alpha(H)}$ on $\mathfrak{g}_\alpha$. Hence, by the criterion just proved, $\exp H$ is central exactly when $H \in P^\vee$. The kernel of exp on $P^\vee$ is $\Lambda \cap P^\vee = \Lambda$. ∎

For the fundamental group we cite the following theorem.

Theorem 7.10 (Fundamental group; H, Chapter 13; BD, Section V.7)**.** *Let K be a compact connected Lie group with maximal torus T. Then*

$$\pi_1(K) \cong \Lambda/Q^\vee = X_*(T)/Q^\vee.$$

Call K *semisimple* if $\mathfrak{k}$ is semisimple. Then Δ spans $\mathfrak{t}^*$, and the simple roots form a basis of $\mathfrak{t}^*$. The simple coroots form a basis of $\mathfrak{t}$ (background, via $h_\alpha = -iH_\alpha$). So all four subgroups are lattices, and the inclusions above have finite index. Weyl's theorem states that a compact connected semisimple group has finite fundamental group and compact universal cover (H, Chapter 13; Knapp, Chapter IV).

7.4 Global forms

Semisimple groups

By Weyl's theorem, the universal cover of our compact connected semisimple K is compact. Let K_{sc} be a compact, connected, simply connected, semisimple Lie group with Lie algebra $\mathfrak{k}$ and maximal torus T_{sc}. The fundamental group theorem, Theorem 7.10, gives $\Lambda_{sc} = Q^\vee$. The torus character criterion and the centre theorem, Theorems 7.5 and 7.9, then give

$$X^*(T_{sc}) = P, \qquad Z(K_{sc}) \cong P^\vee/Q^\vee.$$

Proposition 7.11 (Forms of a semisimple algebra)**.** *(a) Every connected Lie group with Lie algebra $\mathfrak{k}$ is isomorphic to K_{sc}/C for a subgroup $C \subset Z(K_{sc})$.*

(b) For $K = K_{sc}/C$, the image T of T_{sc} is a maximal torus of K, with kernel lattice

$$\Lambda_C = \{H \in \mathfrak{t} : \exp_{sc} H \in C\}, \qquad \Lambda_C/Q^\vee \cong C.$$

The map $C \mapsto \Lambda_C$ is a bijection from subgroups of $Z(K_{sc})$ to lattices Λ with $Q^\vee \subset \Lambda \subset P^\vee$.

(c) $X^*(T) = \{\ell \in P : \chi_\ell(c) = 1 \text{ for all } c \in C\}$.

Proof. (a) By Lecture 4, such a group is isomorphic to K_{sc}/C with C discrete and normal. Such a subgroup C is central.

(b) First, T is a maximal torus with kernel lattice Λ_C. The covering map $p\colon K_{sc} \to K$ is a local isomorphism. So $\operatorname{Lie} T = \mathfrak{t}$ and $\exp_K = p \circ \exp_{sc}$. Hence $\exp_K H = 1$ exactly when $\exp_{sc} H \in C$. A larger torus would have a larger abelian Lie algebra, but $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$ (Lecture 6). So T is maximal.

Next, we prove $Q^\vee \subset \Lambda_C \subset P^\vee$ and $\Lambda_C/Q^\vee \cong C$. Since $C \subset Z(K_{sc})$, the centre theorem, Theorem 7.9, gives $Q^\vee \subset \Lambda_C \subset P^\vee$. The map $\exp_{sc}\colon \Lambda_C \to C$ is onto, since $C \subset T_{sc} = \exp_{sc}(\mathfrak{t})$. Its kernel is $Q^\vee$, so $\Lambda_C/Q^\vee \cong C$.

Conversely, let $Q^\vee \subset \Lambda \subset P^\vee$, and put $C = \exp_{sc}(\Lambda)$. By the centre theorem, Theorem 7.9, C is a subgroup of $Z(K_{sc})$. Clearly $\Lambda \subset \Lambda_C$. For the reverse inclusion, let $H \in \Lambda_C$. Then $\exp_{sc} H \in C = \exp_{sc}(\Lambda)$, so $\exp_{sc} H = \exp_{sc} H'$ for some $H' \in \Lambda$. Hence $H - H'$ lies in the kernel of $\exp_{sc}$ on $\mathfrak{t}$, which is $Q^\vee \subset \Lambda$. So $H \in \Lambda$, and $\Lambda_C = \Lambda$. We also showed $\exp_{sc}(\Lambda_C) = C$ for every C. So $C \mapsto \Lambda_C$ and $\Lambda \mapsto \exp_{sc}(\Lambda)$ are inverse to each other.

(c) A character of $T = T_{sc}/C$ is a character of T_{sc} that is trivial on C. ∎

Remark 7.12. Distinct lattices can give isomorphic groups. Let σ be an automorphism of $\mathfrak{k}$ with $\sigma(\mathfrak{t}) = \mathfrak{t}$. It integrates to an automorphism Σ of K_{sc} (Lecture 4). Since $\Sigma(\exp H) = \exp(\sigma H)$, we get $\Sigma(Z(K_{sc})) = Z(K_{sc})$ and $\Lambda_{\Sigma(C)} = \sigma(\Lambda_C)$. Hence $K_{sc}/C \cong K_{sc}/\Sigma(C)$. Exercise 7.5 gives an example. We do not discuss the converse.

The *simply connected form* is $C = 1$, with $\Lambda = Q^\vee$ and $X^*(T) = P$. The *adjoint form* is $K_{ad} = K_{sc}/Z(K_{sc})$, with $\Lambda = P^\vee$. Its centre $P^\vee/P^\vee$ is trivial by the centre theorem, Theorem 7.9. Write T_{ad} for the image of T_{sc} in K_{ad}. Every root is an integer combination of the simple roots α_i (background). So the lattice $P^\vee$ is defined by the conditions $\alpha_i(H) \in 2\pi\mathbb{Z}$ for the simple roots alone. Since the simple roots form a basis of $\mathfrak{t}^*$, we get $X^*(T_{ad}) = \mathbb{Z}\{\alpha_i\} = Q$.

Let V be the irreducible representation of K_{sc} with dominant highest weight $\lambda \in P$. It exists by the background and Lecture 4. By Schur's lemma, a central element c acts on V by a scalar. On a highest weight vector this scalar is $\chi_\lambda(c)$. By Lecture 4, V descends to K_{sc}/C if and only if

$$\chi_\lambda(c) = 1 \qquad \text{for all } c \in C.$$

Groups with a central torus

If $\mathfrak{k}$ has nonzero centre, then $Z(K)$ is infinite and Δ does not span $\mathfrak{t}^*$. Moreover, $P^\vee$ contains the centre of $\mathfrak{k}$, so $P^\vee$ is not a lattice. The proposition on forms of a semisimple algebra, Proposition 7.11, does not apply. We use a structure theorem instead.

Theorem 7.13 (Global structure; BD, Section V.8; Knapp, Chapter IV)**.** *Let K be compact connected. Let T_0 be the identity component of $Z(K)$, with Lie algebra $\mathfrak{t}_0$. Let K_{sc} be the simply connected group with Lie algebra $[\mathfrak{k},\mathfrak{k}]$. Then T_0 is a torus, $\mathfrak{k} = \mathfrak{t}_0 \oplus [\mathfrak{k},\mathfrak{k}]$, the ideal $[\mathfrak{k},\mathfrak{k}]$ is semisimple, and K_{sc} is compact. Let $\psi\colon K_{sc} \to K$ integrate the inclusion of $[\mathfrak{k},\mathfrak{k}]$. Then*

$$T_0 \times K_{sc} \to K, \qquad (t,g) \mapsto t\,\psi(g),$$

is a surjective homomorphism with finite kernel F. Thus $K \cong (T_0 \times K_{sc})/F$.

Proposition 7.14. *In the global structure theorem, Theorem 7.13, let T_{sc} be a maximal torus of K_{sc} with Lie algebra $\mathfrak{t}_{sc}$, and let T be the image of $T_0 \times T_{sc}$. Identify $\mathfrak{t}^* = \mathfrak{t}_0^* \oplus \mathfrak{t}_{sc}^*$, and let $P_{sc} \subset \mathfrak{t}_{sc}^*$ be the weight lattice of K_{sc}. Then $T \cong (T_0 \times T_{sc})/F$ is a maximal torus of K, and*

$$X^*(T) = \big\{(\ell_0,\ell) \in X^*(T_0) \times P_{sc} : \chi_{\ell_0}(t)\,\chi_\ell(g) = 1 \text{ for all } (t,g) \in F\big\}.$$

Proof. A finite normal subgroup of a connected group is central (Lecture 4). So $F \subset T_0 \times Z(K_{sc}) \subset T_0 \times T_{sc}$ by the centre theorem, Theorem 7.9. Hence $T \cong (T_0 \times T_{sc})/F$. Its Lie algebra $\mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$ is maximal abelian in $\mathfrak{k}$, because $\mathfrak{t}_{sc}$ is maximal abelian in $[\mathfrak{k},\mathfrak{k}]$. So T is a maximal torus.

Each root space satisfies $\mathfrak{g}_\alpha = [\mathfrak{t},\mathfrak{g}_\alpha] \subset [\mathfrak{k},\mathfrak{k}]_{\mathbb{C}}$. So $H_\alpha \in \mathfrak{t} \cap [\mathfrak{k},\mathfrak{k}] = \mathfrak{t}_{sc}$, and the coroots of K are those of K_{sc}. Thus $Q^\vee \subset \mathfrak{t}_{sc}$ is the kernel lattice of T_{sc}. Put

$$\Lambda_0 = \ker\big(\exp|_{\mathfrak{t}_0}\big) = \Lambda \cap \mathfrak{t}_0.$$

The torus $T_0 \times T_{sc}$ has kernel lattice $\Lambda_0 \oplus Q^\vee$. By the torus character criterion, Theorem 7.5, its characters are the pairs with $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$ and $\ell \in P_{sc}$. The characters of T are those trivial on F. ∎

In other words, (ℓ_0, ℓ) is a character of T exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product $\chi_{\ell_0}\chi_\ell$ is trivial on F.

The group F is determined by Λ. The map $T_0 \times T_{sc} \to T$ has differential the identity of $\mathfrak{t} = \mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$. By the torus classification, Theorem 7.3,

$$T_0 \times T_{sc} \cong \mathfrak{t}/(\Lambda_0 \oplus Q^\vee), \qquad T \cong \mathfrak{t}/\Lambda.$$

The map is induced by the identity of $\mathfrak{t}$, so its kernel is

$$F \cong \Lambda/(\Lambda_0 \oplus Q^\vee).$$

7.5 Examples

SU(2) and SO(3)

Take $\mathfrak{t} = \mathbb{R}H_0$. Since $\exp(sH_0) = \operatorname{diag}(e^{is}, e^{-is})$, we have $\Lambda = 2\pi\mathbb{Z}H_0$. Since $[H_0, E_{12}] = 2iE_{12}$, the positive root has $\alpha(H_0) = 2$. So $H_\alpha = H_0$ and $Q^\vee = \Lambda$. Identify ℓ with $m = \ell(H_0)$. Then

$$P = \mathbb{Z}, \qquad Q = 2\mathbb{Z}, \qquad P^\vee = \pi\mathbb{Z}H_0, \qquad Z(\mathrm{SU}(2)) = \exp(\pi\mathbb{Z}H_0) = \{\pm I\}.$$

The adjoint form is $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). It has $\Lambda = \pi\mathbb{Z}H_0$ and

$$X^*(T) = \{m : m\pi \in 2\pi\mathbb{Z}\} = 2\mathbb{Z} = Q, \qquad \pi_1(\mathrm{SO}(3)) \cong \pi\mathbb{Z}/2\pi\mathbb{Z} \cong \mathbb{Z}/2.$$

The module V_m has dimension $m+1$ and weights $m, m-2, \dots, -m$. Physicists call it the spin-$m/2$ representation. Since $-I = \exp(\pi H_0)$, the element $-I$ acts on V_m by $e^{i\pi m} = (-1)^m$. So V_m descends to $\mathrm{SO}(3)$ exactly when m is even.

SU(3) and PSU(3)

For the diagonal torus of $\mathrm{SU}(3)$,

$$\mathfrak{t} = \Big\{\operatorname{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum_j \theta_j = 0\Big\}, \qquad \Lambda = \Big\{2\pi i \operatorname{diag}(k) : k \in \mathbb{Z}^3,\ \sum_j k_j = 0\Big\}.$$

The roots are $\alpha_{jk}(H) = \theta_j - \theta_k$ for $j \neq k$, with coroots $H_{jk} = i(E_{jj} - E_{kk})$. Indeed $[E_{jk}, E_{kj}] = E_{jj} - E_{kk}$ and $\alpha_{jk}(H_{jk}) = 2$. Hence $Q^\vee = \Lambda$. Write $\ell_m(H) = \sum_j m_j\theta_j$ with $m \in \mathbb{R}^3$. Adding a multiple of $(1,1,1)$ to m does not change ℓ_m, since $\sum_j \theta_j = 0$. The condition $\ell_m(\Lambda) \subset 2\pi\mathbb{Z}$ means $m_j - m_k \in \mathbb{Z}$, and this is also the condition for P.

Such a form has a representative $m \in \mathbb{Z}^3$, obtained by subtracting $m_3(1,1,1)$. Two integer representatives differ by an element of $\mathbb{Z}(1,1,1)$, which changes $m_1 + m_2 + m_3$ by a multiple of 3. Using integer representatives,

$$X^*(T) = P \cong \mathbb{Z}^3/\mathbb{Z}(1,1,1),$$
$$Q = \{[m] : m \in \mathbb{Z}^3,\ m_1 + m_2 + m_3 \equiv 0 \bmod 3\}, \qquad P/Q \cong \mathbb{Z}/3.$$

An element of $P^\vee$ has $\theta_j = \theta + 2\pi k_j$ with $3\theta \in 2\pi\mathbb{Z}$. So $Z(\mathrm{SU}(3)) = \{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$. The adjoint form $\mathrm{PSU}(3) = \mathrm{SU}(3)/Z(\mathrm{SU}(3))$ has $X^*(T) = Q$, trivial centre and $\pi_1 \cong P^\vee/Q^\vee \cong \mathbb{Z}/3$.

Use the simple roots α_{12}, α_{23}, with simple coroots H_{12}, H_{23}. The fundamental weights are then $\varpi_1 = \ell_{(1,0,0)}$ and $\varpi_2 = \ell_{(1,1,0)}$. So the highest weight $a\varpi_1 + b\varpi_2$ is $\ell_{(a+b,b,0)}$. Write $V(a,b)$ for the irreducible module with this highest weight. Take

$$H_\omega = \mathrm{diag}\Big(\frac{2\pi i}{3}, \frac{2\pi i}{3}, -\frac{4\pi i}{3}\Big), \qquad \exp H_\omega = \mathrm{diag}(\omega, \omega, \omega^{-2}) = \omega I.$$

Then $\ell_{(a+b,b,0)}(H_\omega) = 2\pi(a+2b)/3$, so ωI acts on $V(a,b)$ by ω^{a+2b}. Thus $V(a,b)$ descends to $\mathrm{PSU}(3)$ if and only if $a \equiv b \bmod 3$. The adjoint module $V(1,1)$ descends. The module $\mathbb{C}^3 = V(1,0)$ does not.

The group $\mathrm{U}(2)$

Let $T_0 = \{zI : z \in \mathrm{U}(1)\}$ and $K_{sc} = \mathrm{SU}(2)$. The map

$$\Psi\colon \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2), \qquad \Psi(z,g) = zg,$$

is onto. Given u, choose z with $z^2 = \det u$. Then $z^{-1}u \in \mathrm{SU}(2)$. If $zg = I$, then $\det(z^{-1}I) = z^{-2} = 1$, so $z = \pm 1$. Hence

$$\mathrm{U}(2) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(2)\big)/F, \qquad F = \{(1, I), (-1, -I)\}.$$

The lattice statements for semisimple groups fail for $\mathrm{U}(2)$ itself. In the semisimple case the roots span $\mathfrak{t}^*$, and Q, P, $Q^\vee$ and $P^\vee$ are lattices. By the computation for diagonal tori, Example 7.6, the diagonal torus of $\mathrm{U}(2)$ has $X^*(T) = \mathbb{Z}^2$. The roots are $\pm\alpha$ with $\alpha(\mathrm{diag}(i\theta)) = \theta_1 - \theta_2$, so $Q = \mathbb{Z}(1,-1)$ spans only a line. The set $P^\vee = \{\theta_1 - \theta_2 \in 2\pi\mathbb{Z}\}$ is not a lattice, and $P^\vee/\Lambda \cong \mathrm{U}(1)$ is the group of scalars.

The fundamental group theorem, Theorem 7.10, still applies. Since $[E_{12}, E_{21}] = \mathrm{diag}(1,-1)$, the coroot is $H_\alpha = i\,\mathrm{diag}(1,-1)$, and

$$\pi_1(\mathrm{U}(2)) \cong \mathbb{Z}^2/\mathbb{Z}(1,-1) \cong \mathbb{Z}, \qquad (k_1, k_2) \mapsto k_1 + k_2.$$

Here $\Lambda_0 = 2\pi i\mathbb{Z}I$, and $\Lambda_0 \oplus Q^\vee$ consists of the $2\pi i\,\mathrm{diag}(k_1, k_2)$ with $k_1 \equiv k_2 \bmod 2$. So $\Lambda/(\Lambda_0 \oplus Q^\vee) \cong \mathbb{Z}/2 \cong F$, as predicted.

Now we compute the central compatibility constraint. Recall from Lecture 4 that Π_m denotes the representation of $\mathrm{SU}(2)$ on V_m. The element $\mathrm{diag}(w, w^{-1})$, with $w \in \mathrm{U}(1)$, acts on a vector of weight j by w^j. For $k \in \mathbb{Z}$ and $m \geq 0$ define

$$\Pi_{k,m}(z,g) = z^k\,\Pi_m(g) \qquad \text{on } V_m.$$

The representation $\Pi_{k,m}$ is irreducible, already on $\mathrm{SU}(2)$. By Lecture 4 it descends to $\mathrm{U}(2)$ exactly when it is trivial on F. Since

$$\Pi_{k,m}(-1,-I) = (-1)^k(-1)^m\ \mathrm{id},$$

the constraint is

$$k \equiv m \pmod 2.$$

The descended representation is $\Pi(u) = z^k \Pi_m(z^{-1}u)$ with $z^2 = \det u$. The two choices of z give the same operator exactly because of the constraint. Exercise 7.3 shows that every irreducible representation of $\mathrm{U}(2)$ arises in this way.

The description of $X^*(T)$ for a central torus, Proposition 7.14, gives the same constraint. Here k is the exponent on scalars, and j runs over the weights of V_m. Put $\ell_0(isI) = ks$ and $\ell(sH_0) = js$. In T_0, the scalar $z = -1$ is the matrix $-I = \exp(i\pi I)$. So $(-1, -I) = (\exp(i\pi I), \exp(\pi H_0))$, and the pair (ℓ_0, ℓ) is a character of T exactly when $k + j$ is even. Every weight j of V_m satisfies $j \equiv m \bmod 2$, so this condition is the constraint found above.

Assume $k \equiv m \pmod 2$, so that $\Pi_{k,m}$ descends to $\mathrm{U}(2)$. Every weight j of V_m satisfies $j \equiv m \pmod 2$, so $k - j$ is even and the exponents below are integers. On $u = \mathrm{diag}(z_1, z_2)$ with $z^2 = z_1 z_2$, a vector of weight j is multiplied by

$$z^k \left(\frac{z_1}{z}\right)^j = (z_1 z_2)^{(k-j)/2} z_1^j = z_1^{(k+j)/2} z_2^{(k-j)/2}.$$

So $\Pi_{k,m}$ has highest weight $\left(\frac{k+m}{2}, \frac{k-m}{2}\right) \in \mathbb{Z}^2$. For example, $\mathbb{C}^2$ is $(k, m) = (1, 1)$, $\det$ is $(2, 0)$, and Ad on the complexification $\mathfrak{su}(2)_{\mathbb{C}}$ is $(0, 2)$.

7.6 Self-checks

1. Does the maximal torus of $\mathrm{SO}(3)$ have a character with $\ell(H_0) = 1$?
   Answer. No. Here $\Lambda = \pi\mathbb{Z}H_0$ and $\ell(\pi H_0) = \pi \notin 2\pi\mathbb{Z}$.
2. Compute $Z(\mathrm{PSU}(3))$ and $\pi_1(\mathrm{PSU}(3))$ from the lattices.
   Answer. Here $\Lambda = P^\vee$. So $Z = P^\vee/P^\vee = 1$ and $\pi_1 = P^\vee/Q^\vee \cong \mathbb{Z}/3$.
3. Is there a representation of $\mathrm{U}(2)$ on which scalars act by z^2 and $\mathrm{SU}(2)$ acts by V_1?
   Answer. No. The element $-I$ is the scalar -1, acting by $(-1)^2 = 1$. The element $-I$ also lies in $\mathrm{SU}(2)$, where it acts on V_1 by -1. The two actions disagree.

7.7 Summary

A compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, and its characters are the ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. For compact connected K, the kernel lattice satisfies $Q^\vee \subset \Lambda \subset P^\vee$. The kernel lattice gives $Z(K) \cong P^\vee/\Lambda$ and, by a cited theorem, $\pi_1(K) \cong \Lambda/Q^\vee$.

For semisimple $\mathfrak{k}$, the connected forms correspond to lattices between $Q^\vee$ and $P^\vee$. Lattices related by an automorphism of $\mathfrak{k}$ preserving $\mathfrak{t}$ give isomorphic groups. With a central torus, write $K \cong (T_0 \times K_{sc})/F$, where $F \cong \Lambda/(\Lambda_0 \oplus Q^\vee)$. A pair (ℓ_0, ℓ) is then a character of the maximal torus exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product is trivial on F.

7.8 Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations* (H), Chapter 13. T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (BD), Sections V.7–8. A. Knapp, *Lie Groups Beyond an Introduction*, Chapter IV. For the discrete-subgroup lemma, Lemma 7.2, see BD, Section I.3, Lemma (3.8).

Lecture 8

Highest weights of representations of compact groups

Objectives. *The Lie algebra course classified finite-dimensional irreducible modules of semisimple algebras by dominant integral weights. This lecture adds the action of the centre of the Lie algebra and decides which of these modules are representations of a given compact connected Lie group K. These are exactly the modules whose highest weight is a character of a maximal torus.*

By the end of this lecture you will be able to:

- Prove that highest weights give a bijection $\mathrm{Irr}(K) \to X^*(T)^+$, assuming cited structure results.
- Decide descent to K_{sc}/F by evaluating a highest weight λ on the elements $H \in \mathfrak{t}$ with $\exp H \in F$.
- Treat a central torus by an integral character and a finite compatibility condition.
- Explain why some modules of a reductive Lie algebra come from no compact group.

8.1 Torus characters and dominance

Throughout, K is a compact connected Lie group with Lie algebra $\mathfrak{k}$. Let T be a maximal torus with Lie algebra $\mathfrak{t}$ (Lecture 6). Put $\Lambda = \ker(\exp\colon \mathfrak{t} \to T)$. A *character* of T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. Its derivative has the form $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$, and

$$\chi(\exp H) = e^{i\ell(H)} \quad (H \in \mathfrak{t}), \qquad \ell(\Lambda) \subset 2\pi\mathbb{Z}. \tag{8.1}$$

Conversely, $\exp\colon \mathfrak{t} \to T$ is a surjective homomorphism with kernel Λ. So each ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$ defines a character. Let $X^*(T)$ be the lattice of these ℓ.

Let Π be a representation of K on V. By Lecture 5, V has a K-invariant inner product. So the operators $\Pi(t)$ with $t \in T$ are commuting unitary operators. Hence these operators have a common orthonormal eigenbasis. For a common eigenvector v, write $\Pi(t)v = c(t)v$. Then c is a character of T, so $c(\exp H) = e^{i\ell(H)}$ for a unique $\ell \in X^*(T)$. Hence V is the direct sum of the weight spaces

$$V_\ell = \{v \in V : \ \Pi(\exp H)v = e^{i\ell(H)}v \ \text{ for all } H \in \mathfrak{t}\}, \qquad \ell \in X^*(T).$$

Differentiating $\Pi(\exp tH)v = e^{it\ell(H)}v$ at $t = 0$ gives $d\Pi(H)v = i\ell(H)v$ for $v \in V_\ell$. The complex-linear extension of $i\ell$ to $\mathfrak{t}_{\mathbb{C}}$ is a weight in the sense of the Lie algebra course. We identify ℓ with this weight.

Roots are the nonzero weights of Ad on $\mathfrak{k}_{\mathbb{C}}$. Thus $[H, X] = i\alpha(H)X$ for $H \in \mathfrak{t}$ and X in the root space of α. In particular, roots lie in $X^*(T)$.

Fix positive roots R^+ as in [I]. Put

$$\mathbb{N}R^+ = \Big\{ \sum_{\alpha \in R^+} n_\alpha \alpha : \ n_\alpha \in \mathbb{Z}_{\geq 0} \Big\}, \qquad \ell' \preceq \ell \iff \ell - \ell' \in \mathbb{N}R^+.$$

There is $H_0 \in \mathfrak{t}$ with $\alpha(H_0) > 0$ for all $\alpha \in R^+$. So every nonzero element of $\mathbb{N}R^+$ is positive at H_0. Hence $\ell' \preceq \ell$ and $\ell \preceq \ell'$ force $\ell = \ell'$. A *highest weight* of Π is a weight ℓ of Π with $\ell' \preceq \ell$ for every weight ℓ' of Π. It is unique if it exists.

Write $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{k}_s$, with $\mathfrak{z}$ the centre and $\mathfrak{k}_s = [\mathfrak{k}, \mathfrak{k}]$ semisimple. Then $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ with $\mathfrak{t}_s = \mathfrak{t} \cap \mathfrak{k}_s$ (Lecture 6). Roots vanish on $\mathfrak{z}$, since $\operatorname{ad} Z = 0$ for $Z \in \mathfrak{z}$.

The algebra $\mathfrak{t}_s$ is maximal abelian in $\mathfrak{k}_s$. Indeed, let $\mathfrak{a} \supset \mathfrak{t}_s$ be an abelian subalgebra of $\mathfrak{k}_s$. Since $\mathfrak{z}$ is central, $\mathfrak{z} \oplus \mathfrak{a}$ is abelian and contains $\mathfrak{t}$. The Lie algebra of a maximal torus is maximal abelian (Lecture 6), so $\mathfrak{z} \oplus \mathfrak{a} = \mathfrak{t}$. Hence $\mathfrak{a} \subset \mathfrak{t} \cap \mathfrak{k}_s = \mathfrak{t}_s$.

The adjoint operators of elements of $\mathfrak{t}_s$ commute and are skew-adjoint, hence simultaneously diagonalisable over $\mathbb{C}$. So the complexification $\mathfrak{t}_{s,\mathbb{C}}$ is toral. It is also maximal toral. If $X \in \mathfrak{k}_s$ commutes with $\mathfrak{t}_s$, then $\mathfrak{t}_s + \mathbb{R}X$ is abelian, so $X \in \mathfrak{t}_s$. Now let $X, Y \in \mathfrak{k}_s$ with $[H, X + iY] = 0$ for all $H \in \mathfrak{t}_s$. The brackets $[H, X]$ and $[H, Y]$ lie in $\mathfrak{k}_s$, so both vanish, and $X, Y \in \mathfrak{t}_s$. Hence $\mathfrak{t}_{s,\mathbb{C}}$ is its own centraliser in $\mathfrak{k}_{s,\mathbb{C}}$. A toral subalgebra $\mathfrak{h} \supset \mathfrak{t}_{s,\mathbb{C}}$ is abelian [I], so $\mathfrak{h}$ lies in this centraliser and $\mathfrak{h} = \mathfrak{t}_{s,\mathbb{C}}$. Thus $\mathfrak{t}_{s,\mathbb{C}}$ is a maximal toral, or Cartan, subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. This is the Cartan subalgebra used in [I].

For a root α, [I] supplies a coroot $h_\alpha \in \mathfrak{t}_{s,\mathbb{C}}$. The corresponding root in the sense of the Lie algebra course is the complex-linear extension of $i\alpha$. This extension takes the value 2 at h_α. By Lecture 7, the real coroot is $H_\alpha = ih_\alpha \in \mathfrak{t}_s$. Since $h_\alpha = -i\,(ih_\alpha)$, the value of the complex-linear extension of $i\alpha$ at h_α equals $\alpha(ih_\alpha)$. So

$$\alpha(ih_\alpha) = 2, \qquad ih_\alpha \in \mathfrak{t}_s.$$

In the same way, the complex-linear extension of $i\ell$ takes the value $\ell(ih_\alpha)$ at h_α. So we put

$$\langle \ell, \alpha^\vee \rangle = \ell(ih_\alpha), \qquad P = \{\lambda \in \mathfrak{t}_s^* : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \ \forall \alpha\},$$

$$X^*(T)^+ = \{\ell \in X^*(T) : \langle \ell, \alpha^\vee \rangle \geq 0 \ \forall \alpha \in R^+\}.$$

By Lecture 7, $2\pi i h_\alpha \in \Lambda$. Hence every $\ell \in X^*(T)$ satisfies

$$\langle \ell, \alpha^\vee \rangle = \tfrac{1}{2\pi}\, \ell(2\pi i h_\alpha) \in \mathbb{Z},$$

so $\ell|_{\mathfrak{t}_s} \in P$. Let Q be the root lattice, the lattice generated by the roots. Roots lie in $X^*(T)$, so for semisimple K we obtain $Q \subset X^*(T) \subset P$. If K is also simply connected, then Λ is generated as a lattice by the $2\pi i h_\alpha$, and $X^*(T) = P$ (Lecture 7).

8.2 The classification theorem

We use the following structure inputs.

(S1) (Lecture 6; torus theorem) In a compact connected Lie group, every element lies in a maximal torus, and all maximal tori are conjugate. The image of exp generates the group. A torus is the image of its Lie algebra under exp. A compact connected abelian Lie group is a torus. The Lie algebra of a maximal torus is a maximal abelian subalgebra.

(S2) (Lecture 7; H Chapter 13; covering theorem) Let $Z_0 = \exp \mathfrak{z}$, a torus. Let K_s be the simply connected Lie group with Lie algebra $\mathfrak{k}_s$. Then K_s is compact. The inclusion $\mathfrak{k}_s \subset \mathfrak{k}$ integrates to $\iota\colon K_s \to K$. The map

$$p\colon \widetilde{K} = Z_0 \times K_s \to K, \qquad p(z,s) = z\,\iota(s),$$

is a surjective homomorphism with finite central kernel F, and $dp = \mathrm{id}_{\mathfrak{k}}$. Hence p is a covering homomorphism by Lecture 4.

(S3) (Lecture 4; integration and descent) Finite-dimensional representations of $\mathfrak{k}_s$ integrate to K_s. Let $\kappa\colon G_1 \to G_2$ be a covering homomorphism with kernel N. A representation $\widetilde{\Pi}$ of G_1 equals $\Pi \circ \kappa$ for a representation Π of G_2 if and only if $\widetilde{\Pi}(N) = \{I\}$.

(S4) ([I], highest-weight theorem) Finite-dimensional $\mathfrak{k}_{s,\mathbb{C}}$-modules are sums of weight spaces with weights in P. An irreducible such module has a unique highest weight λ. Its weight space is a line, every weight lies in $\lambda - \mathbb{N}R^+$, and $\langle \lambda, \alpha^\vee \rangle \geq 0$ for $\alpha \in R^+$. Each dominant $\lambda \in P$ is the highest weight of exactly one irreducible module $V(\lambda)$, up to isomorphism.

Lemma 8.1. *Let G be a compact connected Lie group with a maximal torus S and $\mathfrak{s} = \operatorname{Lie} S$. Every central element f of G equals $\exp H$ for some $H \in \mathfrak{s}$.*

Proof. By the torus theorem (S1), $f \in gSg^{-1}$ for some $g \in G$. As f is central, $f = g^{-1}fg \in S$. Since $\exp\colon \mathfrak{s} \to S$ is onto, $f = \exp H$ with $H \in \mathfrak{s}$. ∎

Lemma 8.2. *Assume the torus theorem (S1) and the covering theorem (S2). Then $\widetilde{T} = \exp_{\widetilde{K}} \mathfrak{t}$ is a maximal torus of $\widetilde{K}$. Every $f \in F$ equals $\exp_{\widetilde{K}} H$ for some $H \in \Lambda$.*

Proof. Since $dp = \mathrm{id}_{\mathfrak{k}}$, naturality of exp (Lecture 2) gives

$$p(\exp_{\widetilde{K}} X) = \exp_K X \qquad (X \in \mathfrak{k}). \tag{8.2}$$

Let T' be the identity component of the closed subgroup $p^{-1}(T)$. Its Lie algebra consists of the X with $\exp_K(tX) \in T$ for all t. This set is $\mathfrak{t}$. As $\mathfrak{t}$ is abelian, $\exp_{\widetilde{K}}|_{\mathfrak{t}}$ is a homomorphism. Its image is a subgroup of T' containing a neighbourhood of 1 (Lecture 2). So the image is open, hence also closed, in the connected group T'. Thus $\widetilde{T} = T'$.

So $\widetilde{T}$ is compact, connected and abelian, hence a torus. Let $S \supset \widetilde{T}$ be a torus. Then $\operatorname{Lie} S$ is abelian and contains $\mathfrak{t}$. Since $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$, $\operatorname{Lie} S = \mathfrak{t}$ and $S = \widetilde{T}$.

Finally, $f \in F$ is central in the compact connected group $\widetilde{K}$. By the central-element lemma, Lemma 8.1, $f = \exp_{\widetilde{K}} H$ with $H \in \mathfrak{t}$. By naturality (8.2), $\exp_K H = p(f) = 1$, so $H \in \Lambda$. ∎

Theorem 8.3. *Assume the structure inputs (S1)–(S4). Each irreducible representation of K has a unique highest weight. It lies in $X^*(T)^+$, and this gives a bijection $\operatorname{Irr}(K) \to X^*(T)^+$.*

Proof. Step 1: the highest weight. Let Π be irreducible on V, and let $Z \in \mathfrak{z}$. Since $[Z, X] = 0$ for every $X \in \mathfrak{k}$, the element $\exp tZ$ commutes with every $\exp X$. The elements $\exp X$ generate K by the torus theorem (S1), so $\exp tZ$ is central. Hence $\Pi(\exp tZ)$ commutes with $\Pi(K)$. Differentiating at $t = 0$, so does $d\Pi(Z)$. By Schur's lemma, $d\Pi(Z)$ is a scalar. The operator $d\Pi(Z)$ is skew-adjoint, so the scalar is imaginary. The scalar depends linearly on Z. Hence

$$d\Pi(Z) = i\mu(Z)I \qquad \text{for some } \mu \in \mathfrak{z}^*.$$

Let $W \subset V$ be a complex subspace invariant under $d\Pi(\mathfrak{k}_s)$. Then W is invariant under $d\Pi(\mathfrak{k})$, and so under every $\Pi(\exp X) = e^{d\Pi(X)}$. The elements $\exp X$ generate K by the torus theorem (S1), so W is K-invariant. Hence V is irreducible under $\mathfrak{k}_s$. The complexification $\mathfrak{k}_{s,\mathbb{C}}$ has the same invariant complex subspaces, so V is also irreducible under $\mathfrak{k}_{s,\mathbb{C}}$.

By the highest-weight theorem (S4), the $\mathfrak{k}_{s,\mathbb{C}}$-module V has a highest weight λ and a highest weight vector v. For $H = Z + H_s$ with $Z \in \mathfrak{z}$ and $H_s \in \mathfrak{t}_s$,

$$d\Pi(H)v = i\big(\mu(Z) + \lambda(H_s)\big)v = i\ell(H)v, \qquad \ell := \mu + \lambda \in \mathfrak{t}^*.$$

Exponentiating gives $\Pi(\exp H)v = e^{i\ell(H)}v$. So v is a common eigenvector of $\Pi(T)$, and $\ell \in X^*(T)$ by the description of characters in § 8.1.

Now let ℓ' be any T-weight of V. The restriction of ℓ' to $\mathfrak{z}$ equals μ. The restriction of ℓ' to $\mathfrak{t}_s$ lies in $\lambda - \mathbb{N}R^+$ by the highest-weight theorem (S4). Roots vanish on $\mathfrak{z}$, so $\ell - \ell' \in \mathbb{N}R^+$. Thus ℓ is the highest weight of Π. It is dominant, since $\langle \ell, \alpha^\vee \rangle = \langle \lambda, \alpha^\vee \rangle \geq 0$.

Step 2: injectivity. Let Π_1, Π_2 be irreducible with the same highest weight ℓ. They share μ and λ. By the uniqueness part of the highest-weight theorem (S4), there is an isomorphism $A \colon V_1 \to V_2$ of $\mathfrak{k}_{s,\mathbb{C}}$-modules. It also intertwines $d\Pi_j(Z) = i\mu(Z)I$. Hence for every $X \in \mathfrak{k}$

$$A\,\Pi_1(\exp X) = A\,e^{d\Pi_1(X)} = e^{d\Pi_2(X)}A = \Pi_2(\exp X)\,A.$$

The elements $\exp X$ generate K by the torus theorem (S1), so A is K-equivariant.

Step 3: existence. Let $\ell \in X^*(T)^+$, with $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. By the integrality computation in § 8.1, $\lambda \in P$, and λ is dominant. The module $V(\lambda)$ is only known to integrate to K_s. Since the map p has kernel F, we must show that F acts trivially. This step uses the hypothesis $\ell \in X^*(T)$, not only $\lambda \in P$.

The kernel $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$ of $\exp|_{\mathfrak{z}}$ lies in Λ. So $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, and

$$\chi_\mu(\exp Z) = e^{i\mu(Z)} \qquad (Z \in \mathfrak{z})$$

is a well-defined character of Z_0. By the highest-weight theorem (S4) and integration (S3), $V(\lambda)$ integrates to an irreducible representation Π_λ of K_s. Define on $V(\lambda)$

$$\widetilde{\Pi}(z, s) = \chi_\mu(z)\,\Pi_\lambda(s).$$

It is irreducible, because Π_λ already is. Let v be a highest weight vector. For $H = Z + H_s$ as above,

$$\widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\mu(Z)}e^{d\Pi_\lambda(H_s)}v = e^{i\mu(Z)}e^{i\lambda(H_s)}v = e^{i\ell(H)}v.$$

Let $f \in F$. By the lifting lemma, Lemma 8.2, $f = \exp_{\widetilde{K}} H$ with $H \in \Lambda$. As f is central, Schur's lemma gives $\widetilde{\Pi}(f) = cI$. Evaluating on v,

$$cv = \widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\ell(H)}v = v,$$

because $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. So F acts trivially. By the descent criterion (S3), $\widetilde{\Pi} = \Pi \circ p$ for a representation Π of K. It is irreducible, since p is onto.

It remains to identify the highest weight of Π. Since $dp = \mathrm{id}_{\mathfrak{t}}$, we have $d\Pi = d\widetilde{\Pi}$. So

$$d\Pi(Z) = i\mu(Z)I \quad (Z \in \mathfrak{z}), \qquad d\Pi|_{\mathfrak{t}_s} = d\Pi_\lambda.$$

Thus V is $V(\lambda)$ as an $\mathfrak{k}_{s,\mathbb{C}}$-module, with highest weight vector v. These are exactly the two data used in Step 1. Step 1 therefore gives the highest weight $\mu + \lambda = \ell$. ∎

Compactness gave the invariant inner product. This yields the weight decomposition of V. It also makes $d\Pi(Z)$ skew-adjoint, so μ is real. Compactness of $\widetilde{K}$ allowed the torus theorem (S1) inside the lifting lemma, Lemma 8.2. Schur's lemma needs only irreducibility. Connectedness entered through the torus theorem (S1).

H §§12.1–12.3 phrase the result with *analytically integral* elements. Hall's kernel lattice is our Λ divided by 2π. Hall obtains existence by analytic methods in Chapter 12. Our proof passes through the covering p.

8.3 Central characters and global forms

Let K_{sc} be compact, connected, simply connected and semisimple, with maximal torus T. Let F be a subgroup of its finite centre.

> **Corollary 8.4.** *Assume the torus theorem (S1), integration and descent (S3), and the highest-weight theorem (S4). Let $\lambda \in P$ be dominant. Then $V(\lambda)$ descends to K_{sc}/F if and only if F acts trivially. Equivalently,*
>
> $$\lambda(H) \in 2\pi\mathbb{Z} \qquad \text{whenever } H \in \mathfrak{t} \text{ and } \exp H \in F.$$

Proof. The first statement is the descent criterion (S3). Let Π be the representation of K_{sc} integrating $V(\lambda)$, which exists by integration (S3). Let $f \in F$. By the central-element lemma, Lemma 8.1, $f = \exp H$ with $H \in \mathfrak{t}$. By Schur's lemma, f acts by a scalar. Evaluating on a highest weight vector gives $\Pi(f) = e^{i\lambda(H)}I$. This is I exactly when $\lambda(H) \in 2\pi\mathbb{Z}$. ∎

Since $f \mapsto \Pi(f)$ is a homomorphism, it suffices to test generators of F.

Now allow a central torus. Here $F = \ker p$ again, as in the covering theorem (S2). Since F is central in $\widetilde{K}$, each $(z, s) \in F$ has s central in K_s.

We recall the notation of Step 3 in the proof of the classification theorem. The kernel of $\exp|_{\mathfrak{z}}$ is $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$. For $\mu \in \mathfrak{z}^*$ with $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, the character χ_μ of Z_0 is given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. For dominant $\lambda \in P$, the representation of K_s integrating $V(\lambda)$ is Π_λ.

Let Π be an irreducible representation of a compact group. By Schur's lemma, each central element acts by a scalar. The map sending a central element to its scalar is a homomorphism from the centre to $\mathbb{C}^\times$, the *central character* of Π. Let c_λ be the central character of Π_λ. So $c_\lambda(s)$ is the scalar by which an element s of the centre of K_s acts on $V(\lambda)$.

Corollary 8.5. *Assume the structure inputs (S1)–(S4). Let $\mu \in \mathfrak{z}^*$ and $\lambda \in \mathfrak{t}_s^*$ satisfy*

(a) $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, so χ_μ is a character of Z_0 (integral character);

(b) $\lambda \in P$ is dominant;

(c) $\chi_\mu(z)\, c_\lambda(s) = 1$ for all $(z,s) \in F$ (finite compatibility).

Then $\rho_{\mu,\lambda}(z,s) = \chi_\mu(z)\Pi_\lambda(s)$ descends to an irreducible representation of K. Every irreducible representation of K arises in this way from exactly one pair (μ,λ).

Proof. By the integrality condition (a), the dominance condition (b) and integration (S3), $\rho_{\mu,\lambda}$ is an irreducible representation of $\widetilde{K}$. Let $f = (z,s) \in F$. As noted before the corollary, s is central in K_s. So $\rho_{\mu,\lambda}(f) = \chi_\mu(z)c_\lambda(s)I$. By the descent criterion (S3), $\rho_{\mu,\lambda}$ descends if and only if the compatibility condition (c) holds. The descended representation is irreducible because p is onto. The computation in Step 3 of the proof of the classification theorem shows that the descended representation has highest weight $\mu + \lambda$.

Conversely, let Π be irreducible with highest weight ℓ. Put $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. Step 3 proves the integrality condition (a) and the dominance condition (b). It also gives Π' with $\Pi' \circ p = \rho_{\mu,\lambda}$ and highest weight ℓ. By the injectivity in Step 2, $\Pi \cong \Pi'$. Since $\Pi' \circ p$ is trivial on F, the compatibility condition (c) holds. The pair is unique because $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ recovers (μ,λ) from $\mu + \lambda$. ∎

Comparing with the classification theorem, Theorem 8.3, the integrality, dominance and compatibility conditions together are equivalent to $\mu + \lambda \in X^*(T)^+$. The U(2) example, Example 8.9, checks this directly.

A module of the reductive algebra $\mathfrak{k}_{\mathbb{C}}$ can fail to come from a representation of K, because compactness constrains the action of the centre.

Proposition 8.6. *Let Π be a representation of K and $Z \in \mathfrak{z}$. Then $d\Pi(Z)$ is diagonalisable with eigenvalues in $i\mathbb{R}$. If $\exp Z = 1$, these eigenvalues lie in $2\pi i\mathbb{Z}$.*

Proof. With respect to an invariant inner product, $\Pi(\exp tZ)$ is unitary, so $d\Pi(Z)$ is skew-adjoint. Hence it is diagonalisable with eigenvalues $i\theta_j$, $\theta_j \in \mathbb{R}$. If $\exp Z = 1$, then $e^{d\Pi(Z)} = \Pi(1) = I$, so $e^{i\theta_j} = 1$. ∎

For $\mathfrak{k} = \mathfrak{u}(1) = i\mathbb{R}$, neither of the following modules comes from U(1):

$$\rho_1(it) = t\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \rho_2(it) = \tfrac{i}{2}t.$$

The first is not semisimple. The second has the wrong period: $\exp(2\pi i) = 1$, but $\rho_2(2\pi i) = \pi i \notin 2\pi i\mathbb{Z}$. Both integrate to the noncompact simply connected group $\mathbb{R}$, where no averaging is available.

The same happens for $\mathfrak{gl}(2,\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}(2,\mathbb{C})$. Letting I act on V_m by any $s \in \mathbb{C}$ gives a module. It comes from U(2) if and only if $s \in \mathbb{Z}$ and $s \equiv m \pmod 2$ (Exercise 8.5). On the semisimple part, Weyl's theorem already gives semisimplicity and weights in P. The remaining constraint is $X^*(T)$, which can be smaller.

8.4 Global forms in examples

Example 8.7 (SU(2) and SO(3)). Let $h = \operatorname{diag}(1,-1)$, so $\mathfrak{t} = \mathbb{R}\,ih$ in $\mathfrak{su}(2)$. Then

$$\exp(\theta\, ih) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \qquad \Lambda = 2\pi\mathbb{Z}\, ih.$$

For the root vector E_{12} we get $[\theta\, ih, E_{12}] = 2i\theta E_{12}$, so $\alpha(ih) = 2$ and $h_\alpha = h$. Thus $\langle \ell, \alpha^\vee \rangle = \ell(ih)$, and

$$X^*(T) = P = \{\ell : \ell(ih) \in \mathbb{Z}\}, \qquad Q = \{\ell : \ell(ih) \in 2\mathbb{Z}\}.$$

The highest weight of V_m has $\ell(ih) = m$. Now $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). The element $-I$ is central, and

$$\exp(\pi ih) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By the quotient criterion, Corollary 8.4, $-I$ acts on V_m by

$$e^{i\ell(\pi ih)} = e^{i\pi m} = (-1)^m.$$

So V_m descends to $\mathrm{SO}(3)$ if and only if m is even. The irreducible representations of $\mathrm{SO}(3)$ have dimensions $1, 3, 5, \ldots$. In physics, V_m describes spin $m/2$. Hence the representations of integer spin descend to the rotation group $\mathrm{SO}(3)$. The representations of half-integer spin are representations of $\mathrm{SU}(2)$ that do not descend.

Let $r\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the covering. Its derivative is an isomorphism, and we identify the Lie algebras through it. Naturality of exp (Lecture 2) gives $\exp_{\mathrm{SO}(3)}(\theta\, ih) = r(\exp(\theta\, ih))$. This element is 1 if and only if $\exp(\theta\, ih) = \pm I$, that is, if and only if $\theta \in \pi\mathbb{Z}$. Hence the maximal torus $r(T)$ of $\mathrm{SO}(3)$ has $X^*(r(T)) = \{\ell : \pi\ell(ih) \in 2\pi\mathbb{Z}\} = Q$. This lattice has index 2 in P.

Example 8.8 (PSU(3)). Let $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum \theta_j = 0\}$ and $\varepsilon_j(H) = \theta_j$. The roots are $\varepsilon_j - \varepsilon_k$ with $j \neq k$, with root vectors E_{jk}. We take the positive roots $\varepsilon_j - \varepsilon_k$ with $j < k$. The simple roots $\varepsilon_1 - \varepsilon_2$ and $\varepsilon_2 - \varepsilon_3$ have the coroots $h_{12} = \operatorname{diag}(1,-1,0)$ and $h_{23} = \operatorname{diag}(0,1,-1)$. Then

$$\varepsilon_1(ih_{12}) = 1, \quad \varepsilon_1(ih_{23}) = 0, \quad (\varepsilon_1+\varepsilon_2)(ih_{12}) = 0, \quad (\varepsilon_1+\varepsilon_2)(ih_{23}) = 1.$$

The fundamental weights ϖ_1 and ϖ_2 are dual to the simple coroots. Thus ϖ_1 takes the values 1 and 0 at ih_{12} and ih_{23}, and ϖ_2 takes the values 0 and 1. So $\varpi_1 = \varepsilon_1$ and $\varpi_2 = \varepsilon_1 + \varepsilon_2$. The label (a,b) denotes the highest weight

$$\ell = a\varpi_1 + b\varpi_2 = (a+b)\varepsilon_1 + b\varepsilon_2.$$

The centre is $\{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$, and

$$\omega I = \exp H_\omega, \qquad H_\omega = \tfrac{2\pi i}{3}\operatorname{diag}(1,1,-2) \in \mathfrak{t}.$$

Then

$$\ell(H_\omega) = (a+b)\tfrac{2\pi}{3} + b\tfrac{2\pi}{3} = \tfrac{2\pi}{3}(a+2b),$$

so ωI acts on the irreducible module with highest weight (a,b) by ω^{a+2b}. Hence this module descends to $\mathrm{PSU}(3) = \mathrm{SU}(3)/\langle \omega I \rangle$ if and only if $3 \mid a+2b$. Since $(a+2b) - (a-b) = 3b$, this means $a \equiv b \pmod 3$. The dimension formula $\frac{1}{2}(a+1)(b+1)(a+b+2)$ [I] gives the following values.

(a,b)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$	$(0,3)$	$(2,2)$
dimension	3	3	6	8	10	10	27
ωI acts by	ω	ω^2	ω^2	1	1	1	1

The adjoint representation $(1,1)$ descends, since $\mathrm{Ad}(\omega I) = \mathrm{id}$.

Example 8.9 (U(2)). Let $p\colon \mathrm{U}(1)\times\mathrm{SU}(2)\to\mathrm{U}(2)$ be given by $p(z,g)=zg$. The map p is onto. Indeed, given u, choose z with $z^2=\det u$ and put $g=z^{-1}u$. For the kernel,

$$zg=I \implies g=z^{-1}I, \quad 1=\det g=z^{-2}, \qquad \text{so} \qquad F=\{(1,I),(-1,-I)\}.$$

Here $\Lambda_{\mathfrak{z}}=2\pi i\mathbb{Z}\,I$, so the integral characters of $Z_0\cong\mathrm{U}(1)$ are $z\mapsto z^q$ with $q\in\mathbb{Z}$. Put $\Pi_{q,m}(z,g)=z^q\Pi_m(g)$. The compatibility condition (c) is

$$\Pi_{q,m}(-1,-I)=(-1)^q(-1)^m I=I \iff q+m \text{ is even.}$$

We cross-check with the classification theorem, Theorem 8.3. The torus $\mathrm{diag}(e^{i\theta_1},e^{i\theta_2})$ has $X^*(T)=\mathbb{Z}^2$, and (n_1,n_2) is dominant if and only if $n_1\ge n_2$. The torus element equals $p(e^{i\psi},\mathrm{diag}(e^{i\phi},e^{-i\phi}))$ when $\theta_1=\psi+\phi$ and $\theta_2=\psi-\phi$. On a highest weight vector of $\Pi_{q,m}$, this torus element acts by $e^{i(q\psi+m\phi)}$, while

$$n_1\theta_1+n_2\theta_2=(n_1+n_2)\psi+(n_1-n_2)\phi.$$

So $q=n_1+n_2$ and $m=n_1-n_2$. The numbers $n_{1,2}=(q\pm m)/2$ are integers exactly when $q+m$ is even. For example, $\mathbb{C}^2$ corresponds to $(q,m)=(1,1)$, and $\det$ corresponds to $(2,0)$.

8.5 Tensor products and consistency checks

Lemma 8.10. *Let f be central in K, acting by scalars c_1 on V_1 and c_2 on V_2. Then f acts by c_1c_2 on $V_1\otimes V_2$ and on each subrepresentation, and by c_1^{-1} on V_1^*.*

Proof. First, $(\Pi_1\otimes\Pi_2)(f)=c_1I\otimes c_2I=c_1c_2\,I$. Invariant subspaces inherit this scalar. On the dual, $\Pi_1^*(f)=\Pi_1(f^{-1})^T=c_1^{-1}I$. ∎

Hence the decompositions from [I] must respect central characters. We check this in the following cases.

- Clebsch–Gordan: $V_m\otimes V_n=\bigoplus_{k=0}^{\min(m,n)}V_{m+n-2k}$. Every summand has parity $m+n$, matching $(-1)^m(-1)^n$. In particular, $V_1\otimes V_1=V_2\oplus V_0$ is a sum of $\mathrm{SO}(3)$-representations, although V_1 is not an $\mathrm{SO}(3)$-representation.
- $\mathrm{SU}(3)$: $(1,0)\otimes(1,0)=(2,0)\oplus(0,1)$, with $9=6+3$ and $\omega\cdot\omega=\omega^2$ on both summands. Also $(1,0)\otimes(0,1)=(1,1)\oplus(0,0)$, with $9=8+1$ and $\omega\cdot\omega^2=1$.
- Duals: $V(a,b)^*\cong V(b,a)$ [I]. The lemma predicts $\omega^{b+2a}=\omega^{-(a+2b)}$. Indeed $(b+2a)+(a+2b)=3(a+b)$.
- Branching from $\mathrm{U}(2)$ to $\mathrm{SU}(2)$: $\Pi_{q,m}$ restricts to V_m. On the torus of $\mathrm{U}(2)$ it has the $m+1$ weights (n_1-j,n_2+j), $0\le j\le m$. Characters z^q multiply, so $\Pi_{1,1}\otimes\Pi_{1,1}=\Pi_{2,2}\oplus\Pi_{2,0}$. Both summands have $q+m$ even.

Self-checks

1. Does $V_3 \otimes V_5$ descend to $\mathrm{SO}(3)$? Decompose it.

   Answer. Yes. The element $-I$ acts by $(-1)^3(-1)^5 = 1$. By Clebsch–Gordan,

   $$V_3 \otimes V_5 = V_8 \oplus V_6 \oplus V_4 \oplus V_2, \qquad 9 + 7 + 5 + 3 = 24 = 4 \cdot 6.$$

2. Which irreducible representations of $\mathrm{PSU}(3)$ have dimension at most 10?

   Answer. The $\mathrm{SU}(3)$-irreducibles of dimension at most 10 are $(0,0)$, $(1,0)$, $(0,1)$, $(2,0)$, $(0,2)$, $(1,1)$, $(3,0)$, $(0,3)$. The next ones, such as $(2,1)$ and $(4,0)$, have dimension 15. The ones that descend to $\mathrm{PSU}(3)$ are those with $a \equiv b \pmod 3$, namely

   $$(0,0),\ (1,1),\ (3,0),\ (0,3),$$

   of dimensions $1, 8, 10, 10$.

3. Let $I \in \mathfrak{gl}(2,\mathbb{C})$ act by 1 on V_2. Does this module come from $\mathrm{U}(2)$?

   Answer. No. It would be $\Pi_{1,2}$, and $1 + 2$ is odd. Directly, compute $\Pi(-I)$ in two ways:

   $$-I = \exp(i\pi I) \implies \Pi(-I) = e^{i\pi \cdot 1} = -1, \qquad -I \in \mathrm{SU}(2) \implies \Pi(-I) = (-1)^2 = 1.$$

   These disagree.

Summary

Let K be a compact connected Lie group, and assume the cited structure inputs. An irreducible representation of K is determined by its highest weight. The possible highest weights are the dominant characters of a maximal torus T. For K_{sc}/F, test F on one highest weight vector. With a central torus, one adds an integral character of the central torus and a compatibility condition on F. In the examples, the descent conditions are m even for $\mathrm{SO}(3)$, $3 \mid a + 2b$ for $\mathrm{PSU}(3)$, and $q + m$ even for $\mathrm{U}(2)$.

Reading

H denotes Hall, *Lie Groups, Lie Algebras, and Representations*. BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

- H §§12.1–12.3: representations of compact groups, analytically integral elements, and the setting of the highest-weight classification for compact connected groups.
- H Chapter 13: fundamental groups of compact groups, and the comparison of algebraic and analytic integrality. This supports (S2).
- BD VI.2: the dominant weights of irreducible representations of compact connected Lie groups.
- Lie Algebras [I]: highest-weight theorem, Weyl's theorem, the dimension formula, Clebsch–Gordan, duals and $\mathfrak{sl}_3$ tensor products.

Lecture 9

Weyl integration and character calculations

Objectives. *On a compact connected group, a class function is determined by its values on a maximal torus. This lecture shows that the Haar integral of a class function can also be computed on the torus. The density is a product over root planes. A factor $1/|W|$ corrects for repeated conjugates. We then use the formula to check character norms and tensor-product multiplicities.*

By the end of this lecture you will be able to:

- State the Weyl integration formula with its measure conventions and its imported steps.
- Derive the density $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$ and the factor $1/|W|$.
- Compute class integrals on $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ and compare them.
- Check character norms and multiplicities with the Weyl character formula on T.

9.1 Measures, roots and the statement

Throughout, K is a compact connected Lie group and $T \subset K$ is a maximal torus. Their Lie algebras are $\mathfrak{t} \subset \mathfrak{k}$. By Lecture 6, every element of K is conjugate to an element of T. Also by Lecture 6, the group $W = N_K(T)/T$ is finite and acts on $X^*(T)$ as the Weyl group of the root system. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$ and set

$$\mathfrak{m} = \mathfrak{t}^{\perp} \subset \mathfrak{k}.$$

For $t \in T$ the map Ad_t is orthogonal and fixes $\mathfrak{t}$ pointwise. Hence Ad_t preserves $\mathfrak{m}$, and $\mathfrak{m} \cong \mathfrak{k}/\mathfrak{t}$ as representations of T.

We use three probability measures.

- dk is the normalised Haar measure on K (Lecture 5).
- dt is the normalised Haar measure on T.
- $d(gT)$ is the unique K-invariant probability measure on K/T.

The existence and uniqueness of $d(gT)$ are cited together with the theorem below. All three measures are given by smooth positive densities.

We now use complex weights. A real character weight $\ell \in X^*(T)$ from Lecture 7 becomes the complex-linear functional $i\ell$ on $\mathfrak{t}_{\mathbb{C}}$. Put

$$\mathcal{X}(T) = \{i\ell : \ell \in X^*(T)\} \subset \mathfrak{t}_{\mathbb{C}}^*.$$

For $\mu = i\ell$, write t^μ for the value at t of the character of T with weight ℓ (Lecture 7). Thus $t^\mu = e^{\mu(H)}$ when $t = \exp H$ with $H \in \mathfrak{t}$. In coordinates, a character keeps the same integer vector whether its weight is viewed in $X^*(T)$ or in $\mathcal{X}(T)$.

With this convention, the root decomposition is

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}_t E = t^\alpha E \quad (E \in \mathfrak{g}_\alpha,\ t \in T). \tag{9.1}$$

Each $\mathfrak{g}_\alpha$ is a complex line. Each root is a weight of the adjoint representation of K. So $t \mapsto t^\alpha$ is a character of T, that is, $\alpha \in \mathcal{X}(T)$. If $t = \exp H$ with $H \in \mathfrak{t}$, then $t^\alpha = e^{\alpha(H)}$ with $\alpha(H) \in i\mathbb{R}$.

Fix a positive system R^+ and let ρ be half the sum of the positive roots. In general $\rho \notin \mathcal{X}(T)$. § 9.4 shows how to avoid ρ as a function on T.

Definition 9.1. An element $t \in T$ is *regular* if $t^\alpha \neq 1$ for every $\alpha \in R$. The regular elements of T form T_{reg}. The regular elements of K form $K_{\mathrm{reg}} = \{gtg^{-1} : g \in K,\ t \in T_{\mathrm{reg}}\}$.

The conjugation map is

$$\psi \colon K/T \times T \to K, \qquad \psi(gT, t) = gtg^{-1}. \tag{9.2}$$

It is well defined because T is abelian. The domain and the codomain of ψ both have dimension $\dim K$.

Theorem 9.2 (Weyl integration formula)**.** *Let K, T, W and the measures be as above. Put*

$$J(t) = \det\bigl(1 - \operatorname{Ad}_t|_{\mathfrak{m}}\bigr) \qquad (t \in T).$$

Then $J \geq 0$, and for every continuous $f \colon K \to \mathbb{C}$,

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T J(t) \int_{K/T} f(gtg^{-1})\,d(gT)\,dt. \tag{9.3}$$

If f is a class function, then

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T f(t)\,J(t)\,dt. \tag{9.4}$$

The theorem is cited from H §11.6, BD IV.1 (1.11) and Knapp, Beyond, VIII.5. We prove two structural parts here: the formula for J and the factor $1/|W|$. Two analytic steps are imported.

(C1) The set K_{reg} is open, and $K \setminus K_{\mathrm{reg}}$ has Haar measure zero. The null-set claim is cited from BD V.2, Proposition (2.6). That proposition covers $K \setminus K_{\mathrm{reg}}$ by the image of a compact manifold of dimension at most $\dim K - 3$. Such an image is null by the null-set fact imported in Lecture 5. BD's regular elements agree with ours by the centraliser-dimension formula of Proposition 9.6. Openness also follows from that proposition. The map ψ is onto (Lecture 6), and $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$. Hence

$K \setminus K_{\mathrm{reg}} = \psi\big(K/T \times (T \setminus T_{\mathrm{reg}})\big)$. The set $T \setminus T_{\mathrm{reg}}$ is a finite union of kernels of characters, so it is closed in T. So $K \setminus K_{\mathrm{reg}}$ is a continuous image of a compact set. Hence it is compact, and therefore closed in K.

(C2) Let M, N be compact manifolds of equal dimension. Give them finite measures defined by smooth positive densities. Let $\psi\colon M \to N$ be smooth and $U \subset N$ open. Assume $N \setminus U$ and $M \setminus \psi^{-1}(U)$ are null. Assume $\psi\colon \psi^{-1}(U) \to U$ is a surjective local diffeomorphism whose fibres all have d points. Then for continuous F on N,

$$\int_M F(\psi(x))\, J_\psi(x)\, dx = d \int_N F(y)\, dy,$$

where J_ψ is the Jacobian of ψ relative to the two densities. Source: Lee, Chapter 16, Propositions 16.40–16.42. Proposition 16.40 is the absolute-Jacobian rule for pulling back a density in coordinates. Lee defines integrals of densities through charts and a partition of unity. Lee does not state (C2) itself, so we derive it from these results.
First, each point $y \in U$ has an open neighbourhood $W \subset U$ whose inverse image consists of d disjoint open sets, each mapped diffeomorphically onto W. To see this, let $x_1, \dots, x_d$ be the preimages of y. Choose disjoint open sets $V_i \ni x_i$ in $\psi^{-1}(U)$ that ψ maps diffeomorphically onto open sets $\psi(V_i)$. The set $C = M \setminus \bigcup_i V_i$ is closed in the compact manifold M, so it is compact. It contains no preimage of y. So $\psi(C)$ is compact, hence closed, and $y \notin \psi(C)$. Put $W = \bigcap_i \psi(V_i) \setminus \psi(C)$. This set is open and contains y. Every preimage of a point of W lies outside C, hence in some V_i. So the inverse image of W is the disjoint union of the sets $V_i \cap \psi^{-1}(W)$. Since $W \subset \psi(V_i)$, the map ψ sends each of these sets diffeomorphically onto W.
Next, suppose F is supported in such a set W. Apply the change-of-variables rule on each of the d sets $V_i \cap \psi^{-1}(W)$ and add the results. A partition of unity on U subordinate to these sets W gives the general case: the partition is countable; sum the pieces by dominated convergence [C], since the partial sums are bounded by $|F \circ \psi| J_\psi$ and $|F|$. The null sets $N \setminus U$ and $M \setminus \psi^{-1}(U)$ do not contribute. BD I.5, (5.19), states the oriented version through the mapping degree.

The class-function version follows from the general formula (9.3). Indeed $f(gtg^{-1}) = f(t)$, and $d(gT)$ has total mass one.

Remark 9.3 (Indicator functions)**.** Probabilities of conjugation-invariant sets need discontinuous f. We extend the class-function formula to one kind of indicator function. Let $c\colon K \to \mathbb{R}$ be a continuous class function and put $E = \{k \in K : c(k) \geq 0\}$. For $n \geq 1$ let $h_n\colon \mathbb{R} \to [0,1]$ be continuous with

$$h_n(x) = 1 \ \ (x \geq 0), \qquad h_n(x) = 0 \ \ (x \leq -1/n), \qquad h_n \text{ linear on } [-1/n, 0].$$

Each $h_n \circ c$ is a continuous class function. For every k, the values $h_n(c(k))$ decrease to the indicator $1_E(k)$. All these functions are bounded by 1, and J is bounded on T. Apply the dominated convergence theorem [C] to both sides of the class-function formula (9.4). This gives

$$\int_K 1_E\, dk = \frac{1}{|W|} \int_T 1_E(t)\, J(t)\, dt.$$

9.2 The root-plane Jacobian and the factor $|W|$

Lemma 9.4 (Root planes). *For $\alpha \in R^+$ put $\mathfrak{m}_\alpha = \mathfrak{k} \cap (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha})$. Then $\mathfrak{m} = \bigoplus_{\alpha \in R^+} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has a real basis consisting of X_α and Y_α, in which*

$$\mathrm{Ad}_t|_{\mathfrak{m}_\alpha} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \text{where } t^\alpha = e^{i\theta}.$$

Consequently, writing $t^\alpha = e^{i\theta_\alpha}$,

$$J(t) = \prod_{\alpha \in R^+} |1 - t^\alpha|^2 = \prod_{\alpha \in R} (1 - t^\alpha) = \prod_{\alpha \in R^+} 4 \sin^2 \frac{\theta_\alpha}{2}. \tag{9.5}$$

Proof. Let σ be complex conjugation on $\mathfrak{k}_{\mathbb{C}}$ with respect to $\mathfrak{k}$. It commutes with the real map Ad_t. For $E \in \mathfrak{g}_\alpha$ we get

$$\mathrm{Ad}_t(\sigma E) = \sigma(t^\alpha E) = \overline{t^\alpha}\, \sigma E = t^{-\alpha} \sigma E,$$

because $|t^\alpha| = 1$. So $\sigma \mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Choose $E \neq 0$ in $\mathfrak{g}_\alpha$ and set

$$X_\alpha = E + \sigma E, \qquad Y_\alpha = i(E - \sigma E).$$

Both are fixed by σ, so they lie in $\mathfrak{k}$. An element $aE + b\,\sigma E$ is fixed by σ exactly when $b = \bar{a}$. If $a = x + iy$, then $aE + \bar{a}\,\sigma E = xX_\alpha + yY_\alpha$. Hence X_α and Y_α form a basis of $\mathfrak{m}_\alpha$. With $t^\alpha = e^{i\theta}$,

$$\begin{aligned} \mathrm{Ad}_t X_\alpha &= e^{i\theta} E + e^{-i\theta} \sigma E = \cos\theta\, X_\alpha + \sin\theta\, Y_\alpha, \\ \mathrm{Ad}_t Y_\alpha &= i e^{i\theta} E - i e^{-i\theta} \sigma E = -\sin\theta\, X_\alpha + \cos\theta\, Y_\alpha. \end{aligned}$$

Taking σ-fixed points in (9.1) gives $\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha > 0} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ is orthogonal to $\mathfrak{t}$. To see this, pick $H \in \mathfrak{t}$ with $\alpha(H) \neq 0$. Then $[H, X_\alpha] = -i\alpha(H) Y_\alpha$ and $[H, Y_\alpha] = i\alpha(H) X_\alpha$, so $\mathfrak{m}_\alpha \subset [\mathfrak{t}, \mathfrak{k}]$. Invariance of the inner product gives $\langle [H, X], H' \rangle = -\langle X, [H, H'] \rangle = 0$ for $H' \in \mathfrak{t}$. Counting dimensions, $\mathfrak{m} = \bigoplus_{\alpha > 0} \mathfrak{m}_\alpha$.

For one plane,

$$\det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = |1 - e^{i\theta}|^2.$$

This equals $4\sin^2(\theta/2)$. Multiply over $\alpha \in R^+$. Finally $(1 - t^\alpha)(1 - t^{-\alpha}) = |1 - t^\alpha|^2$, since $t^{-\alpha} = \overline{t^\alpha}$. ∎

Thus $J \geq 0$, and $J(t) > 0$ exactly when $t \in T_{\mathrm{reg}}$. Also $J(t^{-1}) = J(t)$.

Next we write the differential of ψ in translated coordinates. For $X \in \mathfrak{m}$ and $H \in \mathfrak{t}$ put

$$L_{(gT,t)}(X, H) = \frac{d}{ds}\Big|_{s=0} \psi(gT, t)^{-1}\, \psi\big(g \exp(sX) T,\ t \exp(sH)\big) \in \mathfrak{k}.$$

We identify the tangent space of K/T at gT with $\mathfrak{m}$ by $X \mapsto \frac{d}{ds}|_{s=0}\, g \exp(sX) T$. We identify the tangent spaces of T at t and of K at x with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation. In these identifications, each invariant density is given at every point by its density at the base point. Let $\kappa > 0$ be the ratio of the base-point density of dk to the product of the base-point densities of $d(gT)$ and dt, all evaluated on orthonormal bases. Hence the Jacobian of ψ relative to $d(gT)\, dt$

and dk equals $\kappa\,|\det L_{(gT,t)}|$. Here the determinant is taken with respect to orthonormal bases of $\mathfrak{m} \oplus \mathfrak{t}$ and $\mathfrak{k}$.

The value $|\det L_{(gT,t)}|$ does not depend on the representative g of the coset gT. Indeed, replacing g by gs with $s \in T$ composes L with $\mathrm{Ad}_s|_{\mathfrak{m}}$. This map has determinant 1.

Proposition 9.5. *For all $g \in K$ and $t \in T$, $|\det L_{(gT,t)}| = J(t)$.*

Proof. Multiplying out,

$$\psi(gT,t)^{-1}\psi(ge^{sX}T, te^{sH}) = gt^{-1}e^{sX}t\,e^{sH}e^{-sX}g^{-1} = g\,\exp(s\,\mathrm{Ad}_{t^{-1}}X)\exp(sH)\exp(-sX)\,g^{-1}.$$

The velocity at $s = 0$ of a product of one-parameter subgroups is the sum of their velocities (Lecture 2). Hence

$$L_{(gT,t)}(X,H) = \mathrm{Ad}_g\big((\mathrm{Ad}_{t^{-1}} - 1)X + H\big).$$

Relative to $\mathfrak{k} = \mathfrak{m} \oplus \mathfrak{t}$ this is Ad_g composed with the block map

$$\begin{pmatrix} \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}} - 1 & 0 \\ 0 & 1_{\mathfrak{t}} \end{pmatrix}.$$

The map Ad_g is orthogonal, so $|\det \mathrm{Ad}_g| = 1$. Therefore $|\det L| = |\det(1 - \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}})| = J(t^{-1}) = J(t)$. ∎

Proposition 9.6. *(a) For $t \in T$, $\dim Z_K(t) = \dim T + 2\,\#\{\alpha \in R^+ : t^\alpha = 1\}$. Hence $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.*

(b) If $t, t' \in T_{\mathrm{reg}}$ and $gtg^{-1} = t'$, then $g \in N_K(T)$.

(c) Every $x \in K_{\mathrm{reg}}$ has exactly $|W|$ preimages under ψ.

Proof. (a) The Lie algebra of the closed subgroup $Z_K(t)$ is $\{X \in \mathfrak{k} : \mathrm{Ad}_t X = X\}$ (Lecture 3). A plane rotation by θ fixes a nonzero vector only when $e^{i\theta} = 1$. By the root-plane lemma (Lemma 9.4), the fixed space of Ad_t is $\mathfrak{t} \oplus \bigoplus_{t^\alpha = 1} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has dimension 2, which gives the dimension formula. Centraliser dimensions are unchanged by conjugation. So if t is conjugate to a regular t', then t is regular.

Now let $(gT, t) \in \psi^{-1}(K_{\mathrm{reg}})$. Then gtg^{-1} is conjugate to some $t' \in T_{\mathrm{reg}}$, so t is conjugate to t', and hence $t \in T_{\mathrm{reg}}$. Conversely, $\psi(gT,t) \in K_{\mathrm{reg}}$ for every $t \in T_{\mathrm{reg}}$, by the definition of K_{reg}. These two inclusions give $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.

(b) By the dimension formula in (a), the identity component $Z_K(t')_0$ has Lie algebra $\mathfrak{t}$. It contains the connected group T with the same Lie algebra, so $Z_K(t')_0 = T$ (Lecture 2). The torus gTg^{-1} is connected and centralises $gtg^{-1} = t'$. So $gTg^{-1} \subset T$. Then $\mathrm{Ad}_g\,\mathfrak{t} \subset \mathfrak{t}$, with equality by dimension. Hence $gTg^{-1} \supset \exp\mathfrak{t} = T$, and $g \in N_K(T)$.

(c) Let $x = \psi(gT,t)$ with t regular. Suppose $\psi(hT,s) = x$. By the description of $\psi^{-1}(K_{\mathrm{reg}})$ in (a), s is regular. Put $n = g^{-1}h$. Then $nsn^{-1} = t$. Since s and t are regular, the normaliser statement (b) gives $n \in N_K(T)$. Thus the preimages are the pairs $(gnT,\ n^{-1}tn)$ with $nT \in W$. Each such pair is a preimage. Different classes nT give different cosets gnT. So there are exactly $|W|$ preimages. ∎

Lemma 9.7. $\displaystyle\int_T J(t)\,dt = |W|.$

Proof. The Weyl denominator identity [I] holds in $\mathbb{Z}[\mathcal{X}(T)]$:

$$\prod_{\alpha>0}(1-e^{-\alpha}) = \sum_{w\in W}\operatorname{sgn}(w)\,e^{w\rho-\rho}.$$

Here $\rho - w\rho$ is a sum of positive roots. So $w\rho-\rho \in \mathcal{X}(T)$ even when $\rho \notin \mathcal{X}(T)$. Evaluate both sides at t, and write $D(t)$ for the value of the left side. Then $|D(t)|^2 = J(t)$ by (9.5), so

$$J(t) = \sum_{w,w'\in W}\operatorname{sgn}(w)\operatorname{sgn}(w')\,t^{w\rho-w'\rho}.$$

Distinct characters of T are orthonormal in $L^2(T)$ (Lecture 5). Thus $\int_T t^\mu\,dt = 0$ for $\mu \neq 0$. Since ρ is regular [I], $w\rho = w'\rho$ only when $w = w'$. Hence only the diagonal terms survive, and the integral is $|W|$. ∎

Proof of the Weyl integration formula (Theorem 9.2) from (C1) and (C2). Null sets. The set $K\setminus K_{\mathrm{reg}}$ is null by the regular-set import (C1). The complement of $\psi^{-1}(K_{\mathrm{reg}})$ is $K/T\times(T\setminus T_{\mathrm{reg}})$. Now $T\setminus T_{\mathrm{reg}}$ is the union of the kernels H_α of $t\mapsto t^\alpha$. Each H_α has infinitely many cosets, because t^α maps the connected group T onto the circle. The cosets are disjoint and have equal Haar measure. Since T has finite measure, H_α is null.

Covering. On $K/T\times T_{\mathrm{reg}}$ the Jacobian is $\kappa J(t) > 0$, so ψ is a local diffeomorphism there. By the fibre count in Proposition 9.6, its fibres have $|W|$ points.

Change of variables. Both $K/T\times T$ and K are compact. The change-of-variables import (C2) gives

$$\kappa\int_T J(t)\int_{K/T} f(gtg^{-1})\,d(gT)\,dt = |W|\int_K f(k)\,dk.$$

Take $f = 1$. Then $\kappa\int_T J\,dt = |W|$. Since $\int_T J\,dt = |W|$ by Lemma 9.7, we get $\kappa = 1$. This is (9.3). ∎

9.3 The model: class integrals on $\mathrm{SU}(2)$

Let $K = \mathrm{SU}(2)$ with maximal torus $T = \{t_\theta\}$, where

$$t_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta}\end{pmatrix}, \qquad dt = \frac{d\theta}{2\pi} \text{ on } [0,2\pi).$$

Use the invariant inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. The plane $\mathfrak{m}$ has basis

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0\end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0\end{pmatrix}.$$

A direct multiplication gives

$$t_\theta B t_\theta^{-1} = \begin{pmatrix} 0 & e^{2i\theta} \\ -e^{-2i\theta} & 0\end{pmatrix} = \cos 2\theta\, B + \sin 2\theta\, C, \qquad t_\theta C t_\theta^{-1} = -\sin 2\theta\, B + \cos 2\theta\, C.$$

So $t_\theta^\alpha = e^{2i\theta}$ and

$$J(t_\theta) = |1 - e^{2i\theta}|^2 = 4\sin^2\theta.$$

The Weyl group has order 2. Its nontrivial element is represented by

$$n = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad nt_\theta n^{-1} = t_{-\theta}.$$

Now (9.4) gives, for a continuous class function f,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{1}{\pi}\int_0^{2\pi} f(t_\theta)\sin^2\theta\,d\theta.$$

Since $f(t_{2\pi-\theta}) = f(t_{-\theta}) = f(t_\theta)$ and $\sin^2$ is symmetric about π,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{2}{\pi}\int_0^{\pi} f(t_\theta)\sin^2\theta\,d\theta. \tag{9.6}$$

Every $g \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm i\theta}$ for a unique $\theta \in [0,\pi]$, and $\operatorname{tr} g = 2\cos\theta$. So a class function is a function of $\theta \in [0,\pi]$. For $f = 1$ the right side is $\frac{2}{\pi}\cdot\frac{\pi}{2} = 1$.

Character norms. By the Weyl character formula [I], the representation V_m of dimension $m+1$ has

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin(m+1)\theta}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Then (9.6) gives

$$\int_{\mathrm{SU}(2)} \chi_m\overline{\chi_n}\,dk = \frac{2}{\pi}\int_0^{\pi} \sin(m+1)\theta\,\sin(n+1)\theta\,d\theta = \delta_{mn}.$$

The factor $\sin^2\theta$ cancels both denominators. By Lecture 5, this confirms that each V_m is irreducible and that the representations V_m are pairwise inequivalent.

A tensor-product multiplicity. The multiplicity of V_n in V is $\int_K \chi_V\overline{\chi_n}\,dk$ (Lecture 5). Take $V = V_1^{\otimes 4}$, so $\chi_V(t_\theta) = (2\cos\theta)^4$. For $n = 0$ we need the Wallis integrals

$$\int_0^{\pi}\cos^4\theta\,d\theta = \frac{3\pi}{8}, \qquad \int_0^{\pi}\cos^6\theta\,d\theta = \frac{5\pi}{16}.$$

Write $\sin^2 = 1 - \cos^2$. Then

$$\int_0^{\pi}\cos^4\theta\sin^2\theta\,d\theta = \frac{3\pi}{8} - \frac{5\pi}{16} = \frac{\pi}{16}, \qquad \dim V^{\mathrm{SU}(2)} = \frac{2}{\pi}\cdot 16\cdot\frac{\pi}{16} = 2.$$

So the trivial representation V_0 occurs twice in $V_1^{\otimes 4}$. Exercise 9.3 finds the full decomposition from weights and checks the multiplicity of V_2 by integration.

9.4 Characters on the torus and nonsimply connected groups

Let $\lambda \in \mathcal{X}(T)$ be dominant. Suppose K has an irreducible representation V_λ with highest weight λ. Write χ_λ for its character.

Since K is connected, V_λ is determined by its differential. The complex-linear extension of the differential makes V_λ an irreducible $\mathfrak{k}_{\mathbb{C}}$-module of highest weight λ. The centre of $\mathfrak{k}_{\mathbb{C}}$ acts by a scalar. Hence the semisimple Weyl character formula [I] applies to the derived algebra, with the central weight retained. The resulting formula is an identity in $\mathbb{Z}[\mathcal{X}(T)]$. We use it in the form

$$\chi_\lambda(t)\, D(t) = \sum_{w \in W} \operatorname{sgn}(w)\, t^{w(\lambda+\rho)-\rho}, \qquad D(t) = \prod_{\alpha>0} (1 - t^{-\alpha}). \tag{9.7}$$

Each exponent equals $w\lambda - (\rho - w\rho)$, so it lies in $\mathcal{X}(T)$. Thus (9.7) is an identity of functions on T for every K. If $\rho \in \mathcal{X}(T)$, multiplying by t^ρ gives the usual quotient form.

Proposition 9.8. *Let $\lambda, \mu \in \mathcal{X}(T)$ be highest weights of irreducible representations of K. Then $\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \delta_{\lambda\mu}$.*

Proof. Since $|D|^2 = J$, (9.4) gives

$$\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \frac{1}{|W|} \sum_{w,w'} \operatorname{sgn}(ww') \int_T t^{w(\lambda+\rho)-w'(\mu+\rho)}\, dt.$$

A term survives only if $w(\lambda+\rho) = w'(\mu+\rho)$. Both $\lambda+\rho$ and $\mu+\rho$ are strictly dominant. Each W-orbit meets the dominant chamber exactly once, and strictly dominant weights have trivial stabiliser [I]. So a term survives only if $\lambda = \mu$ and $w = w'$. There are $|W|$ such terms. ∎

This agrees with Schur orthogonality. It is a consistency check, not a new proof. Which dominant $\lambda \in \mathcal{X}(T)$ occur is the subject of Lecture 8. The proposition only needs λ and μ to occur.

Comparison with SO(3). Let $T = \{R_\varphi\}$, where R_φ is the rotation through φ about the z-axis. Exercise 9.4 shows that Ad_{R_φ} rotates $\mathfrak{m}$ through φ. So $R_\varphi^\alpha = e^{i\varphi}$, $J(R_\varphi) = 4\sin^2(\varphi/2)$ and $|W| = 2$. Formal exponentiation of the weight $\rho = \alpha/2$ would give $e^{i\varphi/2}$. This is not a function on T, because φ and $\varphi + 2\pi$ give the same rotation.

The density J is still defined. Conjugation by the rotation $\mathrm{diag}(1,-1,-1)$ reverses the z-axis, so it maps R_φ to $R_{-\varphi} = R_{2\pi-\varphi}$. Hence a class function satisfies $f(R_{2\pi-\varphi}) = f(R_\varphi)$. The factor $\sin^2\frac{\varphi}{2}$ is also unchanged under $\varphi \mapsto 2\pi - \varphi$, and $2\sin^2\frac{\varphi}{2} = 1 - \cos\varphi$. Therefore the class-function formula (9.4) gives

$$\int_{\mathrm{SO}(3)} f\, dk = \frac{1}{\pi} \int_0^{2\pi} f(R_\varphi) \sin^2\frac{\varphi}{2}\, d\varphi = \frac{1}{\pi} \int_0^{\pi} f(R_\varphi)(1 - \cos\varphi)\, d\varphi. \tag{9.8}$$

The irreducible representations of SO(3) are the V_{2j} (Lecture 8). Here $\lambda = j\alpha$, and (9.7) reads

$$\chi_{2j}(R_\varphi)\,(1 - e^{-i\varphi}) = e^{ij\varphi} - e^{-i(j+1)\varphi}.$$

Hence

$$\int |\chi_{2j}|^2 = \frac{1}{2} \int_T |e^{ij\varphi} - e^{-i(j+1)\varphi}|^2 dt = \frac{1}{2} \cdot 2 = 1.$$

9.5 Self-checks

1. Find J for $\mathrm{U}(2)$ with $t = \mathrm{diag}(e^{ia}, e^{ib})$. Is $\rho \in \mathcal{X}(T)$?
   Answer. There is one positive root, $t^\alpha = e^{i(a-b)}$. So $J = |e^{ia} - e^{ib}|^2 = 4\sin^2\frac{a-b}{2}$. Identify $\mathcal{X}(T)$ with $\mathbb{Z}^2$, where (p, q) labels the weight μ with $t^\mu = e^{i(pa+qb)}$. Then $\alpha = (1, -1)$ and $\rho = (\frac{1}{2}, -\frac{1}{2}) \notin \mathcal{X}(T)$.
2. What is the Haar probability that $g \in \mathrm{SU}(2)$ has $\operatorname{tr} g \geq 0$?
   Answer. The remark on indicator functions (Remark 9.3) with $c(g) = \operatorname{tr} g$ lets us use (9.6) for this indicator. The condition on the trace means $\theta \in [0, \pi/2]$. The probability is
   $$\frac{2}{\pi}\int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}.$$
   This matches the symmetry $g \mapsto -g$.
3. Why can $\int_{\mathrm{SU}(2)} |g_{11}|^2 dk$ not be computed from (9.6) with $f(t_\theta) = |e^{i\theta}|^2$?
   Answer. The function $g \mapsto |g_{11}|^2$ is not a class function. Schur orthogonality (Lecture 5) gives the correct value $1/\dim V_1 = \frac{1}{2}$. Using $f(t_\theta) = 1$ in (9.6) would give the wrong value 1. To compute this integral with the Weyl integration formula, one must use the general form (9.3).

Summary

Let K be a compact connected Lie group with maximal torus T. Every regular element of K has exactly $|W|$ preimages under the conjugation map $K/T \times T \to K$. Relative to the normalised measures, the Jacobian of this map at (gT, t) is $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$. This determinant is a product of root-plane factors $|1 - t^\alpha|^2$ over the positive roots α. The Weyl denominator identity fixes the constant. For $\mathrm{SU}(2)$ and a continuous class function f, the Haar integral of f is $\frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\, d\theta$. With the character formula in the form (9.7), norms and multiplicities become Fourier calculations on T. This form never needs ρ to be a character.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., §11.6.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1, V.2 and VI.1.
- Knapp Beyond: *Lie Groups Beyond an Introduction*, VIII.5.
- Lee: *Introduction to Smooth Manifolds*, 2nd ed., Chapter 16, Propositions 16.40–16.42.

Lecture 10

Peter–Weyl and completeness

Objectives. *Lecture 5 proved that matrix coefficients of inequivalent irreducible representations are orthogonal. This lecture proves that finite-dimensional matrix coefficients span a dense subspace of the square-integrable functions on a compact Lie group.*

By the end of this lecture you will be able to:

- Average a rank-one projection to obtain a nonzero positive compact intertwiner.
- Prove that the finite-dimensional invariant subspaces span a dense subspace in every strongly continuous unitary representation of a compact Lie group.
- Identify the finite-dimensional subspaces of $L^2(K)$ invariant under the right translations ρ with spaces of matrix coefficients, and state the Peter–Weyl theorem with multiplicities.
- Compute the coefficient blocks of $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

10.1 Analytic preliminaries

Throughout, K is a compact Lie group. We write $\int_K f(g)\,dg$ for its normalised Haar integral on $C(K)$. We import four facts from Lecture 5.

(H1) The integral is left and right invariant. It is invariant under $g \mapsto g^{-1}$, and $\int_K 1\,dg = 1$.
(H2) If $f \in C(K)$ is nonnegative and not identically zero, then $\int_K f\,dg > 0$.
(H3) A finite-dimensional continuous representation has an invariant inner product. It is then an orthogonal sum of irreducible subrepresentations.
(H4) Schur orthogonality, in the form (10.1) below.

We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5. It states that the two iterated Haar integrals agree.

Let (π, V) be a finite-dimensional continuous unitary representation. Its *matrix coefficients* are the continuous functions

$$c^{\pi}_{u,v}(x) = \langle u, \pi(x)v \rangle, \qquad u, v \in V.$$

The *coefficient space* $\mathscr{R}(K)$ is the span of all such functions, for all such (π, V). Let $\mathrm{Irr}(K)$ be a set of representatives of the isomorphism classes of irreducible finite-dimensional continuous

unitary representations. Write $d_\pi = \dim V_\pi$. For $\pi, \sigma \in \mathrm{Irr}(K)$, Schur orthogonality (H4) reads

$$\int_K \overline{c^{\pi}_{u,v}(x)}\, c^{\sigma}_{u',v'}(x)\, dx = \begin{cases} d_\pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \sigma = \pi, \\ 0, & \sigma \neq \pi. \end{cases} \tag{10.1}$$

Definition 10.1. Let $(H_i)_{i\in I}$ be mutually orthogonal closed subspaces of a Hilbert space H. Their *Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the closure of the algebraic sum $\bigoplus_i H_i$.

The elements of the Hilbert sum are the convergent series $\sum_i v_i$ with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$. For Hilbert spaces H_i not given inside one space, the *external Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the space of families (v_i) with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$.

A bounded operator on a Hilbert space H is *compact* if it is an operator-norm limit of finite-rank operators. A self-adjoint operator T on H is *positive* if $\langle v, Tv\rangle \geq 0$ for all $v \in H$.

We import the spectral theorem for compact self-adjoint operators from Linear Algebra for Physics, Weeks 13–14, in the following form. If T is compact and self-adjoint, then

$$H = \ker T \;\widehat{\oplus}\; \widehat{\bigoplus}_{\mu \neq 0} \ker(T - \mu).$$

Moreover, each $\ker(T-\mu)$ with $\mu \neq 0$ is finite-dimensional. If $T \neq 0$, then T has a nonzero eigenvalue.

Definition 10.2. The space $L^2(K)$ is the Hilbert completion of $C(K)$ for the sesquilinear form $\langle f, h\rangle = \int_K \overline{f(g)}\, h(g)\, dg$.

By positivity (H2), this sesquilinear form is an inner product on $C(K)$. Hence the map $C(K) \to L^2(K)$ is injective. An element of $L^2(K)$ is *continuous* if it lies in $C(K)$. Only continuous elements have values $f(x)$. For continuous functions, $\|f\|_2 \leq \|f\|_\infty = \sup_x |f(x)|$.

Definition 10.3. A *strongly continuous unitary representation* of K on a Hilbert space H is a homomorphism Π from K to the unitary group of H such that $g \mapsto \Pi(g)v$ is continuous for every $v \in H$.

Lemma 10.4. *Let $F\colon K \to H$ be continuous. There is a unique vector $\int_K F(g)\, dg$ in H with*

$$\left\langle w, \int_K F(g)\, dg \right\rangle = \int_K \langle w, F(g)\rangle\, dg \qquad (w \in H).$$

Moreover $\|\int_K F\| \leq \int_K \|F(g)\|\, dg$, and $A\int_K F = \int_K AF$ for bounded A. If each $F(g)$ lies in a closed subspace W, so does $\int_K F$. The integral is unchanged if $F(g)$ is replaced by $F(hg)$.

Proof. The scalar integrand is continuous. The map $w \mapsto \overline{\int_K \langle w, F\rangle}$ is linear with bound $\|w\| \int_K \|F\|$. The Riesz representation theorem for Hilbert spaces, from Linear Algebra for Physics, gives the vector and the norm bound. For bounded A, $\langle w, A\int F\rangle = \int \langle A^* w, F\rangle = \int \langle w, AF\rangle$. If $w \perp W$, the integrand vanishes, so $\int F \in W^{\perp\perp} = W$. Invariance follows from the left invariance in (H1). ∎

Lemma 10.5. *(a) For $\varphi \in C(K)$ and $\varepsilon > 0$ there is an open $U \ni e$ with $|\varphi(yx) - \varphi(x)| < \varepsilon$ and $|\varphi(xy) - \varphi(x)| < \varepsilon$ for all $x \in K$, $y \in U$.*

(b) *For each open $U \ni e$ there is a continuous $\varphi \geq 0$ vanishing outside U with $\varphi(e) > 0$ and $\int_K \varphi = 1$.*

Proof. (a) The function $F(y,x) = |\varphi(yx) - \varphi(x)| + |\varphi(xy) - \varphi(x)|$ is continuous on $K \times K$ and vanishes on $\{e\} \times K$. For each x there are open sets $U_x \ni e$ and $N_x \ni x$ with $F < \varepsilon$ on $U_x \times N_x$. Finitely many $N_{x_1}, \dots, N_{x_r}$ cover K. Take $U = \bigcap_j U_{x_j}$.

(b) Fix a norm $|\cdot|$ on $\mathfrak{k}$, and let B_r be the open ball of radius r about 0. By Lecture 2, exp maps B_r homeomorphically onto an open neighbourhood of e for all sufficiently small $r > 0$. Shrink r so that $\exp(B_r) \subset U$. Put $\psi(\exp X) = \max(0, r/2 - |X|)$ on $\exp(B_r)$ and $\psi = 0$ elsewhere. Then ψ is continuous on $\exp(B_r)$.

The closed ball $\overline{B}_{r/2}$ is compact, so its image under exp is compact and hence closed in the Hausdorff space K. The function ψ vanishes on the open complement $K \setminus \exp(\overline{B}_{r/2})$. The two open sets $\exp(B_r)$ and $K \setminus \exp(\overline{B}_{r/2})$ cover K, so ψ is continuous on K. Since $\psi(e) = r/2 > 0$, positivity (H2) gives $\int \psi > 0$. Take $\varphi = \psi / \int \psi$. Then $\varphi(e) > 0$. ∎

10.2 An averaged compact operator

Theorem 10.6. *Let Π be a strongly continuous unitary representation of K on $H \neq 0$.*

(a) The space H contains a nonzero finite-dimensional invariant subspace.

(b) The sum H_{fin} of all finite-dimensional invariant subspaces is dense in H.

Proof. Fix a unit vector $u \in H$. Let E_u be the rank-one projection given by $E_u v = \langle u, v\rangle u$. Then $\Pi(g) E_u \Pi(g)^{-1} v = \langle \Pi(g)u, v\rangle \Pi(g)u$, which is continuous in g. Define the average

$$T v = \int_K \langle \Pi(g)u, v\rangle \, \Pi(g)u \, dg. \tag{10.2}$$

Step 1: T is bounded, positive and nonzero. The vector-integral lemma, Lemma 10.4, gives $\|Tv\| \leq \|v\|$ and

$$\langle w, Tv\rangle = \int_K \langle \Pi(g)u, v\rangle \langle w, \Pi(g)u\rangle \, dg.$$

Exchanging v, w and conjugating gives the same integral, so $T^* = T$. For $w = v$ the integrand is $|\langle \Pi(g)u, v\rangle|^2 \geq 0$, so T is positive. For $v = w = u$ this integrand is continuous with value 1 at e. By positivity (H2), $\langle u, Tu\rangle > 0$. Hence T is nonzero.

Step 2: T commutes with Π. For $h \in K$,

$$T\Pi(h)v = \int_K \langle \Pi(h^{-1}g)u, v\rangle \, \Pi(g)u \, dg = \int_K \langle \Pi(g)u, v\rangle \, \Pi(h)\Pi(g)u \, dg = \Pi(h) T v.$$

The first equality uses unitarity. The second replaces g by hg. The third takes $A = \Pi(h)$ in the vector-integral lemma, Lemma 10.4.

Step 3: T is compact. The orbit $O = \{\Pi(g)u : g \in K\}$ is compact, since it is a continuous image of K. Let $\varepsilon > 0$. Since O is compact, finitely many open balls of radius ε with centres in O cover O.

Let P be the orthogonal projection onto the span of the centres. Put $x_g = \Pi(g)u$ and $y_g = Px_g$. Each x_g lies within ε of some centre, and y_g is the nearest point of the span of the centres to x_g. Hence $\|x_g - y_g\| < \varepsilon$ for all g. Now

$$\langle x_g, v\rangle\langle w, x_g\rangle - \langle y_g, v\rangle\langle w, y_g\rangle = \langle x_g - y_g, v\rangle\langle w, x_g\rangle + \langle y_g, v\rangle\langle w, x_g - y_g\rangle.$$

Because $\|x_g\| = 1$ and $\|y_g\| \le 1$, the Cauchy–Schwarz inequality shows that this difference has modulus at most $2\varepsilon\|v\|\|w\|$. On the other hand,

$$\langle w, PTPv\rangle = \langle Pw, TPv\rangle = \int_K \langle y_g, v\rangle\langle w, y_g\rangle \, dg.$$

Subtracting gives $|\langle w, (T - PTP)v\rangle| \le 2\varepsilon\|v\|\|w\|$, so $\|T - PTP\| \le 2\varepsilon$. The operator PTP has finite rank, since P has finite rank. Hence T is compact.

Step 4: proof of (a). By Steps 1 and 3, T is compact, self-adjoint and nonzero. Hence the spectral theorem gives an eigenvalue $\mu \neq 0$ of T with finite-dimensional eigenspace $H_\mu \neq 0$. If $Tv = \mu v$, then Step 2 gives $T\Pi(h)v = \Pi(h)Tv = \mu\Pi(h)v$. So H_μ is invariant.

Step 5: proof of (b). A sum of two finite-dimensional invariant subspaces is again one, so H_{fin} is a subspace. Its closure M is invariant. Hence $M^\perp$ is closed and invariant, since $\langle \Pi(g)w, m\rangle = \langle w, \Pi(g)^{-1}m\rangle$. Restricting Π gives a strongly continuous unitary representation on $M^\perp$. If $M^\perp \neq 0$, the existence statement (a), applied to $M^\perp$, gives a finite-dimensional invariant subspace $W \neq 0$ in $M^\perp$. But $W \subset H_{\mathrm{fin}} \subset M$. Thus $W \subset M \cap M^\perp = 0$, a contradiction. ∎

Corollary 10.7. *Let Π be a strongly continuous unitary representation of K on H. Suppose that Π is irreducible, meaning that $H \neq 0$ and that the only closed invariant subspaces are 0 and H. Then $\dim H < \infty$.*

Proof. Since $H \neq 0$, the existence statement, part (a) of Theorem 10.6, gives a nonzero finite-dimensional invariant subspace W. Finite-dimensional subspaces are closed, so $W = H$ by irreducibility. ∎

10.3 The regular representation

For $f \in C(K)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = f(xg).$$

Lemma 10.8. *The translations λ and ρ extend to commuting strongly continuous unitary representations on $L^2(K)$.*

Proof. We treat ρ. The proof for λ is the same, except that it uses left invariance and replaces U below by $U \cap U^{-1}$.

First, ρ is a homomorphism on $C(K)$. For $f \in C(K)$ and $g, h, x \in K$,

$$(\rho(g)\rho(h)f)(x) = (\rho(h)f)(xg) = f(xgh) = (\rho(gh)f)(x).$$

By the right invariance in (H1), $\rho(g)$ preserves $\|\cdot\|_2$. So $\rho(g)$ extends to an isometry of $L^2(K)$ with inverse $\rho(g^{-1})$.

Next, we prove strong continuity at e. Let $f \in L^2(K)$ and $\varepsilon > 0$. Choose $f_0 \in C(K)$ with $\|f - f_0\|_2 < \varepsilon$. Uniform continuity, Lemma 10.5(a), gives an open $U \ni e$ with $\|\rho(g)f_0 - f_0\|_2 \le \|\rho(g)f_0 - f_0\|_\infty < \varepsilon$ for $g \in U$. For $g \in U$, the triangle inequality and the isometry property give

$$\|\rho(g)f - f\|_2 \le \|\rho(g)(f - f_0)\|_2 + \|\rho(g)f_0 - f_0\|_2 + \|f_0 - f\|_2 < 3\varepsilon.$$

Continuity at g_0 follows from $\rho(g)f - \rho(g_0)f = \rho(g_0)\big(\rho(g_0^{-1}g)f - f\big)$, since $\rho(g_0)$ is an isometry.

Finally, for $f \in C(K)$ both $\lambda(g)\rho(h)f$ and $\rho(h)\lambda(g)f$ are the function $x \mapsto f(g^{-1}xh)$. The two bounded operators therefore agree on the dense subspace $C(K)$, so they commute on $L^2(K)$. ∎

For $\varphi \in C(K)$ and $f \in L^2(K)$ put

$$\rho(\varphi)f = \int_K \varphi(g)\rho(g)f \, dg.$$

Lemma 10.9. *For $\varphi \in C(K)$ and $f \in L^2(K)$, the element $\rho(\varphi)f$ is continuous, and*

$$(\rho(\varphi)f)(x) = \langle k_x, f\rangle, \qquad k_x(h) = \overline{\varphi(x^{-1}h)}.$$

Proof. Put $(Sf)(x) = \langle k_x, f\rangle$. Fix $\varepsilon > 0$. Choose U for φ by uniform continuity, Lemma 10.5(a). If $x' \in xU^{-1}$, then $y = x'^{-1}x \in U$. Writing $t = x^{-1}h$ gives $|k_x(h) - k_{x'}(h)| = |\varphi(t) - \varphi(yt)| < \varepsilon$. By the Cauchy–Schwarz inequality, $|Sf(x) - Sf(x')| \le \varepsilon\|f\|_2$. So $Sf \in C(K)$ and $\|Sf\|_2 \le \|Sf\|_\infty \le \|\varphi\|_\infty\|f\|_2$. For $f, w \in C(K)$, Fubini and the substitution $h = xg$ give

$$\langle w, \rho(\varphi)f\rangle = \int_K \varphi(g) \int_K \overline{w(x)}f(xg)\,dx\,dg = \int_K \overline{w(x)} \int_K \varphi(x^{-1}h)f(h)\,dh\,dx = \langle w, Sf\rangle.$$

As $C(K)$ is dense, $\rho(\varphi)f = Sf$ for $f \in C(K)$. Both operators are bounded, so they agree on $L^2(K)$. ∎

Proposition 10.10. *(a) Let $W \subset L^2(K)$ be a finite-dimensional ρ-invariant subspace. Then $W \subset C(K)$. There is $u_W \in W$ with $f(x) = \langle u_W, \rho(x)f\rangle$ for all $f \in W$ and $x \in K$.*

(b) The coefficient space $\mathscr{R}(K)$ is the sum of all finite-dimensional ρ-invariant subspaces. It is dense in $L^2(K)$.

Proof. (a) Choose an orthonormal basis $f_1, \dots, f_n$ of W. Strong continuity gives an open $U \ni e$ with $\|\rho(g)f_i - f_i\| < 1/(2n)$ for all i and $g \in U$. By Lemma 10.5(b), there is a continuous $\varphi \ge 0$ vanishing outside U with $\int_K \varphi = 1$. This normalisation gives $\rho(\varphi)f - f = \int_K \varphi(g)\big(\rho(g)f - f\big)\,dg$. For a unit vector $f = \sum a_i f_i$ we have $|a_i| \le 1$. Hence

$$\|\rho(\varphi)f - f\| \le \sum_i |a_i| \int_K \varphi(g)\|\rho(g)f_i - f_i\|\,dg \le n \cdot \frac{1}{2n} = \frac{1}{2}.$$

So $\|\rho(\varphi)f\| \ge \tfrac{1}{2}$ for every unit vector $f \in W$, and $\rho(\varphi)$ is injective on W. Since W is closed and ρ-invariant, the vector-integral lemma, Lemma 10.4, shows that $\rho(\varphi)$ maps W into W.

As W is finite-dimensional, $\rho(\varphi)W = W$. Every element of W is therefore of the form $\rho(\varphi)f$, and the smoothing lemma, Lemma 10.9, gives $W \subset C(K)$.

Now evaluation $f \mapsto f(e)$ is a linear functional on W. The Riesz representation theorem in W gives u_W with $f(e) = \langle u_W, f\rangle$. For $f \in W$, $\rho(x)f$ lies in W and is the function $h \mapsto f(hx)$. Hence $f(x) = (\rho(x)f)(e) = \langle u_W, \rho(x)f\rangle$.

(b) The restriction of ρ to W is a finite-dimensional continuous unitary representation. By the reproducing formula in (a), each $f \in W$ is a matrix coefficient of this representation. Hence $W \subset \mathscr{R}(K)$. Conversely, $\rho(h)c^{\pi}_{u,w} = c^{\pi}_{u,\pi(h)w}$, so each $c^{\pi}_{u,v}$ lies in the finite-dimensional ρ-invariant space $\{c^{\pi}_{u,w} : w \in V\}$. Density follows from the density statement Theorem 10.6(b), applied to the representation ρ of the regular-representation lemma, Lemma 10.8. ∎

10.4 The Peter–Weyl theorem

The group $K \times K$ acts on $L^2(K)$ by

$$((g,h)\cdot f)(x) = f(g^{-1}xh), \qquad \text{that is, } (g,h) \mapsto \lambda(g)\rho(h).$$

For $\pi \in \mathrm{Irr}(K)$, give $\mathrm{End}(V_\pi)$ the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^*B)$ and the action $(g,h)\cdot A = \pi(h)A\pi(g)^{-1}$.

Let $\overline{V}_\pi$ be the conjugate space, with $\overline{\pi}(g)\bar{u} = \overline{\pi(g)u}$ and inner product $\langle \bar{u}, \bar{v}\rangle = \langle v, u\rangle$. For this inner product, $\overline{\pi}$ is unitary. The representation $\overline{\pi}$ is irreducible, since $W \mapsto \overline{W}$ matches the invariant subspaces of π and $\overline{\pi}$. The coefficients of $\overline{\pi}$ are the conjugate functions:

$$c^{\overline{\pi}}_{\bar{u},\bar{v}}(x) = \big\langle \bar{u}, \overline{\pi(x)v}\big\rangle = \langle \pi(x)v, u\rangle = \overline{c^{\pi}_{u,v}(x)}.$$

The space $\overline{V}_\pi$ is isomorphic to V_π^* by $\bar{u} \mapsto \langle u, \cdot\rangle$. The map $\bar{u} \otimes v \mapsto v\langle u, \cdot\rangle$ identifies $\overline{V}_\pi \otimes V_\pi$ with $\mathrm{End}(V_\pi)$. For $g, h \in K$, unitarity of $\pi(g)$ gives $\pi(h)v\langle u, \pi(g)^{-1}\cdot\rangle = \pi(h)v\langle \pi(g)u, \cdot\rangle$. The left side is the action of (g,h) on $v\langle u,\cdot\rangle$. Hence the pair (g,h) acts on $\overline{V}_\pi \otimes V_\pi$ as $\overline{\pi}(g) \otimes \pi(h)$. Thus the first tensor factor carries the left action λ.

We define

$$\Phi_\pi \colon \mathrm{End}(V_\pi) \to C(K), \qquad \Phi_\pi(A)(x) = \sqrt{d_\pi}\ \mathrm{tr}\big(\pi(x)A\big),$$

and let E_π be its image. For $A = v\langle u,\cdot\rangle$ we get $\Phi_\pi(A) = \sqrt{d_\pi}\, c^{\pi}_{u,v}$.

Theorem 10.11 (Peter–Weyl). *(a) Each Φ_π is a $K \times K$-equivariant isometry from $(\mathrm{End}\, V_\pi, \mathrm{HS})$ onto E_π, and $E_\pi \perp E_\sigma$ for $\pi \neq \sigma$.*

(b) The coefficient space satisfies $\mathscr{R}(K) = \bigoplus_{\pi \in \mathrm{Irr}(K)} E_\pi$, an algebraic direct sum.

(c) The Hilbert completion of $\mathscr{R}(K)$ is $L^2(K) = \widehat{\bigoplus}_{\pi\in\mathrm{Irr}(K)} E_\pi$. The map $\bigoplus_\pi \Phi_\pi$ extends to a unitary $K \times K$-isomorphism

$$\widehat{\bigoplus_\pi}\, \mathrm{End}(V_\pi) \cong \widehat{\bigoplus_\pi}\, \overline{V}_\pi \otimes V_\pi \longrightarrow L^2(K).$$

(d) Under ρ, $E_\pi \cong V_\pi^{\oplus d_\pi}$ and $\dim \mathrm{Hom}_K(V_\pi, L^2(K)) = d_\pi$. Under λ, $E_\pi \cong \overline{V}_\pi^{\oplus d_\pi}$, and again each irreducible occurs with multiplicity equal to its dimension.

Proof. (a) For $A = v\langle u, \cdot\rangle$ and $B = v'\langle u', \cdot\rangle$ we have $A^*B = u\langle v, v'\rangle\langle u', \cdot\rangle$, so $\operatorname{tr}(A^*B) = \langle u', u\rangle\langle v, v'\rangle$. By Schur orthogonality (10.1),

$$\langle \Phi_\pi(A), \Phi_\pi(B)\rangle = d_\pi \cdot d_\pi^{-1}\langle u', u\rangle\langle v, v'\rangle = \langle A, B\rangle_{\mathrm{HS}}.$$

Both sides are sesquilinear and rank-one operators span $\operatorname{End}(V_\pi)$. The same argument gives $E_\pi \perp E_\sigma$. Equivariance is the calculation

$$\Phi_\pi(A)(g^{-1}xh) = \sqrt{d_\pi}\operatorname{tr}\big(\pi(g)^{-1}\pi(x)\pi(h)A\big) = \Phi_\pi\big(\pi(h)A\pi(g)^{-1}\big)(x).$$

(b) Let (σ, V) be a finite-dimensional continuous unitary representation. By complete reducibility (H3), $V = \bigoplus_j V_j$ orthogonally with each V_j irreducible. Write $u = \sum u_j$ and $v = \sum v_j$. Since $\sigma(x)v_k \in V_k$, we get $c^\sigma_{u,v} = \sum_j c^{\sigma_j}_{u_j,v_j}$.

Suppose that $\sigma_j \cong \pi \in \operatorname{Irr}(K)$, and let S be an isomorphism. Since σ_j and π are unitary, S^* is also an intertwiner. By Schur's lemma, $S^*S = \kappa \cdot 1$, and $\kappa > 0$ because S is injective. Hence $U = \kappa^{-1/2}S$ is a unitary intertwiner, and $c^{\sigma_j}_{u_j,v_j} = c^\pi_{Uu_j,Uv_j} \in E_\pi$. Thus $\mathscr{R}(K) = \sum_\pi E_\pi$. The sum is direct by the orthogonality in (a).

(c) Each E_π is finite-dimensional, hence closed. By density of coefficients, Proposition 10.10(b), their sum is dense. So $L^2(K)$ is their Hilbert sum. The map $\bigoplus \Phi_\pi$ is isometric with dense image, so it extends to a unitary. Equivariance passes to the closure by continuity.

(d) Under ρ alone, $A \mapsto \pi(h)A$ acts on each column of A, so $E_\pi \cong V_\pi^{\oplus d_\pi}$. Let $J\colon V_\pi \to L^2(K)$ intertwine ρ. By the Hilbert-sum decomposition (c), $Jv = \sum_\sigma P_\sigma Jv$ in $L^2(K)$. The orthogonal projection P_σ onto E_σ commutes with ρ, because E_σ and $E_\sigma^\perp$ are invariant. For $\sigma \neq \pi$, Schur's lemma gives $P_\sigma J = 0$. So J maps into E_π, and $\dim \operatorname{Hom}_K(V_\pi, L^2(K)) = \dim \operatorname{Hom}_K(V_\pi, V_\pi^{\oplus d_\pi}) = d_\pi$.

Under λ, $A \mapsto A\pi(g)^{-1}$ acts on rows, giving $\overline{V}_\pi^{\oplus d_\pi}$. If $\overline{\pi} \cong \overline{\pi'}$, then conjugating the intertwiner gives $\pi \cong \pi'$. Also $\overline{\overline{\pi}} = \pi$. So conjugation induces a dimension-preserving bijection of the isomorphism classes in $\operatorname{Irr}(K)$. The projections P_σ also commute with λ, so the same Hilbert-sum and Schur arguments apply. ∎

Theorem 10.12. *The coefficient space $\mathscr{R}(K)$ is dense in $C(K)$ for $\|\cdot\|_\infty$.*

Proof. The constant 1 is a coefficient of the trivial representation. Products are coefficients, since $c^\pi_{u,v}c^\sigma_{u',v'} = c^{\pi\otimes\sigma}_{u\otimes u', v\otimes v'}$. Conjugates are coefficients, since $\overline{c^\pi_{u,v}} = c^{\overline{\pi}}_{\bar{u},\bar{v}}$ for the inner product on $\overline{V}_\pi$ fixed above. Hence $\mathscr{R}(K)$ is a subalgebra of $C(K)$ that contains the constants and is closed under conjugation. It remains to prove that $\mathscr{R}(K)$ separates points.

Let $x \neq y$ and suppose $f(x) = f(y)$ for all $f \in \mathscr{R}(K)$. Since $\lambda(g)c^\pi_{u,v} = c^\pi_{\pi(g)u,v}$, the space $\mathscr{R}(K)$ is λ-invariant, so $f(g^{-1}x) = f(g^{-1}y)$ for all g. Put $s = x^{-1}y \neq e$ and $g = xt^{-1}$. Then $f(t) = f(ts)$ for all t. Hence $\rho(s)f = f$ on $\mathscr{R}(K)$. Since $\mathscr{R}(K)$ is dense in $L^2(K)$ by Proposition 10.10(b), and $\rho(s)$ is bounded, $\rho(s) = 1$ on $L^2(K)$.

Since K is Hausdorff, some open $U \ni e$ omits s. By Lemma 10.5(b), some continuous $\varphi \geq 0$ vanishes outside U and satisfies $\varphi(e) > 0 = \varphi(s)$. Then $(\rho(s)\varphi)(e) = \varphi(s) \neq \varphi(e)$, so $\rho(s)\varphi \neq \varphi$ in $C(K)$. Since $C(K) \to L^2(K)$ is injective, $\rho(s)\varphi \neq \varphi$ in $L^2(K)$, which contradicts $\rho(s) = 1$. Hence $\mathscr{R}(K)$ separates points.

We import the complex Stone–Weierstrass theorem. For compact Hausdorff X, it states that a subalgebra of $C(X)$ is uniformly dense if it contains the constants, is closed under conjugation and separates points. We apply it to $\mathscr{R}(K) \subset C(K)$. ∎

10.5 Examples

Example 10.13 (The circle group)**.** Let $K = \mathrm{U}(1)$. Here $\int f = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$. Irreducible representations of an abelian group are one-dimensional by Schur's lemma. By Lecture 7 they are $\chi_n(e^{i\theta}) = e^{in\theta}$, $n \in \mathbb{Z}$. Then $d_{\chi_n} = 1$ and $E_n = \mathbb{C}\chi_n$. Hence

$$\mathscr{R}(\mathrm{U}(1)) = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}\chi_n, \qquad L^2(\mathrm{U}(1)) = \widehat{\bigoplus}_{n\in\mathbb{Z}} \mathbb{C}\chi_n.$$

Every $f \in L^2$ equals $\sum_n \hat{c}_n \chi_n$ with $\hat{c}_n = \langle \chi_n, f\rangle$ and $\|f\|_2^2 = \sum |\hat{c}_n|^2$. This is L^2 completeness of Fourier series.

For the $K \times K$ action,

$$((g,h)\cdot\chi_n)(x) = \chi_n(g^{-1}xh) = \overline{\chi_n(g)}\,\chi_n(h)\,\chi_n(x).$$

So $\mathbb{C}\chi_n \cong \overline{\chi}_n \otimes \chi_n$. Under λ the line $\mathbb{C}\chi_n$ carries χ_{-n}. Under ρ it carries χ_n.

Example 10.14 (The group of unit quaternions)**.** Let $K = \mathrm{SU}(2)$ and write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

By Lecture 8, $\mathrm{Irr}(\mathrm{SU}(2)) = \{V_m : m \geq 0\}$, where V_m has dimension $m+1$. Realise V_m as symmetric tensors in $(\mathbb{C}^2)^{\otimes m}$. Let $e_1^{m-k}e_2^k$ be the average of the $\binom{m}{k}$ words with $m-k$ letters e_1 and k letters e_2. The group acts by $g^{\otimes m}$, and the tensor inner product gives

$$\langle e_1^{m-k}e_2^k, e_1^{m-j}e_2^j\rangle = \delta_{jk}\binom{m}{k}^{-1}.$$

By the Peter–Weyl theorem, Theorem 10.11,

$$L^2(\mathrm{SU}(2)) = \widehat{\bigoplus}_{m\geq 0} E_m, \qquad \dim E_m = (m+1)^2 = 1, 4, 9, 16, \ldots$$

Under ρ, V_m occurs $m+1$ times.

The block $m = 1$. For $V_1 = \mathbb{C}^2$, $c_{e_i,e_j}(g) = g_{ij}$. So $E_1 = \mathrm{span}\{a, -\bar{b}, b, \bar{a}\}$, and the functions $\sqrt{2}\,g_{ij}$ are orthonormal. In particular $\int |a|^2\,dg = \frac{1}{2}$.

The block m. Take the unit vector $u = e_1^m$. Since $ge_1 = ae_1 + be_2$ and the vectors $e_1^{m-k}e_2^k$ are orthogonal, only the $k = 0$ term of the binomial expansion has nonzero inner product with u:

$$c_{u,u}(g) = \left\langle e_1^m, (ae_1 + be_2)^m \right\rangle = a^m.$$

Then Schur orthogonality (10.1) gives $\int |a|^{2m}\,dg = 1/(m+1)$, matching $m = 1$.

Self-checks

1. Why is the operator (10.2) nonzero, although no invariant vector is known?

Answer. By the definition (10.2), $\langle u, Tu\rangle = \int |\langle \Pi(g)u, u\rangle|^2 \, dg$. The integrand is continuous, nonnegative and equal to 1 at e. By positivity (H2), the integral is positive. Hence T is nonzero.

2. Give $\dim E_2 \subset L^2(\mathrm{SU}(2))$ and the multiplicity of V_2 under ρ.

Answer. The block has dimension $\dim E_2 = 9$. The representation V_2 occurs 3 times. Under λ, $E_2 \cong \overline{V}_2^{\oplus 3}$.

3. Why does the coefficient proposition, Proposition 10.10, prove continuity before evaluating at e?

Answer. Only continuous elements of $L^2(K)$ have values. Evaluation is not even bounded on $\mathscr{R}(\mathrm{U}(1))$. For example, the function $f_N = (2N+1)^{-1/2} \sum_{|n|\le N} \chi_n$ has $\|f_N\|_2 = 1$ and $f_N(1) = \sqrt{2N+1}$.

Summary

Let K be a compact Lie group. In a strongly continuous unitary representation of K on $H \neq 0$, an averaged rank-one projection is a nonzero positive compact intertwiner. Its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant. The finite-dimensional invariant subspaces span a dense subspace. In $L^2(K)$, smoothing identifies the finite-dimensional ρ-invariant subspaces with spaces of matrix coefficients. Schur orthogonality then gives the Peter–Weyl theorem. Under ρ, and also under λ, each $\pi \in \mathrm{Irr}(K)$ occurs in $L^2(K)$ with multiplicity d_π. Density in $L^2(K)$ shows that $\mathscr{R}(K)$ separates points. By the Stone–Weierstrass theorem, $\mathscr{R}(K)$ is therefore uniformly dense in $C(K)$.

Reading

- Sepanski, *Compact Lie Groups*, §§3.2–3.3, including Theorem 3.25.
- Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, Sections 1–4.
- Etingof, MIT 18.755 lecture notes, Lectures 9–10.

Lecture 11

Fourier analysis and convolution

Objectives. *On the circle, a function has scalar Fourier coefficients. On a nonabelian compact group, the coefficient at an irreducible representation Π is an operator $\hat{f}(\Pi)$. We derive Plancherel from Peter–Weyl. We compute how convolution acts on these operators. We identify the convolution operators that commute with all translations.*

By the end of this lecture you will be able to:

- Compute $\hat{f}(\Pi)$ for matrix coefficients and derive the Plancherel theorem from Peter–Weyl.
- Prove $(f * h)^{\wedge} = \hat{h}\,\hat{f}$ and check this order by a direct $\mathrm{SU}(2)$ computation.
- Separate L^2 convergence of spectral partial sums from pointwise or uniform convergence.
- Characterise central convolution operators and write the isotypic projectors.

11.1 The operator-valued transform

Throughout, K is a compact Lie group with normalised Haar measure dg, so $\int_K dg = 1$. From Lecture 5 we import the following facts. The measure is invariant under left translation, right translation and inversion $g \mapsto g^{-1}$. Iterated integrals of continuous functions on $K \times K$ may be exchanged. A continuous function $\varphi \geq 0$ with $\int_K \varphi\, dg = 0$ vanishes. This property is strict positivity, which we call *positivity* below.

Every finite-dimensional representation is unitarisable and is a direct sum of irreducible ones. Finally, Schur orthogonality holds in the following form. For irreducible unitary representations Π, Π', vectors $u, v \in V_\Pi$ and vectors $u', v' \in V_{\Pi'}$,

$$\int_K \overline{\langle u, \Pi(g)v\rangle}\, \langle u', \Pi'(g)v'\rangle\, dg = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \not\cong \Pi, \end{cases} \tag{11.1}$$

where $d_\Pi = \dim V_\Pi$.

From Lecture 10 we import two facts. The space $L^2(K)$ is the Hilbert completion of $C(K)$ with respect to the inner product $\langle \varphi, \psi\rangle = \int_K \overline{\varphi}\,\psi\, dg$. By the Peter–Weyl theorem, the span $\mathscr{R}(K)$ of matrix coefficients of finite-dimensional representations is dense in $L^2(K)$.

Fix a set $\widehat{K}$ containing exactly one irreducible unitary representation (Π, V_Π) from each equivalence class. On $\mathrm{End}(V_\Pi)$ we use the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^* B)$. For $u, v \in V_\Pi$, the operator uv^* is given by $w \mapsto \langle v, w\rangle u$.

Definition 11.1. Let $f \in C(K)$ and $\Pi \in \widehat{K}$. The *Fourier transform* of f at Π is

$$\hat{f}(\Pi) = \int_K f(g)\,\Pi(g^{-1})\,dg \in \operatorname{End}(V_\Pi),$$

integrated entry by entry in any basis. For $A \in \operatorname{End}(V_\Pi)$ the *matrix coefficient* c_A is the function given by $c_A(g) = \operatorname{tr}(A\,\Pi(g))$. The space of matrix coefficients of Π is $M_\Pi = \{c_A : A \in \operatorname{End}(V_\Pi)\}$.

A rank-one operator gives the familiar coefficient $c_{uv^*}(g) = \langle v, \Pi(g)u\rangle$. Let Π' be equivalent to Π, say $\Pi'(g) = T\Pi(g)T^{-1}$ with T invertible. Then

$$\operatorname{tr}\big(A\,\Pi'(g)\big) = \operatorname{tr}\big(T^{-1}AT\,\Pi(g)\big) = c_{T^{-1}AT}(g).$$

Hence the matrix coefficients of Π' also lie in M_Π. So M_Π does not depend on the choice of representative. By complete reducibility, $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π.

Lemma 11.2. *For $f \in C(K)$, $\Pi \in \widehat{K}$ and $B \in \operatorname{End}(V_\Pi)$,*

$$\langle c_B, f\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}}. \tag{11.2}$$

Proof. For every operator X we have $\overline{\operatorname{tr} X} = \operatorname{tr} X^*$. Since $\Pi(g)^* = \Pi(g^{-1})$,

$$\overline{c_B(g)} = \operatorname{tr}\big(\Pi(g)^*B^*\big) = \operatorname{tr}\big(B^*\,\Pi(g^{-1})\big).$$

Multiply by $f(g)$ and integrate. The trace is linear, so it passes through the entrywise integral. This gives $\operatorname{tr}(B^*\hat{f}(\Pi))$. ∎

We use (11.2) to extend the transform to $L^2(K)$. The left side of (11.2) is L^2-continuous in f. For fixed f, it is an antilinear functional of B. Hence the finite-dimensional Riesz theorem gives a unique operator $\hat{f}(\Pi)$ satisfying (11.2). For continuous f, this operator agrees with the integral of Definition 11.1, by the pairing lemma, Lemma 11.2.

Lemma 11.3. *Let $\Pi, \Pi' \in \widehat{K}$, $A \in \operatorname{End}(V_\Pi)$ and $B \in \operatorname{End}(V_{\Pi'})$. Then*

$$\langle c_B, c_A\rangle = \begin{cases} d_\Pi^{-1}\langle B, A\rangle_{\mathrm{HS}}, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{11.3}$$

Consequently $\hat{c}_A(\Pi) = d_\Pi^{-1}A$, and $\hat{c}_A(\Pi') = 0$ for $\Pi' \neq \Pi$.

Proof. Both sides are linear in A and antilinear in B. Rank-one operators span, so take $A = uv^*$ and $B = u'v'^*$. Distinct elements of $\widehat{K}$ are inequivalent. By (11.1),

$$\langle c_B, c_A\rangle = \int_K \overline{\langle v', \Pi'(g)u'\rangle}\,\langle v, \Pi(g)u\rangle\,dg = \delta_{\Pi\Pi'}\,d_\Pi^{-1}\langle v, v'\rangle\langle u', u\rangle.$$

On the other hand $(u'v'^*)^* = v'u'^*$, so

$$\langle B, A\rangle_{\mathrm{HS}} = \operatorname{tr}\big(v'u'^*uv^*\big) = \langle u', u\rangle\operatorname{tr}(v'v^*) = \langle u', u\rangle\langle v, v'\rangle.$$

This proves (11.3). By (11.2), $\langle B, \hat{c}_A(\Pi')\rangle_{\mathrm{HS}} = \langle c_B, c_A\rangle$ for every B. Comparing with (11.3) gives the transforms. ∎

11.2 Plancherel from Peter–Weyl

For $f \in L^2(K)$ and a finite set $F \subset \widehat{K}$, the *spectral partial sum* is

$$S_F f(g) = \sum_{\Pi \in F} d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\, \Pi(g)\big).$$

Theorem 11.4 (Plancherel). *Let $f, h \in L^2(K)$.*

(a) The orthogonal projection of $L^2(K)$ onto M_Π is $P_\Pi f = d_\Pi\, c_{\hat{f}(\Pi)}$.

(b) For every $\varepsilon > 0$ there is a finite $F_0 \subset \widehat{K}$ with $\|f - S_F f\| < \varepsilon$ for every finite $F \supseteq F_0$.

(c) $\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$ and $\langle f, h\rangle = \sum_\Pi d_\Pi \langle \hat{f}(\Pi), \hat{h}(\Pi)\rangle_{\mathrm{HS}}$.

(d) If $\hat{f}(\Pi) = 0$ for every Π, then $f = 0$.

Proof. Step 1. By (11.3), $M_\Pi \perp M_{\Pi'}$ for $\Pi \neq \Pi'$, and $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Each M_Π is finite-dimensional, hence closed.

Step 2. Put $f_0 = d_\Pi c_{\hat{f}(\Pi)} \in M_\Pi$. For every $B \in \operatorname{End}(V_\Pi)$, (11.2) and (11.3) give

$$\langle c_B, f - f_0\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} - d_\Pi\, d_\Pi^{-1} \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} = 0.$$

So $f - f_0 \perp M_\Pi$, that is, $f_0 = P_\Pi f$. Step 1 gives $\|P_\Pi f\|^2 = d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$.

Step 3. This step uses the Peter–Weyl theorem. Let $M_F = \bigoplus_{\Pi \in F} M_\Pi$. By Steps 1 and 2, $S_F = \sum_{\Pi \in F} P_\Pi$ is the orthogonal projection onto M_F. Hence $S_F f$ is the point of M_F nearest to f. By the Peter–Weyl theorem, there is $\varphi \in \mathscr{R}(K)$ with $\|f - \varphi\| < \varepsilon$. Since $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π, there is a finite F_0 with $\varphi \in M_{F_0}$. If $F \supseteq F_0$, then $\varphi \in M_F$, so $\|f - S_F f\| \le \|f - \varphi\| < \varepsilon$.

Step 4. Pythagoras and Step 2 give

$$\|f\|^2 = \|S_F f\|^2 + \|f - S_F f\|^2, \qquad \|S_F f\|^2 = \sum_{\Pi \in F} d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2.$$

Hence every finite partial sum of the series in (c) is at most $\|f\|^2$. By Step 3, the partial sum over F exceeds $\|f\|^2 - \varepsilon^2$ once $F \supseteq F_0$. A sum of nonnegative terms is the supremum of its finite partial sums, so the norm identity holds. Polarisation gives the inner product identity. Finally, if $\hat{f}(\Pi) = 0$ for every Π, then the norm identity gives $\|f\| = 0$. This is the uniqueness statement (d). ∎

Corollary 11.5 (Inversion for finite coefficient sums). *Let $f \in \mathscr{R}(K)$. Then $\hat{f}(\Pi) \neq 0$ for only finitely many Π, and for every $g \in K$*

$$f(g) = \sum_{\Pi \in \widehat{K}} d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(g)\big).$$

Proof. Write $f = \sum_{\Pi \in F} c_{A_\Pi}$ with F finite. By the coefficient lemma, Lemma 11.3, $\hat{f}(\Pi) = d_\Pi^{-1} A_\Pi$ for $\Pi \in F$ and $\hat{f}(\Pi) = 0$ otherwise. Substituting returns $\sum_{\Pi \in F} c_{A_\Pi}(g) = f(g)$. ∎

For general $f \in L^2(K)$ we only have the L^2 convergence of spectral partial sums in the Plancherel theorem, Theorem 11.4(b). This statement says nothing about the value of f at a single point.

11.3 Convolution reverses the order

Definition 11.6. For $f, h \in C(K)$ the *convolution* $f * h$ is

$$(f * h)(x) = \int_K f(y)\, h(y^{-1}x)\, dy.$$

Fix x. The map $y \mapsto y^{-1}x$ is inversion followed by right translation, and both preserve dy. So $y \mapsto h(y^{-1}x)$ has L^2 norm $\|h\|$, and Cauchy–Schwarz gives $|(f * h)(x)| \le \|f\|\,\|h\|$.

The function $f * h$ is also continuous. By the uniform-continuity lemma of Lecture 10, h is uniformly continuous on the compact group K. In other words, for every $\varepsilon > 0$ there is a neighbourhood N of e with $|h(z) - h(zw)| < \varepsilon$ for all $z \in K$ and $w \in N$. Hence $|(f * h)(x) - (f * h)(xw)| \le \varepsilon \int_K |f|\, dy$ for $w \in N$.

Theorem 11.7 (Convolution theorem). *For $f, h \in C(K)$ and $\Pi \in \widehat{K}$,*

$$(f * h)^\wedge(\Pi) = \hat{h}(\Pi)\, \hat{f}(\Pi).$$

Proof. Exchange the iterated integrals entrywise:

$$(f * h)^\wedge(\Pi) = \int_K f(y) \Big(\int_K h(y^{-1}x)\, \Pi(x^{-1})\, dx \Big) dy.$$

In the inner integral substitute $x = yz$. Left invariance gives $dx = dz$, and $x^{-1} = z^{-1}y^{-1}$. So the inner integral is

$$\int_K h(z)\, \Pi(z^{-1}) \Pi(y^{-1})\, dz = \hat{h}(\Pi)\, \Pi(y^{-1}).$$

The factor $\Pi(y^{-1})$ coming from f stands on the right, because $x^{-1} = z^{-1}y^{-1}$. Integrating against $f(y)$ and using the definition of $\hat{f}(\Pi)$ gives $\hat{h}(\Pi)\hat{f}(\Pi)$. ∎

Some texts use $\tilde{f}(\Pi) = \int_K f(g)\Pi(g)\, dg$. The same computation then gives $(f * h)^\sim = \tilde{f}\,\tilde{h}$. Check the convention before importing formulas.

Corollary 11.8. *Let $A, B \in \mathrm{End}(V_\Pi)$. Then $c_A * c_B = d_\Pi^{-1} c_{BA}$. If instead $B \in \mathrm{End}(V_{\Pi'})$ with $\Pi' \neq \Pi$, then $c_A * c_B = 0$.*

Proof. By the coefficient lemma and the convolution theorem, Lemma 11.3 and Theorem 11.7, $(c_A * c_B)^\wedge(\Pi) = d_\Pi^{-2} BA$. This is the transform of $d_\Pi^{-1} c_{BA}$. At every other element of $\widehat{K}$ both transforms vanish. The difference φ of the two sides is continuous, and $\|\varphi\| = 0$ by the uniqueness statement of the Plancherel theorem, Theorem 11.4(d). Positivity applied to $|\varphi|^2$ gives $\varphi = 0$. In the second case one of $\hat{c}_A, \hat{c}_B$ vanishes at each element of $\widehat{K}$, so the same argument gives $c_A * c_B = 0$. ∎

For rank-one operators, $(u'v'^*)(uv^*) = \langle v', u\rangle\, u'v^*$. Hence the coefficient-convolution corollary, Corollary 11.8, gives

$$\langle v, \Pi(\cdot)u\rangle * \langle v', \Pi(\cdot)u'\rangle = d_\Pi^{-1} \langle v', u\rangle\, \langle v, \Pi(\cdot)u'\rangle.$$

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π. Write $E_{ij} = e_i e_j^*$ for the matrix unit. Its matrix in this basis has 1 in row i and column j, and 0 elsewhere. By the last display, the matrix

coefficients $D_{ij}(g) = \langle e_i, \Pi(g)e_j\rangle$ satisfy $D_{ij} * D_{kl} = d_\Pi^{-1}\delta_{jk}D_{il}$. So the functions multiply in the natural order. By the coefficient lemma, Lemma 11.3, their transforms are the transposed matrix units $d_\Pi^{-1}E_{ji}$, and these multiply in reverse.

Example 11.9 (Direct check on SU(2)). Write elements of SU(2) as

$$U = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Let Π be the defining representation on $\mathbb{C}^2$, with standard basis e_1, e_2, so $d_\Pi = 2$. Put $f(U) = U_{21} = b$ and $h(U) = U_{12} = -\bar{b}$. Since $\operatorname{tr}(E_{12}U) = U_{21}$ and $\operatorname{tr}(E_{21}U) = U_{12}$, we have $f = c_{E_{12}}$ and $h = c_{E_{21}}$. The convolution theorem predicts the transform of $f * h$. The coefficient-convolution corollary, Corollary 11.8, then gives the function, since $c_{E_{12}} * c_{E_{21}} = \frac{1}{2}c_{E_{21}E_{12}} = \frac{1}{2}c_{E_{22}}$. Hence

$$(f * h)^\wedge(\Pi) = \hat{h}\hat{f} = \tfrac{1}{4}E_{21}E_{12} = \tfrac{1}{4}E_{22}, \qquad \text{so} \quad (f * h)(U) = \tfrac{1}{2}U_{22} = \tfrac{1}{2}\bar{a}.$$

The opposite order would give the transform $\frac{1}{4}E_{11}$ and hence the function $\frac{1}{2}a$.

Now compute directly. Since $a = \langle e_1, Ue_1\rangle$ and $b = \langle e_2, Ue_1\rangle$, (11.1) gives

$$\int_{\mathrm{SU}(2)} |b|^2\, dU = \tfrac{1}{2}, \qquad \int_{\mathrm{SU}(2)} \bar{a}\, b\, dU = \tfrac{1}{2}\langle e_2, e_1\rangle = 0.$$

Let y have entries a, b and let x have entries α, β in the same pattern. Since y is unitary, $y^{-1} = y^*$. Hence

$$y^{-1}x = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} \bar{a}\alpha + \bar{b}\beta & \bar{b}\bar{\alpha} - \bar{a}\bar{\beta} \\ -b\alpha + a\beta & a\bar{\alpha} + b\bar{\beta} \end{pmatrix}.$$

Since $h(y^{-1}x)$ is the $(1,2)$ entry of this matrix, we get

$$(f * h)(x) = \int b\,(\bar{b}\bar{\alpha} - \bar{a}\bar{\beta})\, dy = \bar{\alpha}\int |b|^2\, dy - \bar{\beta}\int \bar{a}\, b\, dy = \tfrac{1}{2}\bar{\alpha}.$$

This matches the prediction. Using $(y^{-1}x)_{21} = -b\alpha + a\beta$, the same method gives $(h * f)(x) = \frac{1}{2}\alpha$. So $f * h \neq h * f$.

11.4 Pointwise and uniform inversion

Lecture 10 also shows that $\mathscr{R}(K)$ is uniformly dense in $C(K)$. This density does not say that the approximants are the partial sums $S_F f$. Uniform convergence of $S_F f$ needs an extra hypothesis.

Proposition 11.10. *Let $f \in C(K)$. Assume*

$$\sum_{\Pi \in \widehat{K}} d_\Pi^{3/2}\, \|\hat{f}(\Pi)\|_{\mathrm{HS}} < \infty.$$

Then the series

$$\sum_{\Pi \in \widehat{K}} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big)$$

converges absolutely and uniformly on K. Its sum is $f(g)$ for every $g \in K$.

Proof. We need two facts about the Hilbert–Schmidt norm. The first is Cauchy–Schwarz for $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$, applied to X^* and Y. The second uses unitarity of $\Pi(g)$:

$$|\operatorname{tr}(XY)| \le \|X\|_{\mathrm{HS}}\, \|Y\|_{\mathrm{HS}}, \qquad \|\Pi(g)\|_{\mathrm{HS}}^2 = \operatorname{tr}\big(\Pi(g)^*\Pi(g)\big) = d_\Pi.$$

Hence, for every $g \in K$,

$$\big| d_\Pi \operatorname{tr}\big(\hat f(\Pi)\Pi(g)\big)\big| \le b_\Pi, \qquad b_\Pi = d_\Pi^{3/2} \|\hat f(\Pi)\|_{\mathrm{HS}}.$$

The numbers b_Π have finite sum by hypothesis. Hence, for each n, only finitely many b_Π exceed $1/n$, so only countably many terms are nonzero. By the Weierstrass test the series converges absolutely and uniformly to a continuous function Φ.

Explicitly, given $\varepsilon > 0$, choose a finite $F_1 \subset \widehat{K}$ with $\sum_{\Pi \notin F_1} b_\Pi < \varepsilon$. Then for every finite $F \supseteq F_1$,

$$\sup_{g \in K} \big|\Phi(g) - S_F f(g)\big| \le \sum_{\Pi \notin F} b_\Pi < \varepsilon.$$

Since dg has mass one, $\|\Phi - S_F f\| < \varepsilon$. Enlarge F to contain the set F_0 given for this ε by the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b). By the triangle inequality, $\|\Phi - f\| < 2\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\|\Phi - f\| = 0$. Positivity applied to $|\Phi - f|^2$ then gives $\Phi = f$. ∎

Remark 11.11 (Cited results)**.** (i) Let K be compact and connected, with maximal torus T of dimension r. Fix a Weyl-invariant inner product on $\mathfrak{t}^*$. By Lecture 8, each $\Pi \in \widehat{K}$ has a highest weight λ_Π. Distinct Π have distinct highest weights, and these lie in the weight lattice of T, a lattice of rank r. Now let $f \in C^\infty(K)$. Then for every N there is C_N with

$$\|\hat f(\Pi)\|_{\mathrm{HS}} \le C_N\, (1 + |\lambda_\Pi|)^{-N} \qquad \text{for all } \Pi \in \widehat{K}.$$

We cite this from M. Sugiura, *Fourier series of smooth functions on compact Lie groups*, Osaka J. Math. 8 (1971), 33–47. The proof uses the Casimir operator.

The Weyl dimension formula, imported from the prerequisite Lie algebras course, gives $d_\Pi \le C(1 + |\lambda_\Pi|)^q$, where q is the number of positive roots. A lattice of rank r has $O(R^r)$ points of norm at most R, so $\sum_\Pi (1 + |\lambda_\Pi|)^{-M} < \infty$ for $M > r$. Taking $N > \tfrac{3}{2}q + r$ gives the hypothesis of the uniform inversion proposition, Proposition 11.10. So the Fourier series of a smooth function on a compact connected Lie group converges absolutely and uniformly to it.

(ii) On $K = \mathrm{U}(1)$ there are continuous functions whose symmetric partial sums diverge at a point. This is du Bois-Reymond's classical example (Grafakos, *Classical Fourier Analysis*, Section 3.3.3, Proposition 3.3.5). So convergence at every point, and hence uniform convergence, can fail for continuous functions.

11.5 Central kernels and isotypic projectors

Left and right translation are given by $(L_g f)(x) = f(g^{-1}x)$ and $(R_g f)(x) = f(xg)$. The character of Π is $\chi_\Pi = \operatorname{tr} \circ \Pi = c_I$. Unitarity gives $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$.

Definition 11.12. A function $k \in C(K)$ is *central* if $k(gxg^{-1}) = k(x)$ for all $g, x \in K$. The *convolution operator* of $k \in C(K)$ is the map $T_k \colon C(K) \to C(K)$ given by $T_k f = f * k$. The operator T_k is *central* if k is central.

Substituting $y = gy'$ in the convolution integral shows that $T_k L_g = L_g T_k$ for every k. The next theorem shows that commutation with right translations characterises the central kernels.

Theorem 11.13. *For $k \in C(K)$ the following are equivalent.*

(a) k is central.

(b) $T_k R_g = R_g T_k$ for every $g \in K$.

(c) Every $\hat{k}(\Pi)$ is a scalar multiple of the identity.

In this case $\hat{k}(\Pi) = \kappa_\Pi I$ with $\kappa_\Pi = d_\Pi^{-1}\langle \chi_\Pi, k\rangle$. For $f \in C(K)$ we have $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. Hence, for finite $F \subset \widehat{K}$, the sums $\sum_{\Pi \in F} \kappa_\Pi P_\Pi f$ converge to $f * k$ in L^2, in the sense of the convergence statement of the Plancherel theorem, Theorem 11.4(b).

Proof. *(a)⇔(c).* Fix g and put $k^g(x) = k(gxg^{-1})$. Substitute $x' = gxg^{-1}$, which preserves dx by two-sided invariance:

$$(k^g)^\wedge(\Pi) = \int_K k(x')\,\Pi(g^{-1}x'^{-1}g)\,dx' = \Pi(g)^{-1}\hat{k}(\Pi)\Pi(g).$$

If k is central, then $k^g = k$. Hence $\hat{k}(\Pi)$ commutes with every $\Pi(g)$, and by Schur's lemma it is a scalar multiple of the identity. Conversely, if every $\hat{k}(\Pi)$ is scalar, then k^g and k have equal transforms. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $k^g = k$.

(a)⇔(b). Right invariance with $y = y'g^{-1}$ gives

$$(T_k R_g f)(x) = \int_K f(y)\,k(gy^{-1}x)\,dy, \qquad (R_g T_k f)(x) = \int_K f(y)\,k(y^{-1}xg)\,dy.$$

If k is central, then $k(y^{-1}xg) = k(gy^{-1}x)$, which is (b). Conversely, assume (b) and set $x = e$. Put $\psi(y) = k(y^{-1}g) - k(gy^{-1})$. Then $\int_K f\psi\,dy = 0$ for all f. Taking $f = \bar{\psi}$ and using positivity gives $\psi = 0$. So $k(zg) = k(gz)$ for all z. Replacing z by zg^{-1} gives $k(z) = k(gzg^{-1})$.

Final claims. We have $\operatorname{tr}\hat{k}(\Pi) = \int_K k(g)\overline{\chi_\Pi(g)}\,dg = \langle \chi_\Pi, k\rangle$. Since $\hat{k}(\Pi) = \kappa_\Pi I$, this trace is $d_\Pi \kappa_\Pi$, which gives the formula for κ_Π. By the convolution theorem, Theorem 11.7, $(f * k)^\wedge(\Pi) = \kappa_\Pi \hat{f}(\Pi)$. Then the projection formula of the Plancherel theorem, Theorem 11.4(a), gives $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. So $S_F(f * k) = \sum_{\Pi \in F} \kappa_\Pi P_\Pi f$, and the L^2 convergence statement, Theorem 11.4(b), applied to $f * k$ gives the convergence. ∎

Corollary 11.14 (Isotypic projectors). *Let $e_\Pi = d_\Pi \chi_\Pi$. For $f \in C(K)$,*

$$P_\Pi f = f * e_\Pi = e_\Pi * f, \qquad (P_\Pi f)(x) = d_\Pi \int_K f(y)\,\chi_\Pi(y^{-1}x)\,dy.$$

Moreover $e_\Pi * e_{\Pi'} = \delta_{\Pi\Pi'} e_\Pi$.

Proof. First, characters are central, and $e_\Pi = d_\Pi c_I$. By the coefficient lemma, Lemma 11.3, $\hat{e}_\Pi(\Pi) = I$ and $\hat{e}_\Pi(\Pi') = 0$ for $\Pi' \neq \Pi$. So the central kernel theorem, Theorem 11.13, applies to $k = e_\Pi$ with $\kappa_{\Pi'} = \delta_{\Pi\Pi'}$. Its sums therefore equal $P_\Pi f$ once $\Pi \in F$, and they converge to

$f * e_\Pi$ in L^2. Hence $\|f * e_\Pi - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f * e_\Pi = P_\Pi f$.

Next, by the convolution theorem, Theorem 11.7, the transforms of $f * e_\Pi$ and $e_\Pi * f$ agree, because $\hat{e}_\Pi$ is scalar at every element of $\widehat{K}$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), the difference of these functions is 0 in $L^2(K)$. Both functions are continuous, so positivity gives $f * e_\Pi = e_\Pi * f$.

Finally, the coefficient-convolution corollary, Corollary 11.8, with $A = B = d_\Pi I$ gives $e_\Pi * e_\Pi = e_\Pi$. Its second case gives $e_\Pi * e_{\Pi'} = 0$ for $\Pi' \neq \Pi$. ∎

The Π-*isotypic part* of the right regular action R is the sum of all finite-dimensional R-invariant subspaces $W \subset C(K)$ with $R|_W \cong \Pi$.

We now identify M_Π as this isotypic part. By cyclicity of the trace, $c_A(xg) = \operatorname{tr}(\Pi(g)A\Pi(x))$. Hence $R_g c_A = c_{\Pi(g)A}$. The map $A \mapsto c_A$ is injective, because $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Let $e_1, \dots, e_{d_\Pi}$ be an orthonormal basis of V_Π. The map $A \mapsto \Pi(g)A$ sends each column Ae_j to $\Pi(g)Ae_j$. So R on M_Π is equivalent to $\Pi^{\oplus d_\Pi}$. The jth copy of Π consists of the functions c_A with $Ae_i = 0$ for $i \neq j$.

Now let $W \subset C(K)$ be finite-dimensional and R-invariant with $R|_W \cong \Pi$. Fix $\Pi' \neq \Pi$. The map $P_{\Pi'}|_W \colon W \to M_{\Pi'}$ intertwines R, since $P_{\Pi'} = T_{e_{\Pi'}}$ with $e_{\Pi'}$ central. Its image is 0 or isomorphic to Π, because W is irreducible. Every irreducible subrepresentation of $M_{\Pi'} \cong \Pi'^{\oplus d_{\Pi'}}$ projects nontrivially to some summand, so Schur's lemma makes it isomorphic to Π'. Hence $P_{\Pi'}W = 0$.

So for $f \in W$ we get $S_F f = P_\Pi f$ whenever $\Pi \in F$. By the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b), $\|f - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f = P_\Pi f \in M_\Pi$. Hence every such W lies in M_Π. Conversely, M_Π is the sum of its d_Π copies of Π. Thus M_Π is the Π-isotypic part of R, and P_Π is the orthogonal projector onto it.

Example 11.15 (Central kernels on SU(2) and the circle)**.** Let V_m be the irreducible representation of SU(2) of dimension $m+1$, and write P_m for P_{V_m}. Let $U \in \mathrm{SU}(2)$ have eigenvalues $e^{i\theta}$ and $e^{-i\theta}$ with $0 < \theta < \pi$. Then Lecture 9 gives $\chi_m(U) = \sin((m+1)\theta)/\sin\theta$. By the isotypic projector corollary, Corollary 11.14, $P_m f = (m+1)\, f * \chi_m$.

Take the kernel $k(U) = |\operatorname{tr} U|^2 = 4\cos^2\theta$. It is central, since the trace is invariant under conjugation. Since χ_1 is real and $V_1 \otimes V_1 \cong V_0 \oplus V_2$, we get $k = \chi_0 + \chi_2$. As a check, $\sin 3\theta/\sin\theta = 4\cos^2\theta - 1$.

Characters are orthonormal, so the central kernel theorem, Theorem 11.13, gives $\kappa_0 = 1$, $\kappa_2 = \frac{1}{3}$ and $\kappa_m = 0$ otherwise. Let $f \in C(\mathrm{SU}(2))$. For $F \supseteq \{V_0, V_2\}$, the sums in that theorem equal $P_0 f + \frac{1}{3}P_2 f$, and $P_0 f$ is the constant $\int_{\mathrm{SU}(2)} f\, dU$. These sums converge to $f * k$ in L^2. Both functions are continuous, so positivity gives

$$f * |\operatorname{tr}|^2 = \int_{\mathrm{SU}(2)} f\, dU + \tfrac{1}{3}P_2 f.$$

In particular $U_{11} * k = 0$, because $U_{11} = c_{E_{11}}$ lies in M_1. At $x = I$ this reads $\int a \cdot 4(\operatorname{Re} a)^2\, dy = 0$. Left translation by $-I$ preserves dy and changes the sign of the integrand. Hence this integral vanishes, as predicted.

On $K = \mathrm{U}(1)$ every $\Pi \in \widehat{K}$ is a character χ_n given by $\chi_n(z) = z^n$ for an integer n, with $d_{\chi_n} = 1$. Here $\hat{f}(\chi_n) = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})e^{-in\theta}d\theta$ is the classical coefficient. We write $\hat{f}(n)$ for $\hat{f}(\chi_n)$ and P_n for P_{χ_n}. For $f, h \in C(\mathrm{U}(1))$, the convolution theorem, Theorem 11.7, reads $(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $f * h = h * f$. The group is abelian, so every function is central, and $P_n f = f * \chi_n = \hat{f}(n)z^n$.

Self-checks

1. On $\mathrm{U}(1)$, why is the reversal in the convolution theorem, Theorem 11.7, invisible?
   Answer. Each Π is one-dimensional. The transforms are numbers, and numbers commute.
2. Let $k(U) = U_{11}$ on $\mathrm{SU}(2)$. Is T_k central?
   Answer. No. Here $k = c_{E_{11}}$ for the defining representation Π, so $\hat{k}(\Pi) = \frac{1}{2}E_{11}$ is not a scalar multiple of the identity. Hence, by the central kernel theorem, Theorem 11.13, k is not central, and so T_k is not central. To check this directly, put
   $$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}).$$
   Then $k(t_\theta) = e^{i\theta}$, while $k(wt_\theta w^{-1}) = e^{-i\theta}$. These differ, for example, at $\theta = \pi/2$.
3. Compute $(\chi_\Pi * \chi_\Pi)(e)$ in two ways.
   Answer. By definition it is $\int_K \chi_\Pi(y)\chi_\Pi(y^{-1})\,dy = \int_K |\chi_\Pi|^2\,dy = 1$. By the identity $e_\Pi * e_\Pi = e_\Pi$ in the isotypic projector corollary, Corollary 11.14, $\chi_\Pi * \chi_\Pi = d_\Pi^{-1}\chi_\Pi$, whose value at e is 1.

Summary

The Fourier coefficient at Π is the operator $\hat{f}(\Pi) = \int f(g)\Pi(g^{-1})\,dg$. Peter–Weyl gives Plancherel. Hence $L^2(K)$ is the Hilbert direct sum of the spaces M_Π, and the orthogonal projection onto M_Π is $P_\Pi f = d_\Pi \operatorname{tr}(\hat{f}(\Pi)\Pi(\cdot))$. The algebraic sum of these spaces is $\mathscr{R}(K)$, which is dense in $L^2(K)$. The transform of a convolution is $(f * h)\hat{} = \hat{h}\hat{f}$. The inversion formula holds exactly for finite coefficient sums. For general $f \in L^2(K)$, the spectral partial sums converge to f in L^2 norm. Uniform inversion needs an extra hypothesis, such as the summability condition of the uniform inversion proposition, Proposition 11.10. For compact connected K, smoothness suffices by the cited estimate in Remark 11.11. Central kernels are exactly those with scalar transforms. On $C(K)$, the isotypic projectors are $f \mapsto d_\Pi f * \chi_\Pi$.

Reading

Sepanski, *Compact Lie Groups*, §3.4, for Fourier theory on compact groups. Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, §§2–4, for representative functions and the Peter–Weyl theorem behind the Plancherel theorem, Theorem 11.4. Compare each source's transform and convolution conventions with Definitions 11.1 and 11.6 before using its formulas.

Lecture 12

Homogeneous spaces and compact induction

Objectives. *Let K be a compact Lie group and H a closed subgroup. Which irreducible representations of K occur in functions on K/H, and how often? We show that each irreducible representation V of K occurs $\dim V^H$ times. Induction from H extends this count to vector-valued functions.*

> By the end of this lecture you will be able to:
>
> - Construct the invariant integral on K/H and identify $L^2(K/H)$ with the right-H-invariant part of $L^2(K)$.
> - Prove that each irreducible V occurs in $L^2(K/H)$ exactly $\dim V^H$ times.
> - Define compact induction and prove Frobenius reciprocity with finite multiplicities.
> - Decompose $L^2(S^2)$ under $\mathrm{SO}(3)$ and relate the summands to spherical harmonics.

Throughout, K is a compact Lie group, not necessarily connected, and $H \subseteq K$ is a closed subgroup. By Lecture 3, K/H is a compact manifold, and the projection $p\colon K \to K/H$, $p(x) = xH$, is continuous and open. The group K acts on K/H by $g \cdot xH = gxH$.

The Haar measures dx on K and dh on H have total mass 1. They are left, right and inversion invariant (Lecture 5). The space $L^2(K)$ is the completion of $C(K)$ for $\langle f_1, f_2\rangle = \int_K \overline{f_1}\, f_2\, dx$, and $C(K) \to L^2(K)$ is injective (Lecture 10).

We use the left and right translation representations

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = f(xg).$$

These commuting representations are unitary and strongly continuous (Lecture 10).

12.1 Functions and integration on K/H

By the definition of the quotient topology, a function φ on K/H is continuous exactly when $\varphi \circ p$ is continuous. A right-H-invariant function on K is constant on cosets. Hence the pullback p^* is a linear bijection

$$p^*\colon C(K/H) \to C(K)^H := \{f \in C(K) : f(xh) = f(x) \text{ for all } x \in K,\ h \in H\}, \qquad p^*\varphi = \varphi \circ p. \tag{12.1}$$

Definition 12.1. The *invariant integral* on K/H is the linear functional

$$\int_{K/H} \varphi(\xi)\, d\xi := \int_K \varphi(p(x))\, dx, \qquad \varphi \in C(K/H).$$

Proposition 12.2. *(a) The integral is positive, $\int_{K/H} 1\, d\xi = 1$, and $\int_{K/H} \varphi(g^{-1}\xi)\, d\xi = \int_{K/H} \varphi(\xi)\, d\xi$ for $g \in K$.*

(b) For $f \in C(K)$, the H-average $f^H(x) = \int_H f(xh)\, dh$ lies in $C(K)^H$, and

$$\int_K f(x)\, dx = \int_K f^H(x)\, dx = \int_{K/H} \Big(\int_H f(xh)\, dh \Big)\, d(xH).$$

Proof. (a) Positivity and normalisation hold for dx. Invariance follows from $\varphi(g^{-1}p(x)) = \varphi(p(g^{-1}x))$ and left invariance of dx.

(b) The map $(x,h) \mapsto f(xh)$ is uniformly continuous on the compact space $K \times H$. So f^H is continuous. For $k \in H$, left invariance of dh gives $f^H(xk) = \int_H f(xkh)\, dh = f^H(x)$. Fubini for continuous functions on $K \times H$ and right invariance of dx give

$$\int_K f^H(x)\, dx = \int_H \int_K f(xh)\, dx\, dh = \int_H \int_K f(x)\, dx\, dh = \int_K f(x)\, dx.$$

By the pullback bijection (12.1), the last expression in (b) is $\int_K f^H\, dx$ written on K/H. ∎

Definition 12.3. The space $L^2(K/H)$ is the completion of $C(K/H)$ for $\langle \varphi, \psi \rangle = \int_{K/H} \overline{\varphi}\, \psi\, d\xi$. Also

$$L^2(K)^H = \{ f \in L^2(K) : \rho(h) f = f \text{ for all } h \in H \}.$$

The form is definite since $\int_{K/H} |\varphi|^2\, d\xi = \|\varphi \circ p\|^2$ and $C(K) \to L^2(K)$ is injective. The same identity shows that p^* is isometric. So p^* extends to an isometry $L^2(K/H) \to L^2(K)$. The subspace $L^2(K)^H$ is closed because each $\rho(h)$ is bounded.

Proposition 12.4. *There is a unique bounded operator A_H on $L^2(K)$ with $A_H f = f^H$ for $f \in C(K)$. It satisfies*

$$\langle f_1, A_H f_2 \rangle = \int_H \langle f_1, \rho(h) f_2 \rangle\, dh \qquad (f_1, f_2 \in L^2(K)). \tag{12.2}$$

It is the orthogonal projection onto $L^2(K)^H$. Consequently p^* maps $L^2(K/H)$ isometrically onto $L^2(K)^H$.

Proof. Let $f \in C(K)$. Cauchy–Schwarz on H gives $|f^H(x)|^2 \le (|f|^2)^H(x)$. Integrate and apply the averaging identity of Proposition 12.2(b) to $|f|^2$:

$$\|f^H\|^2 \le \int_K (|f|^2)^H\, dx = \int_K |f|^2\, dx = \|f\|^2.$$

Since $C(K)$ is dense, A_H exists uniquely, with $\|A_H\| \le 1$.

For continuous f_1, f_2, Fubini shows that both sides of (12.2) equal $\int_K \int_H \overline{f_1(x)} f_2(xh)\, dh\, dx$. Both sides are continuous in (f_1, f_2). On the right, the integrand is continuous in h by strong continuity and bounded by $\|f_1\|\,\|f_2\|$. So (12.2) holds on $L^2(K)$.

Now $\rho(h)^* = \rho(h^{-1})$ and dh is inversion invariant. So the weak formula (12.2) gives $\langle f_1, A_H f_2\rangle = \langle A_H f_1, f_2\rangle$, and A_H is self-adjoint.

Next, A_H fixes $L^2(K)^H$. If $f \in L^2(K)^H$, then the weak formula gives $\langle f_1, A_H f\rangle = \langle f_1, f\rangle$ for all f_1, so $A_H f = f$.

Finally, the range of A_H lies in $L^2(K)^H$. Let $k \in H$. For continuous f, $A_H f = f^H$ lies in $C(K)^H$ by the description of the H-average in Proposition 12.2(b), so $\rho(k)A_H f = A_H f$. Since $C(K)$ is dense and both sides are bounded in f, this identity holds for all f.

Hence A_H is a self-adjoint idempotent with range $L^2(K)^H$.

It remains to identify the image of p^*. Since p^* is an isometry, its image is the closure of $p^*(C(K/H)) = C(K)^H$. This closure lies in the closed subspace $L^2(K)^H$. Conversely, let $f \in L^2(K)^H$ and choose $f_n \in C(K)$ with $f_n \to f$. Then $f_n^H \in C(K)^H$ and $f_n^H = A_H f_n \to A_H f = f$. So the image is $L^2(K)^H$. ∎

Each $\lambda(g)$ commutes with every $\rho(h)$, so $\lambda(g)$ preserves $L^2(K)^H$. Transported by p^*, the operator $\lambda(g)$ acts on $C(K/H)$ by $\varphi \mapsto \varphi(g^{-1}\cdot)$. The resulting unitary representation of K on $L^2(K/H)$ is the *left regular representation*.

12.2 Multiplicities in $L^2(K/H)$

Let (π, V) be a finite-dimensional unitary representation and $u, v \in V$. The *matrix coefficient* is $c_{u,v}(x) = \langle u, \pi(x)v\rangle$. Unitarity gives

$$\lambda(g)c_{u,v} = c_{\pi(g)u,\,v}, \qquad \rho(g)c_{u,v} = c_{u,\,\pi(g)v}. \tag{12.3}$$

For the first identity, $c_{u,v}(g^{-1}x) = \langle u, \pi(g)^{-1}\pi(x)v\rangle = \langle \pi(g)u, \pi(x)v\rangle$.

The map $u \mapsto c_{u,v}$ is antilinear. The *conjugate representation* $\overline{V}$ is the set V with scalar multiplication $t \cdot u = \bar{t}u$, the same operators $\pi(g)$, and inner product $\langle u, u'\rangle_{\overline{V}} = \langle u', u\rangle$. It has the same invariant subspaces as V. So $\overline{V}$ is irreducible when V is, and $(\overline{V})^H = V^H$ as sets.

For each $\sigma \in \mathrm{Irr}(K)$ fix a unitary representative (π_σ, V_σ) of dimension d_σ. Write $\bar{\sigma}$ for the class of $\overline{V_\sigma}$, and let $E_\sigma = \mathrm{span}\{c_{u,v} : u, v \in V_\sigma\} \subseteq C(K)$. We use two earlier results:

(PW1) Schur orthogonality (Lecture 5): $\langle c_{u,v}, c_{u',v'}\rangle = d_\sigma^{-1}\langle u', u\rangle\langle v, v'\rangle$ for $u, v, u', v' \in V_\sigma$.

(PW2) Peter–Weyl (Lecture 10): the E_σ are mutually orthogonal and $L^2(K) = \widehat{\bigoplus}_\sigma E_\sigma$.

By Schur orthogonality (PW1), the linear map $\overline{V_\sigma} \otimes V_\sigma \to E_\sigma$, $u \otimes v \mapsto c_{u,v}$, is $d_\sigma^{-1/2}$ times an isometry. So it is bijective. By the coefficient identities (12.3), it carries $\bar{\pi}_\sigma \otimes 1$ to λ and $1 \otimes \pi_\sigma$ to ρ. Let Q_σ be the orthogonal projection onto E_σ. A unitary operator that preserves a finite-dimensional subspace also preserves its orthogonal complement. Hence Q_σ commutes with all $\lambda(g)$ and $\rho(g)$.

For $\sigma \in \mathrm{Irr}(K)$ let $E_\sigma^H = \mathrm{span}\{c_{u,v} : u \in V_\sigma,\ v \in V_\sigma^H\}$, a subspace of E_σ.

Proposition 12.5. *Let $P_\sigma = \int_H \pi_\sigma(h)\,dh$ be the orthogonal projection of V_σ onto V_σ^H (Lec-*

ture 5). Then $A_H c_{u,v} = c_{u,P_\sigma v}$. Moreover

$$L^2(K/H) \cong L^2(K)^H = \overline{\bigoplus_{\sigma \in \mathrm{Irr}(K)}} E_\sigma^H, \qquad E_\sigma^H \cong \overline{V_\sigma} \otimes V_\sigma^H,$$

where K acts on $\overline{V_\sigma} \otimes V_\sigma^H$ by $\bar\pi_\sigma \otimes 1$.

Proof. The finite-dimensional integral commutes with the functional $w \mapsto \langle u, \pi_\sigma(x) w\rangle$. So

$$(A_H c_{u,v})(x) = \int_H \langle u, \pi_\sigma(x)\pi_\sigma(h) v\rangle \, dh = \left\langle u, \pi_\sigma(x) \int_H \pi_\sigma(h) v \, dh \right\rangle = c_{u, P_\sigma v}(x).$$

Thus $A_H(E_\sigma) = E_\sigma^H$, and A_H fixes E_σ^H. Hence $E_\sigma^H \subseteq L^2(K)^H$. Conversely, let $f \in L^2(K)^H$. By Peter–Weyl (PW2), $f = \sum_\sigma Q_\sigma f$ in L^2. Since A_H is bounded,

$$f = A_H f = \sum_\sigma A_H Q_\sigma f, \qquad A_H Q_\sigma f \in E_\sigma^H.$$

Since E_σ^H lies in E_σ, the spaces E_σ^H are mutually orthogonal by Peter–Weyl (PW2). This proves the decomposition. The isomorphism with $\overline{V_\sigma} \otimes V_\sigma^H$ is the restriction of the bijection above. ∎

Induction in § 12.3 uses vector-valued functions and evaluates them at the identity e. An L^2 class has no value at a point. The next lemma shows that equivariant images of V_σ consist of continuous functions, so these evaluations make sense.

Let W be a finite-dimensional Hilbert space with orthonormal basis $w_1, \dots, w_n$. Then $L^2(K) \otimes W$ is the Hilbert space of functions $\sum_j f_j w_j$ with $f_j \in L^2(K)$. It contains $C(K, W) = C(K) \otimes W$. Let K act by $\lambda \otimes 1$.

Lemma 12.6. *Let $\sigma \in \mathrm{Irr}(K)$. Let $T : V_\sigma \to L^2(K) \otimes W$ be linear with $T\pi_\sigma(g) = (\lambda(g) \otimes 1)T$ for all $g \in K$. Then $T(V_\sigma) \subseteq E_{\bar\sigma} \otimes W$. In particular, every vector in $T(V_\sigma)$ is a continuous function.*

Proof. Fix $\eta \in \mathrm{Irr}(K)$. The operator $Q_\eta \otimes 1$ commutes with $\lambda(g) \otimes 1$. So $(Q_\eta \otimes 1)T$ is a K-map from V_σ into

$$E_\eta \otimes W \cong \overline{V_\eta} \otimes (V_\eta \otimes W),$$

where K acts only on the first factor. This space is a direct sum of $d_\eta \dim W$ copies of $\overline{V_\eta}$. By Schur's lemma, $(Q_\eta \otimes 1)T = 0$ unless $V_\sigma \cong \overline{V_\eta}$, that is, unless $\eta = \bar\sigma$. Apply Peter–Weyl (PW2) in each coordinate. For $v \in V_\sigma$,

$$Tv = \sum_\eta (Q_\eta \otimes 1) Tv = (Q_{\bar\sigma} \otimes 1) Tv \in E_{\bar\sigma} \otimes W.$$ ∎

Thus the values $(Tv)(x)$ are defined for all $v \in V_\sigma$ and $x \in K$.

Let (E, Π) be a unitary representation of K and $\sigma \in \mathrm{Irr}(K)$. The *multiplicity* of σ in E is $m_\sigma(E) = \dim \mathrm{Hom}_K(V_\sigma, E)$. Here Hom_K consists of the linear maps T with $T\pi_\sigma(g) = \Pi(g)T$. These maps are bounded because V_σ is finite-dimensional. The *σ-isotypic subspace* of E is the sum of the images of these maps.

Theorem 12.7. *For every $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2(K/H)\big) = \dim V_\sigma^H.$$

The σ-isotypic subspace is $E_{\bar\sigma}^H \subseteq C(K/H)$, of dimension $d_\sigma \dim V_\sigma^H$. The space $L^2(K/H)$ is the Hilbert sum of its isotypic subspaces.

Proof. Identify $L^2(K/H)$ with $L^2(K)^H$ through p^*, as in Proposition 12.4. Let $T\colon V_\sigma \to L^2(K)^H$ be a K-map. By Lemma 12.6 on equivariant images, with $W = \mathbb{C}$, $T(V_\sigma) \subseteq E_{\bar\sigma}$. A vector of $E_{\bar\sigma}$ fixed by A_H lies in $A_H(E_{\bar\sigma}) = E_{\bar\sigma}^H$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2(K/H)\right) = \operatorname{Hom}_K\left(V_\sigma, E_{\bar\sigma}^H\right).$$

Since $V_{\bar\sigma} \cong \overline{V_\sigma}$, we have $\overline{V_{\bar\sigma}} \cong V_\sigma$. So Proposition 12.5 on the decomposition of $L^2(K)^H$ gives

$$E_{\bar\sigma}^H \cong \overline{V_{\bar\sigma}} \otimes V_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H,$$

with K acting on the first factor only. By Schur's lemma,

$$\operatorname{Hom}_K\left(V_\sigma, V_\sigma \otimes V_{\bar\sigma}^H\right) \cong V_{\bar\sigma}^H.$$

A K-isomorphism $V_{\bar\sigma} \to \overline{V_\sigma}$ carries $V_{\bar\sigma}^H$ onto $(\overline{V_\sigma})^H$. This is the set V_σ^H with conjugated scalars, so its dimension is $\dim V_\sigma^H$. The images of the K-maps $V_\sigma \to E_{\bar\sigma}^H$ span $E_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H$. Finally, $\sigma \mapsto \bar\sigma$ is a bijection of $\operatorname{Irr}(K)$, so the decomposition in Proposition 12.5 gives the Hilbert sum. ∎

Thus the summand E_σ^H of Proposition 12.5 is indexed by σ but is $\bar\sigma$-isotypic. The count is unchanged because $\dim V_\sigma^H = \dim V_{\bar\sigma}^H$. Exercise 12.1 shows this label shift for $K = \mathrm{U}(1)$.

12.3 Compact induction and Frobenius reciprocity

Let (τ, W) be a finite-dimensional unitary representation of H. Averaging over H gives every finite-dimensional representation of H an invariant inner product (Lecture 5).

Definition 12.8. The *induced representation* is

$$\operatorname{Ind}_H^K W = \{F \in C(K, W) : F(xh) = \tau(h)^{-1} F(x) \text{ for all } x \in K,\ h \in H\},$$

with K acting by $(\lambda(g)F)(x) = F(g^{-1}x)$.

Left translation preserves the covariance condition because left and right translations commute. For the trivial representation $W = \mathbb{C}$ of H, we recover $C(K)^H \cong C(K/H)$.

Theorem 12.9 (Frobenius reciprocity, continuous form)**.** *Let (π, V) be a finite-dimensional representation of K. The map*

$$\Phi\colon \operatorname{Hom}_K\left(V, \operatorname{Ind}_H^K W\right) \to \operatorname{Hom}_H(V|_H, W), \qquad \Phi(T)v = (Tv)(e),$$

is a linear bijection. Its inverse sends A to $\Psi(A)$, where $(\Psi(A)v)(x) = A\big(\pi(x)^{-1}v\big)$.

Proof. First, $\Phi(T)$ is H-equivariant. For $h \in H$, equivariance of T and covariance give

$$\Phi(T)\big(\pi(h)v\big) = (\lambda(h)Tv)(e) = (Tv)(h^{-1}) = \tau(h^{-1})^{-1}(Tv)(e) = \tau(h)\,\Phi(T)v.$$

Next, $\Psi(A)v$ is continuous because $x \mapsto \pi(x)^{-1}v$ is continuous. Since A is an H-map, $\Psi(A)v$ is covariant:

$$(\Psi(A)v)(xh) = A\big(\pi(h)^{-1}\pi(x)^{-1}v\big) = \tau(h)^{-1}A\big(\pi(x)^{-1}v\big).$$

The map $\Psi(A)$ is a K-map, since $\pi(x)^{-1}\pi(g) = \pi(g^{-1}x)^{-1}$ gives

$$(\Psi(A)\pi(g)v)(x) = (\Psi(A)v)(g^{-1}x) = (\lambda(g)\Psi(A)v)(x).$$

Finally, $\Phi(\Psi(A))v = A(v)$. In the other order,

$$(\Psi(\Phi(T))v)(x) = \big(T\pi(x^{-1})v\big)(e) = (\lambda(x^{-1})Tv)(e) = (Tv)(x). \qquad \blacksquare$$

Now pass to L^2. For $h \in H$ define $R(h) = \rho(h) \otimes \tau(h)$ on $L^2(K) \otimes W$. These operators are unitary and commute with $\lambda(g) \otimes 1$. For continuous F, $(R(h)F)(x) = \tau(h)F(xh)$. Set

$$L^2\text{-}\operatorname{Ind}_H^K W = \{F \in L^2(K) \otimes W : R(h)F = F \text{ for all } h \in H\}.$$

This closed K-invariant subspace is a unitary representation of K. For the trivial representation $W = \mathbb{C}$, this subspace is $L^2(K)^H$.

Theorem 12.10 (finite multiplicities). *For $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2\text{-}\operatorname{Ind}_H^K W\big) = \dim \operatorname{Hom}_H(V_\sigma|_H, W) \le d_\sigma \dim W.$$

Proof. Let $T : V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$ be a K-map. By Lemma 12.6 on equivariant images, each $F = Tv$ is continuous. Then $R(h)F$ and F are continuous and equal in L^2. Since $C(K) \otimes W \to L^2(K) \otimes W$ is injective, $\tau(h)F(xh) = F(x)$ for all x. So $F \in \operatorname{Ind}_H^K W$. Conversely, continuous covariant functions satisfy $R(h)F = F$. Hence

$$\operatorname{Hom}_K\big(V_\sigma, L^2\text{-}\operatorname{Ind}_H^K W\big) = \operatorname{Hom}_K\big(V_\sigma, \operatorname{Ind}_H^K W\big).$$

Frobenius reciprocity (Theorem 12.9) gives a linear bijection from this space onto $\operatorname{Hom}_H(V_\sigma|_H, W)$, so the dimensions agree. The bound holds because $\operatorname{Hom}_H(V_\sigma|_H, W)$ is a subspace of $\operatorname{Hom}(V_\sigma, W)$, which has dimension $d_\sigma \dim W$. $\blacksquare$

Remark 12.11 (duals and covariance). For $W = \mathbb{C}$ the count is $\dim \operatorname{Hom}_H(V_\sigma, \mathbb{C}) = \dim(V_\sigma^*)^H$. Since $\langle u, \pi_\sigma(h)v\rangle = \langle \pi_\sigma(h)^{-1}u, v\rangle$, the functional $\langle u, \cdot\rangle$ is H-invariant exactly when $u \in V_\sigma^H$. So $u \mapsto \langle u, \cdot\rangle$ is an antilinear bijection $V_\sigma^H \to (V_\sigma^*)^H$. Hence the count agrees with the multiplicity theorem (Theorem 12.7).

The inverse in $\tau(h)^{-1}$ makes the covariance condition compatible with products in H:

$$F(xh_1h_2) = \tau(h_2)^{-1}F(xh_1) = \tau(h_2)^{-1}\tau(h_1)^{-1}F(x) = \tau(h_1h_2)^{-1}F(x).$$

Remark 12.12 (scope). For compact K, we obtained a finite invariant integral. The irreducible representations were finite-dimensional, and Peter–Weyl applied. We proved a count of finite multiplicities, not a general adjunction for Hilbert representations. For noncompact G, the space G/H may carry no invariant measure. Lecture 17 treats that case.

12.4 The sphere $\mathrm{SO}(3)/\mathrm{SO}(2)$

Let $K = \mathrm{SO}(3)$, let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$, and let

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2).$$

The subgroup H is the stabiliser of e_3. Every unit vector is the third column of some rotation, so the orbit map $q(x) = xe_3$ sends K onto S^2. By Lecture 3, q induces a K-equivariant diffeomorphism $K/H \to S^2$. So $F \mapsto F \circ q$ identifies $C(S^2)$ with $C(K)^H$, and $\lambda(g)(F \circ q) = (F \circ g^{-1}) \circ q$. We define $L^2(S^2)$ through this identification.

Let E be a finite-dimensional representation of H. An integer k is a *weight* of E if some nonzero $v \in E$ satisfies $R_\theta v = e^{ik\theta} v$ for all $\theta \in \mathbb{R}$. These vectors and 0 form the *weight-k space*. Suppose E has a basis (v_j) with $R_\theta v_j = e^{ik_j\theta} v_j$ for all θ. Then the weight-k space is spanned by the v_j with $k_j = k$. Indeed, a weight-k vector $\sum_j c_j v_j$ satisfies

$$c_j\big(e^{ik_j\theta} - e^{ik\theta}\big) = 0 \qquad \text{for all } \theta,$$

and this forces $c_j = 0$ when $k_j \neq k$.

From Lecture 8, the irreducibles of $\mathrm{SO}(3)$ are V_ℓ, $\ell = 0, 1, 2, \ldots$, of dimension $2\ell + 1$ and spin ℓ. In this section an unqualified V_ℓ means the $\mathrm{SO}(3)$-module. The $\mathrm{SU}(2)$-modules of Lecture 8 are always named as such. Choose the covering $\phi = \mathrm{Ad}$ in the ordered basis (X_1, X_2, X_3) of Lecture 4. The explicit calculation there gives

$$\phi\big(\mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2})\big) = R_{-\theta}, \qquad \ker\phi = \{\pm I\}.$$

The pullback $\pi_\ell \circ \phi$ is the $\mathrm{SU}(2)$-module $V_{2\ell}$. The diagonal element in this display acts on a basis of $V_{2\ell}$ by the scalars $e^{i(\ell-j)\theta}$, $0 \le j \le 2\ell$. Replacing θ by $-\theta$ therefore gives

$$V_\ell \text{ has a basis } (v_k)_{-\ell \le k \le \ell} \text{ with } \pi_\ell(R_\theta) v_k = e^{ik\theta} v_k \text{ for all } \theta \in \mathbb{R}. \tag{12.4}$$

Theorem 12.13. *Let $\mathcal{Y}_\ell \subseteq L^2(S^2)$ be the V_ℓ-isotypic subspace. Then*

$$L^2(S^2) = \widehat{\bigoplus_{\ell \ge 0}} \mathcal{Y}_\ell, \qquad \mathcal{Y}_\ell \cong V_\ell, \qquad \mathcal{Y}_\ell \subseteq C(S^2).$$

Each integral-spin module occurs exactly once.

Proof. The space V_ℓ^H is the weight-zero space. By the weight basis (12.4) it is the line $\mathbb{C}v_0$, so $\dim V_\ell^H = 1$. Apply the multiplicity theorem (Theorem 12.7). ∎

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$, with $(\lambda(g)f)(v) = f(g^{-1}v)$. Let $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ and $\mathcal{H}_\ell = \{f \in \mathcal{P}_\ell : \Delta f = 0\}$. The chain rule gives $\Delta(f \circ g) = (\Delta f) \circ g$ for orthogonal g, so $\mathcal{H}_\ell$ is K-invariant. Restriction to S^2 is K-equivariant. Restriction is injective on $\mathcal{P}_\ell$, since $f(rv) = r^\ell f(v)$. Elements of $\mathcal{H}_\ell|_{S^2}$ are called *spherical harmonics* of degree ℓ.

Put $z_\pm = x \pm iy$. Since $R_\theta^{-1} = R_{-\theta}$,

$$(\lambda(R_\theta)z_+)(x,y,z) = (\cos\theta\, x + \sin\theta\, y) + i(-\sin\theta\, x + \cos\theta\, y) = e^{-i\theta} z_+.$$

Similarly $\lambda(R_\theta)z_- = e^{i\theta}z_-$ and $\lambda(R_\theta)z = z$. The monomials $z_+^a z_-^b z^c$ with $a+b+c = \ell$ form a basis of $\mathcal{P}_\ell$, and for all θ,

$$\lambda(R_\theta)\, z_+^a z_-^b z^c = e^{i(b-a)\theta}\, z_+^a z_-^b z^c, \qquad |b-a| \le \ell. \tag{12.5}$$

So every weight of $\mathcal{P}_\ell$ satisfies $|k| \le \ell$.

Proposition 12.14. *The space $\mathcal{Y}_\ell$ is spanned by the left translates of $f_\ell = z_+^\ell|_{S^2}$. Hence every element of $\mathcal{Y}_\ell$ is a spherical harmonic of degree ℓ.*

Proof. The polynomial z_+^ℓ is harmonic:

$$\partial_x^2 z_+^\ell + \partial_y^2 z_+^\ell = \ell(\ell-1)z_+^{\ell-2} + i^2\ell(\ell-1)z_+^{\ell-2} = 0.$$

Let U be the span of all $\lambda(g)f_\ell$. Then $U \subseteq \mathcal{H}_\ell|_{S^2}$ is finite-dimensional and K-invariant. Decompose U into irreducibles. A copy of V_j in U lies in the V_j-isotypic subspace $\mathcal{Y}_j$. Since $\mathcal{Y}_j \cong V_j$, this copy equals $\mathcal{Y}_j$. Thus $U = \bigoplus_{j\in S}\mathcal{Y}_j$ for a finite set S. Restriction is injective and equivariant, so the weights of U are weights of $\mathcal{P}_\ell$. Since V_j has weight j, this gives $S \subseteq \{0,\dots,\ell\}$.

Write $f_\ell = \sum_{j\in S} y_j$ with $y_j \in \mathcal{Y}_j$. Each $\lambda(R_\theta)$ preserves each $\mathcal{Y}_j$, and $\lambda(R_\theta)f_\ell = e^{-i\ell\theta}f_\ell$. Since the sum is direct, comparing components gives $\lambda(R_\theta)y_j = e^{-i\ell\theta}y_j$ for all θ. For $j < \ell$ the weight $-\ell$ does not occur in V_j, so $y_j = 0$. Hence $f_\ell \in \mathcal{Y}_\ell$ and $U \subseteq \mathcal{Y}_\ell$. Since $U \neq 0$ and $\mathcal{Y}_\ell$ is irreducible, $U = \mathcal{Y}_\ell$. ∎

Example 12.15 (degrees one and two)**.** We compute $\mathcal{Y}_1$ and $\mathcal{Y}_2$ and their H-fixed lines. Every linear polynomial is harmonic, so $\dim\mathcal{H}_1 = 3$. The map $\Delta\colon \mathcal{P}_2 \to \mathbb{C}$ is onto, since $\Delta(x^2) = 2$. So $\dim\mathcal{H}_2 = 6 - 1 = 5$.

By Proposition 12.14 on spherical harmonics, $\mathcal{Y}_\ell \subseteq \mathcal{H}_\ell|_{S^2}$. Restriction is injective, so $\mathcal{H}_\ell|_{S^2}$ has dimension $2\ell+1$ for $\ell = 1, 2$. By the sphere theorem (Theorem 12.13), also $\dim\mathcal{Y}_\ell = 2\ell+1$. So the inclusion is an equality for $\ell = 1, 2$:

$$\mathcal{Y}_1 = \operatorname{span}\{x,y,z\}|_{S^2}, \qquad \mathcal{Y}_2 = \operatorname{span}\{xy,\ yz,\ zx,\ x^2-y^2,\ 2z^2-x^2-y^2\}|_{S^2}.$$

By the monomial weights (12.5), the weight-zero space of $\mathcal{P}_2$ is spanned by $z_+z_- = x^2+y^2$ and z^2. Since $\Delta\big(a(x^2+y^2)+bz^2\big) = 4a+2b$, its harmonic line is spanned by $2z^2-x^2-y^2$. This function spans $\mathcal{Y}_2^H$, the line predicted by $\dim V_2^H = 1$. Similarly z spans $\mathcal{Y}_1^H$.

Exercise 12.5 proves $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ for all ℓ. That exercise defines the Laplacian of the sphere and computes its eigenvalue on $\mathcal{H}_\ell$. Nothing in this lecture depends on it.

12.5 Self-checks

1. Let K be a finite group, regarded as a compact Lie group of dimension zero, and let $H \le K$ be a subgroup. What does the multiplicity theorem (Theorem 12.7) say?
   Answer. Here $L^2(K/H) = \mathbb{C}[K/H]$ is the permutation representation. An irreducible V occurs $\dim V^H$ times. This is finite-group Frobenius reciprocity for the trivial character of H.
2. Why not define $\Phi(T)v = (Tv)(e)$ directly for $T : V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$?
   Answer. An L^2 class has no value at e. Lemma 12.6 on equivariant images first shows that T takes continuous values.
3. What are the multiplicities for $H = \{e\}$ and for $H = K$?
   Answer. For $H = \{e\}$, σ occurs d_σ times in $L^2(K)$. For $H = K$, a nontrivial irreducible has $V^K = 0$, so only the trivial representation occurs, once, as the constants.

Summary

The invariant integral on K/H is the Haar integral of pullbacks. The averaging operator A_H projects $L^2(K)$ onto $L^2(K)^H \cong L^2(K/H)$. Peter–Weyl then shows that each $\sigma \in \operatorname{Irr}(K)$ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times.

Let V be a finite-dimensional representation of K and W a finite-dimensional unitary representation of H. Evaluation at e gives $\operatorname{Hom}_K(V, \operatorname{Ind}_H^K W) \cong \operatorname{Hom}_H(V|_H, W)$. Isotypic vectors in $L^2\text{-}\operatorname{Ind}_H^K W$ are continuous. So σ occurs in $L^2\text{-}\operatorname{Ind}_H^K W$ exactly $\dim \operatorname{Hom}_H(V_\sigma|_H, W)$ times.

On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ each $\mathrm{SO}(3)$-module V_ℓ occurs once in $L^2(S^2)$, realised by spherical harmonics of degree ℓ.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (1985), Section III.6: induction from closed subgroups and Frobenius reciprocity.
- Sepanski, *Compact Lie Groups* (2007), §2.3 and §§3.3–3.4: the assigned background route for this lecture and the sphere example.
- Knapp, *Lie Groups Beyond an Introduction*, digital second edition (2023), Section IX.2: induced representations of compact groups and Frobenius reciprocity.

Lecture 13

Spin groups and projective representations

Objectives. *A 2π rotation is the identity in $\mathrm{SO}(3)$. On an electron's state vector it acts by -1. We explain this sign using the double cover $\mathrm{Spin}(n) \to \mathrm{SO}(n)$. We then study actions on rays and their lifts to vectors.*

By the end of this lecture you will be able to:

- For $n \geq 2$, construct $\mathrm{Spin}(n)$ and prove that $\rho\colon\ \mathrm{Spin}(n) \to \mathrm{SO}(n)$ is surjective with kernel $\{\pm 1\}$.
- Identify $\mathrm{Spin}(3)$ with $\mathrm{SU}(2)$ and decide which V_m descend to $\mathrm{SO}(3)$.
- Define projective unitary representations, multipliers and central extensions, and lift finite-dimensional projective representations of $\mathrm{SO}(3)$ to $\mathrm{SU}(2)$.
- Use $\mathbb{R}^2$ to explain why a simply connected group can have a nontrivial multiplier.

13.1 Clifford algebras and spin groups

Fix $n \geq 2$ and an orthonormal basis $e_1, \dots, e_n$ of Euclidean $\mathbb{R}^n$. Recall from Lie Algebras that the *Clifford algebra* $Cl(n)$ is the real unital associative algebra generated by $\mathbb{R}^n$ subject to $v^2 = -|v|^2$. Replacing v by $u + v$ gives

$$uv + vu = -2\langle u, v\rangle \qquad (u, v \in \mathbb{R}^n). \tag{13.1}$$

This is the prerequisite convention $uv + vu = 2q(u, v)$ with the negative definite symmetric bilinear form $q(u, v) = -\langle u, v\rangle$.

The products $e_I = e_{i_1} \cdots e_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $Cl(n)$. The *even part* $Cl^0(n)$ is the span of the e_I with $|I|$ even. The *transpose* is the linear map $x \mapsto x^t$ with $(v_1 \cdots v_k)^t = v_k \cdots v_1$. The complexification $Cl(n)_{\mathbb{C}}$ has an irreducible *spinor module* S of dimension $2^{\lfloor n/2 \rfloor}$.

The units $Cl(n)^\times$ form an open set and a Lie group. A unit vector u has $u^2 = -1$ and $u^{-1} = -u$. Let $r_u(v) = v - 2\langle u, v\rangle u$ be reflection in $u^\perp$.

Definition 13.1. $\mathrm{Spin}(n)$ is the set of all products $u_1 u_2 \cdots u_{2k}$ with $k \geq 0$ and unit vectors $u_i \in \mathbb{R}^n$.

Lemma 13.2. *(a)* $\mathrm{Spin}(n)$ *is a subgroup of* $Cl^0(n)^\times$, *and* $x^{-1} = x^t$ *for* $x \in \mathrm{Spin}(n)$.
(b) For a unit vector u *and* $v \in \mathbb{R}^n$ *we have* $uvu^{-1} = -r_u(v)$.

Proof. (a) Even products of vectors lie in $Cl^0(n)$, and the set is closed under products. Let $x = u_1 \cdots u_{2k}$, so $x^t = u_{2k} \cdots u_1$. In

$$x^t x = u_{2k} \cdots u_2 \, (u_1 u_1) \, u_2 \cdots u_{2k}$$

the middle factor is the scalar -1. Next the two factors u_2 become adjacent and give $u_2 u_2 = -1$, and so on. Hence $x^t x = (-1)^{2k} = 1$, and in the same way $x x^t = 1$. Since $x^t \in \mathrm{Spin}(n)$, the set is a subgroup.

(b) By the anticommutation relation (13.1), $uv = -vu - 2\langle u, v\rangle$. Using also $u^{-1} = -u$ and $u^2 = -1$, we get

$$uvu^{-1} = -uvu = (vu + 2\langle u, v\rangle)u = -v + 2\langle u, v\rangle u = -r_u(v). \qquad \blacksquare$$

13.2 The double covering of the rotation group

For $x \in \mathrm{Spin}(n)$ and $v \in \mathbb{R}^n$ put $\rho(x)v = xvx^{-1}$, an element of $Cl(n)$. Write $x = u_1 \cdots u_{2k}$ with unit vectors $u_i \in \mathbb{R}^n$. Apply the conjugation formula of Lemma 13.2(b) once for each of the $2k$ factors of x. The $2k$ signs cancel, and we get

$$\rho(x)v = xvx^{-1} = r_{u_1} \circ \cdots \circ r_{u_{2k}}(v) \qquad (v \in \mathbb{R}^n). \tag{13.2}$$

So $\rho(x)$ maps $\mathbb{R}^n$ to itself, and it is a product of an even number of reflections. Hence $\rho(x) \in \mathrm{SO}(n)$, and ρ is a map from $\mathrm{Spin}(n)$ to $\mathrm{SO}(n)$.

Lemma 13.3 (Cartan–Dieudonné, weak form)**.** *Every $g \in \mathrm{O}(n)$ is a product of at most n reflections. If $\det g = 1$, the number of reflections is even.*

Proof. Induct on n. For $n = 1$ we have $\mathrm{O}(1) = \{1, r_{e_1}\}$. Let $n \geq 2$. If $ge_1 = e_1$, then g preserves $e_1^\perp \cong \mathbb{R}^{n-1}$. By induction $g|_{e_1^\perp}$ is a product of at most $n-1$ reflections of $e_1^\perp$, and each extends to a reflection of $\mathbb{R}^n$ fixing e_1. If $a = ge_1 \neq e_1$, put $u = (a - e_1)/|a - e_1|$. Since $|a| = 1$, we have $\langle a - e_1, a\rangle = \frac{1}{2}|a - e_1|^2$. Therefore

$$r_u(a) = a - \frac{2\langle a - e_1, a\rangle}{|a - e_1|^2}\,(a - e_1) = e_1.$$

So $r_u g$ fixes e_1, and $g = r_u(r_u g)$ uses at most n reflections. Each reflection has determinant -1, which gives the parity. $\blacksquare$

Theorem 13.4. *Let $n \geq 2$.*

(a) $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ *is a continuous surjective homomorphism.*

(b) $\ker \rho = \{\pm 1\}$.

(c) $\mathrm{Spin}(n)$ *is a compact connected Lie group, and ρ is a two-sheeted covering homomorphism.*

(d) For $n \geq 3$, using the cited fact that $\pi_1(\mathrm{SO}(n))$ has order two, the group $\mathrm{Spin}(n)$ is simply connected. So $\mathrm{Spin}(n)$ is the universal covering group of $\mathrm{SO}(n)$.

Proof. (a) Clearly $\rho(xy) = \rho(x)\rho(y)$. Since $x^{-1} = x^t$, the entries of $\rho(x)$ are polynomials in the coordinates of x. Given $g \in \mathrm{SO}(n)$, write $g = r_{w_1} \cdots r_{w_{2k}}$ with $2k \le n$ by the Cartan–Dieudonné lemma, Lemma 13.3. By the reflection formula (13.2), $y = w_1 \cdots w_{2k}$ satisfies $\rho(y) = g$.

(b) Note $-1 = e_1 e_1 \in \mathrm{Spin}(n)$, and ± 1 act trivially. Conversely, let $\rho(x) = \mathrm{id}$, so x commutes with every e_i. Fix i. Sorting the basis elements of x by whether they contain e_i, and moving e_i to the front, gives

$$x = a + e_i b,$$

where a, b are combinations of e_J with $i \notin J$, a even and b odd. By the anticommutation relation (13.1), e_i commutes with a and anticommutes with b. Hence

$$x e_i = e_i a - e_i e_i b, \qquad e_i x = e_i a + e_i e_i b.$$

Equality forces $2e_i^2 b = -2b = 0$. So no basis element containing e_i occurs in x. As i was arbitrary, $x = \lambda 1$ with $\lambda \in \mathbb{R}$. Then $1 = x^t x = \lambda^2$, so $\lambda = \pm 1$.

(c) Let $N = 2\lfloor n/2 \rfloor \ge 2$ and $x \in \mathrm{Spin}(n)$. The surjectivity proof in part (a) gives $y = w_1 \cdots w_{2k}$ with $2k \le N$ and $\rho(y) = \rho(x)$. Append pairs $e_1(-e_1) = 1$ until there are N factors. By the kernel statement (b), $x = \pm y$, so we may replace w_1 by $-w_1$ to get $x = y$. Hence $\mathrm{Spin}(n)$ is the image of the compact connected space $(S^{n-1})^N$ under the continuous map $(w_1, \dots, w_N) \mapsto w_1 \cdots w_N$. So $\mathrm{Spin}(n)$ is compact and connected.

Being compact, the group $\mathrm{Spin}(n)$ is closed in $Cl(n)^\times$, hence a Lie group by the closed subgroup theorem (Lecture 3).

The smooth transitive action of $\mathrm{Spin}(n)$ on $\mathrm{SO}(n)$ through ρ has stabiliser $\{\pm 1\}$. The homogeneous-space theorem (Lecture 3) identifies $\mathrm{Spin}(n)/\{\pm 1\}$ with $\mathrm{SO}(n)$. The quotient by the discrete subgroup $\{\pm 1\}$ is a local diffeomorphism, so $d\rho$ is an isomorphism. The covering criterion of Lecture 4 applies and gives two sheets.

(d) We cite that $\pi_1(\mathrm{SO}(n))$ has order two for $n \ge 3$ (Hall, Chapter 13). So the universal covering $q \colon \tilde{G} \to \mathrm{SO}(n)$ has kernel of order two. Integrate $(d\rho)^{-1} dq$ by the integration theorem of Lecture 4 to a homomorphism $\phi \colon \tilde{G} \to \mathrm{Spin}(n)$. The homomorphisms $\rho \circ \phi$ and q have the same differential on connected $\tilde{G}$, so they agree.

Since dq and $d\rho$ are isomorphisms, so is $d\phi$. Thus $\phi(\tilde{G})$ is an open, hence closed, subgroup, and ϕ is onto by connectedness. Now $\ker q = \phi^{-1}(\{\pm 1\})$ has order $2|\ker \phi|$, so $\ker \phi$ is trivial. A bijective homomorphism of Lie groups is an isomorphism (Lecture 3). ∎

For $n = 2$, Exercise 13.1 shows that $\mathrm{Spin}(2)$ is a circle and ρ is $z \mapsto z^2$. So $\mathrm{Spin}(2)$ is not simply connected.

Example 13.5 (Rotations in a coordinate plane)**.** Fix $a \neq b$ and put

$$x_{ab}(t) = \cos t + \sin t\, e_a e_b = -(\cos t\, e_a + \sin t\, e_b)\, e_a \in \mathrm{Spin}(n).$$

Since $(e_a e_b)^2 = -1$, we have $x_{ab}(t) = \exp(t\, e_a e_b)$, so $e_a e_b \in \mathrm{Lie}(\mathrm{Spin}(n))$. The vectors e_a, e_b anticommute with $e_a e_b$, so $e_a x_{ab}(t)^{-1} = x_{ab}(t) e_a$, and similarly for e_b. Every e_c with $c \neq a, b$ commutes with $e_a e_b$. Using $e_a e_b e_a = e_b$ and $e_a e_b e_b = -e_a$ we get

$$\rho(x_{ab}(t))e_a = x_{ab}(2t)e_a = \cos 2t\, e_a + \sin 2t\, e_b, \qquad \rho(x_{ab}(t))e_b = -\sin 2t\, e_a + \cos 2t\, e_b,$$

and $\rho(x_{ab}(t))e_c = e_c$. So $\rho(x_{ab}(t))$ is the rotation through $2t$ in the (e_a, e_b)-plane. For $t \in [0, \pi]$ the rotations run once around a closed loop. The lift runs from 1 to $x_{ab}(\pi) = -1$, so it is not closed.

Since $\mathrm{Spin}(n) \subset Cl(n)_{\mathbb{C}}$, the spinor module gives a representation Π_S of $\mathrm{Spin}(n)$. The element $-1 \in \mathrm{Spin}(n)$ is the scalar -1 of $Cl(n)$. Hence

$$\Pi_S(-1) = -\,\mathrm{id}_S, \qquad \rho(-1) = \mathrm{id}\,.$$

So Π_S is not of the form $\Pi' \circ \rho$. This gives the opening sign change.

13.3 Spin(3), SU(2) and descent

In $Cl^0(3)$ put $\mathbf{i} = e_2e_3$, $\mathbf{j} = e_3e_1$, $\mathbf{k} = e_1e_2$. For instance $\mathbf{ij} = e_2e_3e_3e_1 = -e_2e_1 = \mathbf{k}$. In the same way

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}.$$

So $Cl^0(3) = \mathrm{span}\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is the quaternion algebra $\mathbb{H}$. Since $\mathbf{i}^t = e_3e_2 = -\mathbf{i}$, the transpose is quaternion conjugation $q \mapsto \bar{q}$. For $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ we have $q\bar{q} = |q|^2 = a^2 + b^2 + c^2 + d^2$. Let $\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}$ be the group of unit quaternions.

Define a real linear map $\Phi\colon \mathbb{H} \to M_2(\mathbb{C})$ by

$$\Phi(1) = I, \quad \Phi(\mathbf{i}) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \Phi(\mathbf{j}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Phi(\mathbf{k}) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}.$$

Direct multiplication shows that these matrices satisfy the quaternion relations. For example, $\Phi(\mathbf{i})\Phi(\mathbf{j}) = \Phi(\mathbf{k})$. Hence Φ is an algebra homomorphism, and

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}, \qquad \det \Phi(q) = |q|^2. \tag{13.3}$$

Proposition 13.6. *(a)* $\mathrm{Spin}(3) = \mathrm{Sp}(1)$.
(b) Φ *restricts to an isomorphism of Lie groups* $\mathrm{Sp}(1) \to \mathrm{SU}(2)$.

Proof. (a) If $q \in \mathrm{Spin}(3)$, then $q\bar{q} = qq^t = 1$ by the inverse formula of Lemma 13.2(a), so $|q| = 1$. Conversely, let $|q| = 1$ and write $q = a + p$ with $p \in \mathrm{span}\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$. If $p = 0$, then $q = \pm 1 \in \mathrm{Spin}(3)$. Otherwise put $w = p/|p|$. Cross terms in w^2 cancel by anticommutation, so $w^2 = -1$. Choose θ with $\cos\theta = a$ and $\sin\theta = |p|$. Then

$$q = \cos\theta + \sin\theta\, w = \exp(\theta w).$$

By the plane-rotation example, Example 13.5, $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathrm{Lie}(\mathrm{Spin}(3))$. The Lie algebra of a closed subgroup is a linear subspace, and exp maps it into the subgroup (Lecture 3). So $\theta w \in \mathrm{Lie}(\mathrm{Spin}(3))$ and $q \in \mathrm{Spin}(3)$.

(b) By the matrix formula (13.3), $\Phi(\bar{q}) = \Phi(q)^*$. So $\Phi(q)\Phi(q)^* = \Phi(q\bar{q}) = I$ for $|q| = 1$, and $\det\Phi(q) = 1$. Conversely, let $g \in \mathrm{SU}(2)$ have first column $(\alpha, \beta)^T$. The second column is a unit vector orthogonal to the first, and $\det g = 1$. This forces it to be $(-\bar{\beta}, \bar{\alpha})^T$. So $g = \Phi(q)$ with $\alpha = a + bi$ and $\beta = c - di$. Thus Φ is a bijective continuous homomorphism. Its inverse is linear, hence continuous. ∎

From now on we identify Spin(3) with SU(2) through Φ. Then $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ has kernel $\{\pm I\}$.

For $m \geq 0$ let V_m be the space of homogeneous polynomials of degree m in $z = (z_1, z_2)$, with

$$(\Pi_m(g)f)(z) = f(g^{-1}z) \qquad (g \in \mathrm{SU}(2)).$$

The space V_m has dimension $m+1$ and spin $m/2$. Every finite-dimensional irreducible representation of SU(2) is isomorphic to exactly one V_m (Lecture 8). Fix an SU(2)-invariant inner product on each V_m (Lecture 5).

The spinor module of $Cl(3)_{\mathbb{C}}$ has dimension two. If its restriction to SU(2) were reducible, complete reducibility would make it $V_0 \oplus V_0$, since V_0 is the only one-dimensional irreducible. That restriction would be trivial, contrary to $\Pi_S(-1) = -\mathrm{id}_S$. Thus the restriction is isomorphic to V_1.

Lemma 13.7 (Descent). *Let $p\colon \tilde{G} \to G$ be a covering homomorphism of connected Lie groups with kernel Z. A representation Π of $\tilde{G}$ has the form $\Pi' \circ p$ for a representation Π' of G if and only if $\Pi(z) = \mathrm{id}$ for all $z \in Z$.*

Proof. One direction is clear. Suppose $\Pi(Z) = \{\mathrm{id}\}$. Then Π is constant on each fibre xZ, so $\Pi'(p(x)) = \Pi(x)$ defines a homomorphism. A covering map is open, so for open O the set $\Pi'^{-1}(O) = p(\Pi^{-1}(O))$ is open. Thus Π' is continuous. ∎

Proposition 13.8. *Π_m factors through $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ if and only if m is even.*

Proof. We have $\ker\rho = \{\pm I\}$ and $(\Pi_m(-I)f)(z) = f(-z) = (-1)^m f(z)$. Apply the descent lemma, Lemma 13.7. ∎

So the irreducible representations of SO(3) are the descents of the V_{2l}, of dimension $2l+1$. The spinor V_1 is not among them.

Example 13.9 (A full turn on V_m). Put

$$u(\theta) = \Phi\Big(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\mathbf{i}\Big) = \mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2}).$$

Since $\mathbf{i} = e_2e_3$, the plane-rotation example, Example 13.5, shows that $\rho(u(\theta))$ is the rotation $R(\theta)$ through θ in the (e_2, e_3)-plane. On the monomial $f_k = z_1^{m-k}z_2^k$ we get

$$\Pi_m(u(\theta))f_k = (e^{-i\theta/2}z_1)^{m-k}(e^{i\theta/2}z_2)^k = e^{i(2k-m)\theta/2}f_k.$$

At $\theta = 2\pi$ every eigenvalue is $e^{i(2k-m)\pi} = (-1)^m$. Hence

$$R(2\pi) = \mathrm{id}, \qquad \Pi_m(u(2\pi)) = (-1)^m\,\mathrm{id}\,.$$

For odd m every vector changes sign. Each ray $\mathbb{C}f$ is unchanged, and so is every transition probability $|\langle f, h\rangle|^2/(\|f\|^2\|h\|^2)$.

Remark 13.10. The trivial character and det of O(2) both have differential 0 on $\mathfrak{so}(2)$, but they differ on a reflection. So for a disconnected group, $d\Pi$ need not determine Π (Exercise 13.4).

13.4 Projective unitary representations

In quantum mechanics a pure state is a ray $\mathbb{C}f$ in a Hilbert space, with $f \neq 0$. Transition probabilities depend only on rays. So a symmetry group acts on rays. Lifting this action to vectors may require choices of phase.

Let $\mathcal{H}$ be a complex Hilbert space of finite dimension $d \geq 1$, and let $\mathbb{T} = \{\lambda \in \mathbb{C} : |\lambda| = 1\}$.

Definition 13.11. The *projective unitary group* is $\mathrm{PU}(\mathcal{H}) = \mathrm{U}(\mathcal{H})/\mathbb{T}$ id with the quotient topology. We write $[U]$ for the class of U.

It is a Lie group (Lecture 3). It acts on rays by $[U] \cdot \mathbb{C}f = \mathbb{C}Uf$.

Definition 13.12. A *projective unitary representation* of a Lie group G on $\mathcal{H}$ is a continuous homomorphism $\bar{\pi} \colon G \to \mathrm{PU}(\mathcal{H})$.

We call $\bar{\pi}$ *irreducible* if 0 and $\mathcal{H}$ are the only subspaces invariant under every U with $[U] \in \bar{\pi}(G)$. A *lift* is any function $U \colon G \to \mathrm{U}(\mathcal{H})$ with $[U(g)] = \bar{\pi}(g)$. No continuity is assumed. Since $[U(g)U(h)] = [U(gh)]$, there is a function $c \colon G \times G \to \mathbb{T}$ with

$$U(g)U(h) = c(g,h)\,U(gh). \tag{13.4}$$

Definition 13.13. A *multiplier* of G is a function $c \colon G \times G \to \mathbb{T}$ such that

$$c(g,h)\,c(gh,k) = c(h,k)\,c(g,hk) \qquad (g,h,k \in G).$$

Multipliers c, c' are *equivalent* if $c'(g,h) = c(g,h)\,b(g)b(h)/b(gh)$ for some function $b \colon G \to \mathbb{T}$. A multiplier is *trivial* if it is equivalent to 1.

Lemma 13.14. *The function c in (13.4) is a multiplier. Its equivalence class does not depend on the lift. It is trivial if and only if $\bar{\pi}$ has a lift that is a homomorphism.*

Proof. Expand $U(g)U(h)U(k)$ in two ways:

$$(U(g)U(h))U(k) = c(g,h)c(gh,k)U(ghk), \qquad U(g)(U(h)U(k)) = c(h,k)c(g,hk)U(ghk).$$

This gives the identity. Any other lift is bU with $b \colon G \to \mathbb{T}$, and its multiplier is $c(g,h)\,b(g)b(h)/b(gh)$. So the class is well defined. The lift bU is a homomorphism exactly when this multiplier equals 1. ∎

Definition 13.15. A *central extension* of G by $\mathbb{T}$ is a Lie group E with a closed central subgroup $Z \cong \mathbb{T}$ and an isomorphism of Lie groups $E/Z \cong G$.

Proposition 13.16. *Let $\bar{\pi}$ be a projective unitary representation of G on $\mathcal{H}$. Then*

$$E_{\bar{\pi}} = \{(g,U) \in G \times \mathrm{U}(\mathcal{H}) : [U] = \bar{\pi}(g)\}$$

is a central extension of G by $Z = \{(1, \lambda\,\mathrm{id}) : \lambda \in \mathbb{T}\}$. The map $(g,U) \mapsto U$ is a unitary representation of $E_{\bar{\pi}}$ in which $(1, \lambda\,\mathrm{id})$ acts by λ.

Proof. The set $E_{\bar\pi}$ is a subgroup of $G \times \mathrm{U}(\mathcal{H})$. It is the preimage of the diagonal under the continuous map $(g, U) \mapsto (\bar\pi(g), [U])$ into the Hausdorff space $\mathrm{PU}(\mathcal{H})^2$. So it is closed, hence a Lie group (Lecture 3). Clearly Z is closed and central. The projection to G is surjective with kernel Z. The projection defines a smooth transitive action of $E_{\bar\pi}$ on G, with stabiliser Z. The homogeneous-space theorem of Lecture 3 therefore identifies $E_{\bar\pi}/Z$ smoothly with G. ∎

Conversely, a unitary representation of a central extension E in which Z acts by $\lambda \mapsto \lambda\,\mathrm{id}$ defines a projective representation of E/Z. In this sense, projective representations of G are ordinary representations of central extensions of G.

13.5 Lifting projective representations

Theorem 13.17. *Let G be a connected Lie group with universal covering homomorphism $q\colon \tilde G \to G$ (Lecture 4). Let $\bar\pi$ be a projective unitary representation of G on a finite-dimensional $\mathcal{H}$. Then there is a unique continuous homomorphism $U\colon \tilde G \to \mathrm{SU}(\mathcal{H})$ with $[U(x)] = \bar\pi(q(x))$.*

Proof. Let $d = \dim \mathcal{H}$. Every $V \in \mathrm{U}(\mathcal{H})$ equals λV_0 with $\lambda^d = \det V$ and $V_0 \in \mathrm{SU}(\mathcal{H})$. So the restriction $p\colon \mathrm{SU}(\mathcal{H}) \to \mathrm{PU}(\mathcal{H})$ of the quotient map is surjective. Its kernel $\{\lambda\,\mathrm{id} : \lambda^d = 1\}$ is finite, and $\mathrm{SU}(\mathcal{H})$ is connected.

The differential of p has kernel $\mathfrak{su}(d) \cap i\mathbb{R}I = 0$. Both $\mathfrak{su}(d)$ and the tangent algebra of $\mathrm{U}(d)/\mathbb{T}I$ have dimension $d^2 - 1$, so dp is an isomorphism. By the covering criterion of Lecture 4, p is a covering homomorphism. For $d = 1$ both groups are trivial.

Since $\tilde G$ is simply connected, the based lifting theorem of Lecture 4 gives a unique continuous lift U with $U(e) = I$ and $p \circ U = \bar\pi \circ q$. It remains to check that U is multiplicative. The two maps $(x, y) \mapsto U(xy)$ and $(x, y) \mapsto U(x)U(y)$ lift the same map on the connected space $\tilde G \times \tilde G$ and agree at (e, e). Uniqueness of covering lifts makes them equal. Thus U is a homomorphism. Any homomorphic lift sends e to I, so uniqueness of the based lift also gives the stated uniqueness. ∎

Corollary 13.18. *Let $\mathcal{H}$ be finite-dimensional.*

(a) For each projective unitary representation $\bar\pi$ of $\mathrm{SO}(3)$ on $\mathcal{H}$, there is a unitary representation U of $\mathrm{SU}(2)$ with $\bar\pi(\rho(x)) = [U(x)]$. Moreover $U(-I) = \pm\,\mathrm{id}$.

(b) If $\bar\pi$ is irreducible, then U is unitarily equivalent to Π_m with $m = \dim \mathcal{H} - 1$.

(c) For each $m \geq 0$, the rule $\bar\pi_m(\rho(x)) = [\Pi_m(x)]$ defines a projective unitary representation of $\mathrm{SO}(3)$. Its multiplier is trivial if and only if m is even.

Proof. (a) The group $\mathrm{SU}(2) \cong S^3$ is simply connected (Lecture 4). Apply the lifting theorem, Theorem 13.17. Since $[U(-I)] = \bar\pi(\mathrm{id})$, $U(-I)$ is a scalar. Its square is $U(I) = \mathrm{id}$, so the scalar is ± 1.

(b) Scalars preserve every subspace, so U is irreducible. By the classification, $U \cong \Pi_m$ with $m + 1 = \dim \mathcal{H}$. By Schur's lemma the isomorphism becomes unitary after rescaling (Lecture 5).

(c) If $\rho(x) = \rho(x')$, then $x' = \pm x$ and $\Pi_m(x') = \pm\Pi_m(x)$. So $\bar\pi_m$ is well defined. It is continuous because ρ is open, as in the descent lemma, Lemma 13.7. For even m, the descent criterion Proposition 13.8 gives a lift that is a homomorphism.

Let m be odd. Put $A = \rho(\Phi(\mathbf{i}))$ and $B = \rho(\Phi(\mathbf{j}))$. Since $\mathbf{i} = x_{23}(\pi/2)$ and $\mathbf{j} = x_{31}(\pi/2)$, the plane-rotation example, Example 13.5, gives

$$A = \operatorname{diag}(1,-1,-1), \qquad B = \operatorname{diag}(-1,1,-1), \qquad AB = BA.$$

Put $P = \Pi_m(\Phi(\mathbf{i}))$ and $Q = \Pi_m(\Phi(\mathbf{j}))$. Since $\mathbf{ij} = -\mathbf{ji}$ and $\Pi_m(-I) = -\mathrm{id}$,

$$PQ = \Pi_m(-I)\,QP = -QP.$$

Suppose $W\colon \mathrm{SO}(3) \to \mathrm{U}(V_m)$ is a homomorphism lifting $\bar\pi_m$, with no continuity assumed. Then $W(A) = \alpha P$ and $W(B) = \beta Q$ with $\alpha, \beta \in \mathbb{T}$. So $W(A)W(B) = -W(B)W(A)$. But $AB = BA$ gives $W(A)W(B) = W(B)W(A)$. Hence $2W(A)W(B) = 0$, which is impossible for invertible operators. By the triviality criterion of Lemma 13.14, the multiplier is nontrivial. ∎

So every finite-dimensional irreducible projective representation of $\mathrm{SO}(3)$ is equivalent to some $\bar\pi_m$. Here equivalence means that a unitary $T : V_m \to \mathcal{H}$ satisfies

$$\bar\pi(\rho(x)) = [T\Pi_m(x)T^{-1}] \qquad (x \in \mathrm{SU}(2)).$$

All spins $m/2$ occur, but only integer spins come from genuine representations. This is the mathematical form of the half-integer spin of the electron.

For separable Hilbert spaces we cite the following result.

Theorem 13.19 (Bargmann; cited). *Let G be a connected Lie group with semisimple Lie algebra $\mathfrak{g}$ and universal covering $q\colon \tilde G \to G$. Let $\mathcal{H}$ be a separable Hilbert space, and let $\bar\pi$ be a homomorphism from G to $\mathrm{U}(\mathcal{H})/\mathbb{T}$ id. Assume that $g \mapsto |\langle f, U(g)h\rangle|$ is continuous for all $f, h \in \mathcal{H}$ and one (hence every) lift U. Then there is a strongly continuous unitary representation $\tilde U$ of $\tilde G$ with $[\tilde U(x)] = \bar\pi(q(x))$ for $x \in \tilde G$.*

Example 13.20 ($\mathbb{R}^2$ and the Heisenberg extension). On $G = \mathbb{R}^2$ put

$$\omega(x,y) = x_1y_2 - x_2y_1, \qquad c(x,y) = e^{i\omega(x,y)/2}.$$

The exponent is bilinear, so c is a multiplier. The factor $b(x)b(y)/b(x+y)$ is symmetric in x, y. So the ratio

$$c(x,y)/c(y,x) = e^{i\omega(x,y)}$$

depends only on the class of c. The ratio is $e^{i} \neq 1$ for $x = (1,0)$ and $y = (0,1)$, so c is nontrivial. Yet $\mathbb{R}^2$ is its own universal cover.

The extension $E = \mathbb{R}^2 \times \mathbb{T}$ of Exercise 13.5 has smooth multiplication. Projection onto $\mathbb{R}^2$ identifies the quotient $E/(\{0\} \times \mathbb{T})$ with $\mathbb{R}^2$, so E is a central extension. The commutator of (x,λ) and (y,μ) in E is $(0, e^{i\omega(x,y)})$.

This does not contradict the lifting theorem, Theorem 13.17. A lift on $\mathbb{C}^d$ with multiplier equivalent to c would satisfy $U(x)U(y) = e^{i\omega(x,y)}U(y)U(x)$. Taking determinants gives $e^{id\,\omega(x,y)} = 1$ for all x, y, which is false. So c occurs only on infinite-dimensional spaces (Lecture 18). Bargmann's theorem does not apply to $\mathbb{R}^2$, because its Lie algebra is abelian rather than semisimple.

Self-check questions

1. In Spin(4), what is $\rho(e_1e_2e_3e_4)$?

   Answer. The element $e_1e_2e_3e_4$ equals $x_{12}(\pi/2)\,x_{34}(\pi/2)$. By the plane-rotation example, Example 13.5, its image under ρ rotates through π in both coordinate planes, so $\rho(e_1e_2e_3e_4) = -\mathrm{id}$. The other lift of $-\mathrm{id}$ is $-e_1e_2e_3e_4$.

2. For odd m, does $\Pi_m(u(2\pi))$ change any transition probability?

   Answer. No. It equals $-\mathrm{id}$. Hence it fixes every ray, and $|\langle -f, -h\rangle| = |\langle f, h\rangle|$.

3. Is there a projective unitary representation of $\mathbb{R}^2$ on $\mathbb{C}^3$ with multiplier $e^{i\omega/2}$?

   Answer. No. The determinant argument of the Heisenberg example, Example 13.20, with $d = 3$ gives $e^{3i\omega(x,y)} = 1$ for all x, y, which is false.

Summary

The group $\mathrm{Spin}(n)$ consists of even products of unit vectors in $Cl(n)$. For $n \geq 2$, conjugation maps it onto $\mathrm{SO}(n)$ with kernel $\{\pm 1\}$. For $n \geq 3$, the cited value of $\pi_1(\mathrm{SO}(n))$ shows that it is the universal cover. Spinor modules send -1 to $-\mathrm{id}$ and do not factor through ρ.

For $n = 3$, $\mathrm{Spin}(3) = \mathrm{Sp}(1) \cong \mathrm{SU}(2)$, and V_m descends to $\mathrm{SO}(3)$ exactly for even m. The lift $u(2\pi)$ of a full turn acts on V_m by $(-1)^m$, which fixes every ray.

In finite dimensions, projective unitary representations of connected Lie groups lift to unitary representations of the universal cover. On separable Hilbert spaces, Bargmann's cited theorem gives strongly continuous lifts for connected groups with semisimple Lie algebra, under its continuity hypothesis. The multiplier on the simply connected group $\mathbb{R}^2$ shows that semisimplicity cannot be dropped.

Reading

- BD, Section I.6: Clifford algebras and $\mathrm{Spin}(n)$. Note the sign convention for v^2.
- Sepanski, Section 1.3: the Clifford-algebra construction of the double cover of $\mathrm{SO}(n)$.
- Kirillov, Elements, §14: projective representations and their multipliers.
- Hall, Chapter 13: fundamental groups of compact Lie groups, including $\mathrm{SO}(n)$.
- V. Bargmann, On unitary ray representations of continuous groups, Ann. of Math. 59 (1954).

Lecture 14

Representative functions and tensor reconstruction

Objectives. *This lecture asks what information about a compact Lie group K is contained in all its representations. We collect all matrix coefficients into one algebra $\mathscr{R}(K)$. The group law becomes three maps on this algebra. We then state how K is recovered, and what is recovered when the unitary structure is forgotten.*

By the end of this lecture you will be able to:

- Prove that $\mathscr{R}(K)$ is a $*$-algebra whose products come from tensor products and whose conjugation comes from conjugate representations.
- Define Δ, ε, S and prove the Hopf algebra identities, also in matrix form.
- State Tannaka–Krein reconstruction with the unitary fibre functor, and contrast it with the complex reconstruction of $K_{\mathbb{C}}$.
- Compute these structures for $\mathrm{U}(1)$ and $\mathrm{SU}(2)$, and explain why $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category when K is infinite.

Throughout, K is a compact Lie group. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ has as objects the finite-dimensional continuous complex representations (Π_V, V). Its morphisms are the K-equivariant linear maps.

14.1 Matrix coefficients

Definition 14.1. Let V be an object of $\mathrm{Rep}_{\mathrm{fd}}(K)$, $\lambda \in V^*$ and $v \in V$. The *matrix coefficient* $c^V_{\lambda,v} \colon K \to \mathbb{C}$ is

$$c^V_{\lambda,v}(g) = \lambda\big(\Pi_V(g)v\big).$$

The space $\mathscr{R}(K)$ of *representative functions* is the linear span of all matrix coefficients of all objects V.

Fix a basis (e_j) of V with dual basis (e^i). The matrix of $\Pi_V(g)$ in this basis is $D^V(g)$, with entries $D^V_{ij}(g) = e^i(\Pi_V(g)e_j)$. Expanding $\lambda = \sum_i \lambda(e_i)e^i$ and $v = \sum_j e^j(v)e_j$ gives

$$c^V_{\lambda,v} = \sum_{i,j} \lambda(e_i)\, e^j(v)\, D^V_{ij}.$$

So every coefficient of V is a linear combination of the D^V_{ij}. For a K-invariant inner product on V (Lecture 5) we may take $\lambda = \langle u, \cdot \rangle$, which gives $c^V_{\lambda,v}(g) = \langle u, \Pi_V(g)v\rangle$.

The *conjugate representation* $\overline{V}$ is the additive group V with scalar multiplication $t \cdot v = \bar{t}v$ and the same operators $\Pi_V(g)$. These operators are linear on $\overline{V}$:

$$\Pi_V(g)(t \cdot v) = \Pi_V(g)(\bar{t}v) = \bar{t}\,\Pi_V(g)v = t \cdot \Pi_V(g)v.$$

For $\lambda \in V^*$ put $\bar{\lambda}(v) = \overline{\lambda(v)}$. The functional $\bar{\lambda}$ is linear on $\overline{V}$:

$$\bar{\lambda}(t \cdot v) = \overline{\bar{t}\,\lambda(v)} = t\,\bar{\lambda}(v).$$

A basis (e_j) of V is also a basis of $\overline{V}$, with dual basis $(\overline{e^i})$. Hence

$$D^{\overline{V}}_{ij}(g) = \overline{e^i}\big(\Pi_V(g)e_j\big) = \overline{D^V_{ij}(g)}, \qquad \text{so} \qquad D^{\overline{V}}(g) = \overline{D^V(g)}.$$

Proposition 14.2. *Let $\lambda \in V^*$, $v \in V$, $\mu \in W^*$, $w \in W$ and $h \in K$. Then:*

(a) $c^V_{\lambda,v} + c^W_{\mu,w} = c^{V\oplus W}_{(\lambda,\mu),(v,w)}$;

(b) $c^V_{\lambda,v}\, c^W_{\mu,w} = c^{V\otimes W}_{\lambda\otimes\mu,\, v\otimes w}$;

(c) $\overline{c^V_{\lambda,v}} = c^{\overline{V}}_{\bar{\lambda},v}$;

(d) $g \mapsto c^V_{\lambda,v}(gh)$ *equals* $c^V_{\lambda,\Pi_V(h)v}$, *and* $g \mapsto c^V_{\lambda,v}(hg)$ *equals* $c^V_{\lambda\circ\Pi_V(h),v}$.

Hence $\mathscr{R}(K)$ is the set of matrix coefficients. It is a subalgebra of $C(K)$ containing the constants, closed under conjugation and translations.

Proof. Matrix coefficients are continuous because $g \mapsto \Pi_V(g)$ is continuous. The sum rule (a) holds because $\Pi_{V\oplus W}(g)$ acts componentwise. For the product rule (b), the tensor product representation gives

$$(\lambda \otimes \mu)\big(\Pi_V(g)v \otimes \Pi_W(g)w\big) = \lambda(\Pi_V(g)v)\,\mu(\Pi_W(g)w).$$

The conjugation rule (c) is the definition of $\bar{\lambda}$. The translation rule (d) uses $\Pi_V(gh) = \Pi_V(g)\Pi_V(h)$ and $\Pi_V(hg) = \Pi_V(h)\Pi_V(g)$.

Scalar multiples are coefficients, since $t\,c^V_{\lambda,v} = c^V_{t\lambda,v}$. By the sum rule (a), sums of coefficients are coefficients. So the span consists of coefficients. By the product, conjugation and translation rules (b)–(d), this set is closed under products, conjugation and translations. The constant 1 is $c^{\mathbb{C}}_{1,1}$ for the trivial representation. ∎

We use the Peter–Weyl theorem for compact Lie groups from Lecture 10. Choose one representative V for each class in $\mathrm{Irr}(K)$ and a basis of each. Then

$$\mathscr{R}(K) = \bigoplus_{[V]\in\mathrm{Irr}(K)} \operatorname{span}\{D^V_{ij}\},$$

an algebraic direct sum whose summand for V has dimension $(\dim V)^2$. Moreover $\mathscr{R}(K)$ is dense in $C(K)$ for the supremum norm and separates points.

14.2 The group law as structure maps

The group law is a map $K \times K \to K$. Pulling functions back along the group law produces functions of two variables. We first show that a tensor of functions is determined by the function of two variables that it defines.

Lemma 14.3. *Let A and B be linear spaces of functions on sets X and Y. The linear map from $A \otimes B$ to functions on $X \times Y$ with $f \otimes h \mapsto \big((x,y) \mapsto f(x)h(y)\big)$ is injective.*

Proof. Every tensor can be written $F = \sum_i f_i \otimes h_i$ with $h_1, \dots, h_n$ linearly independent. Suppose $\sum_i f_i(x)h_i(y) = 0$ for all x, y. Fix x. Then $\sum_i f_i(x)h_i$ is the zero function on Y. The numbers $f_i(x)$ are the coefficients of a vanishing combination of independent functions, so $f_i(x) = 0$ for all i. As x was arbitrary, each $f_i = 0$ and $F = 0$. ∎

Applying the lemma twice handles three factors. So we identify $\mathscr{R}(K)^{\otimes 2}$ and $\mathscr{R}(K)^{\otimes 3}$ with spaces of functions on K^2 and K^3. The identification is an algebra homomorphism for $(f \otimes h)(f' \otimes h') = ff' \otimes hh'$. Indeed both sides give the function $(x,y) \mapsto f(x)f'(x)h(y)h'(y)$.

Definition 14.4. A *Hopf algebra* is a unital algebra A with unital algebra homomorphisms $\Delta \colon A \to A \otimes A$, $\varepsilon \colon A \to \mathbb{C}$ and a linear map $S \colon A \to A$ such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, \qquad (\varepsilon \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon)\Delta,$$
$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

where m is multiplication and $\eta(t) = t1$. A *Hopf $*$-algebra* is a Hopf algebra with an antilinear map $*$ such that $f^{**} = f$, $(fh)^* = h^* f^*$, and Δ, ε preserve $*$. Here $(f \otimes h)^* = f^* \otimes h^*$ and $t^* = \bar{t}$ on $\mathbb{C}$.

For $f \in \mathscr{R}(K)$ and $g, h \in K$ define

$$\Delta f(g,h) = f(gh), \qquad \varepsilon f = f(e), \qquad Sf(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

A priori Δf is only a function on $K \times K$. The first task is to show that it is a tensor in $\mathscr{R}(K)^{\otimes 2}$.

Theorem 14.5. *These maps make $\mathscr{R}(K)$ a commutative Hopf $*$-algebra, and $S(f^*) = (Sf)^*$. For every V and basis (e_j),*

$$\Delta D^V_{ij} = \sum_k D^V_{ik} \otimes D^V_{kj}, \qquad \varepsilon D^V_{ij} = \delta_{ij}, \qquad SD^V_{ij}(g) = \big(D^V(g)^{-1}\big)_{ij}. \tag{14.1}$$

For an orthonormal basis of a K-invariant inner product, $SD^V_{ij} = (D^V_{ji})^*$.

Proof. Codomains. Insert the expansion $v = \sum_k e^k(v)e_k$ for the vector $v = \Pi_V(h)e_j$. This gives

$$D^V_{ij}(gh) = e^i\big(\Pi_V(g)\Pi_V(h)e_j\big) = \sum_k e^i(\Pi_V(g)e_k)\, e^k(\Pi_V(h)e_j).$$

So ΔD^V_{ij} is the function of the tensor $\sum_k D^V_{ik} \otimes D^V_{kj}$. This tensor is unique by the tensor injectivity lemma, Lemma 14.3. The same insertion for a general coefficient gives

$$\Delta c^V_{\lambda,v} = \sum_k c^V_{\lambda,e_k} \otimes c^V_{e^k,v}. \tag{14.2}$$

Hence Δ maps $\mathscr{R}(K)$ into $\mathscr{R}(K)^{\otimes 2}$. Let $\Pi_{V^*}(g)\lambda = \lambda \circ \Pi_V(g)^{-1}$ be the dual representation. Let $\hat{e}_j \in V^{**}$ be evaluation at e_j. Then

$$SD^V_{ij}(g) = e^i\big(\Pi_V(g)^{-1}e_j\big) = \hat{e}_j\big(\Pi_{V^*}(g)e^i\big) = c^{V^*}_{\hat{e}_j,e^i}(g).$$

So S preserves $\mathscr{R}(K)$. Also $\varepsilon D^V_{ij} = e^i(e_j) = \delta_{ij}$, and $f^* \in \mathscr{R}(K)$ by the conjugation rule in Proposition 14.2.

Algebra maps. As functions, $\Delta(ff')(g,h) = f(gh)f'(gh) = (\Delta f\,\Delta f')(g,h)$. The identification is an injective algebra homomorphism, so $\Delta(ff') = \Delta f\,\Delta f'$. Also $\Delta 1 = 1 \otimes 1$. Pointwise products show that ε and S are unital algebra homomorphisms.

Identities. Write $\Delta f = \sum_i f_i \otimes f_i'$, so $\sum_i f_i(g) f_i'(h) = f(gh)$ for $g, h \in K$. As functions of $(g,h,k') \in K^3$,

$$(\Delta \otimes \mathrm{id})\Delta f\,(g,h,k') = \textstyle\sum_i f_i(gh) f_i'(k') = f\big((gh)k'\big),$$
$$(\mathrm{id} \otimes \Delta)\Delta f\,(g,h,k') = \textstyle\sum_i f_i(g) f_i'(hk') = f\big(g(hk')\big).$$

These agree by associativity, so the tensors agree by the tensor injectivity lemma, Lemma 14.3. For the counit, $(\varepsilon \otimes \mathrm{id})\Delta f = \sum_i f_i(e) f_i'$ is the function $h \mapsto f(eh) = f(h)$. For the antipode,

$$m(S \otimes \mathrm{id})\Delta f\,(g) = \sum_i f_i(g^{-1}) f_i'(g) = f(g^{-1}g) = f(e).$$

The mirror identities use $he = h$ and $gg^{-1} = e$. Each group axiom is used once.

$*$-structure. Conjugation is antilinear and involutive, and $\mathscr{R}(K)$ is commutative. The function $\Delta(f^*)(g,h) = \overline{f(gh)}$ is that of $(\Delta f)^*$. Also $\varepsilon(f^*) = \overline{f(e)}$ and $S(f^*)(g) = \overline{f(g^{-1})}$. Finally, if $D^V(g)$ is unitary then $(D^V(g)^{-1})_{ij} = \overline{D^V_{ji}(g)}$. ∎

On matrix coefficients the axioms become matrix identities. We write $D = D^V$ and evaluate each side at group elements:

$$\begin{aligned} \text{coassociativity:} &\quad \textstyle\sum_{k,l} D_{ik}(g) D_{kl}(h) D_{lj}(k') = \big(D(g)D(h)D(k')\big)_{ij}, \\ \text{counit:} &\quad \textstyle\sum_k \delta_{ik} D_{kj}(g) = \big(D(e)D(g)\big)_{ij} = D_{ij}(g), \\ \text{antipode:} &\quad \textstyle\sum_k (SD_{ik})(g)\, D_{kj}(g) = \big(D(g)^{-1}D(g)\big)_{ij} = \delta_{ij}. \end{aligned}$$

So the three identities encode $D(ghk') = D(g)D(h)D(k')$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. In an orthonormal basis the antipode identity reads

$$\sum_k (D_{ki})^* D_{kj} = \delta_{ij}. \tag{14.3}$$

In other words, the columns of $D(g)$ are orthonormal. Hence S and $*$ together encode the unitarity of $D(g)$. Exercise 14.7 checks these identities on the spin 1 representation of SU(2).

14.3 Recovering K

Definition 14.6. A *character* of $\mathscr{R}(K)$ is a unital algebra homomorphism $\psi\colon \mathscr{R}(K) \to \mathbb{C}$. It is a *$*$-character* if $\psi(f^*) = \overline{\psi(f)}$ for all f.

This algebraic use of the word differs from the trace character of a representation and from the characters $T \to \mathrm{U}(1)$ of a torus T (Lecture 7).

For $g \in K$ the evaluation $\mathrm{ev}_g(f) = f(g)$ is a $*$-character.

Proposition 14.7. *The characters form a group $\mathcal{X}(K)$ with*

$$\psi_1\psi_2 = (\psi_1 \otimes \psi_2) \circ \Delta, \qquad 1 = \varepsilon, \qquad \psi^{-1} = \psi \circ S.$$

The $*$-characters form a subgroup $\mathcal{X}_*(K)$, and $g \mapsto \mathrm{ev}_g$ is an injective homomorphism $K \to \mathcal{X}_*(K)$.

Proof. Since Δ, $\psi_1 \otimes \psi_2$ and S are algebra homomorphisms, $\psi_1\psi_2$ and $\psi \circ S$ are characters. Applying $\psi_1 \otimes \psi_2 \otimes \psi_3$ to coassociativity gives $(\psi_1\psi_2)\psi_3 = \psi_1(\psi_2\psi_3)$. The counit identity gives $\varepsilon\psi = \psi\varepsilon = \psi$. Multiplicativity means $\psi \circ m = m_{\mathbb{C}} \circ (\psi \otimes \psi)$, so

$$(\psi \circ S)\psi = \psi \circ m \circ (S \otimes \mathrm{id}) \circ \Delta = \psi \circ \eta \circ \varepsilon = \varepsilon.$$

The mirror identity gives $\psi(\psi \circ S) = \varepsilon$. The relations $\Delta(f^*) = (\Delta f)^*$ and $S(f^*) = (Sf)^*$ show that products and inverses of $*$-characters are $*$-characters. Finally $(\mathrm{ev}_g \otimes \mathrm{ev}_h)\Delta f = f(gh)$. Injectivity holds because $\mathscr{R}(K)$ separates points. ∎

In other words, Δ, ε and S give the multiplication, the identity and the inverses in $\mathcal{X}(K)$. To recover K we must know that every $*$-character is an evaluation. We now describe the same question in categorical terms.

Definition 14.8. The *fibre functor* $\omega\colon \mathrm{Rep}_{\mathrm{fd}}(K) \to \mathrm{Vect}_{\mathbb{C}}$ sends each object to its underlying space and each morphism to itself. A *tensor automorphism* of ω is a family $\theta = (\theta_V)$ with $\theta_V \in \mathrm{GL}(V)$ for each object V, such that

$$\theta_W T = T\theta_V \quad (T\colon V \to W \text{ a morphism}), \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

A tensor automorphism θ is *unitary* if θ_V is unitary for every K-invariant inner product on every V.

Tensor automorphisms form a group $\mathrm{Aut}_{\otimes}(\omega)$ under $(\theta\theta')_V = \theta_V\theta'_V$, with identity (id_V) and inverse (θ_V^{-1}). Indeed, products of natural families are natural, and naturality $\theta_W T = T\theta_V$ gives $T\theta_V^{-1} = \theta_W^{-1}T$. Products and inverses also preserve the tensor and unit conditions. Products and inverses of unitary operators are unitary, so the unitary tensor automorphisms form a subgroup $\mathrm{Aut}^u_{\otimes}(\omega)$.

The data of the unitary fibre functor are the following.

(i) The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ with its tensor product and unit object $\mathbb{C}$.
(ii) The functor ω, which remembers the underlying vector spaces and linear maps.
(iii) On each $\omega(V)$, the set of K-invariant inner products. On the algebra side these inner products correspond to the $*$-structure.

Each $g \in K$ gives the family with $\theta_V = \Pi_V(g)$. This family is natural because morphisms are equivariant. It is monoidal by the definition of $V \otimes W$. It is unitary because the inner products are K-invariant.

Tensor automorphisms and characters are related by the map $\theta \mapsto \psi_\theta$ given by

$$\psi_\theta\big(c^V_{\lambda,v}\big) = \lambda(\theta_V v). \tag{14.4}$$

Exercises 14.4 and 14.5 show that $\theta \mapsto \psi_\theta$ is a group isomorphism $\mathrm{Aut}_\otimes(\omega) \to \mathcal{X}(K)$. They also show that θ is unitary exactly when ψ_θ is a $*$-character. In any basis of V, the matrix of θ_V is $\left(\psi_\theta(D^V_{ij})\right)_{ij}$.

So a $*$-character ψ gives one operator θ_V on each V, determined by $\psi = \psi_\theta$. The operator θ_V is unitary for every K-invariant inner product on V. Hence its matrix in an orthonormal basis is unitary, while its matrix in a general basis need not be. The hard step of reconstruction is to find a single $g \in K$ whose operators $\Pi_V(g)$ equal the operators θ_V for all V at once. The SU(2) model, Example 14.12, finds g directly. In general we cite this step.

Theorem 14.9 (Tannaka–Krein duality; cited from BD III.7)**.** *Let K be a compact Lie group. Every $*$-character of $\mathscr{R}(K)$ is ev_g for a unique $g \in K$. Equivalently, $g \mapsto (\Pi_V(g))_V$ is a group isomorphism $K \to \mathrm{Aut}^u_\otimes(\omega)$.*

Theorem 14.10 (Complexification; cited from BD III.8)**.** *Let K be a compact Lie group and put $K_\mathbb{C} = \mathcal{X}(K)$, with the topology of pointwise convergence on $\mathscr{R}(K)$. Then $K_\mathbb{C}$ carries a complex manifold structure that makes it a complex Lie group with Lie algebra $\mathfrak{k}_\mathbb{C} = \mathfrak{k} \otimes_\mathbb{R} \mathbb{C}$. The map $g \mapsto \mathrm{ev}_g$ identifies K with a maximal compact subgroup of $K_\mathbb{C}$. Every object of $\mathrm{Rep}_{\mathrm{fd}}(K)$ extends uniquely to a holomorphic representation of $K_\mathbb{C}$. In algebraic language, $K_\mathbb{C}$ is a complex algebraic group with reductive identity component and coordinate ring $\mathscr{R}(K)$. For connected K, it is a connected complex reductive group.*

Thus plain complex tensor automorphisms recover $\mathrm{Aut}_\otimes(\omega) \cong K_\mathbb{C}$, not K. The unitary condition, equivalently the $*$-structure, cuts $K_\mathbb{C}$ down to K.

14.4 Two models

Example 14.11 (U(1))**.** The irreducible representations of U(1) are the lines $\mathbb{C}_n$, $n \in \mathbb{Z}$, on which z acts by z^n (characters of tori, Lecture 7). The functions z^n are orthonormal for Haar measure (Lecture 5), hence independent. So

$$\mathscr{R}(\mathrm{U}(1)) = \mathbb{C}[z, z^{-1}], \qquad \Delta z = z \otimes z, \quad \varepsilon z = 1, \quad Sz = z^{-1}, \quad z^* = z^{-1}.$$

A character ψ is determined by $w = \psi(z)$. Since $\psi(z)\psi(z^{-1}) = 1$, we have $w \neq 0$. Every $w \in \mathbb{C}^\times$ occurs, via $z^n \mapsto w^n$. The $*$-condition reads $\bar{w} = \psi(z^*) = w^{-1}$, that is $|w| = 1$. Thus $\mathcal{X} = \mathbb{C}^\times$ and $\mathcal{X}_* = \mathrm{U}(1)$.

The categorical computation agrees. Let $\theta \in \mathrm{Aut}_\otimes(\omega)$, and let w_n be the scalar $\theta_{\mathbb{C}_n}$. Every V is the sum of its weight spaces

$$V(n) = \{v \in V : \Pi_V(z)v = z^n v \text{ for all } z \in \mathrm{U}(1)\}.$$

For $v \in V(n)$, the map $T\colon \mathbb{C}_n \to V$, $1 \mapsto v$, is a morphism, so

$$\theta_V v = \theta_V T 1 = T\theta_{\mathbb{C}_n} 1 = w_n v.$$

The isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \to \mathbb{C}_{m+n}$, $1 \otimes 1 \mapsto 1$, gives $w_{m+n} = w_m w_n$, and $w_0 = 1$. Hence θ_V acts by w^n on $V(n)$, with $w = w_1 \in \mathbb{C}^\times$. Weight spaces are orthogonal for every invariant inner product, so θ is unitary exactly when $|w| = 1$. Conversely, each $w \in \mathbb{C}^\times$ occurs, through the character $z^n \mapsto w^n$ and Exercise 14.5. So $\mathrm{Aut}_\otimes(\omega) \cong \mathbb{C}^\times = \mathrm{U}(1)_\mathbb{C}$ and $\mathrm{Aut}^u_\otimes(\omega) \cong \mathrm{U}(1)$.

Example 14.12 (SU(2))**.** Define functions a, b, c, d on $\mathrm{SU}(2)$ by

$$g = \begin{pmatrix} a(g) & b(g) \\ c(g) & d(g) \end{pmatrix} = \begin{pmatrix} \alpha & -\bar\beta \\ \beta & \bar\alpha \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

They are the coefficients D_{ij} of $\mathbb{C}^2$ in the standard basis, and

$$ad - bc = 1, \qquad a^* = d, \qquad b^* = -c.$$

Since $\det g = 1$, the inverse matrix is

$$g^{-1} = \begin{pmatrix} d(g) & -b(g) \\ -c(g) & a(g) \end{pmatrix}.$$

So the matrix formulas (14.1) give

$$\begin{aligned} \Delta a &= a \otimes a + b \otimes c, & \Delta b &= a \otimes b + b \otimes d, & Sa &= d, & Sb &= -b, \\ \Delta c &= c \otimes a + d \otimes c, & \Delta d &= c \otimes b + d \otimes d, & Sc &= -c, & Sd &= a. \end{aligned}$$

Also ε sends a, d to 1 and b, c to 0. One checks $SD_{ij} = D^*_{ji}$; for example $Sb = -b = c^*$.

The functions a, b, c, d generate $\mathscr{R}(\mathrm{SU}(2))$ as an algebra. Indeed every irreducible representation is $V_m \cong \mathrm{Sym}^m(\mathbb{C}^2)$ (Lecture 8). In it g sends $e_1^p e_2^{m-p}$ to $(ae_1 + ce_2)^p (be_1 + de_2)^{m-p}$. So each $D_{ij}^{V_m}$ is a polynomial of degree m in a, b, c, d. By the Peter–Weyl theorem these coefficients span $\mathscr{R}(\mathrm{SU}(2))$, which proves the claim.

Let ψ be a $*$-character. Collect its values on the generators in the matrix

$$A = \begin{pmatrix} \psi(a) & \psi(b) \\ \psi(c) & \psi(d) \end{pmatrix}.$$

The relations give

$$\psi(d) = \overline{\psi(a)}, \qquad \psi(c) = -\overline{\psi(b)}, \qquad |\psi(a)|^2 + |\psi(b)|^2 = \psi(ad - bc) = 1.$$

So $A \in \mathrm{SU}(2)$. The algebra maps ψ and ev_A agree on generators, so $\psi = \mathrm{ev}_A$. This verifies Tannaka–Krein duality, Theorem 14.9, for $\mathrm{SU}(2)$ directly.

Without the $*$-condition we only learn $\det A = 1$. Conversely every $A \in \mathrm{SL}(2, \mathbb{C})$ gives a character $\psi_A \colon p(a, b, c, d) \mapsto p(A)$ (Exercise 14.3). So $\mathcal{X}(\mathrm{SU}(2)) = \mathrm{SL}(2, \mathbb{C}) = \mathrm{SU}(2)_{\mathbb{C}}$. For $A = \mathrm{diag}(2, \frac{1}{2})$ we get $\psi_A(a) = 2$ but $\psi_A(a^*) = \psi_A(d) = \frac{1}{2} \neq \bar 2$. The corresponding tensor automorphism has $\theta_{\mathbb{C}^2} = A$, which is not unitary.

14.5 The representation category

We use the notions of a rigid and of a symmetric monoidal category from the prerequisite course. Rigidity requires duals, with evaluation and coevaluation maps that satisfy the zigzag identities. A symmetry is a natural isomorphism $\tau_{V,W} \colon V \otimes W \to W \otimes V$ with $\tau_{W,V}\tau_{V,W} = \mathrm{id}$, compatible with associativity.

Definition 14.13. A *fusion category* is a $\mathbb{C}$-linear semisimple rigid monoidal category with finite-dimensional morphism spaces, simple unit object, and finitely many isomorphism classes of simple objects.

The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is semisimple (Lecture 5) with simple unit $\mathbb{C}$ and finite-dimensional morphism spaces. It is rigid, with the dual representation and

$$\mathrm{ev}_V(\lambda \otimes v) = \lambda(v), \qquad \mathrm{coev}_V(1) = \sum_i e_i \otimes e^i.$$

It is symmetric with $\tau(v \otimes w) = w \otimes v$. Exercise 14.6 checks these claims.

Proposition 14.14. *If K is infinite, then $\mathrm{Irr}(K)$ is infinite and $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category. In particular, both conclusions hold when K is connected and nontrivial.*

Proof. We first show that $\mathscr{R}(K)$ is infinite-dimensional. Let $x_1, \dots, x_n \in K$ be distinct. Since $\mathscr{R}(K)$ separates points, for $i \neq j$ we can choose $f_{ij} \in \mathscr{R}(K)$ with $f_{ij}(x_i) \neq f_{ij}(x_j)$. Then

$$h_i = \prod_{j \neq i} \frac{f_{ij} - f_{ij}(x_j)}{f_{ij}(x_i) - f_{ij}(x_j)} \in \mathscr{R}(K).$$

At x_i every factor equals 1. At x_k with $k \neq i$ the factor $j = k$ vanishes. So $h_i(x_k) = \delta_{ik}$. A relation $\sum_i t_i h_i = 0$ evaluated at x_k gives $t_k = 0$. So $h_1, \dots, h_n$ are independent and $\dim \mathscr{R}(K) \geq n$ for every n. If $\mathrm{Irr}(K)$ were finite, Peter–Weyl would give

$$\dim \mathscr{R}(K) = \sum_{[V] \in \mathrm{Irr}(K)} (\dim V)^2 < \infty.$$

So $\mathrm{Irr}(K)$ is infinite. The simple objects of $\mathrm{Rep}_{\mathrm{fd}}(K)$ are the irreducible representations, and a fusion category has only finitely many isomorphism classes of simple objects. So $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category.

Finally, a finite Hausdorff space is discrete. Hence a connected Hausdorff space with at least two points is infinite, and the first part applies to connected nontrivial K. ∎

For a finite group the representation category is a fusion category (Exercise 14.2).

The algebra $\mathscr{R}(K)$ is commutative. Its coproduct is cocommutative only for abelian K. Indeed cocommutativity means

$$f(gh) = f(hg) \quad \text{for all } f \in \mathscr{R}(K),\ g, h \in K.$$

Since $\mathscr{R}(K)$ separates points, this forces $gh = hg$ for all $g, h \in K$.

A later course on quantum groups deforms this classical structure into noncommutative Hopf $*$-algebras. Lecture 28 returns to this comparison. Nothing in this lecture depends on that course.

Self-checks

1. In $\mathscr{R}(\mathrm{U}(1))$ let $f(z) = \mathrm{Re}\, z$. Compute Δf, εf and Sf.
   Answer. $f = \frac{1}{2}(z + z^{-1})$, so $\Delta f = \frac{1}{2}(z \otimes z + z^{-1} \otimes z^{-1})$, $\varepsilon f = 1$ and $Sf = f$. Note $\Delta f \neq f \otimes f$.
2. On $\mathrm{SU}(2)$ write $g \mapsto |g_{11}|^2$ in terms of a, b, c, d. Evaluate ψ_A on it for $A = \mathrm{diag}(2, \frac{1}{2})$.
   Answer. $|\alpha|^2 = aa^* = ad$, and $\psi_A(ad) = 2 \cdot \frac{1}{2} = 1$, while $|A_{11}|^2 = 4$. A non-$*$ character does not evaluate absolute values.
3. For $\mathrm{U}(1)$ let $\theta \in \mathrm{Aut}_{\otimes}(\omega)$ with $\theta_{\mathbb{C}_1} = 3$. Find $\theta_{\mathbb{C}_{-2}}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}}$. Is θ unitary?
   Answer. $\theta_{\mathbb{C}_{-2}} = \frac{1}{9}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}} = \mathrm{diag}(3, \frac{1}{3})$. It is not unitary, since $|3| \neq 1$.

Summary

Matrix coefficients of all finite-dimensional continuous representations of the compact Lie group K form the commutative $*$-algebra $\mathscr{R}(K)$. Tensor products give its products and conjugate representations give its involution. The maps $\Delta f(g,h) = f(gh)$, $\varepsilon f = f(e)$ and $Sf(g) = f(g^{-1})$ make it a Hopf $*$-algebra. On coefficients they are $D(gh) = D(g)D(h)$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. The $*$-characters, or unitary tensor automorphisms of ω, recover K. All characters, or all tensor automorphisms, give $K_{\mathbb{C}}$. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is rigid and symmetric. When K is infinite, it has infinitely many simple objects and is not a fusion category.

Reading

BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

BD III.1 for representative functions and their algebra structure. BD III.7 for Tannaka–Krein duality. BD III.8 for the complexification of compact Lie groups. The tensor-category terms are from the prerequisite course.

Part II

Noncompact representations, geometry and physics

Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. Noncompact groups behave differently. The representations studied in Part II are infinite-dimensional, and the derivative may be defined only on a proper subspace. The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g\rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We use one fact about $L^2(\mathbb{R})$. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\,dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and inner products of these vectors are given by the same integral. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Linear interpolation across the jumps of f gives functions $g_n \in C_c(\mathbb{R})$ with $\int_{\mathbb{R}} |f - g_n|^2\,dx \to 0$. The sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. This vector does not depend on the approximating sequence, and it depends linearly on f. For two such functions f and $\tilde f$ we have $\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx$ and $\langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx$. The preliminary note in the solution of Exercise 15.1 proves these statements. It constructs the functions g_n explicitly and uses the Cauchy–Schwarz inequality for Riemann integrals.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot 1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f \cdot 1_{[-m,m]} - f \cdot 1_{[-n,n]} \right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m > n \to \infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n \to \infty$ in the formulas for the truncations gives

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx, \qquad \langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx.$$

The second improper integral converges absolutely, since $\int_{-n}^{n} |\bar f \tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t) f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t) f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1) f$ is the derivative of $t \mapsto \lambda(t) f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t) f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.15, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.4. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.5. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.

(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.

(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.

(d) For $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v \, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v \, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v \, dg.$$

Now h appears only inside the smooth scalar function $(h,g)\mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h\mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n\ge 0$, $\int_G \varphi_n\,dg=1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v-v\|\le \sup_{g\in \operatorname{supp}\varphi_n}\|\Pi(g)v-v\|\longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G=\mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.6. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A)\to\mathcal{H}$ defined on a subspace $D(A)\subset\mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B\subset A$ if $D(B)\subset D(A)$ and $Bv=Av$ for $v\in D(B)$.

Definition 15.7. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w\in\mathcal{H}$ for which some $u\in\mathcal{H}$ satisfies

$$\langle w, Av\rangle=\langle u,v\rangle\qquad\text{for all } v\in D(A),$$

and then $A^*w=u$. The operator A is *skew-symmetric* if $\langle Av,w\rangle=-\langle v,Aw\rangle$ for all $v,w\in D(A)$. It is *skew-adjoint* if $A^*=-A$, including $D(A^*)=D(A)$, and *self-adjoint* if $A^*=A$, including $D(A^*)=D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A\subset A^*$.

Suppose $B\subset A$ and B is densely defined. Then $D(A)\supset D(B)$ is dense too, so both adjoints exist. In this case $A^*\subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v\in D(B)$.

Lemma 15.8. *Let B be densely defined and skew-symmetric. If $v_n\in D(B)$, $v_n\to 0$ and $Bv_n\to u$, then $u=0$.*

Proof. For every $w\in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w,u\rangle=\lim_n\langle w,Bv_n\rangle=-\lim_n\langle Bw,v_n\rangle=0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u=0$. ∎

Definition 15.9. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar B$ is the set of $v\in\mathcal{H}$ for which there are $v_n\in D(B)$ with $v_n\to v$ and (Bv_n) convergent. Then $\bar Bv=\lim_n Bv_n$.

The value $\bar Bv$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.8.

Definition 15.10. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t \to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.11 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.

(c) Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.

We cite most of this theorem and prove only the invariance statement in (a). For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\mathrm{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\mathrm{Hall}} = -iA$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in (b).

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$ and $H_{\mathrm{Hall}} = -H$.

Corollary 15.12. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.5, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.11(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.11(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, we have $B_1 = A$. With $\hbar = 1$, the operator $A_1 = iA$ is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R}\to\mathbb{C}$ such that

$$\sup_{x\in\mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j,k\ge 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.13. *Let $f\in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)|\le C(1+y^2)^{-1}$ for all y. Then $f\in D(A)$ and $Af=-f'$.*

Proof. Fix $0<|h|\le 1$. Taylor's formula with integral remainder, for $s\mapsto f(x-sh)$ on $[0,1]$, gives

$$f(x-h)=f(x)-hf'(x)+h^2\int_0^1 (1-s)f''(x-sh)\,ds.$$

So the error function is

$$q_h(x)=\frac{f(x-h)-f(x)}{h}+f'(x)=h\int_0^1 (1-s)f''(x-sh)\,ds,$$

and $|q_h(x)|\le \frac{|h|}{2}\sup_{|y-x|\le 1}|f''(y)|$. If $|y-x|\le 1$, then $x^2\le (|y|+1)^2\le 2y^2+2$, so $1+x^2\le 3(1+y^2)$. Therefore

$$|q_h(x)|\le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2\le \frac{9C^2h^2}{4}\int_{\mathbb{R}}\frac{dx}{(1+x^2)^2}=\frac{9\pi C^2h^2}{8}.$$

Thus $\left\|\frac{\lambda(h)f-f}{h}-(-f')\right\|=\|q_h\|\to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)|\le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.14. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f\in\mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f=-f'$. The closure of $B=-d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y=x-t$. Expanding $x^k=(y+t)^k$ bounds $|x^k f^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^i f^{(j)}(y)|$.

Smoothness. Each $f\in\mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.13. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''|\le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R})\subset L^2(\mathbb{R})$. By the lemma, $t\mapsto\lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f-\lambda(t_0)f}{h}=\lambda(t_0)\,\frac{\lambda(h)f-f}{h}\longrightarrow \lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t\mapsto\lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f'\in\mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t\mapsto\lambda(t)\big((-1)^k f^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \geq 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R,R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \leq \sup_{|t|\leq 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.11(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise.

Example 15.15. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2-2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \leq 6(1+x^2)$, we get $|w''(x)| \leq 6(1+x^2)^{-2} \leq 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.13, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.14. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.5(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \leq 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \leq t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.12, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

15.6 Self-checks

1. Let $\dim \mathcal{H} < \infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x) = xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f \in \mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.14, gives $Af = -f' = (x^2 - 1)e^{-x^2/2}$. Then $\langle f, Af\rangle = \int_{\mathbb{R}} x(x^2 - 1)e^{-x^2}\,dx = 0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f, Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0, \infty)$. Then $D \subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.11(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]} \neq 0$. It is not invariant either. Given $t < 0$, choose a nonzero $f \in D$ supported in $(0, |t|)$. Then $\lambda(t)f$ is supported in $(t, 0)$, so $\lambda(t)f \notin D$.

15.7 Summary

Unitary representations are required to be strongly continuous, not norm continuous. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

15.8 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
- Etingof, MIT 18.757 lecture notes, Lectures 1–7: an alternative introduction to continuous representations of Lie groups.

Lecture 16

Noncompact Haar Measure and the Modular Function

Objectives. *Lecture 5 integrated over compact groups. This lecture drops compactness. It asks which translation-invariant integrals a Lie group carries and how left and right integration differ. Every Lie group has a left-invariant and a right-invariant density. With equal normalisations at the identity, the left density divided by the right density is the modular function. It controls right translation, inversion and the right regular representation.*

By the end of this lecture you will be able to:

- Construct the left and right Haar densities of a Lie group and prove that each is unique up to a positive factor.
- Prove the right-translation and inversion identities for $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.
- Compute both Haar densities and Δ for the affine group $ax + b$.
- Construct the left and right regular representations as strongly continuous unitary representations on $L^2(G)$.

Throughout, G is a Lie group of dimension n with Lie algebra $\mathfrak{g} = T_eG$. We write left translation, right translation, conjugation and inversion as

$$L_g(x) = gx, \qquad R_g(x) = xg, \qquad C_g = L_g \circ R_{g^{-1}}, \qquad \iota(x) = x^{-1}.$$

Then $R_g \circ R_h = R_{hg}$ and $d(C_g)_e = \mathrm{Ad}(g)$.

16.1 Densities and integration

Under a linear map T of $\mathbb{R}^n$, Lebesgue measure scales by $|\det T|$. A density imposes this rule on each tangent space.

Definition 16.1. Let V be a real vector space of dimension n. A *positive density* on V is a nonzero function $\delta\colon V^n \to [0,\infty)$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \qquad (T \in \mathrm{End}\,V).$$

Let $e_1,\dots,e_n$ be a basis and $v_j = Te_j$. Then $\delta(v_1,\dots,v_n) = |\det T|\,\delta(e_1,\dots,e_n)$. So δ is fixed by the number $\delta(e_1,\dots,e_n)$, which is positive since $\delta \neq 0$. Two positive densities on V therefore differ by a positive factor.

Definition 16.2. A *smooth positive density* μ on G assigns a positive density μ_x on T_xG to each $x \in G$ such that, in every chart with coordinates y, the coefficient $m(y) = \mu_x(\partial_1, \dots, \partial_n)$ is smooth.

For a diffeomorphism F of G and a smooth function $\varphi > 0$ put

$$(F^*\mu)_x(v_1, \dots, v_n) = \mu_{F(x)}(dF_x v_1, \dots, dF_x v_n), \qquad (\varphi\mu)_x = \varphi(x)\,\mu_x.$$

The chain rule gives $(F \circ H)^* = H^*F^*$. The definitions give $F^*(\varphi\mu) = (\varphi \circ F)\,F^*\mu$.

Let $C_c(G)$ be the space of continuous functions $f : G \to \mathbb{C}$ with compact support.

If $\operatorname{supp} f$ lies in one chart, set $\int_G f\mu = \int_{\mathbb{R}^n} f(y)\,m(y)\,dy$. A change of chart multiplies the coefficient m by the absolute Jacobian determinant. Hence, by the change of variables formula, the integral is well defined.

For general $f \in C_c(G)$ we use a partition of unity, adapted from Lecture 5. Here a *partition of unity* on $\operatorname{supp} f$ is a finite family of smooth functions $\psi_i \geq 0$ on G, each with compact support in one chart, such that $\sum_i \psi_i = 1$ on $\operatorname{supp} f$. On a compact manifold, Lecture 5 requires the sum to equal 1 everywhere. Since f vanishes outside $\operatorname{supp} f$, we have $f = \sum_i \psi_i f$. We add the chart integrals of the functions $\psi_i f$.

We cite this construction and the following three facts. Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII, carries out the same construction with differential forms.

(I1) $f \mapsto \int_G f\mu$ is linear, and $\int_G f\mu > 0$ if $f \geq 0$ and $f \neq 0$.
(I2) $\int_G (f \circ F)\,F^*\mu = \int_G f\mu$ for every diffeomorphism F of G.
(I3) $\int_G f\,(\varphi\mu) = \int_G (f\varphi)\,\mu$.

In a chart, $F^*\mu$ has coefficient $m(F(y))\,|\det DF(y)|$. So (I2) is the change of variables formula.

16.2 Left and right Haar densities

Theorem 16.3 (Haar densities)**.** *Fix a positive density μ_e on $\mathfrak{g}$.*

(a) There is exactly one smooth positive density μ_L with $(\mu_L)_e = \mu_e$ and $L_g^*\mu_L = \mu_L$ for all $g \in G$. Every left-invariant smooth positive density is a positive multiple of μ_L.

(b) There is exactly one smooth positive density μ_R with $(\mu_R)_e = \mu_e$ and $R_g^*\mu_R = \mu_R$ for all $g \in G$. Every right-invariant smooth positive density is a positive multiple of μ_R.

(c) Write $\int_G f(x)\,d_Lx$ for $\int_G f\mu_L$, and similarly d_Rx. For $f \in C_c(G)$ and $g \in G$,

$$\int_G f(gx)\,d_Lx = \int_G f(x)\,d_Lx, \qquad \int_G f(xg)\,d_Rx = \int_G f(x)\,d_Rx.$$

Proof. Uniqueness. Let ν be a left-invariant smooth positive density. Evaluate $L_{x^{-1}}^*\nu = \nu$ at x:

$$\nu_x(v_1, \dots, v_n) = \nu_e\big(d(L_{x^{-1}})_x v_1, \dots, d(L_{x^{-1}})_x v_n\big). \tag{16.1}$$

So ν is determined by ν_e. Two positive densities on $\mathfrak{g}$ differ by a positive factor. Thus $\nu_e = c\mu_e$ for some $c > 0$. Once μ_L is constructed below, the formula gives $\nu = c\mu_L$.

Existence. Define μ_L by (16.1) with $\nu_e = \mu_e$. Linear maps act entrywise on tuples $v = (v_1, \dots, v_n)$. Since $L_{(gx)^{-1}} \circ L_g = L_{x^{-1}}$, the chain rule gives

$$(L_g^* \mu_L)_x(v) = \mu_e\big(d(L_{(gx)^{-1}})_{gx}\, d(L_g)_x v\big) = \mu_e\big(d(L_{x^{-1}})_x v\big) = (\mu_L)_x(v).$$

Smoothness. This step uses the smooth group law. Choose a basis $X_1, \dots, X_n$ of $\mathfrak{g}$. The left-invariant fields $\tilde{X}_i(x) = d(L_x)_e X_i$ are smooth (Lecture 1). Since $L_{x^{-1}} \circ L_x = \mathrm{id}$, we have $d(L_{x^{-1}})_x \tilde{X}_i(x) = X_i$. Hence

$$(\mu_L)_x\big(\tilde{X}_1(x), \dots, \tilde{X}_n(x)\big) = \mu_e(X_1, \dots, X_n) =: c_0 > 0.$$

In a chart write $\tilde{X}_i = \sum_j P_{ji}(y)\, \partial_j$. The matrix $P(y)$ is smooth. It is invertible because $d(L_x)_e$ is an isomorphism. The density rule gives $c_0 = |\det P(y)|\, m(y)$. So $m = c_0\, |\det P|^{-1}$ is smooth.

Right translations. Use $\nu_x(v) = \nu_e(d(R_{x^{-1}})_x v)$. The key identity is $R_{(xg)^{-1}} \circ R_g = R_{x^{-1}}$, since $z\, g\, g^{-1} x^{-1} = z x^{-1}$. The fields $d(R_x)_e X_i$ replace $\tilde{X}_i$.

Integrals. Apply the change of variables (I2) with $F = L_g$ or $F = R_g$. ∎

On a compact group one normalises the total mass to 1, as in Lecture 5. For noncompact G there is no canonical normalisation. We use the same density μ_e for both μ_L and μ_R.

16.3 The modular function

Definition 16.4. The *modular function* of G is $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. The group G is *unimodular* if $\Delta \equiv 1$.

Since $\mathrm{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism (Lecture 2), Δ is a smooth homomorphism into the multiplicative group $(0, \infty)$.

Theorem 16.5 (Integration identities). *Let $f \in C_c(G)$ and $g \in G$.*

(i) $R_g^* \mu_L = \Delta(g)\, \mu_L$. *Hence* $\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x$.

(ii) $\mu_R = \Delta^{-1} \mu_L$, *that is,* $\int_G f(x)\, d_R x = \int_G f(x)\, \Delta(x)^{-1}\, d_L x$.

(iii) $\int_G f(x^{-1})\, d_L x = \int_G f(x)\, d_R x$ *and* $\int_G f(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G f(x)\, d_L x$.

Proof. (i) Left and right translations commute. Hence

$$L_h^* R_g^* \mu_L = (R_g \circ L_h)^* \mu_L = (L_h \circ R_g)^* \mu_L = R_g^* L_h^* \mu_L = R_g^* \mu_L.$$

So $R_g^* \mu_L$ is left-invariant. By the uniqueness in Theorem 16.3(a), it equals $c(g) \mu_L$ for some $c(g) > 0$. We find $c(g)$ at e. Since $L_{g^{-1}} \circ R_g = C_{g^{-1}}$,

$$(R_g^* \mu_L)_e(v) = \mu_e\big(d(L_{g^{-1}})_g\, d(R_g)_e v\big) = \mu_e\big(\mathrm{Ad}(g^{-1}) v\big) = |\det \mathrm{Ad}(g)|^{-1} \mu_e(v).$$

Thus $c(g) = \Delta(g)$. Now the change of variables (I2) with $F = R_g$ and the multiplication rule (I3) give

$$\int_G f \mu_L = \int_G (f \circ R_g)\, R_g^* \mu_L = \Delta(g) \int_G f(xg)\, d_L x.$$

(ii) Use $F^*(\varphi\mu) = (\varphi \circ F)F^*\mu$, the right-translation identity (i) and the homomorphism property of Δ:

$$\big(R_g^*(\Delta^{-1}\mu_L)\big)_x = \Delta(xg)^{-1}\Delta(g)\,(\mu_L)_x = \Delta(x)^{-1}(\mu_L)_x.$$

So $\Delta^{-1}\mu_L$ is right-invariant with value μ_e at e. By uniqueness in Theorem 16.3(b) it equals μ_R. Then the multiplication rule (I3) gives the integral form.

(iii) First, $d\iota_e = -\mathrm{id}$. Indeed, the multiplication map $\mathrm{mult}\colon G\times G \to G$ satisfies $\mathrm{mult}(x,e) = x$ and $\mathrm{mult}(e,x) = x$ for $x \in G$. So $d(\mathrm{mult})_{(e,e)}(v,w) = v + w$ for $v, w \in \mathfrak{g}$. Differentiating $x\,\iota(x) = e$ at e gives $v + d\iota_e v = 0$. Therefore

$$(\iota^*\mu_L)_e(v_1,\dots,v_n) = \mu_e(-v_1,\dots,-v_n) = |\det(-\mathrm{id})|\,\mu_e(v_1,\dots,v_n) = \mu_e(v_1,\dots,v_n).$$

Next, $(xg)^{-1} = g^{-1}x^{-1}$ says $\iota \circ R_g = L_{g^{-1}} \circ \iota$. Hence

$$R_g^*\iota^*\mu_L = (\iota \circ R_g)^*\mu_L = (L_{g^{-1}} \circ \iota)^*\mu_L = \iota^* L_{g^{-1}}^*\mu_L = \iota^*\mu_L.$$

By the uniqueness in Theorem 16.3(b), $\iota^*\mu_L = \mu_R$. Apply the change of variables (I2) with $F = \iota$ to $f \circ \iota$. This gives $\int_G f\,d_R x = \int_G f(x^{-1})\,d_L x$. For the second identity apply the first to $h = f\Delta \in C_c(G)$ and use the density relation (ii):

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_L x = \int_G h(x)\,d_R x = \int_G f(x)\Delta(x)\Delta(x)^{-1}\,d_L x. \qquad \blacksquare$$

Remark 16.6. The right-translation identity (i) is equivalent to $\int_G f(xg^{-1})\,d_L x = \Delta(g)\int_G f\,d_L x$. For Borel sets E, using the Radon measure of § 16.6, it reads $\mu_L(Eg) = \Delta(g)\mu_L(E)$. This is Folland's convention. Some texts call $|\det \mathrm{Ad}(g)|$ the modular function. When comparing sources, check which defining identity is used.

Corollary 16.7. *(a) Every compact Lie group is unimodular.*

(b) If G is connected, then $\Delta(\exp X) = e^{-\operatorname{tr}\operatorname{ad} X}$. So G is unimodular if and only if $\operatorname{tr}\operatorname{ad} X = 0$ for all $X \in \mathfrak{g}$. This holds when $\mathfrak{g}$ is abelian, nilpotent or semisimple.

Proof. (a) Since G is compact and Δ is continuous, the image $\Delta(G)$ is a compact subgroup of $(0,\infty)$. If $t \neq 1$ lies in it, then so does the unbounded set $\{t^k : k \in \mathbb{Z}\}$. So $\Delta(G) = \{1\}$.

(b) We have $\mathrm{Ad}(\exp X) = e^{\operatorname{ad} X}$ (Lecture 2) and $\det e^A = e^{\operatorname{tr} A}$. This gives the formula. If all traces vanish, then $\Delta = 1$ on $\exp\mathfrak{g}$. This set contains a neighbourhood U of e (Lecture 2). The subgroup H generated by U is open, since $hU \subset H$ for $h \in H$. It is closed, since its complement is a union of open cosets. By connectedness $H = G$, so $\Delta \equiv 1$. Conversely, $\Delta \equiv 1$ gives $e^{-t\operatorname{tr}\operatorname{ad} X} = 1$ for all t, so $\operatorname{tr}\operatorname{ad} X = 0$. For nilpotent $\mathfrak{g}$ each $\operatorname{ad} X$ is nilpotent. For semisimple $\mathfrak{g}$ we have $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$ and $\operatorname{tr}\operatorname{ad}[X,Y] = \operatorname{tr}[\operatorname{ad} X, \operatorname{ad} Y] = 0$. $\blacksquare$

16.4 The affine group

The *$ax + b$ group* is

$$G = \left\{ g(a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} :\ a > 0,\ b \in \mathbb{R} \right\}.$$

It acts on $\mathbb{R}$ by $x \mapsto ax+b$. The pair (a,b) is a global chart onto $(0,\infty)\times\mathbb{R}$. So integrals over G are ordinary double integrals. Matrix multiplication gives

$$g(a,b)\,g(a',b') = g(aa',\ ab'+b), \qquad g(a,b)^{-1} = g(a^{-1}, -a^{-1}b).$$

A tangent vector at g is a matrix $\begin{pmatrix}\alpha & \beta\\ 0 & 0\end{pmatrix}$, which is $\alpha\partial_a + \beta\partial_b$ in the chart. The Lie algebra has basis

$$X = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, \qquad Y = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \qquad [X,Y] = Y.$$

Translations are restrictions of linear maps on matrices. Hence $d(L_g)_e Z = gZ$ and $d(R_g)_e Z = Zg$. We fix the density μ_e on $\mathfrak{g}$ by $\mu_e(X,Y) = 1$.

Left density. For $g = g(a,b)$,

$$gX = \begin{pmatrix}a & 0\\ 0 & 0\end{pmatrix} = a\,\partial_a, \qquad gY = \begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix} = a\,\partial_b.$$

So the matrix P from the smoothness step in the proof of Theorem 16.3 is

$$P = \begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}.$$

Then $m_L = |\det P|^{-1} = a^{-2}$.

Right density. Similarly,

$$Xg = \begin{pmatrix}a & b\\ 0 & 0\end{pmatrix} = a\,\partial_a + b\,\partial_b, \qquad Yg = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} = \partial_b, \qquad P = \begin{pmatrix}a & 0\\ b & 1\end{pmatrix}.$$

Then $m_R = a^{-1}$. In summary,

$$d_L x = a^{-2}\,da\,db, \qquad d_R x = a^{-1}\,da\,db.$$

Modular function. With $g^{-1} = \begin{pmatrix}a^{-1} & -a^{-1}b\\ 0 & 1\end{pmatrix}$ we compute

$$gXg^{-1} = \begin{pmatrix}1 & -b\\ 0 & 0\end{pmatrix} = X - bY, \qquad gYg^{-1} = \begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix} = aY.$$

In the basis (X,Y),

$$\mathrm{Ad}(g) = \begin{pmatrix}1 & 0\\ -b & a\end{pmatrix}, \qquad \det \mathrm{Ad}(g) = a, \qquad \Delta(g(a,b)) = a^{-1}.$$

This matches the density relation in Theorem 16.5(ii): $\Delta^{-1} m_L = a \cdot a^{-2} = a^{-1} = m_R$. It also matches the trace formula in Corollary 16.7(b). Indeed $\operatorname{ad} X$ fixes Y and kills X, so $\operatorname{tr}\operatorname{ad} X = 1$. And $\exp tX = g(e^t, 0)$ has $\Delta = e^{-t}$. Hence the $ax+b$ group is not unimodular.

The connectedness hypothesis in Corollary 16.7(b) is needed. Let H be the subgroup of elements $g(a,b)$ with $a \in 2^{\mathbb{Z}}$ and $b \in \mathbb{R}$. It is closed, since $2^{\mathbb{Z}}$ is closed in $(0,\infty)$. Its Lie algebra $\mathbb{R}Y$ is abelian. Yet $\mathrm{Ad}(g)Y = gYg^{-1} = aY$ for $g = g(a,b) \in H$. So the modular function Δ_H of H satisfies $\Delta_H(g(a,b)) = a^{-1}$, which is not identically 1.

Direct check of the right-translation identity in Theorem 16.5(i). Let $x = g(a', b')$ and $g = g(a, b)$. Then $xg = g(a'a,\ a'b + b')$. Substitute $u = a'a$, $v = a'b + b'$. Then $du\,dv = a\,da'\,db'$ and $a'^{-2} = a^2 u^{-2}$. So

$$\int_G f(xg)\,d_L x = \iint f(a'a,\ a'b+b')\,\frac{da'\,db'}{a'^2} = \iint f(u,v)\,\frac{a^2}{u^2}\,\frac{du\,dv}{a} = a\int_G f\,d_L x.$$

This is $\Delta(g)^{-1}\int_G f\,d_L x$.

Direct check of the second inversion identity in Theorem 16.5(iii). Here $x^{-1} = g(1/a, -b/a)$ and $\Delta(x^{-1}) = a$. Substitute $u = 1/a$, $v = -b/a$. The Jacobian determinant of $(a,b) \mapsto (u,v)$ is

$$\det\begin{pmatrix} -a^{-2} & 0 \\ ba^{-2} & -a^{-1}\end{pmatrix} = a^{-3}, \qquad \text{so} \qquad da\,db = u^{-3}\,du\,dv.$$

Since $a \cdot a^{-2} = a^{-1} = u$,

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_L x = \iint f\!\left(\tfrac{1}{a}, -\tfrac{b}{a}\right)\frac{a\,da\,db}{a^2} = \iint f(u,v)\,u\,u^{-3}\,du\,dv = \int_G f\,d_L x.$$

16.5 Regular representations

Definition 16.8. The space $L^2(G)$ is the Hilbert completion of $C_c(G)$ for the inner product

$$\langle f, h\rangle = \int_G \overline{f(x)}\,h(x)\,d_L x.$$

By the positivity statement (I1), this inner product is positive definite. For $f \in C_c(G)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

Lemma 16.9. *Let $f \in C_c(G)$ and $\varepsilon > 0$. There is a neighbourhood $V = V^{-1}$ of e with $|f(yx) - f(x)| < \varepsilon$ and $|f(xy) - f(x)| < \varepsilon$ for all $x \in G$ and $y \in V$.*

Proof. We treat $f(yx)$. The other case is symmetric. If U_L works for $f(yx)$ and U_R for $f(xy)$, put $V' = U_L \cap U_R$ and $V = V' \cap V'^{-1}$.

Fix a compact neighbourhood W of e. Choose a decreasing neighbourhood basis $V_1 \supset V_2 \supset \cdots$ of e inside W. Suppose no V_k works. Then there are $y_k \in V_k$ and x_k with $|f(y_k x_k) - f(x_k)| \geq \varepsilon$. One of x_k and $y_k x_k$ lies in $K = \operatorname{supp} f$. So x_k lies in the compact set $K \cup W^{-1}K$. Since G is second countable and Hausdorff, compact subsets are sequentially compact. So a subsequence has $x_k \to x$. As $y_k \to e$, also $y_k x_k \to x$. Continuity gives $f(y_k x_k) - f(x_k) \to 0$, which is a contradiction. ∎

Proposition 16.10. *The maps λ and ρ extend to strongly continuous unitary representations of G on $L^2(G)$, and $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.*

Proof. Both are homomorphisms on $C_c(G)$. For example,

$$\rho(g)\rho(h)f(x) = \Delta(g)^{1/2}\Delta(h)^{1/2} f(xgh) = \rho(gh)f(x).$$

Both operators preserve norms. By the left invariance in Theorem 16.3(c) and the right-translation identity in Theorem 16.5(i), applied to $|f|^2$,

$$\|\lambda(g)f\|^2 = \int_G |f(g^{-1}x)|^2 \, d_L x = \|f\|^2, \qquad \|\rho(g)f\|^2 = \Delta(g) \int_G |f(xg)|^2 \, d_L x = \|f\|^2.$$

Each operator is a bijection of $C_c(G)$, with inverse $\lambda(g^{-1})$ or $\rho(g^{-1})$. An isometric bijection of a dense subspace extends to an isometry with closed, dense range, hence to a unitary operator. The homomorphism property passes to the extensions by continuity. The operators commute, by the computation

$$\lambda(g)\rho(h)f(x) = \Delta(h)^{1/2} f(g^{-1}xh) = \rho(h)\lambda(g)f(x).$$

Strong continuity. Fix $f \in C_c(G)$ with support K, and a compact neighbourhood $W = W^{-1}$ of e. For $g \in W$, the functions $\lambda(g)f - f$ and $f(\cdot\, g) - f$ vanish outside $WK \cup KW$. A finite sum of chart bump functions gives $\varphi \in C_c(G)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on $WK \cup KW$. Fix $\varepsilon > 0$. Take the neighbourhood supplied by the uniform-continuity statement of Lemma 16.9 and intersect it with W to obtain $V \subset W$. For $g \in V$,

$$\|\lambda(g)f - f\|^2 \leq \varepsilon^2 \int_G \varphi \, d_L x.$$

The function φ was fixed before ε, and $\varepsilon > 0$ is arbitrary. Hence $\lambda(g)f \to f$ as $g \to e$. For ρ use

$$\|\rho(g)f - f\| \leq \Delta(g)^{1/2} \, \|f(\cdot\, g) - f\| + |\Delta(g)^{1/2} - 1| \, \|f\|$$

and the continuity of Δ. The squared norm $\|f(\cdot\, g) - f\|^2$ obeys the same bound as $\|\lambda(g)f - f\|^2$, by the second estimate of Lemma 16.9. Hence $\rho(g)f \to f$ as $g \to e$.

For general $h \in L^2(G)$, choose $f \in C_c(G)$ close to h. Then

$$\|\lambda(g)h - h\| \leq 2\|h - f\| + \|\lambda(g)f - f\|.$$

This gives continuity at e. Continuity at g_0 follows from $\lambda(g)h - \lambda(g_0)h = \lambda(g_0)\big(\lambda(g_0^{-1}g)h - h\big)$. The same holds for ρ. ∎

Without the factor $\Delta(g)^{1/2}$, right translation multiplies squared norms by $\Delta(g)^{-1}$. On the $ax + b$ group the factor $\Delta(g)^{-1}$ equals a for $g = g(a, b)$. Exercise 16.5 shows that λ and ρ are unitarily equivalent.

Right translation of a rectangle in the $ax + b$ group

In the $ax + b$ group the modular function can be computed from weighted areas, without differentiating a matrix. For a region E in the (a, b)-chart we write

$$\mu_L(E) = \iint_E a^{-2} \, da \, db$$

and call it the *left Haar measure* of E. Now let

$$E = \{(a, b) : 1 \le a \le 2,\ 0 \le b \le 1\}.$$

Its left Haar measure is

$$\mu_L(E) = \int_1^2 \int_0^1 a^{-2}\, db\, da = \left[-a^{-1}\right]_1^2 = \frac{1}{2}.$$

For $h = (s, t)$, right translation sends (a, b) to $(as, b + at)$. Its Jacobian in (a, b) is s, while the new density coefficient is $(as)^{-2}$. Hence

$$\mu_L(Eh) = \int_E (as)^{-2} s\, da\, db = s^{-1}\mu_L(E) = \Delta(h)\mu_L(E).$$

The shear parameter t changes the shape of the region but not this scale factor. Left translation by h has Jacobian s^2 and density coefficient $(sa)^{-2}$, so it preserves $\mu_L(E)$.

For $f \in C_c(G)$, the right-translation identity in Theorem 16.5(i), applied to $|f|^2$, gives $\|f(\cdot\, h)\|_2^2 = \Delta(h)^{-1}\|f\|_2^2 = s\,\|f\|_2^2$.

The indicator $\mathbf{1}_E$ of E is not continuous, so it is not in $C_c(G)$. It lies in $L^2(G)$ through the Radon-measure interpretation of § 16.6, where $\mu_L(E)$ is the Radon measure of E. The uncorrected right translation of $\mathbf{1}_E$ is $\mathbf{1}_E(xh) = \mathbf{1}_{Eh^{-1}}(x)$. Therefore

$$\|\mathbf{1}_E(\,\cdot\, h)\|_2^2 = \mu_L(Eh^{-1}) = s\,\mu_L(E).$$

Multiplication by $\Delta(h)^{1/2} = s^{-1/2}$ restores the norm in both cases. The inverse in the translated set Eh^{-1} explains why the integral identity has $\Delta(h)^{-1}$, whereas the region identity has $\Delta(h)$.

16.6 Locally compact groups

A topological group is *locally compact* if e has a compact neighbourhood. We cite the following result (Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2). Let G be a locally compact Hausdorff group. A linear functional I on $C_c(G)$ is *positive* if $I(f) \ge 0$ whenever $f \ge 0$. There is a nonzero positive linear functional I on $C_c(G)$ with $I(f(g\,\cdot)) = I(f)$ for all $f \in C_c(G)$ and $g \in G$. It is unique up to a positive factor. There is a unique continuous homomorphism $\Delta\colon G \to (0, \infty)$ satisfying $I(f(\cdot\, g^{-1})) = \Delta(g)I(f)$.

A general locally compact group has no normalisation at e that compares left and right integrals. The cited result instead contains two identities. They replace the density relation in Theorem 16.5(ii) and the inversion identities in part (iii). First, the functional $f \mapsto I(f\Delta^{-1})$ is right-invariant. Second, I takes the same value on $f \in C_c(G)$ and on the function $x \mapsto f(x^{-1})\Delta(x^{-1})$.

The Riesz representation theorem realises I as a Radon measure. We cite Folland, *Real Analysis*, second edition, Theorem 7.2 in §7.1.

For a Lie group, uniqueness makes I a multiple of $f \mapsto \int_G f\, d_L x$. So the two modular functions agree. Since $C_c(G)$ is dense in L^2 of a Radon measure (Folland, *Real Analysis*, Proposition 7.9 in §7.2), our $L^2(G)$ is the usual one.

Self-checks

1. In the $ax+b$ group let $g = g(2,5)$. Suppose $\int_G f\, d_L x = 7$. Find $\Delta(g)$ and $\int_G f(xg)\, d_L x$.

   Answer. $\Delta(g) = 1/2$. By the right-translation identity in Theorem 16.5(i), the integral is $\Delta(g)^{-1} \cdot 7 = 14$.

2. Suppose some left-invariant smooth positive density on G is also right-invariant. Must G be unimodular?

   Answer. Yes. By the uniqueness in Theorem 16.3(a), the density is $c\mu_L$ for some $c > 0$, so $R_g^* \mu_L = \mu_L$ for all $g \in G$. By the right-translation identity in Theorem 16.5(i), this means $\Delta(g) = 1$ for all $g \in G$.

3. On the $ax+b$ group, by what factor does $f \mapsto f(\cdot\, g(3,1))$ multiply $\|f\|^2$ in $L^2(G)$?

   Answer. By the right-translation identity in Theorem 16.5(i), applied to $|f|^2$, the factor is $\Delta(g(3,1))^{-1} = 3$. Hence the map is not unitary without the factor $\Delta^{1/2}$.

Summary

Translating a fixed positive density on $\mathfrak{g}$ gives the left and right Haar densities. Each is unique up to a positive factor. Right translation scales μ_L by $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. When μ_L and μ_R have the same value at e, we have $\mu_R = \Delta^{-1}\mu_L$, and inversion exchanges μ_L and μ_R. For the $ax+b$ group, $d_L x = a^{-2}\, da\, db$, $d_R x = a^{-1}\, da\, db$ and $\Delta = a^{-1}$. The factor $\Delta^{1/2}$ makes right translation unitary on $L^2(G)$.

Reading

- Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII "Integration", sections "Differential Forms and Measure Zero" and "Haar Measure for Lie Groups". Compare its normalisation of the modular function with ours.
- Kirillov, *Elements of the Theory of Representations*, §9, from the course reference route. Check its modular-function convention.
- Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, sections "Haar Measure" and "The Modular Function", for general locally compact groups.

Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h) F(xh)\, dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1} \operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h') F(xh')\, dh' = \sigma(h_0)^{-1} P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

17.2 Quotient measures

As in Lecture 16, a *measure* on a locally compact space X here means a positive linear functional $\varphi \mapsto \int_X \varphi\, d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi \rangle = \int_X \bar{\varphi}\psi\, d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\, dx = \Delta_G(g)^{-1} \int_G F(x)\, dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\, dh. \tag{17.2}$$

It is well defined by left invariance of dh.

> **Lemma 17.2.** *If a measure μ on G/H satisfies $\int_G F\, dx = \int_{G/H} F^H\, d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1} \int_G F\, dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\, dh = \Delta_H(h_0)^{-1} F^H(xH).$$

Hence $\Delta_G(h_0)^{-1} \int_G F\, dx = \Delta_H(h_0)^{-1} \int_G F\, dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\, dx > 0$. Dividing both sides by $\int_G F\, dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax + b$ group and its scaling subgroup A, we will see $\Delta_G(a, 0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

Theorem 17.3 (Quotient measures; cited)**.** *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, "Measures on Homogeneous Spaces", cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. Repeating the computation in the modular compatibility lemma, Lemma 17.2, with $F\rho$ in place of F confirms that the transformation law (17.3) is compatible with our modular convention. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1g_2, p) = \lambda(g_1, g_2p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1g_2, p) = \frac{\rho(g_1g_2x)}{\rho(g_2x)} \cdot \frac{\rho(g_2x)}{\rho(x)} = \lambda(g_1, g_2p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g,\cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle \, d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f \, d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1} f(g^{-1}x)$. Also $q(\operatorname{supp} \Pi(g)f) = g \cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1}, p)^{1/2} \, \lambda(g_2^{-1}, g_1^{-1}p)^{1/2} \, f(g_2^{-1} g_1^{-1} x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1 g_2)^{-1}, p)^{1/2}$. Also $\lambda(e, p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g,\cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1}, p)\, n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\, \lambda(g^{-1}, gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\, d\mu = \int n_f(p)\, \lambda(g^{-1}, gp)\, \lambda(g, p)\, d\mu(p) = \int n_f \, d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G \sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation $\operatorname{Ind}_H^G \sigma$ is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B \cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1}, p)^{1/2} f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g, p) \mapsto \lambda(g^{-1}, p)$ is jointly continuous and equals one at $g = e$. Also $(g, p) \mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B \times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p \in S_i$ as $g \to e$. There are only finitely many S_i, so the same holds on S.

The measure μ is finite on compact sets, by its Radon-measure interpretation in Lecture 16. So if the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \le \varepsilon^2 \mu(S).$$

This proves norm continuity on $\mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\,\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\, \lambda(g, \cdot)\, d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim, and we do not use it.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a, b)(a', b') = (aa', b + ab'), \qquad (a, b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a, b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $x \mapsto ax + b$. By Lecture 16, $dx = a^{-2}\, da\, db$ is a left Haar measure and $\Delta_G(a, b) = 1/a$. Let $N = \{n_c = (1, c)\}$ be the translations and $A = \{(a, 0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b+ac)$, the coset $(a,b)N$ is $\{a\}\times\mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a,ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_L(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_L} = da.$$

For the Lebesgue choice, take $g = (a_0,b_0)$ and $x = (a,0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_L(g,a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\,da = a_0\int_0^\infty \varphi(u)\,du$. The cocycle identity reads $a_1 a_2 = a_1\cdot a_2$, because $g_1 g_2$ has first entry $a_1 a_2$.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a,b+ac) = e^{-i\xi c} f(a,b)$. Take $b = 0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f\circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c}e^{-i\xi b/a}$. So $f\mapsto\varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2\rho(a)\,da/a$.

Let $g = (a_0,b_0)$. Then $g^{-1}(a,0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x = (a,0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty),da/a), \tag{17.9}$$

$$(\Pi^L_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty),da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g = (a_0,b_0)$, $g_j = (a_j,b_j)$ and $\varphi,\psi\in C_c((0,\infty))$.*

(a) $\|\Pi^L_\xi(g)\psi\| = \|\psi\|$, while $\psi\mapsto e^{i\xi b_0/a}\psi(a/a_0)$ multiplies $\|\psi\|^2$ by a_0.

(b) $\Pi^L_\xi(g_1)\Pi^L_\xi(g_2) = \Pi^L_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^L_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^L_\xi(g_1)\Pi^L_\xi(g_2)\psi)(a) &= a_1^{-1/2} e^{i\xi b_1/a}\,(\Pi^L_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1a_2}\Big)\\ &= (a_1a_2)^{-1/2}e^{i\xi(b_1+a_1b_2)/a}\,\psi\Big(\frac{a}{a_1a_2}\Big) = (\Pi^L_\xi(g_1g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_L is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^L_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_L)^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a, 0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let* $\kappa > 0$. *The operator* $(U\varphi)(a) = \varphi(a/\kappa)$ *is unitary on* $L^2((0,\infty), da/a)$, *and* $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ *for all* $g \in G$.

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$\begin{aligned}(U\Pi_\xi(g)\varphi)(a) &= (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),\\ (\Pi_{\kappa\xi}(g)U\varphi)(a) &= e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).\end{aligned}$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. $\blacksquare$

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0, \infty) \cong G/N$, composed with the quotient map $G \to G/N$. Irreducibility is not discussed here.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a, b)\rho(a, b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}} \left(\int_0^\infty F(a, t)\,\frac{da}{a} \right) dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1, t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0} \right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\,dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.

2. Is $\rho(a,b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a,0)n_1) = \rho(a,a) = a + a^2 \neq a = \rho(a,0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

$$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, “Measures on Homogeneous Spaces”: quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements, Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.

Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. We then state the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient and use it to contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a,b,c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a,b,c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a,b,c)(a',b',c') = (a+a',\ b+b',\ c+c'+ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H)=\{(0,0,c)\,:\,c\in\mathbb{R}\}\cong\mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z,\qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ)=I+W+\tfrac12 W^2=\big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1}=(a+s,b,c)(-a,-b,-c+ab)=(s,0,-sb)=\exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X=X-bZ$. In the same way $\mathrm{Ad}(g)Y=Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g)=\begin{pmatrix}1&0&0\\0&1&0\\-b&a&1\end{pmatrix},\qquad \det D_{\mathrm{Ad}}(g)=1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle=\int_{\mathbb{R}}\overline{f(t)}\,h(t)\,dt$. We use the Lebesgue-integration inputs from Lecture 16: completeness of L^2, Fubini–Tonelli, dominated convergence and change of variables. In particular, dominated convergence justifies differentiation under the integrals below. We also use two standard facts. First, Lebesgue integrals are invariant under $t\mapsto t+a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a,b,c)f)(t) = e^{i\lambda(c+bt)} f(t+a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t+a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a,b,c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a,b,c) = (0,0,c)(0,b,0)(a,0,0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15.

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0,0,c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a,b,c)$ and $g' = (a',b',c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t+a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t+a+a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t+a+a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0,0,0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t+a) = e^{i\lambda b(t+a)} f(t+a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s) - f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R + 2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f - h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g\to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab'-a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

Remark 18.4. Let $A = \{(0,b,c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0,b,c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s\to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X,Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R,R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left|\frac{f(t+s)-f(t)}{s} - f'(t)\right| \le \tfrac{1}{2}|s|\,\|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left|\frac{e^{i\lambda st}-1}{s} - i\lambda t\right| |f(t)| \le \tfrac{1}{2}\lambda^2|s|R^2|f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s}-1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q,P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. We state without proof that a commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c}I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U,V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann). *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We cite the Stone–von Neumann theorem, Theorem 18.9, from Kirillov, *Lectures on the Orbit Method*, Chapter 2.

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

A standard proof, going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \left\langle f, \rho(\exp(aX+bY))h \right\rangle da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a,b,c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on bounded v-intervals by an integrable Gaussian bound, including the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2 - (t+a)^2/2 + i\lambda b t}\, dt.$$

Complete the square with $w = \frac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model. This computation verifies its normalisation. It is not a proof of the general theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Cauchy–Schwarz in t and translation invariance give

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus the Fubini–Tonelli theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4 - (t+a/2)^2} f(t + a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u - t)^2}{4} - \frac{(t + u)^2}{4} = -\frac{t^2 + u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t) \langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central

character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof. Here its rank-one integral kernel is visible.

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a,b,c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.

3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi\rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi\rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. The Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.

Lecture 19

Coadjoint orbits and the nilpotent orbit method

Objectives. *A representation acts on vectors. The coadjoint action of a Lie group instead acts on linear functionals on its Lie algebra. Each orbit of this action has a natural symplectic form. For connected simply connected nilpotent groups, these orbits label all irreducible strongly continuous unitary representations up to unitary equivalence. We prove the geometric construction and work through the Heisenberg example. The general classification is an explicit cited theorem.*

By the end of this lecture you will be able to:

- Compute a coadjoint action and its stabilisers.
- Construct the orbit symplectic form and check its signs.
- State the nilpotent orbit theorem with its global hypotheses.
- Choose a polarisation for a functional on a two-dimensional Heisenberg orbit and recover the Schrödinger representation by induction.

19.1 The dual action

Throughout, G is a finite-dimensional real Lie group with Lie algebra $\mathfrak{g}$. Its real dual is $\mathfrak{g}^* = \mathrm{Hom}_{\mathbb{R}}(\mathfrak{g}, \mathbb{R})$. We pair a functional ℓ and a vector X by $\ell(X)$. This dual is different from the complex dual of a representation space.

Definition 19.1. The *coadjoint action* of G on $\mathfrak{g}^*$ is

$$(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X).$$

The *coadjoint orbit* through ℓ is $O_\ell = \{g \cdot \ell : g \in G\}$. Its stabiliser is $G_\ell = \{g : g \cdot \ell = \ell\}$.

The inverse is needed for the action law. Indeed,

$$(g \cdot (h \cdot \ell))(X) = \ell(\mathrm{Ad}(h^{-1})\,\mathrm{Ad}(g^{-1})X) = \ell(\mathrm{Ad}((gh)^{-1})X) = ((gh) \cdot \ell)(X).$$

The stabiliser is closed by continuity. It is therefore an embedded Lie subgroup by the closed-subgroup theorem of Lecture 3. Differentiating the action gives

$$(\mathrm{ad}^*(X)\ell)(Y) = -\ell([X,Y]), \qquad \mathfrak{g}_\ell = \{X : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}. \tag{19.1}$$

The identity $\exp(tX)\cdot\ell = e^{t\,\mathrm{ad}^*(X)}\ell$ shows that $\exp(tX)\in G_\ell$ for every t exactly when $\mathrm{ad}^*(X)\ell = 0$. Thus $\mathrm{Lie}(G_\ell) = \mathfrak{g}_\ell$. For example, a central X lies in every $\mathfrak{g}_\ell$. Central directions therefore do not generate motion on an orbit.

We give $\mathcal{O}_\ell$ the smooth structure of G/G_ℓ. This uses the homogeneous-space theorem from Lecture 3. The natural map

$$G/G_\ell \longrightarrow \mathfrak{g}^*, \qquad gG_\ell \longmapsto g\cdot\ell$$

is injective and an immersion. At the identity coset, its derivative is induced by $X\mapsto \mathrm{ad}^*(X)\ell$, whose kernel is precisely $\mathfrak{g}_\ell$. At other points, equivariance gives the same rank. The orbit dimension is consequently $\dim\mathfrak{g} - \dim\mathfrak{g}_\ell$.

For a general Lie group, an orbit need not be an embedded submanifold of $\mathfrak{g}^*$. All tangent-space statements below use the quotient smooth structure. The Heisenberg orbits will be ordinary affine planes and points, so this distinction disappears there.

Write

$$\xi_X(\eta) = \left.\frac{d}{dt}\right|_0 \exp(tX)\cdot\eta = \mathrm{ad}^*(X)\eta, \qquad \eta\in\mathcal{O}_\ell.$$

The vectors $\xi_X(\eta)$ span the tangent space of $\mathcal{O}_\ell$ at η. Two choices of X give the same vector exactly when their difference belongs to $\mathfrak{g}_\eta$.

19.2 A symplectic form on each orbit

We need a small amount of differential-form language. A smooth 2-form ω assigns an alternating bilinear form ω_x to every tangent space, smoothly in local coordinates. On $\mathbb{R}^2$, for example,

$$(dp\wedge dq)((u_1,u_2),(v_1,v_2)) = u_1v_2 - u_2v_1.$$

A form ω is *nondegenerate* if, at every point x, the only tangent vector u at x with $\omega_x(u,v)=0$ for every tangent vector v at x is $u=0$.

In coordinates $x_1,\dots,x_d$, put $\omega_{ij} = \omega(\partial_i,\partial_j) = -\omega_{ji}$. Then $\omega = \frac{1}{2}\sum_{i,j}\omega_{ij}\,dx_i\wedge dx_j$. The form ω is *closed* when

$$\partial_i\omega_{jk} + \partial_j\omega_{ki} + \partial_k\omega_{ij} = 0 \quad \text{for all } i,j,k.$$

Equivalently, closedness means $d\omega = 0$. A closed nondegenerate 2-form is called a *symplectic form*. These are coordinate-independent conditions. We import this elementary differential-form calculus, including pullbacks and the exterior derivative d, from Warner, Chapter 2. We will state the identities used in the proof.

Theorem 19.2 (Orbit symplectic form). *Every coadjoint orbit has a G-invariant symplectic form ω defined by*

$$\omega_\eta(\xi_X(\eta),\xi_Y(\eta)) = \eta([X,Y]). \tag{19.2}$$

This is the Kirillov–Kostant–Souriau form, with the sign convention of the infinitesimal action (19.1).

Proof. Well-definedness. If X changes by $A \in \mathfrak{g}_\eta$, then $\eta([A,Y]) = 0$. Changing Y has the same effect by antisymmetry. Thus the defining formula (19.2) depends only on the two tangent vectors. It is alternating and bilinear.

Nondegeneracy. If $\omega_\eta(\xi_X, \xi_Y) = 0$ for every Y, then $X \in \mathfrak{g}_\eta$ by the stabiliser formula (19.1). Hence $\xi_X(\eta) = 0$.

Smoothness. The fields ξ_X are smooth and span every tangent space. Choose a local frame from these fields. The defining formula (19.2) gives smooth coefficients in that frame, so ω is smooth.

Invariance. For $g \in G$, let g_* be the differential of the map $\eta \mapsto g \cdot \eta$ on the orbit. Since $g \exp(tX) = \exp(t \operatorname{Ad}(g)X)g$, differentiating $g \cdot (\exp(tX) \cdot \eta)$ at $t = 0$ gives

$$g_* \xi_X(\eta) = \xi_{\operatorname{Ad}(g)X}(g \cdot \eta).$$

Using that $\operatorname{Ad}(g)$ preserves brackets, we get

$$\begin{aligned} \omega_{g\cdot\eta}(g_*\xi_X, g_*\xi_Y) &= (g \cdot \eta)([\operatorname{Ad}(g)X, \operatorname{Ad}(g)Y]) \\ &= \eta([X,Y]). \end{aligned}$$

So the form is invariant.

Closedness. Let $Q : G \to \mathcal{O}_\ell$ be the orbit map, given by $Q(g) = g \cdot \ell$. Put $\beta = Q^*\omega$. Invariance makes β a left-invariant form on G, and $\beta_e(X,Y) = \ell([X,Y])$. Exterior differentiation commutes with pullback. For a left-invariant 2-form, the differential at e is

$$(d\beta)_e(X,Y,Z) = -\beta_e([X,Y],Z) + \beta_e([X,Z],Y) - \beta_e([Y,Z],X).$$

We have now stated the two identities imported from Warner, Chapter 2. They are the commutation of d with pullback and the preceding formula for a left-invariant 2-form. Substitution gives

$$(d\beta)_e(X,Y,Z) = -\ell([[X,Y],Z]) + \ell([[X,Z],Y]) - \ell([[Y,Z],X]) = 0$$

by the Jacobi identity. Left invariance gives $d\beta = 0$ everywhere. Therefore $Q^*(d\omega) = 0$. Since Q is a submersion, every tuple of tangent vectors on the orbit lifts to G. This forces $d\omega = 0$. ∎

A nondegenerate alternating form exists only in even dimension. To see this, let A be the matrix of such a form in a basis. Then $A^T = -A$. If the size d of A were odd, then transposition invariance of the determinant would give $\det A = \det(-A) = -\det A$. Hence $\det A = 0$, and the form would be degenerate. In particular, every coadjoint orbit has even dimension. This is a useful first check on a proposed calculation.

Reading the form through linear functions

There is a useful way to remember the sign in the defining formula (19.2). For $X \in \mathfrak{g}$, restrict the linear function $\mu_X(\eta) = \eta(X)$ to an orbit. Its derivative in a tangent direction is

$$(d\mu_X)_\eta(\xi_Y) = (\operatorname{ad}^*(Y)\eta)(X) = -\eta([Y,X]) = \eta([X,Y]).$$

Thus $d\mu_X = \iota_{\xi_X}\omega$, where contraction means $(\iota_V\omega)(W) = \omega(V, W)$. This identifies the infinitesimal action using only the orbit form and the linear functions.

For a smooth real function f, nondegeneracy determines a unique tangent vector field X_f by $\iota_{X_f}\omega = df$. It is called the Hamiltonian vector field in this convention. The preceding calculation says $X_{\mu_X} = \xi_X$. Consequently the bracket

$$\{f, h\} = \omega(X_f, X_h)$$

satisfies $\{\mu_X, \mu_Y\} = \mu_{[X,Y]}$. We only need this identity for linear functions. It says that the Lie bracket is still visible after passing to an orbit.

A central element $W \in \mathfrak{g}$ has $\xi_W = 0$. Thus $d\mu_W = \iota_{\xi_W}\omega = 0$, so μ_W is locally constant on each orbit. It is constant when G is connected. Such a constant can still be the value of a bracket of linear functions. For the Heisenberg basis X, Y and Z of the next section, the identity above gives $\{\mu_X, \mu_Y\} = \mu_Z$, and μ_Z takes one constant value on each orbit.

A change of convention for Hamiltonian vector fields would change some of these signs. Here the definitions, rather than a remembered physics formula, determine them. This calculation concerns finite-dimensional smooth manifolds. It does not differentiate an unbounded operator or assert a quantisation rule.

19.3 Heisenberg planes and points

Let H be the Heisenberg group of Lecture 18, with Lie algebra $\mathfrak{h}$ spanned by X, Y and Z. We use exactly its matrix coordinates:

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab'), \qquad [X, Y] = Z.$$

In particular, c is a matrix entry. It is not the central coefficient in the logarithm, which is $c - ab/2$.

Write $\ell = (p, q, \lambda)$ when $\ell(X) = p$, $\ell(Y) = q$ and $\ell(Z) = \lambda$. The adjoint formulas of Lecture 18 are

$$\mathrm{Ad}(a, b, c)X = X - bZ, \qquad \mathrm{Ad}(a, b, c)Y = Y + aZ, \qquad \mathrm{Ad}(a, b, c)Z = Z.$$

The inverse of $g = (a, b, c)$ has first coordinates $(-a, -b)$. Hence $\mathrm{Ad}(g^{-1})X = X + bZ$ and $\mathrm{Ad}(g^{-1})Y = Y - aZ$. Applying ℓ to these vectors gives

$$(a, b, c) \cdot (p, q, \lambda) = (p + \lambda b, q - \lambda a, \lambda). \tag{19.3}$$

For $\lambda \neq 0$, this reaches every point of the plane with fixed last coordinate. Its stabiliser is the centre. For $\lambda = 0$, every point is fixed. Thus the orbit list consists of planes for nonzero λ and individual points $(p, q, 0)$.

On a plane, p and q are coordinates. Differentiating the action formula (19.3) gives

$$\xi_X = -\lambda\partial_q, \qquad \xi_Y = \lambda\partial_p, \qquad \xi_Z = 0.$$

The defining value of the form is $\omega(\xi_X, \xi_Y) = \ell(Z) = \lambda$. Since $(dp \wedge dq)(\xi_X, \xi_Y) = \lambda^2$, we obtain

$$\omega_\lambda = \frac{1}{\lambda}\, dp \wedge dq. \tag{19.4}$$

The coefficient is $1/\lambda$, not λ, because the vectors ξ_X and ξ_Y already carry a factor λ.

Point orbits have zero-dimensional tangent spaces. The zero form on such a tangent space is nondegenerate, since its kernel is the zero space. The orbit method will associate $(p, q, 0)$ with the one-dimensional character

$$\chi_{p,q}(a, b, c) = e^{i(pa+qb)}.$$

These are different representations for different (p, q). The trivial central character, with $\lambda = 0$, does not mean that the representation is trivial. Only $(0, 0, 0)$ gives the trivial character.

19.4 Polarisations and the orbit theorem

A Lie algebra is *nilpotent* if its lower central series terminates:

$$\mathfrak{g}_1 = \mathfrak{g}, \qquad \mathfrak{g}_{r+1} = [\mathfrak{g}, \mathfrak{g}_r], \qquad \mathfrak{g}_s = 0 \text{ for some } s.$$

For the Heisenberg algebra, $\mathfrak{g}_2 = \mathbb{R}Z$ and $\mathfrak{g}_3 = 0$. Nilpotence is stronger than solvability.

We import a global fact: for a connected simply connected nilpotent group, $\exp : \mathfrak{g} \to G$ is a diffeomorphism, and BCH is a finite polynomial. Every Lie subalgebra $\mathfrak{m}$ gives a closed subgroup $M = \exp \mathfrak{m}$. Indeed, finite BCH keeps products in $\mathfrak{m}$, and the exponential diffeomorphism takes the closed vector subspace $\mathfrak{m}$ to a closed subset. See Knapp, *Lie Groups Beyond an Introduction*, Section I.16.

Definition 19.3. A *real polarisation* for $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{m} \subset \mathfrak{g}$ such that

$$\ell([\mathfrak{m}, \mathfrak{m}]) = 0, \qquad \dim \mathfrak{m} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

A subspace $A \subseteq \mathfrak{g}$ is called *isotropic* for ℓ when

$$\ell([A, A]) = 0.$$

A real polarisation is thus an isotropic Lie subalgebra of the displayed dimension. This dimension says that the image of $\mathfrak{m}$ in $\mathfrak{g}/\mathfrak{g}_\ell$ is as large as the image of an isotropic subspace can be. In fact, the two displayed conditions force $\mathfrak{g}_\ell \subset \mathfrak{m}$, because an isotropic subspace of maximal dimension contains $\mathfrak{g}_\ell$. The subsection on the dimension of a polarisation below proves both statements.

For such a subalgebra, put

$$\chi_\ell(\exp W) = e^{i\ell(W)}, \qquad W \in \mathfrak{m}.$$

This is well-defined because exp is bijective. It is a character of M. Every BCH correction is a bracket of elements of $\mathfrak{m}$, so ℓ kills it and $\chi_\ell(\exp U \exp V) = \chi_\ell(\exp U)\chi_\ell(\exp V)$.

Theorem 19.4 (Nilpotent orbit theorem; cited)**.** *Let G be connected, simply connected and nilpotent. Every $\ell \in \mathfrak{g}^*$ has a real polarisation. The unitary induced representation*

$$\Pi_\ell = \operatorname{Ind}_{\exp \mathfrak{m}}^{G} \chi_\ell$$

is irreducible. Its equivalence class does not depend on the polarisation. Two such represen-

> *tations are equivalent exactly when their functionals lie on the same coadjoint orbit. Every irreducible strongly continuous unitary representation of G is unitarily equivalent to some Π_ℓ.*

This is Kirillov's theorem, imported from *Lectures on the Orbit Method*, Chapter 3. Induction means unitary induction with the quotient measure convention of Lecture 17. The statement concerns complex Hilbert representations, not just finite-dimensional modules.

The global assumptions matter. A central quotient can restrict which representations descend. A solvable group need not be nilpotent. Apart from the trivial group, no compact group and no semisimple group satisfies this theorem's hypotheses. We will meet related orbit constructions in Lecture 27, with a different integrality condition and a different representation model.

The dimension of a polarisation

We now isolate the linear algebra behind a polarisation. Define the alternating form

$$B_\ell(X, Y) = \ell([X, Y])$$

on $\mathfrak{g}$. Its kernel is $\mathfrak{g}_\ell$. Suppose $\dim O_\ell = 2d$ and $\dim \mathfrak{g}_\ell = k$. Then $\dim \mathfrak{g} = 2d + k$. The form B_ℓ induces a nondegenerate alternating form on the quotient $\mathfrak{g}/\mathfrak{g}_\ell$, which has dimension $2d$. A vector space with a nondegenerate alternating form is called a *symplectic vector space*.

Let L be a subspace of this quotient. Its *orthogonal* $L^\perp$ consists of the vectors whose induced pairing with every vector of L is zero. The subspace L is *isotropic* when the induced form vanishes on pairs of vectors of L, that is, when $L \subseteq L^\perp$. Nondegeneracy gives $\dim L^\perp = 2d - \dim L$, so an isotropic L has $\dim L \leq d$. Let $A \subset \mathfrak{g}$ be an isotropic subspace for ℓ. Its image in the quotient is isotropic, so this image has dimension at most d. Therefore

$$\dim A = \dim(A \cap \mathfrak{g}_\ell) + \dim\big((A + \mathfrak{g}_\ell)/\mathfrak{g}_\ell\big) \leq k + d.$$

Equality forces $\mathfrak{g}_\ell \subset A$. It also forces the image of A to be a d-dimensional isotropic subspace. The dimension in the definition of a polarisation is exactly $k + d$.

There is one extra demand: A must be a Lie subalgebra. A maximal isotropic vector subspace need not have this property. The existence of a suitable subalgebra is a substantial part of the nilpotent orbit theorem. The definition alone does not prove existence.

This count also predicts the number of variables in an induced model. If $\mathfrak{m}$ is a polarisation, then

$$\dim(G/M) = \dim \mathfrak{g} - \dim \mathfrak{m} = d = \tfrac{1}{2} \dim O_\ell.$$

The Heisenberg plane is two-dimensional, but its representation acts on functions of one real variable. The number of these variables is half the orbit dimension. The orbit does not single out which variables to use. A polarisation makes that choice.

19.5 Induction recovers the Schrödinger formula

Take $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Its stabiliser algebra is $\mathbb{R}Z$, so a polarisation must have dimension two. Choose

$$\mathfrak{m} = \operatorname{span}\{Y, Z\}, \qquad M = \{(0, b, c)\}, \qquad \chi_\ell(0, b, c) = e^{i\lambda c}.$$

The subalgebra $\mathfrak{m}$ is abelian and two-dimensional, so it satisfies both polarisation conditions. A section of $H/M \cong \mathbb{R}$ is $s(t) = (t, 0, 0)$. The quotient measure is dt, invariant under the induced translation action.

An induced function satisfies $f(gm) = \chi_\ell(m)^{-1} f(g)$. Write $\varphi(t) = f(s(t))$. Since

$$(a, b, c) = s(a)(0, b, c - ab),$$

its values are

$$f(a, b, c) = e^{-i\lambda(c-ab)}\varphi(a).$$

Left translation acts by $(\Psi_\lambda(g)f)(h) = f(g^{-1}h)$. For $g = (a, b, c)$,

$$g^{-1}s(t) = (t - a, -b, -c + ab).$$

Insert this into the preceding covariance formula. Its central expression becomes $-c + ab - (t - a)(-b) = -c + bt$. Therefore

$$(\Psi_\lambda(a, b, c)\varphi)(t) = e^{i\lambda(c-bt)}\varphi(t - a). \tag{19.5}$$

The operator is unitary: the phase has absolute value one and translation preserves dt. The general unitary induction construction of Lecture 17 supplies the representation law and strong continuity. Both can also be checked from this formula.

The sign of the translation differs from the Schrödinger formula of Lecture 18 because our section uses the left action on H/M. It gives an equivalent model. Let $R\varphi(x) = \varphi(-x)$. Then $R^{-1} = R$, and

$$(R\Psi_\lambda(a, b, c)R^{-1}u)(x) = e^{i\lambda(c+bx)}u(x + a) = \Pi_\lambda(a, b, c)u(x).$$

Thus the orbit theorem reproduces the Schrödinger representation with the precise convention already used. The plane's parameter λ determines its nonzero central character.

For a point $(p, q, 0)$, the polarisation is all of $\mathfrak{h}$. Induction from H to itself gives $\chi_{p,q}$. The plane and point calculations therefore recover the complete Heisenberg classification from Lecture 18. The general orbit theorem extends that pattern to every connected simply connected nilpotent group.

What the Heisenberg orbit list explains

The orbit calculation separates two phenomena. For $\lambda \neq 0$, changing p or q stays within one plane. It cannot change the equivalence class of the representation. For $\lambda = 0$, each point is an orbit of its own, so p and q both survive as representation labels. The discontinuous change in orbit dimension matches the change from infinite-dimensional representations to characters.

One can check the zero-charge conclusion without the general orbit theorem. In an irreducible unitary representation, every central group element acts by a scalar, by the unitary Schur lemma used in Lecture 18. If that scalar character is trivial, the representation factors through $H/Z(H) \cong \mathbb{R}^2$. An irreducible unitary representation of this abelian group is one-dimensional. Its continuous characters are $e^{i(pa+qb)}$, obtained by applying the one-parameter character classification to each coordinate.

For nonzero central character, the Stone–von Neumann theorem gives uniqueness directly. The orbit theorem recovers that result and explains its geometric label. It also shows what to try for a larger nilpotent group: calculate ad^*, find orbit invariants, determine the stabiliser dimension, choose a real polarisation for ℓ, and induce its character. The last exercise carries out the geometric steps in one four-dimensional example.

Self-checks

1. Why must the stabiliser algebra contain the centre?

Answer. If X is central, then $[X, Y] = 0$ for every Y. The stabiliser formula (19.1) puts X in $\mathfrak{g}_\ell$ for every ℓ.

2. Does the formula $\omega_\lambda = \lambda\, dp \wedge dq$ have the right normalisation?

Answer. No. Its value on $(-\lambda\partial_q, \lambda\partial_p)$ would be λ^3. The required value is λ, so the coefficient is $1/\lambda$.

3. Can one apply the orbit theorem directly to a Heisenberg group with circle centre?

Answer. No. That group is not simply connected. Apply the theorem to its simply connected cover. Then use the Descent theorem of Lecture 4. Only representations trivial on the covering kernel descend.

Summary

The coadjoint action turns $\mathfrak{g}^*$ into a union of even-dimensional symplectic orbits. For connected simply connected nilpotent groups, a polarisation converts an orbit into an irreducible induced representation. For the Heisenberg group, planes give Schrödinger representations and points give characters. The cited classification is exact in this setting. Its hypotheses are part of the result.

Reading

- Kirillov, *Lectures on the Orbit Method*, Chapters 2–3: Heisenberg representations, polarisations and the nilpotent correspondence.
- Knapp, *Lie Groups Beyond an Introduction*, Section I.16: simply connected nilpotent groups.

- Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Chapter 2: differential forms and exterior differentiation. The proof above uses only the two differential-form identities stated in that proof.

Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k \exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad} X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad} H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

The Lie Algebras course supplies both types [I]. The real span of a Chevalley basis is a split form. Weyl's unitary trick turns the same basis into a compact form; see Knapp Beyond, Section VI.1.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X,Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X,\theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X,Y) = \kappa_\theta(Y,X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k},\mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X,Y] = [-X,-Y] = [X,Y]$. Every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism [C: Knapp Beyond, Section VI.2].

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n,\mathbb{R})$ and $\theta(X) = -X^T$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n,\mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n,\mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^T Y^T - Y^T X^T = (YX - XY)^T = \theta[X,Y].$$

On $\mathfrak{sl}(n,\mathbb{C})$ we import $\kappa(X,Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X,X) = 2n \operatorname{tr}(XX^T) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n,\mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^* = X^T - iY^T$. So $Z^* = -Z$ exactly when $X^T = -X$ and $Y^T = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n,\mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z,Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^*Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n,\mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form [C: Knapp Beyond, Sections VI.1–VI.2]. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X, Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The real rank does not depend on the choices of θ and $\mathfrak{a}$ [C: Knapp Beyond, Chapter VI]. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Next fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker \lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^T$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

Theorem 20.8 (Cartan and Iwasawa decompositions; cited)**.** *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) *A and N are closed subgroups, and $K \times A \times N \to G$, $(k,a,n) \mapsto kan$, is a diffeomorphism.*

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

20.3 The rank-one model $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n,\mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E-F)$ and $\mathfrak{p} = \operatorname{span}(H, E+F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$). *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^{t} = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^{2}}.$$

Proof. The first column of g is nonzero, so $r > 0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta - \beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^{t} & e^{t}x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^{t} = r$ and $x = (\alpha\beta+\gamma\delta)/r^{2}$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^{T}$ with $p>0$ and unit length, so $p = 1$. The second column is a unit vector orthogonal to $(1,0)^{T}$ with positive second entry, so it is $(0,1)^{T}$. Hence $k = k'$ and $an = a'n'$. Comparing diagonals gives $a = a'$, and then $n = n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r > 0$. $\blacksquare$

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u) = \sum_{k\ge 0} \frac{u^k}{(2k)!}, \qquad \mathcal{S}(u) = \sum_{k\ge 0} \frac{u^k}{(2k+1)!}.$$

For $u = s^2 \ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s = 0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y \in \mathfrak{p}$, set $u = -\det Y \ge 0$. Then $Y^2 = uI$ and*

$$\exp Y = C(u)I + \mathcal{S}(u)Y.$$

The map $\exp\colon \mathfrak{p} \to \mathcal{P}$ is a bijection. Its inverse is

$$L(Q) = \frac{Q - cI}{\mathcal{S}(C^{-1}(c))}, \qquad c = \tfrac{1}{2}\operatorname{tr} Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ among symmetric matrices.

Proof. Write $Y = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$. Then $Y^2 = (p^2+q^2)I = uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y = (\exp \frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det \exp Y = e^{\operatorname{tr} Y} = 1$.

Every coefficient of the series for C' is positive, so $C' > 0$ on $(0,\infty)$. Also, $C'(0) = \frac{1}{2}$. So $C' > 0$ on some interval $(-\varepsilon, \infty)$. Thus C is a strictly increasing bijection $[0,\infty) \to [1,\infty)$, with smooth inverse near $[1,\infty)$. Also $\mathcal{S} > 0$ near $[0,\infty)$.

For injectivity, let $Q = \exp Y$. Then $\operatorname{tr} Q = 2C(u)$, which determines u because C is strictly increasing on $[0,\infty)$. The formula for $\exp Y$ then gives $Y = (Q - C(u)I)/\mathcal{S}(u) = L(Q)$.

For surjectivity, let $Q \in \mathcal{P}$ have eigenvalues $\mu, \mu^{-1} > 0$. Then $c = \frac{1}{2}(\mu + \mu^{-1}) \ge 1$. Put $u = C^{-1}(c)$ and $Y = L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2 = 2cQ - I$, so $(Q - cI)^2 = (c^2-1)I$. Since $u\,\mathcal{S}(u)^2 = C(u)^2 - 1 = c^2 - 1$, we get $Y^2 = uI$. Therefore

$$\exp Y = C(u)I + \mathcal{S}(u)Y = cI + (Q - cI) = Q. \qquad \blacksquare$$

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2,\mathbb{R})$)**.** *The map $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k\exp X$, is a diffeomorphism. Its inverse is*

$$g \mapsto \bigl(g\exp(-X),\, X\bigr), \qquad X = \tfrac{1}{2}L(g^{T}g).$$

Proof. Suppose first that $g = k\exp X$ with $k \in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^{T}g = \exp(X)k^{T}k\exp(X) = \exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X = L(g^{T}g)$, so X and then k are forced. Conversely, given g, the matrix $g^{T}g$ lies in $\mathcal{P}$. Define X and $k = g\exp(-X)$ as stated. Then

$$k^{T}k = \exp(-X)\exp(2X)\exp(-X) = I, \qquad \det k = 1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. ∎

> **Corollary 20.12** ($G = KAK$). *Every $g \in G$ can be written as $g = k_1 a_t k_2$ with $k_1, k_2 \in K$ and $t \geq 0$, and t is unique. The number e^{2t} is the larger eigenvalue of $g^{T} g$.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk')a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^{T} g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. ∎

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^{t} = \sqrt{2}$ and $x = \tfrac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^{T} g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \frac{3}{2}$. The equation $C(u) = \cosh \sqrt{u} = \frac{3}{2}$ gives $\sqrt{u} = 2 \log \tau$, since $\tau^{2} + \tau^{-2} = 3$. Also $\sinh \sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2} L(Q) = \frac{2 \log \tau}{\sqrt{5}} \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log \tau}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log \tau)^{2}$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g \exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm \log \tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log \tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{ z \in \mathbb{C} : \operatorname{Im} z > 0 \}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma) \operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

Proof. We have $a_t \cdot z = e^{2t} z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \tfrac{1}{2} \log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

So Iwasawa coordinates are Cartesian coordinates (x,y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t}z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n,\mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n,\mathbb{R})$ is $n-1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p}=0$, so $A=N=\{I\}$ and the Iwasawa decomposition reads $G=K$.
2. Find the Iwasawa decomposition of $g=\begin{pmatrix}0&1\\-1&2\end{pmatrix}$.
   Answer. Here $r=1$, $\cos\varphi=0$ and $\sin\varphi=-1$, so $k=k_{-\pi/2}$ and $a=I$. Also $x=(0\cdot 1+(-1)\cdot 2)/1=-2$. Indeed
   $$\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}1&-2\\0&1\end{pmatrix}=\begin{pmatrix}0&1\\-1&2\end{pmatrix}.$$
3. Where does $-I$ sit in $G=KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G=KAN$, the element $-I$ has factors $k=k_{\pi}$ and $a=n=I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g}=\mathfrak{k}\oplus\mathfrak{p}$. The compact form $\mathfrak{k}\oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n,\mathbb{R})$ has real rank $n-1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G=K\exp\mathfrak{p}$ and $G=KAN$. For $\mathrm{SL}(2,\mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t\geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G=KAN$ to construct the principal series.

Lecture 21

Smooth Vectors, K-Types and $(\mathfrak{g}, K)$-Modules

Objectives. *A representation of a noncompact group usually acts on an infinite-dimensional Hilbert space. We use its derivatives on smooth vectors. This lecture extracts an algebraic object from such a representation. The object consists of the smooth vectors that are finite under a compact subgroup K. On these vectors $\mathfrak{g}$ and K act together, subject to compatibility conditions. The model is $\mathrm{SL}(2,\mathbb{R})$ with $K = \mathrm{SO}(2)$.*

By the end of this lecture you will be able to:

- Define smooth vectors, K-finite vectors, K-types and admissibility.
- Prove that the K-finite smooth vectors of a strongly continuous representation on a Hilbert space form a $(\mathfrak{g}, K)$-module.
- Use the compatibility conditions to show that E and F shift the circle K-types of $\mathrm{SL}(2,\mathbb{R})$ by ± 2.
- State which global questions the $(\mathfrak{g}, K)$-module leaves open, and which cited theorems address them.

21.1 Smooth vectors and K-finite vectors

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$. The subgroup $K \subseteq G$ is compact, with Lie algebra $\mathfrak{k} \subseteq \mathfrak{g}$. The representation Π of G on a Hilbert space $\mathcal{H}$ is strongly continuous. Here each operator $\Pi(g)$ is bounded, linear and invertible. The map $g \mapsto \Pi(g)v$ is norm continuous for every $v \in \mathcal{H}$. Unitarity is assumed only where stated.

Definition 21.1. A vector $v \in \mathcal{H}$ is *smooth* if the orbit map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is smooth. This means that in every chart all iterated partial derivatives exist as norm limits and are continuous. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ we set

$$d\Pi(X)v = \left.\frac{d}{dt}\right|_{t=0} \Pi(\exp tX)v.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, because a linear combination of smooth orbit maps is smooth.

For unitary representations, the following facts were imported in Lecture 15. We now also import the density and invariance facts (F1)–(F2) for strongly continuous representations by bounded operators. A source for this more general smooth-vector theorem is Knapp RT,

Sections III.3–4. Here Knapp RT is A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (2001).

(F1) *Density* (Gårding). The space $\mathcal{H}^\infty$ is dense in $\mathcal{H}$.

(F2) *Invariance.* Each operator $d\Pi(X)$ maps $\mathcal{H}^\infty$ into itself, and $X \mapsto d\Pi(X)$ is a Lie algebra representation of $\mathfrak{g}$ on $\mathcal{H}^\infty$.

(F3) *Skew-symmetry.* If Π is unitary, then $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for all $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$.

We extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$ by $d\Pi(X + iY) = d\Pi(X) + i\, d\Pi(Y)$ for $X, Y \in \mathfrak{g}$. By the invariance property (F2), this is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{H}^\infty$.

One elementary fact is proved here. If $v \in \mathcal{H}^\infty$ and $h \in G$, then $\Pi(h)v \in \mathcal{H}^\infty$. Indeed, the orbit map of $\Pi(h)v$ is $g \mapsto \Pi(gh)v$. This is the orbit map of v composed with the smooth right translation $g \mapsto gh$.

Definition 21.2. A vector $v \in \mathcal{H}$ is *K-finite* if $\operatorname{span}\{\Pi(k)v : k \in K\}$ is finite-dimensional. We write $\mathcal{H}_K$ for the set of K-finite vectors.

The set $\mathcal{H}_K$ is a subspace, because for $a, b \in \mathbb{C}$ the span of the vectors $\Pi(k)(av + bw)$, $k \in K$, lies in $\operatorname{span}\Pi(K)v + \operatorname{span}\Pi(K)w$.

Let $\widehat{K}$ be the set of equivalence classes of irreducible continuous representations of K. Each class δ has a finite-dimensional representative space V_δ (Lecture 10). Let $\operatorname{Hom}_K(V_\delta, \mathcal{H})$ be the space of linear maps $V_\delta \to \mathcal{H}$ that commute with K.

Definition 21.3. The *multiplicity* of $\delta \in \widehat{K}$ in Π is $m_\delta(\Pi) = \dim \operatorname{Hom}_K(V_\delta, \mathcal{H})$. The class δ is a *K-type* of Π if $m_\delta(\Pi) \geq 1$. The representation Π is *admissible* if $m_\delta(\Pi) < \infty$ for every $\delta \in \widehat{K}$.

Let $v \in \mathcal{H}_K$ and $W_v = \operatorname{span}\Pi(K)v$. Then W_v is finite-dimensional and K-invariant. All norms on W_v are equivalent, so strong continuity makes K act continuously on W_v. By complete reducibility (Lecture 5), W_v is a direct sum of irreducible subspaces. Hence every K-finite vector is a finite sum of vectors from K-types.

For the circle we write k_θ, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, with $k_\theta k_\phi = k_{\theta+\phi}$. Its irreducible continuous representations are the characters $\chi_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 7). We label the K-type χ_n by n.

Example 21.4. Let $G = K = \mathrm{SO}(2)$ and $\mathcal{H} = L^2(\mathbb{R}/2\pi\mathbb{Z})$, with

$$\langle f, h\rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{f(x)}\, h(x)\, dx, \qquad (\Pi(k_\theta)f)(x) = f(x + \theta).$$

This representation is unitary and strongly continuous. The functions $e_n(x) = e^{inx}$ form an orthonormal basis (Lecture 10). Put $\mathcal{H}(n) = \{v : \Pi(k_\theta)v = e^{in\theta}v \text{ for all } \theta\}$. Then $e_n \in \mathcal{H}(n)$.

We show that $\mathcal{H}(n) = \mathbb{C}e_n$. Let $a_m(v) = \langle e_m, v\rangle$. For continuous v, the substitution $y = x + \theta$ and periodicity give

$$a_m(\Pi(k_\theta)v) = \frac{1}{2\pi}\int_0^{2\pi} e^{-im(y-\theta)}v(y)\, dy = e^{im\theta}a_m(v).$$

Both sides are continuous linear functionals of v, so density extends the identity to all L^2 vectors. If $v \in \mathcal{H}(n)$, the left side is also $e^{in\theta}a_m(v)$. For $m \neq n$ take $\theta = \pi/(m-n)$. Then $e^{im\theta} = -e^{in\theta}$, so $a_m(v) = 0$. Hence $v = a_n(v)e_n$.

So $\mathcal{H}_K$ is the space of trigonometric polynomials $f = \sum_{|n|\le N} c_n e_n$. Each such f is smooth, because its orbit map

$$\theta \mapsto \Pi(k_\theta)f = \sum_{|n|\le N} c_n\, e^{in\theta}\, e_n$$

is a finite sum of smooth scalar functions times fixed vectors. Thus every K-finite vector is smooth. Every multiplicity is 1, so Π is admissible. The space $\mathcal{H}_K$ is dense but proper. For example, $\sum_n 2^{-|n|}e_n$ is not K-finite.

A K-finite vector need not be smooth. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by translation, and let $K = \{0\}$. Every vector is K-finite. The indicator function of $[0,1]$ is not smooth (Exercise 21.5). So we impose both conditions. We write $\mathcal{H}_K^\infty$ for the space of smooth K-finite vectors:

$$\mathcal{H}_K^\infty = \mathcal{H}^\infty \cap \mathcal{H}_K.$$

21.2 The compatibility conditions

Definition 21.5. A $(\mathfrak{g}, K)$-*module* is a complex vector space V with a Lie algebra representation of $\mathfrak{g}$ on V, written $v \mapsto X \cdot v$, and a group homomorphism $K \to \mathrm{GL}(V)$, written $v \mapsto k \cdot v$, such that:

(a) (*finiteness condition*) every $v \in V$ lies in a finite-dimensional K-invariant subspace W on which K acts continuously;

(b) (*derivative condition*) for $X \in \mathfrak{k}$ and $v \in V$, with W as in the finiteness condition (a),

$$X \cdot v = \frac{d}{dt}\Big|_{t=0} \exp(tX) \cdot v \qquad \text{(derivative computed in } W\text{)};$$

(c) (*covariance condition*) for $k \in K$, $X \in \mathfrak{g}$ and $v \in V$,

$$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v).$$

A *morphism* of $(\mathfrak{g}, K)$-modules is a linear map that commutes with both actions. A $(\mathfrak{g}, K)$-module V is *admissible* if $\dim \mathrm{Hom}_K(V_\delta, V) < \infty$ for every $\delta \in \widehat{K}$.

The derivative in the derivative condition (b) does not depend on the choice of W. Indeed, two choices W and W' both lie in the finite-dimensional space $W + W'$, and all norms on $W + W'$ are equivalent.

The $\mathfrak{g}$-action extends complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Both sides of the covariance condition (c) are complex-linear in X, so the covariance condition also holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

The derivative condition (b) involves only elements $\exp(tX)$, which lie in the identity component of K. Hence, if K is disconnected, the K-action carries information that $\mathfrak{k}$ cannot detect (Exercise 21.7).

The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. In § 21.3 it gives the shifts of the circle K-types for $\mathrm{SL}(2,\mathbb{R})$.

Proposition 21.6. *Let Π be a strongly continuous representation of G on $\mathcal{H}$, and let $K \subseteq G$ be compact. Then $\mathcal{H}_K^\infty$ is invariant under $\Pi(k)$ for $k \in K$ and under $d\Pi(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. With these actions $\mathcal{H}_K^\infty$ is a $(\mathfrak{g}, K)$-module. Let Π' be a strongly continuous representation of G on a Hilbert space $\mathcal{H}'$. Let $T : \mathcal{H} \to \mathcal{H}'$ be bounded and linear, with $T\Pi(g) = \Pi'(g)T$ for all $g \in G$. Then T maps $\mathcal{H}_K^\infty$ into $\mathcal{H}_K'^\infty$, and its restriction is a morphism.*

Proof. Step 1: the K-action. Let $v \in \mathcal{H}_K^\infty$ and $k \in K$. Then $\Pi(k)v$ is smooth by the right-translation argument in § 21.1. Since $Kk = K$,

$$\operatorname{span} \Pi(K)\Pi(k)v = \operatorname{span} \Pi(Kk)v = \operatorname{span} \Pi(K)v,$$

which is finite-dimensional. So $\Pi(k)v \in \mathcal{H}_K^\infty$. In particular, $W_v = \operatorname{span} \Pi(K)v$ is contained in $\mathcal{H}_K^\infty$.

Step 2: the covariance identity. Let $k \in K$, $X \in \mathfrak{g}$ and $w \in \mathcal{H}^\infty$. We claim

$$\Pi(k)\, d\Pi(X)w = d\Pi(\operatorname{Ad}(k)X)\, \Pi(k)w. \tag{21.1}$$

By the conjugation formula for exponentials (Lecture 2), $k \exp(tX)k^{-1} = \exp(t \operatorname{Ad}(k)X)$. Hence, for $t \neq 0$,

$$\Pi(k)\Big[\frac{\Pi(\exp tX)w - w}{t}\Big] = \frac{\Pi(\exp(t \operatorname{Ad}(k)X))\, \Pi(k)w - \Pi(k)w}{t}.$$

Let $t \to 0$. On the left, the bracket tends to $d\Pi(X)w$, and $\Pi(k)$ is bounded. So the left side tends to $\Pi(k)\, d\Pi(X)w$. On the right, $\Pi(k)w$ is smooth, so the quotient tends to $d\Pi(\operatorname{Ad}(k)X)\Pi(k)w$.

Step 3: $\mathfrak{g}$ preserves K-finiteness. This is the main step. The difficulty is that $d\Pi(X)$ does not commute with $\Pi(k)$. So the span of $\Pi(K)\, d\Pi(X)v$ is not obviously finite-dimensional. The covariance identity (21.1) replaces commutation. It moves X to $\operatorname{Ad}(k)X$, which still lies in the finite-dimensional space $\mathfrak{g}$.

Let $v \in \mathcal{H}_K^\infty$ and $W = W_v$. By Step 1, $W \subseteq \mathcal{H}^\infty$, so $d\Pi(X)w$ is defined for $w \in W$. Choose bases $X_1, \dots, X_d$ of $\mathfrak{g}$ and $w_1, \dots, w_r$ of W. Put

$$U = \operatorname{span}\{d\Pi(X_i)w_j : 1 \le i \le d,\ 1 \le j \le r\}.$$

Then $\dim U \le dr$. Let $X \in \mathfrak{g}$, $w \in W$ and $k \in K$. Expand

$$\operatorname{Ad}(k)X = \sum_{i=1}^{d} a_i(k)X_i, \qquad \Pi(k)w = \sum_{j=1}^{r} b_j(k)w_j,$$

with scalars $a_i(k)$ and $b_j(k)$. The second expansion uses $\Pi(k)w \in W$. By the covariance identity (21.1) and linearity of $d\Pi$ in both arguments,

$$\Pi(k)\, d\Pi(X)w = d\Pi(\operatorname{Ad}(k)X)\, \Pi(k)w = \sum_{i=1}^{d}\sum_{j=1}^{r} a_i(k)\, b_j(k)\, d\Pi(X_i)w_j \in U.$$

In particular $\Pi(K)\, d\Pi(X)v \subseteq U$. So $d\Pi(X)v$ is K-finite. It is smooth by the invariance property (F2). Complex combinations stay in the subspace $\mathcal{H}_K^\infty$, so the same holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

Step 4: the axioms. By the invariance property (F2) and Step 3, $d\Pi$ is a representation of $\mathfrak{g}$ on $\mathcal{H}_K^\infty$. The finiteness condition (a) holds with $W = W_v$, by Step 1 and the continuity of

the K-action on W_v shown in § 21.1. For the derivative condition (b), let $X \in \mathfrak{k}$. The curve $t \mapsto \Pi(\exp tX)v$ lies in W_v. Its derivative in $\mathcal{H}$ is $d\Pi(X)v$. A finite-dimensional subspace is closed, and W_v carries the restricted norm. So the difference quotients converge in W_v to $d\Pi(X)v$. The covariance condition (c) is the covariance identity (21.1).

Step 5: intertwiners. Let $v \in \mathcal{H}_K^\infty$. The orbit map $g \mapsto \Pi'(g)Tv = T\Pi(g)v$ is smooth, because T is bounded and linear. Also $\operatorname{span}\Pi'(K)Tv = T(\operatorname{span}\Pi(K)v)$ is finite-dimensional. Differentiating $T\Pi(\exp tX)v = \Pi'(\exp tX)Tv$ at $t = 0$ gives $T\,d\Pi(X)v = d\Pi'(X)Tv$. Commutation with K is the hypothesis. ∎

Definition 21.7. The $(\mathfrak{g}, K)$-*module of* Π is $\mathcal{H}_K^\infty$ with the actions given in the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6. Two representations are *infinitesimally equivalent* if their $(\mathfrak{g}, K)$-modules are isomorphic.

21.3 Circle K-types for $\mathrm{SL}(2,\mathbb{R})$

Now $G = \mathrm{SL}(2,\mathbb{R})$ and $K = \mathrm{SO}(2)$. Put

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

The matrix k_θ here is $k_{-\theta}$ in the notation of Lecture 20. Then $\mathfrak{k} = \mathbb{R}W$, and $\theta \mapsto k_\theta$ identifies $\mathbb{R}/2\pi\mathbb{Z}$ with K. So K is connected.

Let V be a $(\mathfrak{g}, K)$-module. For $n \in \mathbb{Z}$ put

$$V(n) = \{v \in V : k_\theta \cdot v = e^{in\theta}v \text{ for all } \theta\}.$$

By the finiteness condition (a) and complete reducibility (Lecture 5), every $v \in V$ is a finite sum of vectors from the spaces $V(n)$. The sum is direct, because the functions $e^{in\theta}$ are linearly independent. So $V = \bigoplus_{n\in\mathbb{Z}} V(n)$, and $\dim V(n)$ is the multiplicity of the K-type n.

Let $H = \operatorname{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then H, S, W is a basis of $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$. Multiplying matrices gives

$$[H,S] = 2W, \qquad [W,H] = -2S, \qquad [W,S] = 2H.$$

In $\mathfrak{g}_\mathbb{C} = \mathfrak{sl}(2,\mathbb{C})$ we set

$$H' = -iW = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}(H + iS) = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}(H - iS) = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}.$$

The real brackets give

$$\begin{aligned} [H', E] &= -\tfrac{i}{2}\big([W,H] + i[W,S]\big) = -\tfrac{i}{2}(-2S + 2iH) = H + iS = 2E, \\ [H', F] &= -\tfrac{i}{2}\big([W,H] - i[W,S]\big) = -\tfrac{i}{2}(-2S - 2iH) = -(H - iS) = -2F, \\ [E, F] &= \tfrac{1}{4}\big(-i[H,S] + i[S,H]\big) = -\tfrac{i}{2}[H,S] = -iW = H'. \end{aligned}$$

Thus (H', E, F) is an $\mathfrak{sl}_2$-triple. The $\mathfrak{sl}_2$ theory of the prerequisite Lie Algebras course applies to it. The element H' spans $\mathfrak{k}_\mathbb{C}$, whereas $H \notin \mathfrak{k}$. Also $F = \overline{E}$, where $\overline{X + iY} = X - iY$ for $X, Y \in \mathfrak{g}$.

Lemma 21.8. *For every $\theta \in \mathbb{R}$, the complex-linear extension of $\mathrm{Ad}(k_\theta)$ to $\mathfrak{g}_{\mathbb{C}}$ satisfies $\mathrm{Ad}(k_\theta)H' = H'$, $\mathrm{Ad}(k_\theta)E = e^{2i\theta}E$ and $\mathrm{Ad}(k_\theta)F = e^{-2i\theta}F$.*

Proof. Since $W = iH'$, we get

$$\mathrm{ad}(W)E = i[H', E] = 2iE, \qquad \mathrm{ad}(W)F = i[H', F] = -2iF, \qquad \mathrm{ad}(W)H' = 0.$$

By the relation between Ad and ad in Lecture 2, $\mathrm{Ad}(\exp \theta W) = e^{\theta \,\mathrm{ad}\, W}$, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. On an eigenvector of $\mathrm{ad}\, W$ with eigenvalue μ this operator is multiplication by $e^{\theta\mu}$. The eigenvalues 0, $2i$ and $-2i$ of $\mathrm{ad}\, W$ on H', E and F give the three formulas. ∎

Exercise 21.1 checks the adjoint lemma by matrix multiplication.

Proposition 21.9. *Let V be a $(\mathfrak{g}, K)$-module for $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. If $v \in V(n)$, then*

$$H' \cdot v = nv, \qquad E \cdot v \in V(n+2), \qquad F \cdot v \in V(n-2).$$

Proof. Let $v \in V(n)$. Since $\exp(\theta W) \cdot v = e^{in\theta}v$, the derivative condition (b) with $X = W$ gives

$$W \cdot v = \frac{d}{d\theta}\Big|_{\theta=0} e^{in\theta}v = in\, v, \qquad \text{so} \qquad H' \cdot v = -i\, W \cdot v = nv.$$

By the covariance condition (c), extended to $X \in \mathfrak{g}_{\mathbb{C}}$, and the adjoint lemma, Lemma 21.8,

$$k_\theta \cdot (E \cdot v) = (\mathrm{Ad}(k_\theta)E) \cdot (k_\theta \cdot v) = e^{2i\theta}E \cdot (e^{in\theta}v) = e^{i(n+2)\theta}\, E \cdot v.$$

So $E \cdot v \in V(n+2)$. The argument for F is the same with $e^{-2i\theta}$, and gives $F \cdot v \in V(n-2)$. ∎

In this proof the shift by 2 comes from the covariance condition (c). For this connected group K, the shift also follows from the derivative condition (b) and the bracket $[H', E] = 2E$. Indeed, this bracket and $H' \cdot v = nv$ show that H' multiplies $E \cdot v$ by $n+2$. The operator H' acts on each summand $V(m)$ of $V = \bigoplus_{m\in\mathbb{Z}} V(m)$ by m, so $E \cdot v \in V(n+2)$. The bracket $[H', F] = -2F$ gives $F \cdot v \in V(n-2)$ in the same way.

Let Ω be the element

$$\Omega = H'^2 + 2EF + 2FE$$

of the universal enveloping algebra $U(\mathfrak{g}_{\mathbb{C}})$. It is the Casimir element of the triple. By the prerequisite Lie Algebras course, it is central in $U(\mathfrak{g}_{\mathbb{C}})$. Substituting $EF = FE + H'$ gives

$$\Omega = H'^2 + 2H' + 4FE, \qquad \Omega \cdot v = n(n+2)v + 4F \cdot E \cdot v \quad (v \in V(n)). \tag{21.2}$$

By the shift proposition, Proposition 21.9, $F \cdot E \cdot v \in V(n)$. So Ω preserves each $V(n)$.

Example 21.10. The group $\mathrm{SL}(2, \mathbb{R})$ acts on $\mathbb{C}^2$ by matrix multiplication, and hence on $V_m = \mathrm{Sym}^m \mathbb{C}^2$, of dimension $m+1$. The orbit maps are polynomial in the matrix entries, so every vector is smooth. Since V_m is finite-dimensional, every vector is also K-finite. Hence V_m is its own $(\mathfrak{g}, K)$-module. The Lie algebra acts on products by the Leibniz rule. Put $u_\pm = (1, \pm i)^T$. Then

$$k_\theta u_+ = \begin{pmatrix} \cos\theta + i\sin\theta \\ -\sin\theta + i\cos\theta \end{pmatrix} = e^{i\theta}u_+, \qquad k_\theta u_- = e^{-i\theta}u_-.$$

Direct multiplication (Exercise 21.2) gives $Eu_- = u_+$, $Eu_+ = 0$, $Fu_+ = u_-$ and $Fu_- = 0$. Let $w_j = u_+^j u_-^{m-j}$ for $0 \le j \le m$. The group acts on each factor, so

$$k_\theta w_j = (e^{i\theta}u_+)^j (e^{-i\theta}u_-)^{m-j} = e^{i(2j-m)\theta} w_j.$$

Hence the K-types of V_m are $-m, -m+2, \dots, m$, each with multiplicity one. The Leibniz rule gives

$$E \cdot w_j = (m-j) w_{j+1} \quad (0 \le j < m), \qquad F \cdot w_j = j w_{j-1} \quad (0 < j \le m).$$

The endpoint actions are $Ew_m = 0$ and $Fw_0 = 0$, in agreement with the shift proposition, Proposition 21.9. Since $E \cdot w_m = 0$ and $w_m \in V(m)$, the Casimir formula (21.2) gives $\Omega \cdot w_m = m(m+2) w_m$. The K-types are the familiar $\mathfrak{sl}_2$-weights, read with respect to the compact element H' rather than the diagonal element H.

If Π is unitary and Ω acts on $\mathcal{H}_K^\infty$ by a scalar λ, the skew-symmetry (F3) turns the Casimir formula (21.2) into formulas for $\|d\Pi(E)v\|^2$ and $\|d\Pi(F)v\|^2$. Exercise 21.4 derives them. Exercise 21.3 builds infinite-dimensional $(\mathfrak{g}, K)$-modules from the same raising and lowering pattern.

21.4 What the $(\mathfrak{g}, K)$-module does not record

A $(\mathfrak{g}, K)$-module has no topology. It does not record the norm or the action of elements of G outside K. We cite theorems that relate it back to representations. None of them is proved or used later in this lecture.

Admissibility. Let G be a connected linear reductive group with maximal compact subgroup K (Lecture 20). The following two statements are in Knapp RT, Chapter VIII.

(i) If a strongly continuous Hilbert representation Π is admissible and $\Pi|_K$ is unitary, then every K-finite vector is smooth, so $\mathcal{H}_K^\infty = \mathcal{H}_K$.

(ii) (Harish-Chandra) Every irreducible unitary representation of G is admissible.

The circle example, Example 21.4, shows the conclusion of the smoothness statement (i) directly in the case $G = K$.

Globalization. Passing from a $(\mathfrak{g}, K)$-module to a continuous group representation is called globalization. A Harish-Chandra module is an admissible $(\mathfrak{g}, K)$-module finitely generated over $U(\mathfrak{g}_\mathbb{C})$. A positive definite Hermitian form on a Harish-Chandra module is invariant if the K-action is unitary and the $\mathfrak{g}$-action is skew-symmetric for real $X \in \mathfrak{g}$.

For connected linear reductive G with maximal compact K, an irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization. It is unique up to unitary equivalence. This theorem is cited from P. Etingof's MIT 18.757 notes. The globalization is Theorem 7.5, printed p. 38, and its uniqueness is Proposition 7.1, printed p. 37. The theorem supplies an action of G on a Hilbert space completing the module, together with its strong continuity. A completion of the vector space alone supplies neither.

One $(\mathfrak{g}, K)$-module can arise from completions of different kinds. Take $G = K = \mathrm{SO}(2)$ as in the circle example, Example 21.4. Translation also acts continuously on the Banach

space $C(\mathbb{R}/2\pi\mathbb{Z})$ with the supremum norm, which is not a Hilbert space. With the same definitions, the smooth K-finite vectors of $C(\mathbb{R}/2\pi\mathbb{Z})$ are again the trigonometric polynomials. So $L^2(\mathbb{R}/2\pi\mathbb{Z})$ and $C(\mathbb{R}/2\pi\mathbb{Z})$ have the same $(\mathfrak{g}, K)$-module.

This common module has infinitely many distinct circle types and is not finitely generated over $U(\mathfrak{k}_{\mathbb{C}})$. Thus this example is outside the irreducible Harish-Chandra theorem. The more general Casselman–Wallach globalization theorem belongs to further study.

Integration. The identities among $d\Pi(E)$, $d\Pi(F)$ and $d\Pi(H')$ were valid because Π was given. Formal commutation relations among unbounded operators do not by themselves produce a group representation. For the modules of Exercise 21.3, the question of which ones come from unitary representations is taken up in Lecture 22 and Lecture 23.

21.5 Self-checks

Check 1. In Step 3 of the proof of the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6, why must all of W_ν consist of smooth vectors?

Answer. The space U is spanned by vectors $d\Pi(X_i)w_j$ with $w_j \in W_\nu$. These are defined only for smooth w_j. Step 1 shows that each $\Pi(k)v$ is smooth.

Check 2. What are the K-types of Ad on $\mathfrak{g}_{\mathbb{C}}$? Which V_m has the same K-types?

Answer. By the adjoint lemma, Lemma 21.8, the elements E, H' and F have K-types 2, 0, -2, each with multiplicity one. The module V_2 has the same K-types.

Check 3. Could the eigenvalues of $H = \operatorname{diag}(1, -1)$ serve as K-type labels?

Answer. No. The derivative condition (b) relates the K-action only to elements of $\mathfrak{k}_{\mathbb{C}} = \mathbb{C}H'$. Since $H \notin \mathfrak{k}_{\mathbb{C}}$, the action of H is not determined by the K-action. In the V_m example, Example 21.10, the element $H = E + F$ moves $w_j \in V(n)$, where $n = 2j - m$, into $V(n+2) \oplus V(n-2)$.

Summary

Let K be a compact subgroup of G. For a strongly continuous representation of G on a Hilbert space, the smooth K-finite vectors carry an action of $\mathfrak{g}$ and of K. The derivative condition (b) relates the two actions on $\mathfrak{k}$. The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. The proof that $\mathfrak{g}$ preserves K-finiteness uses the covariance condition.

For the circle, the K-finite vectors are the trigonometric polynomials. For $\mathrm{SL}(2, \mathbb{R})$ with $K = \mathrm{SO}(2)$, the covariance condition (c) shows that E and F shift the circle K-types by 2 and -2, respectively. The derivative condition (b) shows that H' acts on the K-type n by multiplication by n. For this connected K, the brackets $[H', E] = 2E$ and $[H', F] = -2F$ then give the same shifts.

For connected linear reductive G with maximal compact subgroup K, three results are cited. Irreducible unitary representations are admissible. In an admissible strongly continuous

Hilbert representation Π with $\Pi|_K$ unitary, K-finite vectors are smooth. An irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization, unique up to unitary equivalence.

Reading

- Knapp RT, Sections III.3–4: smooth vectors and the facts (F1)–(F3).
- Knapp RT, Chapter VIII: admissible representations and Harish-Chandra's admissibility theorem.
- P. Etingof, MIT 18.757 *Representations of Lie Groups*, lecture notes, Lectures 2 and 5–7: a second account of this material.
- Etingof, Lecture 7, Theorem 7.5 and Proposition 7.1: the unitary globalization result stated here and its uniqueness. For further study, Wallach, *Real Reductive Groups II*, develops Casselman–Wallach globalization.

Lecture 22

The principal series of $\mathrm{SL}(2,\mathbb{R})$

Objectives. *This lecture builds the principal series of $G = \mathrm{SL}(2,\mathbb{R})$. The principal series is induced from characters of the upper triangular subgroup. We realise it on functions on a circle. We prove unitarity when the parameter is purely imaginary. Finally we compute how the Lie algebra moves vectors between K-types.*

By the end of this lecture you will be able to:

- State the inducing data (ε, ν) and the role of the modular function.
- Pass between induced functions, homogeneous functions on the plane and the circle model.
- Prove that $\Pi_{\varepsilon,\nu}$ is unitary for $\nu \in i\mathbb{R}$.
- Compute the ladder of K-types and locate the reducibility points.

22.1 Inducing data

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$. We use the matrices

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Let $K = \{k_\theta\}$, $A = \{a_t\}$, $N = \{n_x\}$ and $M = \{\pm I\}$. The group M is the centre of G.

Definition 22.1. The *minimal parabolic subgroup* of G is the group $P = MAN$ of upper triangular matrices in G.

Each $p \in P$ has a unique form $p = m a_t n_x$. Its diagonal determines m and t; its upper-right entry then determines x. The subgroup N is normal in P, so $m a_t n_x \mapsto m a_t$ is a homomorphism. Note that $K \cap P = M$.

The tangent algebra of P has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Direct multiplication gives

$$\mathrm{Ad}(a_t)H = H, \qquad \mathrm{Ad}(a_t)E = e^{2t}E, \qquad \mathrm{Ad}(n_x)H = H - 2xE, \qquad \mathrm{Ad}(n_x)E = E,$$

and $\mathrm{Ad}(-I) = \mathrm{id}$. Since Ad and det are multiplicative, these formulas give $|\det \mathrm{Ad}(m a_t n_x)| = e^{2t}$. We use the modular convention of Lecture 16, namely

$$\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx.$$

With this convention $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. Therefore

$$\Delta_P(ma_tn_x) = e^{-2t}. \tag{22.1}$$

The group G itself is unimodular because it is connected with semisimple Lie algebra, by Lecture 16.

Definition 22.2. Let $\varepsilon \in \{0,1\}$ and $\nu \in \mathbb{C}$. The *inducing character* $\chi_{\varepsilon,\nu} : P \to \mathbb{C}^\times$ is

$$\chi_{\varepsilon,\nu}(ma_tn_x) = \sigma_\varepsilon(m)\, e^{\nu t}, \qquad \sigma_\varepsilon(\pm I) = (\pm 1)^\varepsilon.$$

It is a continuous character, since it factors through $P \to MA$. It is unitary exactly when $\nu \in i\mathbb{R}$.

Definition 22.3. The space $I_{\varepsilon,\nu}$ consists of the continuous $f : G \to \mathbb{C}$ with

$$f(gp) = \Delta_P(p)^{1/2}\chi_{\varepsilon,\nu}(p)^{-1} f(g) \qquad (g \in G,\ p \in P).$$

For $g \in G$, the operator $\Pi_{\varepsilon,\nu}(g)$ on $I_{\varepsilon,\nu}$ is given by $(\Pi_{\varepsilon,\nu}(g)f)(h) = f(g^{-1}h)$ for $h \in G$.

By the modular formula (22.1), the defining condition reads

$$f(g\, ma_tn_x) = \sigma_\varepsilon(m)\, e^{-(\nu+1)t} f(g). \tag{22.2}$$

Left and right translations commute, so $\Pi_{\varepsilon,\nu}(g)$ preserves $I_{\varepsilon,\nu}$. Also $\Pi(g_1g_2) = \Pi(g_1)\Pi(g_2)$. We drop subscripts when (ε,ν) is fixed.

Remark 22.4. The factor $\Delta_P^{1/2}$ gives another form of the normalisation in Lecture 17. In Lecture 17 the square root appears in the action.

Let q be the rho-function of Lecture 17. It is a positive continuous function on G with $q(gp) = \Delta_P(p)q(g)$ for $g \in G$ and $p \in P$. Let f satisfy the unitary covariance condition of Lecture 17, and set $F(g) = \sqrt{q(g)}f(g)$. By the transformation law of q, the square root moves into the covariance condition. The action becomes plain left translation. For $\nu \in i\mathbb{R}$ the resulting condition is

$$|F(gp)|^2 = \Delta_P(p)|F(g)|^2.$$

This condition allows integration over G/P. Below we give the circle integral and prove its invariance directly. Nothing else from Lecture 17 is used.

The shift $e^{-t} = \Delta_P(a_t)^{1/2}$ is often written $e^{-\rho(tH)}$. Here ρ is the linear functional on $\mathbb{R}H$ with $\rho(H) = 1$, half the eigenvalue 2 of $\mathrm{ad}\, H$ on E.

22.2 The plane model and the circle model

Let $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$ and $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}} = k_\theta \mathbf{e}_1$.

Lemma 22.5. *The map $g \mapsto g\mathbf{e}_1$ induces a homeomorphism $G/N \to \mathbb{R}^2 \setminus \{0\}$.*

Proof. If $g\mathbf{e}_1 = \mathbf{e}_1$, the first column of g is $\mathbf{e}_1$, and $\det g = 1$ forces $g \in N$. So the fibres of $g \mapsto g\mathbf{e}_1$ are the cosets gN. The map is onto, with continuous section

$$s(v) = \begin{pmatrix} v_1 & -v_2/|v|^2 \\ v_2 & v_1/|v|^2 \end{pmatrix}, \qquad s(v)\mathbf{e}_1 = v, \qquad \det s(v) = 1.$$

Thus $v \mapsto s(v)N$ is a continuous inverse. ∎

Proposition 22.6. *The rule $F(g\mathbf{e}_1) = f(g)$ is a bijection from $I_{\varepsilon,\nu}$ onto the continuous functions F on $\mathbb{R}^2 \setminus \{0\}$ satisfying*

$$F(cv) = \operatorname{sgn}(c)^{\varepsilon}\,|c|^{-(\nu+1)}F(v) \qquad (c \in \mathbb{R}^{\times}). \tag{22.3}$$

Under it, $\Pi_{\varepsilon,\nu}(g)$ becomes $F \mapsto F(g^{-1}\cdot)$.

Proof. Taking $p = n_x$ in the covariance law (22.2) shows $f(gn_x) = f(g)$. By the plane identification (Lemma 22.5), F is well defined, and $F = f \circ s$ is continuous. For $c = \pm e^t$ we have $g(\pm a_t)\mathbf{e}_1 = c\,g\mathbf{e}_1$. So the covariance law with $p = \pm a_t$ gives the homogeneity rule (22.3).

Conversely, given F, put $f(g) = F(g\mathbf{e}_1)$. Since $g\,ma_tn_x\mathbf{e}_1 = \pm e^t g\mathbf{e}_1$ for $m = \pm I$, the homogeneity rule gives the covariance law. Finally, $(\Pi(g)f)(h) = f(g^{-1}h) = F(g^{-1}h\mathbf{e}_1)$ for $h \in G$. So $\Pi(g)$ becomes $F \mapsto F(g^{-1}\cdot)$. ∎

A homogeneous function F is determined by its values on the unit circle. Put $\varphi(\theta) = F(u_\theta) = f(k_\theta)$. Then

$$F(ru_\theta) = r^{-(\nu+1)}\varphi(\theta) \quad (r > 0), \qquad \varphi(\theta+\pi) = (-1)^{\varepsilon}\varphi(\theta).$$

The second rule is the homogeneity rule (22.3) with $c = -1$. Let C_ε be the space of continuous 2π-periodic functions with this parity. For every ν, restriction to K identifies $I_{\varepsilon,\nu}$ with C_ε. This identification is the *compact picture*. In the compact picture the space C_ε does not depend on ν.

Definition 22.7. For $h \in G$, the map τ_h of $\mathbb{R}/2\pi\mathbb{Z}$ is defined by $hu_\theta = |hu_\theta|\,u_{\tau_h(\theta)}$.

Apply the plane model (Proposition 22.6) with $g^{-1}u_\theta = |g^{-1}u_\theta|\,u_{\tau_{g^{-1}}(\theta)}$. This gives the circle model:

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\,\varphi\big(\tau_{g^{-1}}(\theta)\big). \tag{22.4}$$

The parameter ν enters only through the scalar factor.

Example 22.8. For rotations $k_\alpha^{-1}u_\theta = u_{\theta-\alpha}$, so

$$(\Pi(k_\alpha)\varphi)(\theta) = \varphi(\theta-\alpha), \qquad \Pi(-I)\varphi = (-1)^{\varepsilon}\varphi.$$

The second formula uses $-I = k_\pi$. Thus the centre M acts by σ_ε for every ν. We call ε the central parity.

For the diagonal subgroup, $a_s^{-1}u_\theta = (e^{-s}\cos\theta,\ e^{s}\sin\theta)^{\mathsf{T}}$. Hence

$$r_s(\theta)^2 := |a_s^{-1}u_\theta|^2 = e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta, \qquad \tan\tau_{a_s^{-1}}(\theta) = e^{2s}\tan\theta.$$

When $\varepsilon = 0$ the constant function $e_0 = 1$ lies in C_0, and $\Pi_{0,\nu}(a_s)e_0 = r_s^{-(\nu+1)}$.

22.3 Unitarity on the imaginary axis

Let $\mathcal{H}_\varepsilon$ be the closure of C_ε in $L^2(\mathbb{R}/2\pi\mathbb{Z})$, with inner product

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \overline{\varphi(\theta)}\, \psi(\theta)\, d\theta.$$

By Fourier analysis on the circle (Lecture 10), $\mathcal{H}_\varepsilon$ is the closed span of the functions $e_n(\theta) = e^{in\theta}$ with $n \equiv \varepsilon \pmod 2$. The key step towards unitarity is a change of variables on the circle.

Lemma 22.9. *Each τ_h is an orientation-preserving diffeomorphism. Moreover $\tau_{h_1 h_2} = \tau_{h_1} \circ \tau_{h_2}$ and*

$$\tau_h'(\theta) = |hu_\theta|^{-2}.$$

Proof. Linear maps commute with positive scaling, so $h_1 h_2 u_\theta$ is a positive multiple of $h_1 u_{\tau_{h_2}(\theta)}$. This gives the composition law, and $\tau_{h^{-1}}$ inverts τ_h. Locally $\tau_h(\theta)$ is a smooth branch of the argument of the nonvanishing smooth curve hu_θ. So τ_h is smooth.

For the derivative, write $r = |hu_\theta|$ and $\tau = \tau_h(\theta)$. Differentiate $hu_\theta = r\, u_\tau$ in θ:

$$hu_\theta' = r' u_\tau + r\tau' u_\tau', \qquad u_\theta' = (-\sin\theta, \cos\theta)^{\mathsf{T}}.$$

Now compute the determinant of the pair (hu_θ, hu_θ') in two ways. By multiplicativity of the determinant it equals $\det h \cdot \det(u_\theta, u_\theta') = 1$. By the first display it also equals

$$\det(ru_\tau,\ r' u_\tau + r\tau' u_\tau') = r^2 \tau' \det(u_\tau, u_\tau') = r^2 \tau'.$$

Hence $\tau' = r^{-2} > 0$. The inverse $\tau_{h^{-1}}$ is smooth by the same argument. ∎

As a check, take $h = a_s^{-1}$ and differentiate $\tan\tau = e^{2s}\tan\theta$ from the diagonal example (Example 22.8):

$$\tau' = \frac{e^{2s}\sec^2\theta}{1 + e^{4s}\tan^2\theta} = \frac{e^{2s}}{\cos^2\theta + e^{4s}\sin^2\theta} = \frac{1}{e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta} = \frac{1}{r_s(\theta)^2}.$$

Proposition 22.10. *For $\nu \in \mathbb{C}$, $\varphi, \psi \in C_\varepsilon$ and $g \in G$,*

$$\langle \Pi_{\varepsilon,-\bar\nu}(g)\varphi,\ \Pi_{\varepsilon,\nu}(g)\psi \rangle = \langle \varphi, \psi \rangle.$$

Proof. Put $h = g^{-1}$, $r(\theta) = |hu_\theta|$ and $\theta' = \tau_h(\theta)$. Since $r > 0$, we have $\overline{r^{-(-\bar\nu+1)}} = r^{\nu-1}$. By the circle-model formula (22.4), the integrand is

$$\overline{(\Pi_{\varepsilon,-\bar\nu}(g)\varphi)(\theta)}\,(\Pi_{\varepsilon,\nu}(g)\psi)(\theta) = r^{\nu-1} r^{-\nu-1}\,\overline{\varphi(\theta')}\psi(\theta') = \overline{\varphi(\theta')}\psi(\theta')\, r(\theta)^{-2}.$$

By the Jacobian lemma (Lemma 22.9), $r(\theta)^{-2}\, d\theta = d\theta'$. As θ runs once around the circle, so does θ', in the same direction. So the integral equals $\int_0^{2\pi} \overline{\varphi(\theta')}\psi(\theta')\, d\theta'$. After division by 2π, this is $\langle \varphi, \psi \rangle$. ∎

Theorem 22.11. *Let $\nu \in i\mathbb{R}$. Each $\Pi_{\varepsilon,\nu}(g)$ extends uniquely from C_ε to a unitary operator on $\mathcal{H}_\varepsilon$. The map $g \mapsto \Pi_{\varepsilon,\nu}(g)$ is a strongly continuous unitary representation of G.*

Proof. For $\nu \in i\mathbb{R}$ we have $-\bar{\nu} = \nu$. So the pairing identity (Proposition 22.10) says that $\Pi(g)$ preserves norms on C_ε. As C_ε is dense, $\Pi(g)$ extends uniquely to an isometry of $\mathcal{H}_\varepsilon$.

The relation $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$ holds on C_ε by the plane model (Proposition 22.6). By continuity it holds on $\mathcal{H}_\varepsilon$. So $\Pi(g^{-1})$ inverts $\Pi(g)$. An invertible isometry is unitary.

For strong continuity, fix $\varphi \in C_\varepsilon$ and $g_0 \in G$. By the circle-model formula (22.4), the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is continuous. It is uniformly continuous on $U \times [0, 2\pi]$ for a compact neighbourhood U of g_0. Hence $\sup_\theta |(\Pi(g)\varphi - \Pi(g_0)\varphi)(\theta)| \to 0$ as $g \to g_0$. The L^2 norm is at most this supremum.

For general $\psi \in \mathcal{H}_\varepsilon$ and $\delta > 0$, choose $\varphi \in C_\varepsilon$ with $\|\psi - \varphi\| < \delta$. Since every $\Pi(g)$ is unitary,

$$\|\Pi(g)\psi - \Pi(g_0)\psi\| \le 2\delta + \|\Pi(g)\varphi - \Pi(g_0)\varphi\|,$$

and the right side is below 3δ for g near g_0. ∎

Example 22.12. Unitarity can fail off the imaginary axis. For example, take $\varepsilon = 0$ and $\nu \in \mathbb{R}$. By the diagonal example (Example 22.8),

$$\|\Pi_{0,\nu}(a_s)e_0\|^2 = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^{\nu+1}}, \qquad A = e^{-2s},\ B = e^{2s}.$$

For $A, B > 0$, the substitution $u = \tan\theta$ on each quarter period gives

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = 4\int_0^{\infty} \frac{du}{A + Bu^2} = \frac{2\pi}{\sqrt{AB}}.$$

Applying $-\partial_A - \partial_B$ to this identity gives

$$\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^2} = \frac{\pi(A+B)}{(AB)^{3/2}}.$$

Here $AB = 1$. For $\nu = 0$ the norm is 1, as the unitarity theorem (Theorem 22.11) predicts. For $\nu = 1$ the squared norm is $\frac{1}{2}(e^{2s} + e^{-2s}) = \cosh 2s$, which exceeds 1 for $s \neq 0$. Exercise 22.6 gives the general skew-symmetry test.

For general $\nu \in \mathbb{C}$ the substitution in the proof of the pairing identity (Proposition 22.10) still gives

$$\|\Pi_{\varepsilon,\nu}(g)\varphi\|^2 \le \max_\theta |g^{-1}u_\theta|^{-2\operatorname{Re}\nu}\,\|\varphi\|^2.$$

So each $\Pi_{\varepsilon,\nu}(g)$ is bounded, locally uniformly in g. The proof of the unitarity theorem (Theorem 22.11) then gives a strongly continuous representation on $\mathcal{H}_\varepsilon$. In its density step, 2δ is replaced by $2C\delta$, where C bounds $\|\Pi_{\varepsilon,\nu}(g)\|$ on a compact neighbourhood of g_0. By the pairing identity (Proposition 22.10), the adjoint is $\Pi_{\varepsilon,\nu}(g)^* = \Pi_{\varepsilon,-\bar{\nu}}(g^{-1})$.

22.4 K-types and the Lie algebra action

By the rotation example (Example 22.8), $\Pi(k_\alpha)e_n = e^{-in\alpha}e_n$. So the character $k_\alpha \mapsto e^{-in\alpha}$ of K occurs exactly once when $n \equiv \varepsilon \pmod 2$, and not at all otherwise. This matches the centre: $k_\pi = -I$ acts on e_n by $(-1)^n = (-1)^\varepsilon$. Let

$$V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\} \qquad \text{(finite sums)}.$$

For $\nu \in i\mathbb{R}$ this is the space of K-finite vectors of Lecture 21.

For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ put

$$d\Pi(X)\varphi = \frac{d}{ds}\Big|_{s=0} \Pi_{\varepsilon,\nu}(\exp sX)\varphi.$$

By the circle-model formula (22.4), the function $(s,\theta) \mapsto (\Pi(\exp sX)\varphi)(\theta)$ is smooth. So the difference quotients converge uniformly in θ. Hence they converge in $\mathcal{H}_\varepsilon$.

More generally, the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is jointly smooth. On compact sets in a group chart, the difference quotients for its derivatives converge uniformly in θ. Hence φ is a smooth vector in the Hilbert representation, in the sense used in Lecture 21.

For $\nu \in i\mathbb{R}$ this derivative agrees with the derivative on smooth vectors in Lecture 15. We extend $d\Pi$ complex linearly to $\mathfrak{g}_{\mathbb{C}}$.

Lemma 22.13. *For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g}$, using the dot product of $\mathbb{R}^2$,*

$$(d\Pi(X)\varphi)(\theta) = (\nu+1)\,(u_\theta \cdot Xu_\theta)\,\varphi(\theta) - \det(u_\theta, Xu_\theta)\,\varphi'(\theta).$$

Proof. Let $v(s) = \exp(-sX)u_\theta = r(s)u_{\tau(s)}$. Then $r(0)=1$, $\tau(0)=\theta$ and $v'(0) = -Xu_\theta$. As in the proof of the Jacobian lemma (Lemma 22.9), $\det(v,v') = r^2\tau'$, so $\tau'(0) = -\det(u_\theta, Xu_\theta)$. Also $(r^2)' = 2\,v\cdot v'$, so $r'(0) = -u_\theta \cdot Xu_\theta$. Now differentiate $r(s)^{-(\nu+1)}\varphi(\tau(s))$ at $s=0$. The product rule with these values of $r'(0)$ and $\tau'(0)$ gives the stated formula. ∎

We use the basis

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with $\exp(\alpha W) = k_\alpha$. We have $Wu_\theta = u_\theta'$, $Hu_\theta = (\cos\theta, -\sin\theta)^{\mathsf T}$ and $Qu_\theta = (\sin\theta, \cos\theta)^{\mathsf T}$. The derivative formula (Lemma 22.13) gives the table:

X	$u_\theta \cdot Xu_\theta$	$\det(u_\theta, Xu_\theta)$	$d\Pi(X)\varphi$
W	0	1	$-\varphi'$
H	$\cos 2\theta$	$-\sin 2\theta$	$(\nu+1)\cos 2\theta\,\varphi + \sin 2\theta\,\varphi'$
Q	$\sin 2\theta$	$\cos 2\theta$	$(\nu+1)\sin 2\theta\,\varphi - \cos 2\theta\,\varphi'$

Set $X_\pm = H \pm iQ \in \mathfrak{g}_{\mathbb{C}}$. Since $\sin 2\theta \mp i\cos 2\theta = \mp i e^{\pm 2i\theta}$,

$$d\Pi(X_\pm)\varphi = e^{\pm 2i\theta}\big((\nu+1)\varphi \mp i\varphi'\big).$$

The element W used here is the negative of the W in Lecture 21, so k_α here is $k_{-\alpha}$ there. In the notation of Lecture 21, $E = X_+/2$, $F = X_-/2$ and $H' = iW$. Thus the Fourier vector e_n has type n in that lecture's parametrisation, even though its character here is $e^{-in\alpha}$.

Proposition 22.14. *For $n \equiv \varepsilon \pmod 2$,*

$$d\Pi(W)e_n = -in\,e_n, \qquad d\Pi(X_+)e_n = (\nu+1+n)\,e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu+1-n)\,e_{n-2}.$$

Proof. Insert $e_n' = in\,e_n$ into the table row for W and into the display for $d\Pi(X_\pm)$. For example, $(\nu+1) - i(in) = \nu+1+n$. ∎

The ladder formulas also show that $d\Pi$ respects brackets, for every $\nu \in \mathbb{C}$. On e_n the eigenvalues of $d\Pi(W)$ give

$$[d\Pi(W), d\Pi(X_\pm)]e_n = \big(-i(n\pm 2) + in\big)(\nu+1\pm n)\,e_{n\pm 2} = \mp 2i\,d\Pi(X_\pm)e_n.$$

The products of the two ladder operators are

$$d\Pi(X_\mp)d\Pi(X_\pm)e_n = (\nu+1\pm n)(\nu-1\mp n)\,e_n = \big(\nu^2 - (1\pm n)^2\big)e_n.$$

Subtracting the two cases gives

$$[d\Pi(X_+), d\Pi(X_-)]e_n = \big((1+n)^2 - (1-n)^2\big)e_n = 4n\,e_n = 4i\,d\Pi(W)e_n.$$

In $\mathfrak{g}_{\mathbb{C}}$ the brackets $[W,H] = 2Q$, $[W,Q] = -2H$ and $[H,Q] = -2W$ give $[W, X_\pm] = \mp 2iX_\pm$ and $[X_+, X_-] = 4iW$ (Exercise 22.4). The elements W, X_+ and X_- span $\mathfrak{g}_{\mathbb{C}}$, and $d\Pi$ is complex linear. Hence $d\Pi([Y,Z]) = [d\Pi(Y), d\Pi(Z)]$ for all $Y, Z \in \mathfrak{g}_{\mathbb{C}}$. So $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $V_{\varepsilon,\nu}$ for every $\nu \in \mathbb{C}$.

Proposition 22.15. *A subspace $U \subseteq V_{\varepsilon,\nu}$ is stable under $\Pi(K)$ and $d\Pi(\mathfrak{g})$ exactly when $U = \operatorname{span}\{e_n : n \in S\}$ for a set S such that the following two conditions hold. If $n \in S$ and $\nu+1+n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu+1-n \neq 0$, then $n-2 \in S$. Hence the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is irreducible if and only if ν is not an integer with $\nu \equiv \varepsilon + 1 \pmod 2$.*

Proof. Let U be K-stable and let $u = \sum c_n e_n \in U$ be a finite sum. The e_n with $c_n \neq 0$ span a finite-dimensional K-stable space F. The continuous curve $\alpha \mapsto e^{in\alpha}\Pi(k_\alpha)u$ lies in the closed subspace $U \cap F$ of F. Hence

$$\frac{1}{2\pi}\int_0^{2\pi} e^{in\alpha}\,\Pi(k_\alpha)u\,d\alpha = c_n e_n$$

lies in U. Thus U is spanned by the e_n it contains. Since H and Q are linear combinations of $X_\pm$, and $d\Pi(W)$ preserves every span of the e_n, stability under $d\Pi(\mathfrak{g})$ means stability under $d\Pi(X_\pm)$. By the ladder formulas (Proposition 22.14), this is the stated property of S.

If no coefficient $\nu+1\pm n$ vanishes, every nonempty admissible S contains all $n \equiv \varepsilon$. If $\nu+1+n_0 = 0$, then $S = \{n \le n_0\}$ is admissible and proper. If $\nu+1-n_0 = 0$, use $S = \{n \ge n_0\}$. Finally, $\nu = -(n_0+1)$ or $\nu = n_0 - 1$ with $n_0 \equiv \varepsilon$ means exactly that $\nu \in \mathbb{Z}$ and $\nu \equiv \varepsilon + 1 \pmod 2$. ∎

The passage from $(\mathfrak{g}, K)$-submodules to closed invariant subspaces is Harish-Chandra's correspondence for admissible representations. We do not prove it. We cite the outcome.

Theorem 22.16 (cited). *Let $\nu \in \mathbb{C}$. The representation $\Pi_{\varepsilon,\nu}$ on $\mathcal{H}_\varepsilon$ has no closed invariant subspace other than 0 and $\mathcal{H}_\varepsilon$ exactly when $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$. In that case $V_{\varepsilon,\nu} \cong V_{\varepsilon,-\nu}$ as*

> $(\mathfrak{g}, K)$*-modules. For* $\nu \in i\mathbb{R}$*, the representations* $\Pi_{\varepsilon,\nu}$ *and* $\Pi_{\varepsilon,-\nu}$ *are unitarily equivalent, and all* $\Pi_{\varepsilon,\nu}$ *are irreducible except* $\Pi_{1,0}$*. The space* $\mathcal{H}_1$ *of* $\Pi_{1,0}$ *is the orthogonal sum of the closed spans of* $\{e_n : n > 0\}$ *and* $\{e_n : n < 0\}$*, and each summand is irreducible. See Knapp,* Representation Theory of Semisimple Groups*, §II.5, and Etingof's 18.757 notes, Lecture 19.*

The pieces at the reducibility points return in Lecture 23.

The reducible module at $(\varepsilon, \nu) = (1, 0)$

The simplest reducibility point is $(\varepsilon, \nu) = (1, 0)$. The Fourier labels are the odd integers, and the ladder formulas become

$$d\Pi(X_+)e_n = (1+n)e_{n+2}, \qquad d\Pi(X_-)e_n = (1-n)e_{n-2}.$$

At $n = -1$, the upward coefficient vanishes. At $n = 1$, the downward coefficient vanishes. Thus the space $V_{1,0}$ is the direct sum

$$V_{1,0} = \operatorname{span}\{e_1, e_3, e_5, \ldots\} \;\oplus\; \operatorname{span}\{e_{-1}, e_{-3}, e_{-5}, \ldots\}.$$

Each summand is preserved by both ladder operators and by K. Within either chain, the remaining connecting coefficients are nonzero. The weight-projection argument in the proof of the submodule criterion (Proposition 22.15) shows that each chain is algebraically irreducible.

The cited irreducibility theorem (Theorem 22.16) gives the two closed irreducible summands. Their group invariance is part of that cited theorem, beyond the algebraic calculation above. Lecture 23 identifies them as limits of discrete series and explains their norms. The full odd ladder breaks at the gap between -1 and 1.

For comparison, $(\varepsilon, \nu) = (0, 0)$ has only even labels. Neither equation $1 + n = 0$ nor $1 - n = 0$ has an even solution. Its algebraic module is therefore irreducible.

We compare the two modules through the Casimir element $\Omega = H^2 + Q^2 - W^2$ of $U(\mathfrak{g}_{\mathbb{C}})$. Exercise 22.4 shows $\Omega = X_+X_- - 2iW - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Since $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$, it extends to $U(\mathfrak{g}_{\mathbb{C}})$, and this identity holds for the operators. By the ladder formulas (Proposition 22.14), the three terms act on e_n by $\nu^2 - (1-n)^2$, $-2n$ and n^2. Hence Ω acts on $V_{\varepsilon,\nu}$ by the scalar

$$\nu^2 - (1-n)^2 - 2n + n^2 = \nu^2 - 1.$$

At $(\varepsilon, \nu) = (1, 0)$ and at $(0, 0)$ this scalar is -1. The two modules nevertheless have different central parity and different reducibility. A Casimir value alone is not a classification label.

22.5 Self-checks

1. What is $\Pi_{\varepsilon,\nu}(-I)$?
   Answer. $(-I)^{-1}u_\theta = u_{\theta+\pi}$ has length 1. So $(\Pi(-I)\varphi)(\theta) = \varphi(\theta+\pi) = (-1)^\varepsilon\varphi(\theta)$. The centre acts by σ_ε for every ν.

2. For which (ε,ν) is e_0 fixed by G?
   Answer. We need $e_0 \in C_\varepsilon$, so $\varepsilon = 0$. We also need $d\Pi(X_+)e_0 = (\nu+1)e_2 = 0$, so $\nu = -1$. Conversely, for $\nu = -1$ the factor in the circle-model formula (22.4) is 1, so $\Pi(g)e_0 = e_0$.
3. Why is $V_{1,0}$ reducible?
   Answer. Here $\nu + 1 + n = 0$ at $n = -1$ and $\nu + 1 - n = 0$ at $n = 1$. So $d\Pi(X_+)e_{-1} = 0$ and $d\Pi(X_-)e_1 = 0$. The sets $\{n \geq 1\}$ and $\{n \leq -1\}$ are admissible, and $V_{1,0}$ is the direct sum of their spans.

22.6 Summary

The principal series $\Pi_{\varepsilon,\nu}$ is induced from the character $\sigma_\varepsilon \otimes e^{\nu t}$ of $P = MAN$, shifted by $\Delta_P^{1/2}$. Restriction to K identifies every $I_{\varepsilon,\nu}$ with the continuous parity-ε functions on the circle. The group acts on these functions by the circle-model formula (22.4). The identity $\tau'_h = |hu_\theta|^{-2}$ makes the pairing of ν with $-\bar{\nu}$ invariant. Hence, for $\nu \in i\mathbb{R}$, the operators $\Pi_{\varepsilon,\nu}(g)$ extend to a strongly continuous unitary representation on $\mathcal{H}_\varepsilon$.

Each K-type occurring in $V_{\varepsilon,\nu}$ occurs once and is spanned by one e_n with $n \equiv \varepsilon \pmod 2$. The elements $X_\pm$ move n by ± 2 with coefficients $\nu + 1 \pm n$. We proved that the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is reducible exactly at integers ν of parity opposite to ε. The cited irreducibility theorem gives the same criterion for closed invariant subspaces of $\mathcal{H}_\varepsilon$.

22.7 Further reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (Princeton, 2001), §II.5 for $\mathrm{SL}(2,\mathbb{R})$, and §§VII.1–2 for induction from parabolic subgroups in general.
- P. Etingof, lecture notes for MIT 18.757, *Representations of Lie Groups*, Lectures 9 and 19.

Lecture 23

Discrete series and the rank one unitary picture

Objectives. *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules. Finally we state the full list of irreducible unitary representations.*

By the end of this lecture you will be able to:

- Prove that Π_n is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its K-types $n, n+2, \dots$ and the action of H', E, F.
- Decide which lowest weights λ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification and derive the positivity bounds behind it.

23.1 The disk model

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation Π, a vector v has *K-type k* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all θ. If such a v is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for a, b, c, d shows that φ is an isomorphism of Lie groups from G onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by γ is complex linear on all 2×2 matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}, \qquad j(h, z) = \bar\beta z + \bar\alpha.$$

Lemma 23.1. *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

(a) $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

(b) h *maps* D *into* D *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

(c) $d(h \cdot z)/dz = j(h,z)^{-2}$.

Proof. (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so h maps D into D. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So h maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar\beta_1(\alpha_2 z + \beta_2) + \bar\alpha_1(\bar\beta_2 z + \bar\alpha_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z, 1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z, 1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar\beta z + \bar\alpha) - \bar\beta(\alpha z + \beta) = 1$. ∎

Definition 23.2. Let $n \geq 2$ be an integer and $dA = dx\,dy$. For f holomorphic on D put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1-|z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic f on D with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in D with smooth boundary.

Lemma 23.3. *If $f(z) = \sum_{m \geq 0} a_m z^m$ is holomorphic on D, then*

$$\|f\|_n^2 = \sum_{m \geq 0} c_m |a_m|^2, \qquad c_m = \|z^m\|_n^2 = \frac{m!\,(n-1)!}{(n+m-1)!} = \binom{n+m-1}{m}^{-1}.$$

So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m = c_m^{-1/2} z^m$. Moreover $\|g\|_n \leq \sup_D |g|$ for bounded holomorphic g.

Proof. On $|z| = s < 1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi} |f(se^{i\theta})|^2 d\theta = 2\pi \sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi} \int_{|z| \leq r} |f|^2 (1-|z|^2)^{n-2} dA = \sum_m |a_m|^2\, (n-1) \int_0^r 2s^{2m+1} (1-s^2)^{n-2} ds.$$

The terms are nonnegative and increase with r, so we may let $r \to 1$ termwise. With $t = s^2$ and repeated integration by parts,

$$(n-1) \int_0^1 t^m (1-t)^{n-2} dt = (n-1) \frac{m!\,(n-2)!}{(n+m-1)!} = c_m.$$

Conversely, let $(a_m)_{m \geq 0}$ be complex numbers with $C = \sum_m c_m |a_m|^2 < \infty$. Then $|a_m|^2 \leq C \binom{n+m-1}{m} \leq C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on D. By the power-series import, its sum f is holomorphic on D. The formula above then gives $\|f\|_n^2 = C$. Hence $f \mapsto (c_m^{1/2} a_m)_m$ is an isometry of $\mathcal{A}_n$ onto ℓ^2. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2 \leq \sup |g|^2\, \|1\|_n^2 = \sup |g|^2$, since $c_0 = 1$. ∎

Theorem 23.4. *For $h \in \mathrm{SU}(1,1)$ and $f \in \mathcal{A}_n$ define*

$$(\Pi_n(h) f)(z) = j(h^{-1}, z)^{-n}\, f(h^{-1} \cdot z). \tag{23.5}$$

Then $g \mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of G on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.

Proof. Homomorphism. Apply the cocycle identity of Lemma 23.1(b) to h_2^{-1} and h_1^{-1}:

$$(\Pi_n(h_1)\Pi_n(h_2) f)(z) = j(h_1^{-1}, z)^{-n}\, j(h_2^{-1}, h_1^{-1} \cdot z)^{-n} f(h_2^{-1} h_1^{-1} \cdot z) = (\Pi_n(h_1 h_2) f)(z).$$

Isometry. Substitute $z = h\cdot w$. The cocycle identity with $h^{-1}h = I$ gives $j(h^{-1}, h\cdot w) = j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\ &\quad= |j(h,w)|^{2n}|f(w)|^2\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R \subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over R.

The map $w \mapsto h\cdot w$ is a bijection of D with inverse $w \mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n = \|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

Continuity. Let f, g be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n \geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood U of I. So $h \mapsto \langle g, \Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\| = 1$, so an $\varepsilon/3$ argument extends this to all f, g. Then

$$\|\Pi_n(h)f - f\|_n^2 = 2\|f\|_n^2 - 2\,\mathrm{Re}\langle f, \Pi_n(h)f\rangle_n \to 0 \qquad (h \to I).$$

Unitarity moves this to every other point. ∎

Remark 23.5. Put $\delta = \mathrm{diag}(1,-1)$. Since $\bar\gamma = \gamma\delta$, the matrix $\bar\gamma$ in place of γ gives $\Pi_n^-(g) = \Pi_n(\delta g\delta^{-1})$. Conjugation by δ preserves G, although $\delta \notin G$. Since $\delta k_\theta \delta^{-1} = k_{-\theta}$, the K-types of Π_n^- are the negatives of those of Π_n. Which one is called holomorphic depends on the Cayley matrix and the sign of the K-angle. We label them by K-types.

23.2 K-types and Lie algebra operators

Proposition 23.6. *For all θ and $m \geq 0$,*

$$\Pi_n(k_\theta)z^m = e^{i(n+2m)\theta}z^m.$$

Hence the K-types of Π_n are $n, n+2, n+4, \dots$, each with multiplicity one, and the K-finite vectors are the polynomials.

Proof. By the formula (23.2), the element $h = \varphi(k_\theta)$ has $h^{-1} = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. So $h^{-1}\cdot z = e^{2i\theta}z$ and $j(h^{-1}, z) = e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer k define

$$P_k f = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m = e_m$ if $k = n + 2m$, and $P_k e_m = 0$ otherwise. Being bounded, P_k is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of K-type k.

If v is K-finite, its orbit spans a finite-dimensional space U. Every $P_k v$ lies in U, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of v is therefore a polynomial. Conversely, every polynomial is a finite sum of K-types. ∎

Lemma 23.7. *Polynomials are smooth vectors for Π_n. For $Y = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathfrak{su}(1,1)$ and a polynomial f,*

$$d\Pi_n(Y) f = n(rz + s) f + \bigl(rz^2 + (s - p)z - q\bigr) f'. \tag{23.6}$$

Proof. By the lower bound for $|j(h, z)|$ in Lemma 23.1(a), $(h, z) \mapsto (\Pi_n(h)f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in h of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So f is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y)f$ is the pointwise t-derivative at 0 of $(\Pi_n(h_t)f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz + s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1 - tp)z - tq}{1 - t(rz + s)} = z + t\bigl(rz^2 + (s - p)z - q\bigr) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz + s) + O(t^2)$, and the product and chain rules give (23.6). ∎

Proposition 23.8. *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H') z^m &= (n + 2m) z^m, \\ d\Pi_n(E) z^m &= (n + m) z^{m+1}, \\ d\Pi_n(F) z^m &= -m\, z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F) 1 = 0. \end{aligned}$$

The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n - 2)$.

Proof. The right side of (23.6) is complex linear in (p, q, r, s). So it also computes $d\Pi_n$ on the matrices in (23.3). For H' we have $p = -1$, $s = 1$, giving $nf + 2zf'$. For E we have $r = 1$, giving $nzf + z^2 f'$. For F we have $q = 1$, giving $-f'$. Apply these to z^m. Since $[E, F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On z^m its eigenvalue is

$$(n + 2m)^2 + 2(n + 2m) - 4(n + m)(m + 1) = n^2 - 2n.$$ ∎

For a unitary representation Π and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices H_0, S and W are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators H_0 and S. It also turns the coefficient i of S into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors u and v satisfy

$$\langle d\Pi(E)u, v\rangle = -\langle u, d\Pi(F)v\rangle, \qquad \langle d\Pi(H')u, v\rangle = \langle u, d\Pi(H')v\rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for c_m in Lemma 23.3. Conversely, distinct K-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

Theorem 23.9. *Π_n is irreducible for every integer $n \geq 2$.*

Proof. Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each P_k maps V into V, since its Riemann sums do and V is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some e_m lies in V. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in V. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in V. By the operator formulas of Proposition 23.8, E sends z^m to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, F sends z^m to $-mz^{m-1}$ with $m \neq 0$. So every z^k lies in V. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on G. The family Π_n, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of e_0.

23.3 Descent conditions

Proposition 23.10. *Let Π be a strongly continuous representation of G on a Hilbert space. Let $v \neq 0$ have K-type k.*

(a) *$k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such Π has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its K-finite part.*

(b) *$\Pi(-I)v = (-1)^k v$.*

(c) *$\Pi_n(-I) = (-1)^n \mathrm{id}$. Hence Π_n factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if n is even.*

Proof. (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight K-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of K-type λ, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of K-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I, z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So Π_n factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every K-type $n+2m$ of Π_n has the parity of n. The integrality argument of Proposition 23.10(a) uses the period 2π of K in G. On the universal covering group the lift of K is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

Remark 23.11. The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure μ on D gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on D. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

23.4 Comparison and the unitary picture

Let V be a $(\mathfrak{g}, K)$-module of G (Lecture 21) on which Ω acts by a scalar c. Write v_k for a vector of K-type k. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

Example 23.12. Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In Π_n the K-types are $n, n+2, \ldots$. F kills z^0, and E is injective.

(ii) Put $d = n - 2$ and replace n by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends f to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most d. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills z^d. The K-types are $-d, \ldots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a K-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n - 1$, and with ε chosen so that its K-types have the parity of n. By Lecture 22, Exercise 22.4, Ω acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$. So the module is reducible. It has a filtration whose subquotients have K-types $k \leq -n$, $2 - n \leq k \leq n - 2$ and $k \geq n$. These subquotients have the K-types and Casimir value of Π_n^-, of the finite-dimensional module Π_{-d} with $d = n - 2$, and of Π_n.

Proposition 23.13. *Let V be an irreducible $(\mathfrak{g}, K)$-module of G. Assume each K-type has multiplicity at most one and Ω acts by c. Suppose V has a positive definite inner product satisfying (23.7) for the module action. If k and $k+2$ are K-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

Consequently:

(i) if the K-types are $\lambda, \lambda+2, \lambda+4, \ldots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;

(ii) if the K-types are $2\mathbb{Z}$, then $c < 0$;

(iii) if the K-types are $2\mathbb{Z}+1$, then $c < -1$.

Proof. By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that c is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the K-type spaces of that parity is a submodule. By irreducibility, all K-types have

the parity of k. Since E raises and F lowers K-types by 2, the span of the v_j with $j \le k$ is invariant. It is nonzero and misses v_{k+2}, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda - 2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda} v_\lambda$, which makes λ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even k and -1 over odd k. ∎

Twisting by δ from Remark 23.5 interchanges E, F, sends H' to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \ge 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \dots$.

Theorem 23.14 (Bargmann; cited). *Every irreducible unitary representation of G is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its K-types and its Casimir value c. Here Π_1 means formula (23.5) at $n = 1$ on the completion of polynomials in the norm $\sum |a_m|^2$. The unitarity and irreducibility of Π_1 are part of this cited theorem.*

Family	*K-types*	c
trivial	$\{0\}$	0
even principal series	$2\mathbb{Z}$	$c \le -1$
odd principal series	$2\mathbb{Z} + 1$	$c < -1$
complementary series	$2\mathbb{Z}$	$-1 < c < 0$
Π_n, $n \ge 1$	$n, n+2, \dots$	$n(n-2)$
Π_n^-, $n \ge 1$	$-n, -n-2, \dots$	$n(n-2)$

For $n \ge 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.

Proposition 23.13 proves the necessity of the c-ranges for these K-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its K-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z} + 1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the L^2 norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its K-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one K-types on which Ω is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of G. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

Since G is simple and noncompact, its only finite-dimensional unitary irreducible representation is trivial. Exercise 23.3 checks this directly for the finite-dimensional modules Π_{-d} of Example 23.12(ii).

23.5 Self-checks

1. For $n = 3$, compute c_1 and c_2. Check (23.7) for $u = z$ and $v = z^2$.

   Answer. $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does Π_5 factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   Answer. No, since $\Pi_5(-I) = -\mathrm{id}$. By the K-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with K-types $1, 3, 5, \dots$ and $c = -\frac{1}{2}$?

   Answer. No. By the cited Harish-Chandra facts after Theorem 23.14, the K-finite vectors of such a representation satisfy the hypotheses of Proposition 23.13. Its lowest-weight case (i) shows that the lowest K-type 1 forces $c = 1 \cdot (1-2) = -1$.

23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation Π_n on $\mathcal{A}_n$. Its K-types are $n, n+2, \dots$, each once. The operator E raises the K-type by 2 and F lowers it by 2. The Casimir element Ω acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral K-types. The representation Π_n descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even n. In an irreducible unitary module with K-types $\lambda, \lambda+2, \dots$, the lowest weight is an integer $\lambda \geq 1$, and Ω acts by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.

Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w \rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x \rangle = \langle w, x \rangle = w_0 x_0 - \vec{w} \cdot \vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x \rangle = \langle x, x \rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det \Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det \Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T \eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim \mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1, 2, 3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(t K_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

Parity $\mathrm{diag}(1,-1,-1,-1)$ and time reversal $\mathrm{diag}(-1,1,1,1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x\rangle, \qquad x_0 = \tfrac{1}{2}\operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2,\mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4,\mathbb{R})$ by

$$X(\varphi(A)x) = A\,X(x)\,A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements)**.** Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2(x_0+x_3) & (a/\bar a)(x_1 - ix_2) \\ (\bar a/a)(x_1+ix_2) & |a|^{-2}(x_0-x_3)\end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + ix_2 \mapsto e^{i\theta}(x_1+ix_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp tiX_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j}|A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W \operatorname{diag}(\lambda, \lambda^{-1}) W^*$ with W unitary and $\lambda > 0$. Put $P = W \operatorname{diag}(\lambda^{1/2}, \lambda^{-1/2}) W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^{\perp} = \operatorname{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \operatorname{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k \sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W' \sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W' \operatorname{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\operatorname{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW' \exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\operatorname{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk} X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk} J_k, \qquad [J_i, K_j] = \epsilon_{ijk} K_k, \qquad [K_i, K_j] = -\epsilon_{ijk} J_k.$$

24.3 Complexifying the Lorentz algebra

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

Proposition 24.6. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\, z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\, zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\, \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On SU(2), V_m is the spin-$m/2$ representation.

Definition 24.7. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L,j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L+1)(2j_R+1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.8. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L,j_R)}(-I) = (-1)^{2j_L+2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L+2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L,j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. We import from the Lie algebra background: these modules are irreducible and pairwise non-isomorphic, since their highest weights (m,n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m,n). This is the complexified differential of $D^{(m/2,n/2)}$, so $\Pi \cong D^{(m/2,n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon)S^n(U)S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the background gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

Only $(0,0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0,0)$. Exercise 24.6 gives the proof.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.9 (Weyl spinors). The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.8, the two representations are inequivalent. A direct check uses traces. For $A = \mathrm{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2,\mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.10 (Four-vectors). Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2,\mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.8. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.8(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0)\oplus(0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on SU(2)	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac12,0)$, $(0,\frac12)$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac12,\frac12)$	4	$V_0\oplus V_2$	yes	four-vector
$(1,\frac12)$	6	$V_1\oplus V_3$	no	–

24.6 Self-checks

(1) Compute $\varphi(\operatorname{diag}(2,\frac12))$.

(2) Find all $A\in\mathrm{SL}(2,\mathbb{C})$ with $\varphi(A)=R_3(\pi)$.

(3) Does $(\frac32,\frac12)$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to SU(2).

Answer. (1) By the diagonal formula of Example 24.4 with $a=2$, we get $x_0\pm x_3\mapsto 4^{\pm1}(x_0\pm x_3)$. Since $a/\bar a=1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t=4$, so $\cosh t=\frac{17}{8}$ and $\sinh t=\frac{15}{8}$. Thus

$$\varphi\big(\operatorname{diag}(2,\tfrac12)\big)=\begin{pmatrix}\frac{17}{8}&0&0&\frac{15}{8}\\0&1&0&0\\0&0&1&0\\\frac{15}{8}&0&0&\frac{17}{8}\end{pmatrix}.$$

Check: $17^2-15^2=64$.

(2) The element $\exp(\pi X_3)=\operatorname{diag}(-i,i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A=\pm\operatorname{diag}(-i,i)$.

(3) Here $2j_L+2j_R=4$ is even, so it descends. Its dimension is $4\cdot2=8$. The restriction is $V_2\oplus V_4$, of dimension $3+5=8$.

24.7 Summary

The action $X\mapsto AXA^*$ on Hermitian matrices preserves $\det X=\langle x,x\rangle$. It gives a two-sheeted covering $\varphi:\mathrm{SL}(2,\mathbb{C})\to\mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L,j_R), given by $S^{2j_L}(A)\otimes S^{2j_R}(\bar A)$. Exactly those with $2j_L+2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\frac12,0)$ and $(0,\frac12)$, and four-vectors are $(\frac12,\frac12)$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.

Lecture 25

Poincaré representations and little groups

Objectives. *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where x_0 is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the 2×2 Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g) \colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\, \Phi(x)\, g^*.$$

Lemma 25.1. *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of g. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

Proof. The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on x. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on h. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of g, an entry of $\Phi(x)$ and the complex conjugate of an entry of g. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of g.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, Λ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

Definition 25.2. The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a,g)(a',g') = \big(a + \Lambda(g)a',\; gg'\big). \tag{25.2}$$

Proposition 25.3. *G is a Lie group of dimension 10. Its identity is $(0,I)$ and $(a,g)^{-1} = \big(-\Lambda(g)^{-1}a,\; g^{-1}\big)$. The translations $A = \{(a,I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0,g)(a,I)(0,g)^{-1} = \big(\Lambda(g)a,\; I\big). \tag{25.3}$$

The map $(a,g) \mapsto (a,\Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0,\pm I)\}$.

Proof. Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and Λ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a,g)\big(-\Lambda(g)^{-1}a,\, g^{-1}\big) = (a-a,\; I), \qquad \big(-\Lambda(g)^{-1}a,\, g^{-1}\big)(a,g) = \big(-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\; I\big).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since g^{-1} is the adjugate of g. Hence G is a Lie group. The projection $(a,g) \mapsto g$ is a continuous homomorphism with kernel A, so A is closed and normal. Next, $(0,g)(a,I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0,g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of Λ. ∎

From now on, the Poincaré group means G. The central element $z = (0,-I)$ maps to the identity.

25.2 Momentum characters and mass shells

Let U be a strongly continuous unitary representation of G on a Hilbert space. On A it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

Lemma 25.4. *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation χ_p and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and p_0 is the energy.

A *projection-valued measure* E on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that E is *concentrated* on a Borel set S when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure E on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

The *support* of E is the complement of the union of open sets O with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set F with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0\, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2\, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of E on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all a, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. The same calculation with E, together with the uniqueness of E in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

Definition 25.5. The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

Theorem 25.6. *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, Y_κ and $\{0\}$.*

Proof. Invariants. The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1 \mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g\Phi g^* v = (g^* v)^* \Phi (g^* v)$ and g is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

Transitivity. Write $\Phi(p) = u \operatorname{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts u in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their Φ-images are $\operatorname{diag}(m, m)$, $\operatorname{diag}(2, 0)$ and $\operatorname{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \operatorname{diag}(s, s^{-1})$. Then

$$d_s \operatorname{diag}(\alpha, \beta)\, d_s^* = \operatorname{diag}(s^2 \alpha,\ s^{-2}\beta). \tag{25.5}$$

On X_m^+ we have $\mu_1, \mu_2 > 0$ and $\mu_1 \mu_2 = m^2$. Take $s^2 = \mu_1 / m$. Then $s^{-2} m = \mu_2$, so $p = \Lambda(u d_s) k_m$. On C^+ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1 / 2$. On Y_κ we have $\mu_1 > 0 > \mu_2$ and $\mu_1 \mu_2 = -\kappa^2$; take $s^2 = \mu_1 / \kappa$. Since $\Lambda(g)$ is linear, X_m^- and C^- are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by s^2 and $p_0 - p_3$ by s^{-2}. It is a boost along the x_3-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, C^+ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with r rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

25.3 Little groups and Mackey's classification

Definition 25.7. The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2, \mathbb{C}) : \Lambda(g) p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix}.$$

Proposition 25.8. *(a)* $H_{k_m} = \mathrm{SU}(2)$.

(b) $H_k = \{n(b) r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda) n(b) r(\lambda)^{-1} = n(\lambda^2 b)$.

(c) $\Lambda(r(e^{-i\theta/2}))$ *rotates the* (x_1, x_2)*-plane by the angle* $+\theta$ *and fixes* x_0 *and* x_3.

(d) $H_{\ell_\kappa} = \{g : g \sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1, 1)$, *where* $\sigma_3 = \operatorname{diag}(1, -1)$. *Also* $H_0 = \mathrm{SL}(2, \mathbb{C})$.

Proof. (a) The condition is $g(mI)g^{*} = mI$, that is, $gg^{*} = I$.

(b) Since $\Phi(k) = 2e_1e_1^{*}$, the condition is $vv^{*} = e_1e_1^{*}$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar{v}_2 \\ v_2\bar{v}_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar{\lambda} \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such g fixes k. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \bar{\lambda} & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar{\lambda}^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.8(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call N the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing k, not spacetime translations.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.8(c), this generates positive-angle rotations about x_3, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

Theorem 25.9 (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where H is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^{*}$ put $\chi_p(a) = e^{-ip(a)}$. Let H act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let H_p be the stabiliser of p. For a strongly continuous unitary representation U of G, let E be the projection-valued measure on $(\mathbb{R}^n)^{*}$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

(a) If U is irreducible, then E is concentrated on one orbit Ω.

(b) Let $p \in \Omega$ and let σ be a strongly continuous irreducible unitary representation of H_p. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\operatorname{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure E is concentrated on Ω.

(c) Every strongly continuous irreducible unitary representation whose measure E is concentrated on Ω is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations σ are unitarily equivalent.

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our G it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^{*}$ of the theorem with the vectors p used above. The remarks after the orbit theorem, Theorem 25.6, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

Corollary 25.10. *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of G with E concentrated on X_m^+ are labelled by $n \geq 0$. The label is the $\mathrm{SU}(2)$-representation V_n, of dimension $n+1$ and spin $n/2$.*

(b) Up to unitary equivalence, those with E concentrated on C^+ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.

Proof. Combine Mackey's theorem, Theorem 25.9, with the little groups computed in Proposition 25.8. In the massive case, strongly continuous irreducible unitary representations of the compact group $\mathrm{SU}(2)$ are finite-dimensional by Lecture 10. Hence they are the V_n of Lecture 8. ∎

25.4 Massless sectors: helicity and continuous spin

Proposition 25.11. *Let σ be a finite-dimensional continuous unitary representation of H_k on V. Then $\sigma(n(b)) = I$ for all b. If σ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

Proof. The operators $\sigma(n(b))$ are unitary and commute. So V is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let V_ζ be the joint eigenspace for ζ. Let S be the finite set of ζ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.8(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\, b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some λ^2. If some $\zeta \neq 0$ lay in S, the whole circle $|\zeta'| = |\zeta|$ would lie in S. This contradicts finiteness. So $S = \{0\}$, and N acts trivially.

Then σ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto $\mathrm{U}(1)$, since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If σ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real ν. Periodicity in θ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where σ is a finite-dimensional continuous irreducible unitary representation of H_k. By Proposition 25.11, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

Remark 25.12. Mackey's theorem, Theorem 25.9, also applies to $H_k = N \rtimes T$. The T-orbits on the characters of N are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the σ_ν. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of H_k that are nontrivial on N (Exercise 25.7). Inducing them to G gives the *continuous-spin* representations. Their spectral measure is concentrated on C^+, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has E concentrated on C^+. Its closed support meets C^+ and is Lorentz-invariant by covariance of E. Transitivity then puts all of C^+ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes X_m^- and C^-, where $p_0 < 0$. It also excludes Y_κ. Indeed, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of S cover Y_κ. By second countability, countably many of these translates already cover Y_κ. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of E makes every translate of S an E-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of E on that orbit in a nonzero representation.

25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let D_n be the irreducible unitary representation of $\mathrm{SU}(2)$ on V_n. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure μ_m on X_m^+ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p(mI)h_p^* = \Phi(p)$, so $\Lambda(h_p)k_m = p$. The *Wigner rotation* is

$$W(g,p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

Proposition 25.13. *(a)* $W(g,p) \in \mathrm{SU}(2)$.
(b) $W(gg',p) = W(g,p)\,W(g',\Lambda(g)^{-1}p)$.
(c) $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
(d) If μ_m is Λ-invariant, then each $U(a,g)$ is unitary.
(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.

Proof. (a) Put $q = \Lambda(g)^{-1}p$. Then h_q maps k_m to q, g maps q to p, and h_p^{-1} maps p to k_m. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With q as above, $W(g,p)W(g',q) = h_p^{-1} g h_q \, h_q^{-1} g' h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q \cdot a' = p \cdot \Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and D_n is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2 \, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary g, the matrix gh_qg^{-1} is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

By the rotation case of the Wigner rotation, Proposition 25.13(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum k_m the little group acts through D_n, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by t acts by multiplication by e^{-itp_0}. Its self-adjoint generator P_0 is multiplication by p_0, with domain

$$\mathrm{Dom}(P_0) = \{\psi \in \mathcal{H}_{m,n} : p_0\psi \in \mathcal{H}_{m,n}\}.$$

Since $p_0 \geq m$ on X_m^+, we get $P_0 \geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of μ_m.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$. Rotations fix p_0 and act orthogonally on $\vec{p}$. The boost d_s needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported ψ, the formula is jointly continuous in (a,g,p). The positive square root h_p is continuous by matrix functional calculus. For (a,g) in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all ψ.
3. *Identification with induction.* The section map is

   $$f \longmapsto \psi_f, \qquad \psi_f(p) = f(0, h_p).$$

   It carries the induced representation from $A \rtimes \mathrm{SU}(2)$, with inducing data $\chi_{k_m} \otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.9(b), shows that the model is irreducible.

Remark 25.14. The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

Self-checks

1. Is $(-1,0,0,0)$ in the orbit of k_1?
   Answer. No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of H_k on $\mathbb{C}^2$ preserve some inner product?
   Answer. No. By Proposition 25.11, every finite-dimensional continuous unitary representation of H_k is trivial on N. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   Answer. $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.13(e).

Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by Λ. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. Mackey's theorem reduces the classification to little groups. Up to unitary equivalence, irreducible unitary representations with E concentrated on X_m^+ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.

Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Induced representations act on equivariant functions. The same functions can be read as sections of a vector bundle. In the Borel–Weil setting, holomorphic sections of an associated line bundle form a finite-dimensional representation. We compute these holomorphic sections completely for the projective line. The general Borel–Weil theorem is a cited extension.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting $\mathrm{SU}(2)$ representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. The local-section theorem [C: Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26] therefore gives a smooth local section near every point of G/H. We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

The bundle $E = G \times_H V$ has this structure. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1}s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h) f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1} g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

We also need a holomorphic version of this construction. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*. A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that the local sections s_i are holomorphic. Assume the U_i cover G/H.

If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

In this case $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. In this case E is a *holomorphic* vector bundle. The local trivialisations given by the sections s_i are its *holomorphic bundle charts*. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1} s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense of the first section.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \leq k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1, z) = f_0(z)$ and $F(w, 1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k}v_1^k$,

$$\Pi_m(t_\theta)F_k = e^{i(2k-m)\theta}F_k, \qquad 0 \le k \le m.$$

The weights are $-m, -m+2, \dots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis (v_0^2, v_0v_1, v_1^2), expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\,v_0^2 + (|a|^2 - |b|^2)v_0v_1 + a\bar{b}\,v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \ge 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta}f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section. A supplementary example in the solution of Exercise 26.5 identifies $O(m)$ with $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$, where T acts on $\mathbb{C}_{-m}$ by $e^{-im\theta}$.

26.5 The general Borel–Weil theorem

Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G, so we first identify the two groups.

A faithful representation ρ of K exists [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]. Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N,\mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N,\mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra (Sepanski, Definition 7.48). By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto the *flag manifold* G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X,Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0,0)$ is the complex-linear isomorphism $(X,Y) \mapsto b(X+Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. By Sepanski, Lemma 7.49(d), each $b \in B$ is uniquely a product $t \exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$. These factors depend smoothly on b.

Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)).

In our labels, Sepanski, Definition 7.52, defines $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp \mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with $f(gb) = \chi_\lambda(b) f(g)$. Let $s_i : U_i \to G$ be holomorphic local sections of $\pi : G \to G/B$ whose domains cover G/B. By the first section, they and the holomorphic character χ_λ give holomorphic bundle charts of L_λ. A section is holomorphic in that sense exactly when its function $f : G \to \mathbb{C}$ is holomorphic. The proof is the argument of the first section, with holomorphic in place of smooth. In it, the map $q(g) = s_i(\pi(g))^{-1}g$ on $\pi^{-1}(U_i)$ is now holomorphic into B by the slice-chart property. The covariance writes $f(g)$ on $\pi^{-1}(U_i)$ as the product of $\chi_\lambda(q(g))$ and $f(s_i(\pi(g)))$, and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil; cited). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

We import the theorem from Sepanski, Theorem 7.58, for the groups G and B above. Sepanski's Definition 7.54 calls a section holomorphic when its function on G is holomorphic.

By the equivalence above, his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, and this change of index produces the dual. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2,\mathbb{C})$ (Lecture 14). Thus $G = \mathrm{SL}(2,\mathbb{C})$, and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}.

The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

The preliminary note in the solution of Exercise 26.5 compares the two section spaces. With Theorem 26.3, it follows that $F \mapsto f_F$, with $f_F(g) = F(g(1,0)^T)$ for $g \in G$, is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$. For $m < 0$, both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$. For $\mathrm{SU}(2)$ these weights coincide; for general groups they need not.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Borel–Weil extends this construction to flag manifolds.

Lecture 27

Compact orbits and geometric quantisation

Objectives. *A compact coadjoint orbit has a symplectic form. Not every such orbit supplies a representation of the chosen global group. We calculate the obstruction on a sphere. The symplectic area of the sphere divided by $2\pi\hbar$ must be integral. An equivariant lift imposes a further condition when the group is* $\mathrm{SO}(3)$. *Holomorphic sections then connect the orbit to the representations of the previous lecture.*

By the end of this lecture you will be able to:

- Compute the coadjoint-orbit form of $\mathrm{SU}(2)$ with its radius normalisation.
- Define a unitary connection locally and prove the sphere integrality condition.
- Distinguish an ordinary prequantum bundle from an equivariant one.
- Recover spin $m/2$ by holomorphic sections and keep this construction separate from shifted character formulas.

27.1 The orbit sphere and its area

Use the basis of Lecture 4,

$$X_j = -\frac{i}{2}\sigma_j, \qquad [X_1, X_2] = X_3$$

with cyclic versions of the bracket. The Pauli matrices satisfy

$$\sigma_i\sigma_j = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k.$$

Identify $\mathfrak{su}(2)^*$ with $\mathbb{R}^3$ by $x_j = \ell(X_j)$. For $u = (u_1, u_2, u_3)$ put $X_u = \sum_j u_j X_j$. Then $[X_u, X_v] = X_{u\times v}$.

The adjoint action is the rotation cover from Lecture 4. Its dual action is the same rotation action in these orthonormal coordinates. Thus its orbits are the spheres

$$S_r^2 = \{x : |x| = r\} \quad (r > 0)$$

and the origin. The infinitesimal action is $\xi_u(x) = u \times x$. Indeed, pairing it with v gives $(u \times x)\cdot v = -x \cdot (u \times v)$, which is exactly the dual derivative from Lecture 19.

Proposition 27.1. *Let $r > 0$. The orbit form at $x \in S^2_r$ is*

$$\omega_x(v, w) = \frac{x \cdot (v \times w)}{r^2}, \qquad v, w \in T_x S^2_r.$$

If Ω is the outward-oriented area form of the sphere, then $\omega = \Omega/r$. In spherical coordinates,

$$\omega = r \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2_r} \omega = 4\pi r. \tag{27.1}$$

Proof. The KKS formula of Lecture 19 requires

$$\omega_x(u \times x, v \times x) = x \cdot (u \times v).$$

At $x = re_3$, the two tangent vectors are $r(u_2, -u_1, 0)$ and $r(v_2, -v_1, 0)$. Substitution in the stated formula gives $r(u_1 v_2 - u_2 v_1)$, the required value. Rotation invariance gives the formula at every x.

The outward area form is $\Omega_x(v, w) = x \cdot (v \times w)/r$. Thus $\omega = \Omega/r$. The usual sphere parametrisation has area form $r^2 \sin\theta \, d\theta \wedge d\phi$, giving the spherical formula (27.1). Integration over $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$ gives $4\pi r$. ∎

Write $\xi_j = \xi_{e_j}$ for the standard basis vectors e_j. At the north pole, $\xi_1 = -r\partial_{x_2}$ and $\xi_2 = r\partial_{x_1}$, and $\omega(\xi_1, \xi_2) = r$. This is a convenient sign check. The origin must be treated separately: it is a point orbit, and none of the formulas dividing by r applies there.

Fix a positive constant $\hbar$. It sets the scale at which we compare symplectic area with phases. Keeping it explicit will make the integrality condition easier to track.

27.2 Connections described in local frames

A Hermitian line bundle is a complex line bundle with a smoothly varying positive inner product on its fibres. A local *unitary frame* is a section of norm one that never vanishes. In such a frame, a section is described by a complex coefficient function u.

Definition 27.2. A *unitary connection* differentiates local coefficients by

$$\nabla u = du + Au,$$

where A is an imaginary-valued 1-form. Let unitary frames e_i on U_i and e_j on U_j have connection forms A_i and A_j. If $e_j = g_{ij} e_i$, with $g_{ij} : U_i \cap U_j \to \mathrm{U}(1)$, the connection forms are required to satisfy

$$A_j = A_i + g_{ij}^{-1} dg_{ij}.$$

The *curvature* F_∇ is the imaginary-valued 2-form given on U_i by $F_\nabla = dA_i$.

Here du is the ordinary differential, evaluated on tangent vectors. A section s has coefficients u_i and u_j in the two frames. Since $e_j = g_{ij} e_i$, these coefficients satisfy $u_i = g_{ij} u_j$, or equivalently $u_j = g_{ij}^{-1} u_i$. Differentiating $u_i = g_{ij} u_j$ gives

$$du_i + A_i u_i = g_{ij}\big(du_j + (A_i + g_{ij}^{-1} dg_{ij}) u_j\big).$$

The left side is the coefficient of ∇s in the frame e_i. In the frame e_j this coefficient is $du_j + A_j u_j$, and coefficients change by the factor g_{ij}. Hence frame independence of ∇s for every section is exactly the rule for A_j in the definition.

Since g_{ij} is scalar, $d(g_{ij}^{-1} dg_{ij}) = 0$ locally. Indeed, write $g_{ij} = e^{i\alpha}$ with real α, so this form is $i\,d\alpha$. Hence $dA_i = dA_j$ and the curvature is global.

The same connection can be written in any nonvanishing frame. The change-of-frame formulas remain valid for $g_{ij} \in \mathbb{C}^\times$, with complex-valued A_i. Only unitary frames require imaginary-valued forms.

Along a path, the equation for parallel transport is $du + Au = 0$. In one frame its solution multiplies u by $\exp(-\int A)$. This is a phase because A is imaginary. Curvature measures the resulting phase around small loops.

Definition 27.3. A *prequantum line bundle* for (M, ω) at scale $\hbar$ is a Hermitian line bundle with unitary connection satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega. \tag{27.2}$$

The sign is part of our convention. A prequantum bundle is an initial geometric input. Its space of all smooth sections is usually far larger than the desired representation. A polarisation, such as holomorphicity, supplies a further selection.

We cite the general prequantisation theorem. A closed real 2-form ω on a smooth manifold admits a prequantum line bundle at scale $\hbar$ exactly when its periods divided by $2\pi\hbar$ are integers. A *period* of ω is the integral of ω over a closed oriented surface mapped smoothly into the manifold, or over an integer combination of such surfaces. Equivalently, $[\omega/(2\pi\hbar)]$ is an integral cohomology class. See Kirillov, Chapter 1, Section 2.4.

We prove the sphere case directly.

27.3 The integral sphere bundle

Put

$$m = \frac{2r}{\hbar}.$$

Take a northern chart excluding the south pole and a southern chart excluding the north pole. Define

$$A_N = -\frac{im}{2}(1 - \cos\theta)d\phi, \qquad A_S = \frac{im}{2}(1 + \cos\theta)d\phi. \tag{27.3}$$

Although $d\phi$ is singular at each pole, A_N is smooth at the north pole and A_S at the south pole. For example, in a northern complex coordinate $z = \tan(\theta/2)e^{i\phi}$,

$$A_N = \frac{m}{2}\frac{z\,d\bar{z} - \bar{z}\,dz}{1 + |z|^2}.$$

This expression is smooth at $z = 0$. The southern formula has the same property in its own coordinate.

Differentiation gives

$$dA_N = dA_S = -\frac{im}{2}\sin\theta\, d\theta \wedge d\phi = -\frac{i}{\hbar}\omega.$$

On the overlap, $A_S - A_N = im\, d\phi$. Thus A_N and A_S satisfy the change-of-frame formula if the frame transition is

$$e_S = e^{im\phi} e_N.$$

This is a single-valued function on the overlap exactly when $m \in \mathbb{Z}$. For positive radius this means m is a positive integer.

Proposition 27.4. *The sphere S^2_r, $r > 0$, admits a prequantum line bundle if and only if $2r/\hbar \in \mathbb{Z}$. For $m = 2r/\hbar$, the preceding construction supplies one.*

Proof. Sufficiency is the explicit construction. For necessity, suppose that S^2_r carries a prequantum line bundle.

Step 1 (unitary frames on the two disks). Use the two open disk charts obtained by deleting the opposite poles. On each chart, parallel transport a unit vector at its origin along radial segments. Cover each radial segment by finitely many local unitary frames. The change-of-frame law makes transport agree on overlaps. The resulting finite composition of ODE solutions depends smoothly on the endpoint and gives a smooth unitary frame. Each chart contains a neighbourhood of its closed hemisphere, so these frames are valid up to and beyond the equator. Call them e_N and e_S, with connection forms A_N and A_S.

Step 2 (the transition function and its lift). Write $e_S = g e_N$ near the equator, with $g : S^1 \to \mathrm{U}(1)$. Orient the equator by increasing ϕ, as the boundary of the northern disk.

Lift $g(e^{i\phi})$ to a real phase $\alpha(\phi)$ for $0 \le \phi \le 2\pi$, using the circle covering from Lecture 4. Since the endpoints give the same value, $\alpha(2\pi) - \alpha(0) = 2\pi k$ for an integer k.

Step 3 (Stokes). By the change-of-frame law, $A_S = A_N + g^{-1}dg$. Moreover $g^{-1}dg = i\, d\alpha$, since $g = e^{i\alpha}$ along the equator. Let D_N and D_S be the closed northern and southern hemispheres. We apply the calculus Stokes formula for a 1-form on each oriented disk. The southern disk induces the opposite orientation on the equator. Hence

$$\int_{S^2} F_\nabla = \int_{D_N} dA_N + \int_{D_S} dA_S = \int_{S^1} (A_N - A_S) = -\int_{S^1} g^{-1}dg = -2\pi i k.$$

By the prequantum condition (27.2) and the spherical formula (27.1),

$$k = \frac{i}{2\pi}\int F_\nabla = \frac{1}{2\pi\hbar}\int \omega = \frac{2r}{\hbar}.$$

Hence $2r/\hbar \in \mathbb{Z}$. ∎

Since $m > 0$, the curvature has no global primitive on the sphere. If $F = dA$ globally, Stokes' theorem would give $\int_{S^2} F = 0$. The period is nonzero because $m \neq 0$. The two local potentials are therefore essential.

27.4 Holomorphic sections and equivariance

Identify the sphere with $\mathbb{CP}^1$ by

$$z \longmapsto \frac{r}{1+|z|^2}\bigl(2\operatorname{Re} z, 2\operatorname{Im} z, 1-|z|^2\bigr).$$

The north pole is $z=0$, and $z=\tan(\theta/2)e^{i\phi}$ is the northern coordinate used above. Recall from Lecture 26 that $\mathrm{SU}(2)$ acts on lines by $g\cdot[v]=[gv]$. The identification map is equivariant for this action and the coadjoint action. Indeed, the line of $v=(1,z)$ corresponds to the traceless Hermitian matrix proportional to

$$vv^*/\|v\|^2 - I/2.$$

Conjugation by g sends this matrix to the matrix for the line of gv, and it rotates the Pauli coordinates of the matrix. With $|z|=\tan(\theta/2)$, we have $d|z| = \frac{1}{2}(1+|z|^2)\,d\theta$ and $r\sin\theta = 2r|z|/(1+|z|^2)$. Substitution in the spherical formula (27.1) gives the sphere form

$$\omega = \frac{2ir\,dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Since $i\,dz\wedge d\bar z = 2\,d(\operatorname{Re} z)\wedge d(\operatorname{Im} z)$, this form has the same positive orientation as the spherical formula (27.1).

The holomorphic frame transition of $\mathcal{O}(m)$ in Lecture 26 is $e_S^{\mathrm{hol}} = z^m e_N^{\mathrm{hol}}$. The Hermitian metric with

$$\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-m}$$

gives unitary frames $e_N = (1+|z|^2)^{m/2}e_N^{\mathrm{hol}}$ and e_S whose transition is $(z/|z|)^m = e^{im\phi}$. This is the transition of the integral sphere bundle, so the potentials (27.3) define a unitary connection on $\mathcal{O}(m)$.

The metric is $\mathrm{SU}(2)$-invariant. To see this, we describe it fibre by fibre. A point f of the fibre over a line $[v]$ is a function on that line with $f(\lambda v)=\lambda^m f(v)$ for $\lambda\in\mathbb{C}^\times$. In this model, the holomorphic sections of Lecture 26 are the homogeneous polynomials of degree m, restricted to each line. On the fibre over $[v]$ put

$$\|f\|^2 = \frac{|f(v)|^2}{\|v\|^{2m}}.$$

Replacing v by λv multiplies the numerator and the denominator by $|\lambda|^{2m}$. Hence the norm does not depend on the chosen vector v. An element $g\in\mathrm{SU}(2)$ sends f to the point gf over $[gv]$ with $(gf)(gv)=f(v)$. Since g is unitary, $\|gv\|=\|v\|$, and so $\|gf\|=\|f\|$. The northern frame e_N^{hol} takes the value 1 at $v=(1,z)$, as the polynomial v_0^m does. Since $\|(1,z)\|^2 = 1+|z|^2$, this norm agrees with the metric displayed above.

A connection is *compatible* with the holomorphic structure when its form in a holomorphic frame has no $d\bar z$ term. Apply the change-of-frame law to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-m/2}e_N$ and use the coordinate formula for A_N above. The form of this connection in the frame e_N^{hol} is then

$$A_N^{\mathrm{hol}} = A_N - \frac{m}{2}\,d\log(1+|z|^2) = -m\frac{\bar z\,dz}{1+|z|^2}.$$

Exercise 27.4 carries out this frame change in coordinates. The form A_N^{hol} has only a dz term, and its curvature is

$$dA_N^{\mathrm{hol}} = m\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2} = -\frac{i}{\hbar}\omega.$$

Thus the prequantum bundle can be taken to be $O(m)$, with its usual holomorphic structure.

For a given metric, the unitary connection compatible with the holomorphic structure is unique. Let e be a holomorphic frame and put $h = \|e\|^2$. Let A be the connection form in the frame e. The unitary frame $h^{-1/2}e$ has an imaginary-valued connection form A_u. The frame e is $h^{1/2}$ times this unitary frame, so the change-of-frame law gives $A = A_u + \frac{1}{2}d\log h$. Moreover $\bar{A}_u = -A_u$, and $\log h$ is real. Hence

$$A + \bar{A} = d\log h.$$

For a function f, write $\partial f = (\partial f/\partial z)\,dz$ and $\bar{\partial} f = (\partial f/\partial \bar{z})\,d\bar{z}$, so that $df = \partial f + \bar{\partial} f$. Suppose that A has no $d\bar{z}$ term. Then $\bar{A}$ has no dz term. Comparing the dz terms on the two sides gives

$$A = \partial \log h.$$

Thus the metric determines the compatible unitary connection. For $h = (1+|z|^2)^{-m}$ this formula gives $-m\bar{z}\,dz/(1+|z|^2)$, which is A_N^{hol} again.

When $m = 2r/\hbar$ is a positive integer, the *holomorphic quantisation* of S_r^2 is the section space

$$Q(S_r^2) = H^0(\mathbb{CP}^1, O(m)) \cong V_m.$$

By the section computation of Lecture 26, this space has dimension $m+1$ and spin $j = m/2 = r/\hbar$. An invariant positive inner product comes from compact averaging. The point orbit uses the trivial line and gives the trivial representation, without a radius denominator.

Definition 27.5. Let K be a Lie group with Lie algebra $\mathfrak{k}$, and let $M \subset \mathfrak{k}^*$ be a coadjoint orbit with its KKS form ω. An *equivariant prequantum bundle* for K is a prequantum line bundle for (M,ω) at scale $\hbar$ with a smooth K-action on its total space. The action must cover the coadjoint action on M, be linear and unitary on fibres, preserve the connection, and satisfy the *orbit normalisation*. This normalisation requires that, at each $\ell \in M$, the stabiliser K_ℓ act on the fibre over ℓ with differential

$$X \mapsto i\ell(X)/\hbar, \qquad X \in \mathrm{Lie}(K_\ell).$$

The orbit normalisation fixes the infinitesimal lift. Without it, merely asking for a bundle action would not specify its relation to the orbit functional.

At the north pole, the stabiliser in SU(2) is generated by $\exp(tX_3)$, of period 4π. In an equivariant prequantum bundle, this stabiliser acts on the fibre by the character

$$\exp(tX_3) \longmapsto e^{irt/\hbar} = e^{imt/2}.$$

This is a character exactly when $m \in \mathbb{Z}$. It is the action on the fibre of $O(m)$ from Lecture 26: $t_\theta = \exp(-2\theta X_3)$ acts there by $e^{-im\theta}$. The natural SU(2) action is linear on fibres and preserves the invariant metric. Each $g \in \mathrm{SU}(2)$ acts by a holomorphic bundle map. It sends holomorphic frames to holomorphic frames. Its holomorphic coordinate change sends forms with only a dz term to forms of the same kind. Hence g transports the connection to a unitary

connection compatible with the holomorphic structure. By uniqueness, the transported connection is the original one. Therefore the natural action supplies the required lift.

For SO(3) the same stabiliser rotation has period 2π. Its required character is single-valued only when $e^{2\pi i r/\hbar} = 1$, or m is even. Conversely, for even m the centre $-I$ acts trivially on every fibre of $O(m)$, so the action descends. Therefore

ordinary prequantum bundle	$m \in \mathbb{Z}$
SU(2)-equivariant prequantum bundle	$m \in \mathbb{Z}$
SO(3)-equivariant prequantum bundle	$m \in 2\mathbb{Z}$

for the positive-radius sphere, with $m > 0$. An ordinary integral sphere with odd m does exist. It does not quantise equivariantly for the global rotation group SO(3).

Phase changes and the odd-degree obstruction

One might try to change the SU(2) lift by phases so that $-I$ acts trivially. We show that, for this sphere and its fixed connection, every such phase change is trivial.

Throughout this subsection, an action on $O(m)$ means a smooth action that is linear and unitary on fibres, as in the definition of an equivariant prequantum bundle. Suppose two such SU(2) actions cover the same sphere action and preserve the connection. For fixed g, compare their fibre maps. Their ratio is a bundle automorphism covering the identity. It is linear and unitary on each fibre, and each fibre is a complex line. Hence the ratio is multiplication by a smooth function $a_g : S^2 \to \mathrm{U}(1)$. Since both actions preserve the connection,

$$a_g^{-1} da_g = 0.$$

The sphere is connected, so a_g is constant. Call this constant $\chi(g)$. The group-action law then implies

$$\chi(gh) = \chi(g)\chi(h).$$

Thus changing the lift while preserving the connection can only twist it by a continuous character of SU(2).

Every such character is trivial. Its differential is a real Lie algebra map from $\mathfrak{su}(2)$ to the abelian algebra $i\mathbb{R}$, so it vanishes on brackets. Since

$$[\mathfrak{su}(2), \mathfrak{su}(2)] = \mathfrak{su}(2),$$

the differential vanishes. Since the connected group SU(2) is generated by exponentials (Lecture 2), the character is identically one. Consequently a phase change of this kind cannot alter the action of the centre. Hence the natural action of Lecture 26 is the only such SU(2) action on $O(m)$ that covers the sphere action and preserves the connection.

Now suppose that SO(3) acted on $O(m)$ in this way. Composition with the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ would give such an SU(2) action on which $-I$ acts trivially. By uniqueness, that action would be the natural one, where $-I$ acts on every fibre by $(-1)^m$. Hence for odd m no SO(3) action on $O(m)$ that is linear and unitary on fibres covers the rotation action and preserves the connection. This conclusion does not use the orbit normalisation.

This phase calculation explains the relation of the stabiliser-character test to projective representations in Lecture 13: a cover acts on the bundle and on sections, while its central kernel may remain visible as a phase.

A spin-one comparison with functions on the sphere

For $r = \hbar$, the integer is $m = 2$. The quantisation is the three-dimensional space of homogeneous quadratic polynomials in (v_0, v_1)

$$F(v_0, v_1) = Av_0^2 + Bv_0v_1 + Cv_1^2.$$

There is another spin-one model in Lecture 12: the linear coordinate functions on the sphere. The two models are isomorphic as representations, although a bundle section and a scalar function are different geometric objects.

We now identify the two models explicitly. Write $F(v) = v^T Qv$ and set

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T(Q) = \epsilon^{-1}Q = \begin{pmatrix} -B/2 & -C \\ A & B/2 \end{pmatrix}.$$

This is an invertible linear map from symmetric matrices to traceless matrices. The polynomial action sends Q to $g^{-T}Qg^{-1}$. Since $g^T\epsilon g = \epsilon$ for $g \in \mathrm{SL}(2,\mathbb{C})$,

$$T(g^{-T}Qg^{-1}) = gT(Q)g^{-1}.$$

Thus T intertwines the quadratic section representation with the complexified adjoint representation.

For $Y \in \mathfrak{su}(2)_{\mathbb{C}}$, let ℓ_x be the functional with coordinates $x_j = \ell_x(X_j)$ and define the linear function $f_Y(x) = \ell_x(Y)$ on the sphere, extending ℓ_x complex-linearly. Under pullback by g^{-1},

$$f_Y(g^{-1} \cdot x) = \ell_x(\mathrm{Ad}(g)Y) = f_{\mathrm{Ad}(g)Y}(x).$$

Hence the map sending Y to f_Y is another intertwiner. The three basis elements X_j give the independent functions x_j, so this map is an isomorphism onto their span. Since $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, the map $Q \mapsto f_{T(Q)}$ identifies the two spin-one models explicitly. The normalised period is still $m = 2$, the dimension is three, and the action descends to $\mathrm{SO}(3)$.

27.5 The compact construction and the shift

The sphere illustrates the general compact picture. Let K be compact, connected, simply connected and semisimple. Let λ be a dominant integral weight in the real convention $d\chi = i\lambda$. Recall the positive Borel subalgebra fixed in Lecture 26. Our sign conventions follow that choice. Extend λ by zero on the root directions, and take the orbit through $\ell = -\hbar\lambda$. By the orbit normalisation, the stabiliser of ℓ acts on the fibre over ℓ by a character with differential $-i\lambda$.

For strictly dominant λ, the stabiliser is a maximal torus T. The orbit is K/T and the bundle is L_λ of Lecture 26. Choose its invariant complex structure from the positive Borel subalgebra. The Borel–Weil theorem of Lecture 26 then gives V_λ^*. For a weight on a chamber wall, the stabiliser K_ℓ is larger. The orbit is then the partial flag manifold K/K_ℓ. The analogous line bundle and holomorphic-section result are cited from Kirillov, Chapter 5, Sections 2–3. We do not prove that general geometric construction here.

For a different global form of K, the stabiliser character must exist for the actual stabiliser group. This is the equivariant integrality test. It reproduces the character-lattice restriction of Lecture 8. An ordinary integral period condition alone does not encode the global form.

No ρ shift was used in this construction. On the sphere, the orbit label is m, and the uncorrected section space has dimension $m+1$. The character nevertheless contains $m+1$:

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin((m+1)\theta)}{\sin\theta} \qquad (\sin\theta \neq 0).$$

This is the Weyl character formula, with its familiar $\lambda + \rho$ numerator. It does not say that the prequantum orbit above had label $m+1$. Half-form corrections are another construction and also require a convention. Optional Exercise 27.6 shows one such choice on the sphere, and nothing in this lecture depends on it.

Self-checks

1. If $r = 3\hbar/2$, find the normalised period and the quantised representation.

Answer. The period divided by $2\pi\hbar$ is $2r/\hbar = 3$. The bundle is $O(3)$ and its sections give V_3, of dimension four and spin $3/2$.

2. Does that sphere admit an SO(3)-equivariant prequantum lift with the orbit normalisation?

Answer. No. Its stabiliser character would be $e^{3it/2}$ on a rotation of period 2π. It takes the value -1 at $t = 2\pi$.

3. Why can the connection have local potentials although no global potential exists?

Answer. The difference of two local potentials is the logarithmic differential of a transition function. A nonzero curvature period prevents a single global primitive but is consistent with these local potentials.

Summary

The SU(2) orbit of radius $r > 0$ has symplectic area $4\pi r$. Prequantisation requires $m = 2r/\hbar$ to be integral. Holomorphic sections of $O(m)$ give spin $m/2$. Equivariance for SO(3) further requires even m. The ordinary period, the global equivariant lift and the holomorphic selection are separate steps.

Reading

Kirillov, *Lectures on the Orbit Method*, Chapter 1, Section 2.4, for integrality, and Chapter 5, Sections 2–3, for compact orbits, holomorphic constructions and character formulas. Sepanski, *Compact Lie Groups*, Section 7.4, supplies the Borel–Weil statement used in Lecture 26.

Lecture 28

Reconstruction and the boundary of quantum groups

Objectives. *The group, its coefficient algebra and its tensor representations encode the same global information in different forms. Differentiation connects these descriptions to the enveloping algebra. It loses information, however, unless the allowed group representations are retained. We make that connection explicit and identify the structures that a later course on quantum groups will deform.*

By the end of this lecture you will be able to:

- Construct the pairing between $U(\mathfrak{k}_{\mathbb{C}})$ and representative functions, with the correct multiplication order.
- For connected K, explain why representative functions form a restricted part of an algebraic dual.
- Recover compact and complex groups from the appropriate tensor automorphisms.
- Use the circle and the $\mathrm{SU}(2)/\mathrm{SO}(3)$ pair to identify what global data survive differentiation.

28.1 The two algebras

Let K be a compact Lie group and $\mathfrak{k}$ its real Lie algebra. As in Lecture 14, $\mathcal{R}(K)$ is the span of the coefficients

$$c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)v), \qquad \lambda \in V^*,\ v \in V,$$

of finite-dimensional continuous complex representations. Every sum of coefficients is one coefficient of a direct sum. Its Hopf structure is

$$\Delta_{\mathcal{R}} f(g,h) = f(gh), \qquad \varepsilon_{\mathcal{R}} f = f(e), \qquad S_{\mathcal{R}} f(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

These formulas and their targets were proved in Lecture 14. We use the ordinary algebraic tensor product. Thus $\Delta_{\mathcal{R}} f$ is a finite sum of tensors, not an infinite expansion.

On the other side, put $U = U(\mathfrak{k}_{\mathbb{C}})$, the universal enveloping algebra from Lie Algebras. It is generated by $\mathfrak{k}_{\mathbb{C}}$ with $XY - YX = [X,Y]$. Its standard Hopf maps are determined by

$$\Delta_U X = X \otimes 1 + 1 \otimes X, \qquad \varepsilon_U X = 0, \qquad S_U X = -X. \tag{28.1}$$

Here Δ_U and ε_U extend as algebra homomorphisms. The antipode S_U extends so that it reverses products. This extension is well defined because reversing the defining relation

sends $XY - YX - [X,Y]$ to $YX - XY + [X,Y]$, which is again zero in U. We check that Δ_U respects the defining relation:

$$[\Delta_U X, \Delta_U Y] = [X,Y] \otimes 1 + 1 \otimes [X,Y] = \Delta_U [X,Y].$$

Mixed tensor factors commute, which is why the cross terms vanish. Coassociativity and the counit identities follow by checking the generators, since the maps are algebra homomorphisms.

The antipode identities hold on X because $-X + X = 0$. They extend to products. Suppose that they hold on u and v. Then

$$\begin{aligned} \sum S_U(u_{(1)}v_{(1)})u_{(2)}v_{(2)} &= \sum S_U(v_{(1)})S_U(u_{(1)})u_{(2)}v_{(2)} \\ &= \varepsilon_U(u) \sum S_U(v_{(1)})v_{(2)} = \varepsilon_U(u)\varepsilon_U(v)1. \end{aligned}$$

The first equality uses that S_U reverses products. The second uses the identity for u, and the third uses the identity for v. The other antipode identity is proved the same way.

The two algebras have opposite kinds of commutativity. The coefficient algebra is commutative. It is cocommutative exactly when K is abelian, because coefficients separate points. The enveloping algebra is cocommutative, as the generator formulas (28.1) show, but is generally noncommutative.

The compact real form also defines an involution on U by $X^* = -X$ for real $X \in \mathfrak{k}$, extended antilinearly with $(uv)^* = v^*u^*$. For real X, Y, applying $*$ to the defining relation gives $YX - XY + [X,Y]$, which is again zero. Thus $*$ is well defined. Applying it twice fixes the generators, so it is an involution.

In every finite-dimensional unitary representation,

$$d\Pi(u^*) = d\Pi(u)^*.$$

This follows first for real generators, whose operators are skew-adjoint, then for sums and products. The involution remembers the compact real form inside its complexification.

28.2 Differentiating coefficients

For $X \in \mathfrak{k}$, let

$$(R_X f)(g) = \left.\frac{d}{dt}\right|_0 f(g \exp(tX)).$$

This is the left-invariant vector field generated by X, acting on functions. The letter R records multiplication on the right in the argument. It is distinct from the right regular group action, although differentiating that action gives this operator. By Lecture 1, $[R_X, R_Y] = R_{[X,Y]}$. Hence, by the universal property of U, these operators extend to R_u for $u \in U$.

Definition 28.1. The *coefficient pairing* is

$$\langle u, f \rangle = (R_u f)(e), \qquad u \in U,\ f \in \mathcal{R}(K).$$

The coefficient pairing is complex bilinear. Its angle brackets denote a pairing, not a Hermitian inner product. For a coefficient, direct differentiation gives

$$R_X c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)d\Pi_V(X)v), \qquad \langle u, c^V_{\lambda,v}\rangle = \lambda(d\Pi_V(u)v). \tag{28.2}$$

The second identity follows for products of generators and then by linearity. Since the pairing was defined on functions by differential operators, it does not depend on the chosen coefficient expression.

The order matters. For two real generators $X, Y \in \mathfrak{k}$,

$$\langle XY, f\rangle = \partial_t\partial_s|_0\, f(\exp(tX)\exp(sY)).$$

Indeed, for $f = c^V_{\lambda,v}$ we have $R_X R_Y f(g) = \lambda(\Pi_V(g)d\Pi_V(X)d\Pi_V(Y)v)$. Interchanging X and Y would reverse the matrix product. Subtracting the pairing of YX with f from the pairing of XY with f gives $\langle [X,Y], f\rangle$.

Proposition 28.2 (Hopf pairing identities). *Write finite coproducts as $\Delta_U u = \sum u_{(1)} \otimes u_{(2)}$ and $\Delta_{\mathcal{R}} f = \sum f_{(1)} \otimes f_{(2)}$. Then*

$$\begin{aligned}
\langle ab, f\rangle &= \sum \langle a, f_{(1)}\rangle\langle b, f_{(2)}\rangle,\\
\langle u, fh\rangle &= \sum \langle u_{(1)}, f\rangle\langle u_{(2)}, h\rangle,\\
\langle 1, f\rangle &= \varepsilon_{\mathcal{R}} f, \qquad \langle u, 1\rangle = \varepsilon_U u,\\
\langle S_U u, f\rangle &= \langle u, S_{\mathcal{R}} f\rangle.
\end{aligned}$$

Proof. Choose a basis (e_j) of V with dual (e^j). For $f = c^V_{\lambda,v}$,

$$\Delta_{\mathcal{R}} f = \sum_j c^V_{\lambda,e_j} \otimes c^V_{e^j,v}.$$

Applying the right side of the first identity gives

$$\sum_j \lambda(d\Pi(a)e_j)\, e^j(d\Pi(b)v) = \lambda(d\Pi(a)d\Pi(b)v).$$

This is $\langle ab, f\rangle$ by the coefficient formula (28.2).

For the second identity, $R_X(fh) = (R_X f)h + f(R_X h)$ is the ordinary Leibniz rule. Iterating it gives exactly the coproduct (28.1). For instance,

$$\Delta_U(XY) = XY \otimes 1 + X \otimes Y + Y \otimes X + 1 \otimes XY.$$

Thus evaluation at e proves the claim for every product and hence for every u. The identity $\langle 1, f\rangle = \varepsilon_{\mathcal{R}} f$ follows because R_1 is the identity. Applying derivatives to constants gives $\langle u, 1\rangle = \varepsilon_U u$.

Finally, write $\Pi^\vee$ for the dual representation on V^*, given by $\Pi^\vee(g)\mu = \mu \circ \Pi_V(g)^{-1}$. The function $S_{\mathcal{R}} c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)^{-1}v)$ is the coefficient of $\Pi^\vee$ with vector λ and with the functional given by evaluation at v. On V^*, a real generator acts by $d\Pi^\vee(X)\mu = -\mu \circ d\Pi(X)$. For every $u \in U$,

$$d\Pi^\vee(u)\mu = \mu \circ d\Pi(S_U u).$$

This holds for generators and extends multiplicatively because S_U reverses products. Apply the coefficient formula (28.2) to this coefficient of $\Pi^\vee$, and then to $c^V_{\lambda,v}$. Both sides of the last identity equal $\lambda(d\Pi(S_U u)v)$, which proves it. ∎

Hence the pairing makes the product of each algebra dual to the coproduct of the other. In this algebraic form, it relates the infinitesimal action to group multiplication. The proposition does not assert that one of these algebras is the entire linear dual of the other.

A noncommutative calculation

Take $K = \mathrm{SU}(2)$, so $\mathfrak{k}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. We test the order identity in its defining representation. Put

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are complexified generators. The derivative is extended complex-linearly; we are not claiming that $\exp(tE)$ belongs to $\mathrm{SU}(2)$ for real t. Let $D_{ij}(g) = g_{ij}$ be coefficients of its defining representation. Then

$$EF = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad FE = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows that $\langle EF, D_{11}\rangle = 1$, whereas $\langle FE, D_{11}\rangle = 0$. Their difference is $\langle H, D_{11}\rangle = 1$. The pairing therefore distinguishes EF from FE in U, even though its other argument belongs to a commutative function algebra.

The coefficient coproduct gives the same answer:

$$\Delta_{\mathcal{R}} D_{11} = D_{11} \otimes D_{11} + D_{12} \otimes D_{21}.$$

Pairing the first tensor factor with E and the second with F yields

$$E_{11}F_{11} + E_{12}F_{21} = 0 + 1 = 1.$$

Pairing the first factor with F and the second with E yields zero. Thus a commutative algebra can have a coproduct that encodes a noncommutative group law. Commutativity of the product does not by itself decide cocommutativity of the coproduct.

The regular actions on coefficients

Let $L_g f(x) = f(g^{-1}x)$ and $R_g f(x) = f(xg)$, as in the compact harmonic-analysis lectures. Here R_g denotes the group action. The symbols R_X and R_u denote differential operators. The pairings of the translates of a coefficient are

$$\langle u, L_g c_{\lambda,v}\rangle = \lambda\big(\Pi(g^{-1})d\Pi(u)v\big),$$

$$\langle u, R_g c_{\lambda,v}\rangle = \lambda\big(d\Pi(u)\Pi(g)v\big).$$

For the first formula, $L_g c$ is the coefficient with functional $\lambda \circ \Pi(g^{-1})$. For the second, $R_g c$ has vector $\Pi(g)v$. Inserting these into the coefficient formula (28.2) proves both identities without any operator-domain question.

Notice the positions of the group matrix and the derivative matrix. Left translation changes the dual variable. Right translation changes the vector variable. This is the same tensor order $V^* \otimes V$ that appeared in the Peter–Weyl theorem. The enveloping-algebra pairing continues that convention rather than introducing a second one.

28.3 What the dual contains

For any unital algebra A, its *finite dual* A° consists of the functionals that vanish on a two-sided ideal of finite codimension. Equivalently they are matrix coefficients of finite-dimensional A-modules. Here is the elementary equivalence.

A finite-dimensional module gives a homomorphism $A \to \mathrm{End}(V)$ with finite-codimensional kernel; every coefficient vanishes on that kernel. Conversely, if φ vanishes on such an ideal I, use the finite-dimensional regular module A/I, vector $1 + I$ and functional $\bar{\varphi}$. Then $\varphi(a) = \bar{\varphi}(a(1 + I))$.

By the coefficient formula (28.2), the pairing therefore maps $\mathcal{R}(K)$ into U°. Only the modules that integrate to the chosen compact group occur in this image. A finite-dimensional representation of $\mathfrak{k}_{\mathbb{C}}$ need not respect the periods of K or the restrictions imposed by a central quotient.

Proposition 28.3 (Injectivity for connected groups). *If K is connected, the map $\mathcal{R}(K) \to U^\circ$ from the pairing is injective.*

Proof. Suppose $\langle u, f\rangle = 0$ for every u. Since every sum of coefficients is one coefficient of a direct sum, we may write $f = c^V_{\lambda,v}$. For any finite list $X_1, \dots, X_s \in \mathfrak{k}$, expand the finite-dimensional matrix exponentials in

$$f(\exp(t_1X_1)\cdots\exp(t_sX_s)) = \lambda\big(e^{t_1 d\Pi(X_1)}\cdots e^{t_s d\Pi(X_s)}v\big).$$

These power series converge absolutely for every real or complex tuple $(t_1, \dots, t_s)$. The Taylor coefficient of $t_1^{a_1}\cdots t_s^{a_s}$ is

$$\frac{\langle X_1^{a_1}\cdots X_s^{a_s}, f\rangle}{a_1!\cdots a_s!} = 0.$$

Thus the displayed function vanishes for all real tuples. A connected Lie group is generated by its exponentials (Lecture 2), so $f = 0$ on K. ∎

For disconnected K, the same argument detects only the identity component. A coefficient may vanish on that component and be nonzero elsewhere. Thus connectedness cannot simply be removed.

Example 28.4 (The circle). Let $K = \mathrm{U}(1)$ and choose X with $\exp(tX) = e^{it}$. Then

$$U = \mathbb{C}[X], \qquad \mathcal{R}(K) = \mathbb{C}[z, z^{-1}], \qquad \langle X^k, z^n\rangle = (in)^k.$$

The image consists of finite linear combinations of the evaluation functionals $p(X) \mapsto p(in)$, $n \in \mathbb{Z}$. The finite dual is larger. Evaluation at $X = 1$ belongs to U° but is not in this image. If it were a combination supported on $n_1, \dots, n_s$, the polynomial $p(X) = \prod_j (X - in_j)$ would have zero value under that combination, but $p(1) \neq 0$.

The full linear dual is larger still. It contains the functional L given by $L(X^k) = k!$. By the equivalence above, every element of U° is a matrix coefficient of a finite-dimensional module. Suppose that X acts on that module by the operator A and that the coefficient uses the vector v and the functional λ. Then the coefficient takes the value $\lambda(A^k v)$ on X^k. Choose a norm on the module, with the corresponding operator and dual norms. This value is then bounded in absolute value by $C\|A\|^k$, where $C = \|\lambda\|\|v\|$. The sequence $k!$ has no such exponential bound. Thus L is not even in U°.

28.4 Global forms and tensor reconstruction

The groups SU(2) and SO(3) have isomorphic real Lie algebras, hence the same complex enveloping algebra after this identification. Their coefficient algebras retain different representations. Pullback through $p : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives

$$p^*\mathcal{R}(\mathrm{SO}(3)) = \{f \in \mathcal{R}(\mathrm{SU}(2)) : f(-g) = f(g)\}. \tag{28.3}$$

Indeed, a representation descends exactly when $-I$ acts trivially. On V_m it acts by $(-1)^m$. The algebraic Peter–Weyl decomposition therefore identifies the right side with precisely the coefficient blocks for even m. Odd blocks have been excluded, not annihilated by differentiation. Both coefficient algebras inject into the same finite dual, with different images.

To recover the group, keep the category of finite-dimensional continuous complex representations of K, all its intertwiners, and its forgetful functor ω. This functor sends each representation to its underlying vector space and each intertwiner to its linear map.

A *tensor automorphism* of ω is a family of maps $\theta_V \in \mathrm{GL}(V)$, one for each representation V, such that

$$\theta_W T = T\theta_V \quad \text{for every intertwiner } T : V \to W, \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

Here $\mathbb{C}$ is the trivial representation. A tensor automorphism is *unitary* if every θ_V is unitary for every invariant inner product on V. Write $\mathrm{Aut}_{\otimes}(\omega)$ for the group of tensor automorphisms and $\mathrm{Aut}^u_{\otimes}(\omega)$ for its subgroup of unitary tensor automorphisms.

Each $g \in K$ supplies the unitary tensor automorphism $\theta_V = \Pi_V(g)$. A tensor automorphism θ corresponds to the algebra homomorphism $\mathcal{R}(K) \to \mathbb{C}$ given by $c^V_{\lambda,v} \mapsto \lambda(\theta_V v)$. Unitary tensor automorphisms correspond to $*$-homomorphisms. Exercises 14.4–14.5 prove this correspondence.

Let $K_{\mathbb{C}}$ be the complexification of K introduced in Lecture 14. It is a complex Lie group, and its identity component is reductive. The reconstruction theorem is cited from Bröcker–tom Dieck, Sections III.7–8. It identifies

$$K \cong \mathrm{Aut}^u_{\otimes}(\omega), \qquad K_{\mathbb{C}} \cong \mathrm{Aut}_{\otimes}(\omega).$$

The compact group has the topology of pointwise convergence on the representation spaces. In words, every unitary tensor automorphism is $\Pi_V(g)$ for a unique $g \in K$. Every tensor automorphism is given by a unique element of $K_{\mathbb{C}}$, acting through the holomorphic extensions of the representations Π_V to $K_{\mathbb{C}}$. The unitary condition distinguishes the compact target from its complexification.

For the circle, write $\mathbb{C}_n$ for the one-dimensional representation $z \mapsto z^n$ of U(1), where $n \in \mathbb{Z}$. Every finite-dimensional representation of U(1) is a direct sum of copies of these representations. Naturality makes θ act on every copy of $\mathbb{C}_n$ by one scalar $a_n \in \mathbb{C}^\times$. The tensor rule, the isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \cong \mathbb{C}_{m+n}$ and the unit rule give $a_{m+n} = a_m a_n$ and $a_0 = 1$. Therefore $a_n = a^n$ with $a = a_1 \in \mathbb{C}^\times$. Every such choice of a defines a tensor automorphism. This tensor automorphism is unitary exactly when $|a| = 1$. This recovers $\mathbb{C}^\times$ and U(1) respectively.

For SU(2), let $V = \mathbb{C}^2$ be the defining representation. The alternating tensor $e_1 \otimes e_2 - e_2 \otimes e_1$ is invariant. Since $\theta_{V\otimes V} = \theta_V \otimes \theta_V$ and $\theta_{\mathbb{C}} = 1$, naturality with the intertwiner from $\mathbb{C}$ given by this tensor forces $\theta_V \otimes \theta_V$ to preserve it. Because $\theta_V \otimes \theta_V$ multiplies this tensor by $\det \theta_V$, we get $\det \theta_V = 1$. Every irreducible representation is a symmetric power of V, and direct sums and intertwiners determine the rest of the tensor automorphism.

Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ acts on these tensor constructions and gives a tensor automorphism, using the complexification from Lecture 14. Hence the complex target is $\mathrm{SL}(2,\mathbb{C})$. Imposing unitarity on V gives SU(2). For the even-weight category the central element $-I$ acts as the identity tensor automorphism, so reconstruction instead yields the quotient global group.

A circle quotient and its representation lattice

There is a smaller example of global information in a coefficient subalgebra. Let $\mu_N \subset \mathrm{U}(1)$ be the Nth roots of unity. The quotient is again a circle, but keep the quotient map and the identification of Lie algebras fixed. A character z^n descends through this map exactly when

$$\zeta^n = 1 \quad \text{for every } \zeta \in \mu_N, \qquad \text{or equivalently } N \mid n.$$

Thus the pulled-back coefficient algebra is

$$\mathbb{C}[z^N, z^{-N}] \subset \mathbb{C}[z, z^{-1}].$$

Under the same infinitesimal pairing, its evaluation points are $iN\mathbb{Z}$, instead of $i\mathbb{Z}$. The real Lie algebras of these two circles are isomorphic. The quotient map, however, has changed which periods a representation must respect. Rescaling an abstract Lie algebra isomorphism does not erase the specified quotient map.

The subalgebra remains closed under the Hopf operations. For its generator $w = z^N$,

$$\Delta w = w \otimes w, \qquad \varepsilon w = 1, \qquad Sw = w^{-1}, \qquad w^* = w^{-1}.$$

Reconstruction from this algebra and its unitary structure recovers the quotient circle together with its representation lattice. The same calculation with even highest weights gives the SU(2) to SO(3) distinction. In both cases, keeping a Lie algebra while changing the allowed coefficient subalgebra changes the global representation problem.

28.5 What a quantum-group course changes

The structures above give a concrete starting point for deformation. The coefficient algebra encodes the multiplication of group elements in its coproduct. The enveloping algebra encodes the infinitesimal bracket in its associative product. The pairing between them relates these two operations. A deformation can change these operations while retaining the Hopf identities.

Classical object	Product	Coproduct
$\mathcal{R}(K)$	commutative	usually not cocommutative
$U(\mathfrak{k}_{\mathbb{C}})$	usually not commutative	cocommutative

A noncommutative coefficient algebra is no longer an algebra of scalar-valued functions on an ordinary group. A noncocommutative coproduct changes how tensor products of representations carry an action. The order of tensor factors can then matter. Associativity of the tensor product of representations still follows from coassociativity.

A braiding is additional structure, not an automatic property of a Hopf algebra. In suitable quasitriangular examples on the enveloping side, an R-matrix supplies an intertwiner between the two tensor orders. The coefficient-side formulation uses the corresponding coquasitriangular structure. The future course will define these notions and prove their compatibility identities. They are not needed for any theorem here.

The $*$-structure also remains substantive. An arbitrary Hopf $*$-algebra is not automatically a compact quantum group; positivity and analytic completion require further hypotheses. Similarly, a representation category does not automatically become a fusion category. For an infinite compact Lie group there are infinitely many irreducibles. Finite fusion categories in quantum-group examples require particular constructions and assumptions.

The useful preparation is therefore the classical bookkeeping developed in this course: distinguish an algebraic coefficient space from a completion, track coproduct order, keep the real or unitary structure, and retain the global representation lattice. These are the data a deformation must replace or preserve. The theory of deformation parameters, $U_q(\mathfrak{g})$ and roots of unity begins in the later course.

Self-checks

1. For the circle, compute $\langle X^2, z^{-3} \rangle$.

Answer. It is $(-3i)^2 = -9$. The pairing is bilinear, so there is no complex conjugation in this calculation.

2. Does $U(\mathfrak{sl}(2,\mathbb{C}))$ alone decide whether V_1 (spin $1/2$) is allowed?

Answer. No. It is allowed for $\mathrm{SU}(2)$ and excluded for $\mathrm{SO}(3)$. The chosen coefficient subalgebra, or equivalently the allowed global representations, records that distinction.

3. For the circle tensor automorphism $a = 2$, what are $\theta_{\mathbb{C}_2}$ and $\theta_{\mathbb{C}_{-1}}$? Is it unitary?

Answer. They are 4 and $1/2$. It is a complex tensor automorphism, but it is not unitary and does not represent an element of the compact circle.

Summary

Differentiation pairs representative functions with the enveloping algebra. For connected K, the pairing embeds them in the finite dual. There the global group selects a specific subalgebra. Tensor reconstruction recovers K when unitary structure is kept and $K_{\mathbb{C}}$ when it is forgotten. The coefficient and enveloping Hopf structures provide the classical starting point for quantum groups.

Reading

Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Sections III.1 and III.7–8: coefficient algebras, reconstruction and complexification. The enveloping-algebra and tensor-product inputs are from the prerequisite Lie Algebras course. Lecture 14 contains the coefficient Hopf identities used here. The quantum-group paragraph is an outlook, not an additional classification theorem.